

The Elements

ΣΤΟΙΧΕΙΑ

Euclid

translated by Thomas L. Heath

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*The **Elements** (Στοιχεῖα) of Euclid of Alexandria (fl. c. 300 BC) — the most influential textbook ever written, and the foundation of deductive geometry. Across thirteen books it builds, from a handful of definitions, postulates, and common notions, the whole edifice of plane and solid geometry, proportion, and number theory. English translation by Sir Thomas L. Heath (1908), with the classical figures. Use the A / B switch to read the original Greek of Heiberg alongside.*

BOOK I. DEFINITIONS.

1. A POINT IS that which has no part.
2. A line is breadthless length.
3. The extremities of a line are points.
4. A straight line is a line which lies evenly with the points on itself.
5. A surface is that which has length and breadth only.
6. The extremities of a surface are lines.
7. A plane surface is a surface which lies evenly with the straight lines on itself.
8. A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.
9. And when the lines containing the angle are straight, the angle is called rectilineal.
10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.
11. An obtuse angle is an angle greater than a right angle.
12. An acute angle is an angle less than a right angle.

- 13.** A boundary is that which is an extremity of anything.
- 14.** A figure is that which is contained by any boundary or boundaries.
- 15.** A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another; **16.** And the point is called the centre of the circle.
- 17.** A diameter of the circle is any straight line drawn through the centre and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.
- 18.** A semicircle is the figure contained by the diameter and the circumference cut off by it. And the centre of the semicircle is the same as that of the circle.
- 19.** Rectilineal figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.
- 20.** Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.
- 21.** Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acuteangled triangle that which has its three angles acute.
- 22.** Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither

equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

23. Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

POSTULATES.

1. Let the following be postulated: To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

COMMON NOTIONS.

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

PROPOSITIONS.

Proposition 1. ON A GIVEN finite straight line to construct an equilateral triangle.

Let AB be the given finite straight line.

Thus it is required to construct an equilateral triangle on the straight line AB.

With centre A and distance AB let the circle BCD be described; {Post. 3} again, with centre B and distance BA let the circle ACE be described; {Post. 3} and from the point C, in which the circles cut one another, to the points A, B let the straight lines CA, CB be joined. {Post. 1} Now, since the point A is the centre of the circle CDB, AC is equal to AB. {Def. 15} Again, since the point B is the centre of the circle CAE, BC is equal to BA. {Def. 15} But CA was also proved equal to AB; therefore each of the straight lines CA, CB is equal to AB.

And things which are equal to the same thing are also equal to one another; {C.N. 1} therefore CA is also equal to CB.

Therefore the three straight lines CA, AB, BC are equal to one another.

Therefore the triangle ABC is equilateral; and it has been constructed on the given finite straight line AB.

(Being) what it was required to do.

Proposition 2. To place at a given point (as an extremity) a straight line equal to a given straight line.

Let A be the given point, and BC the given straight line.

Thus it is required to place at the point A (as an extremity) a straight line equal to the given straight line BC.

From the point A to the point B let the straight line AB be joined; {Post. 1} and on it let the equilateral triangle DAB be constructed. {I. 1} Let the straight lines AE, BF be produced in a straight line with DA, DB; {Post. 2} with centre B and distance BC let the circle CGH be described; {Post. 3} and again, with centre D and distance DG let the circle GKL be described. {Post. 3} Then,

since the point B is the centre of the circle CGH, BC is equal to BG.

Again, since the point D is the centre of the circle GKL, DL is equal to DG.

And in these DA is equal to DB; therefore the remainder AL is equal to the remainder BG. {C.N. 3} But BC was also proved equal to BG; therefore each of the straight lines AL, BC is equal to BG.

And things which are equal to the same thing are also equal to one another; {C.N. 1} therefore AL is also equal to BC.

Therefore at the given point A the straight line AL is placed equal to the given straight line BC.

(Being) what it was required to do.

Proposition 3. Given two unequal straight lines, to cut off from the greater a straight line equal to the less.

Let AB, C be the two given unequal straight lines, and let AB be the greater of them.

Thus it is required to cut off from AB the greater a straight line equal to C the less.

At the point A let AD be placed equal to the straight line C; {I. 2} and with centre A and distance AD let the circle DEF be described. {Post. 3} Now, since the point A is the centre of the circle DEF, AE is equal to AD. {Def. 15} But C is also equal to AD. Therefore each of the straight lines AE, C is equal to AD; so that AE is also equal to C. {C.N. 1} Therefore, given the two straight lines AB, C, from AB the greater AE has been cut off equal to C the less.

(Being) what it was required to do.

Proposition 4. If two triangles have the two sides equal to two sides respectively, and have the angles contained by the equal

straight lines equal, they will also have the base equal to the base, the triangle will be equal to the triangle, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend.

Let ABC , DEF be two triangles having the two sides AB , AC equal to the two sides DE , DF respectively, namely AB to DE and AC to DF , and the angle BAC equal to the angle EDF .

I say that the base BC is also equal to the base EF , the triangle ABC will be equal to the triangle DEF , and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend, that is, the angle ABC to the angle DEF , and the angle ACB to the angle DFE .

For, if the triangle ABC be applied to the triangle DEF , and if the point A be placed on the point D and the straight line AB on DE , then the point B will also coincide with E , because AB is equal to DE .

Again, AB coinciding with DE , the straight line AC will also coincide with DF , because the angle BAC is equal to the angle EDF ; hence the point C will also coincide with the point F , because AC is again equal to DF .

But B also coincided with E ; hence the base BC will coincide with the base EF .

{For if, when B coincides with E and C with F , the base BC does not coincide with the base EF , two straight lines will enclose a space: which is impossible.

Therefore the base BC will coincide with EF } and will be equal to it. {C.N. 4} Thus the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it.

And the remaining angles will also coincide with the remaining angles and will be equal to them, the angle ABC to the angle DEF , and the angle ACB to the angle DFE .

Therefore etc.

(Being) what it was required to prove.

Proposition 5. In isosceles triangles the angles at the base are equal to one another, and, if the equal straight lines be produced further, the angles under the base will be equal to one another.

Let ABC be an isosceles triangle having the side AB equal to the side AC; and let the straight lines BD, CE be produced further in a straight line with AB, AC. {Post. 2} I say that the angle ABC is equal to the angle ACB, and the angle CBD to the angle BCE.

Let a point F be taken at random on BD; from AE the greater let AG be cut off equal to AF the less; {I. 3} and let the straight lines FC, GB be joined. {Post. 1} Then, since AF is equal to AG and AB to AC, the two sides FA, AC are equal to the two sides GA, AB, respectively; and they contain a common angle, the angle FAG.

Therefore the base FC is equal to the base GB, and the triangle AFC is equal to the triangle AGB, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend, that is, the angle ACF to the angle ABG, and the angle AFC to the angle AGB. {I. 4} And, since the whole AF is equal to the whole AG, and in these AB is equal to AC, the remainder BF is equal to the remainder CG.

But FC was also proved equal to GB; therefore the two sides BF, FC are equal to the two sides CG, GB respectively; and the angle BFC is equal to the angle CGB, while the base BC is common to them; therefore the triangle BFC is also equal to the triangle CGB, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend; therefore the angle FBC is equal to the angle GCB, and the angle BCF to the angle CBG.

Accordingly, since the whole angle ABG was proved equal to the angle ACF, and in these the angle CBG is equal to the angle BCF, the remaining angle ABC is equal to the remaining angle ACB; and they are at the base of the triangle ABC.

But the angle FBC was also proved equal to the angle GCB; and they are under the base.

Therefore etc.

Q. E. D.

Proposition 6. If in a triangle two angles be equal to one another, the sides which subtend the equal angles will also be equal to one another.

Let ABC be a triangle having the angle ABC equal to the angle ACB; I say that the side AB is also equal to the side AC.

For, if AB is unequal to AC, one of them is greater.

Let AB be greater; and from AB the greater let DB be cut off equal to AC the less; let DC be joined.

Then, since DB is equal to AC, and BC is common, the two sides DB, BC are equal to the two sides AC, CB respectively; and the angle DBC is equal to the angle ACB; therefore the base DC is equal to the base AB, and the triangle DBC will be equal to the triangle ACB, the less to the greater: which is absurd. Therefore AB is not unequal to AC; it is therefore equal to it.

Therefore etc.

Q. E. D.

Proposition 7. Given two straight lines constructed on a straight line (from its extremities) and meeting in a point, there cannot be constructed on the same straight line (from its extremities), and on the same side of it, two other straight lines meeting in another point and equal to the former two

respectively, namely each to that which has the same extremity with it.

For, if possible, given two straight lines AC, CB constructed on the straight line AB and meeting at the point C, let two other straight lines AD, DB be constructed on the same straight line AB, on the same side of it, meeting in another point D and equal to the former two respectively, namely each to that which has the same extremity with it, so that CA is equal to DA which has the same extremity A with it, and CB to DB which has the same extremity B with it; and let CD be joined.

Then, since AC is equal to AD, the angle ACD is also equal to the angle ADC; {I. 5} therefore the angle ADC is greater than the angle DCB; therefore the angle CDB is much greater than the angle DCB.

Again, since CB is equal to DB, the angle CDB is also equal to the angle DCB.

But it was also proved much greater than it: which is impossible.

Therefore etc.

Q. E. D.

Proposition 8. If two triangles have the two sides equal to two sides respectively, and have also the base equal to the base, they will also have the angles equal which are contained by the equal straight lines.

Let ABC, DEF be two triangles having the two sides AB, AC equal to the two sides DE, DF respectively, namely AB to DE, and AC to DF; and let them have the base BC equal to the base EF; I say that the angle BAC is also equal to the angle EDF.

For, if the triangle ABC be applied to the triangle DEF, and if the point B be placed on the point E and the straight line BC on EF,

the point C will also coincide with F, because BC is equal to EF.

Then, BC coinciding with EF, Ba, ac will also coincide with ED, DF; for, if the base BC coincides with the base EF, and the sides BA, AC do not coincide with ED, DF but fall beside them as EG, GF, then, given two straight lines constructed on a straight line (from its extremities) and meeting in a point, there will have been constructed on the same straight line (from its extremities), and on the same side of it, two other straight lines meeting in another point and equal to the former two respectively, namely each to that which has the same extremity with it. But they cannot be so constructed. {I. 7} Therefore it is not possible that, if the base BC be applied to the base EF, the sides BA, AC should not coincide with ED, DF; they will therefore coincide, so that the angle BAC will also coincide with the angle EDF, and will be equal to it.

If therefore etc.

Q. E. D.

Proposition 9. To bisect a given rectilineal angle.

Let the angle BAC be the given rectilineal angle.

Thus it is required to bisect it.

Let a point D be taken at random on AB; let AE be cut off from AC equal to AD; {I. 3} let DE be joined, and on DE let the equilateral triangle DEF be constructed; let AF be joined.

I say that the angle BAC has been bisected by the straight line AF.

For, since AD is equal to AE, and AF is common, the two sides DA, AF are equal to the two sides EA, AF respectively.

And the base DF is equal to the base EF; therefore the angle DAF is equal to the angle EAF. {I. 8} Therefore the given rectilineal angle BAC has been bisected by the straight line AF.

Q. E. F.

Proposition 10. To bisect a given finite straight line.

Let AB be the given finite straight line.

Thus it is required to bisect the finite straight line AB.

Let the equilateral triangle ABC be constructed on it, {I. 1} and let the angle ACB be bisected by the straight line CD; {I. 9} I say that the straight line AB has been bisected at the point D.

For, since AC is equal to CB, and CD is common, the two sides AC, CD are equal to the two sides BC, CD respectively; and the angle ACD is equal to the angle BCD; therefore the base AD is equal to the base BD. {I. 4} Therefore the given finite straight line AB has been bisected at D.

Q. E. F.

Proposition 11. To draw a straight line at right angles to a given straight line from a given point on it.

Let AB be the given straight line, and C the given point on it.

Thus it is required to draw from the point C a straight line at right angles to the straight line AB.

Let a point D be taken at random on AC; let CE be made equal to CD; {I. 3} on DE let the equilateral triangle FDE be constructed, {I. 1} and let FC be joined; I say that the straight line FC has been drawn at right angles to the given straight line AB from C the given point on it.

For, since DC is equal to CE, and CF is common, the two sides DC, CF are equal to the two sides EC, CF respectively; and the base DF is equal to the base FE; therefore the angle DCF is equal to the angle ECF; {I. 8} and they are adjacent angles.

But, when a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is

right; {Def. 10} therefore each of the angles DCF, FCE is right.

Therefore the straight line CF has been drawn at right angles to the given straight line AB from the given point C on it.

Q. E. F.

Proposition 12. To a given infinite straight line, from a given point which is not on it, to draw a perpendicular straight line.

Let AB be the given infinite straight line, and C the given point which is not on it; thus it is required to draw to the given infinite straight line AB, from the given point C which is not on it, a perpendicular straight line.

For let a point D be taken at random on the other side of the straight line AB, and with centre C and distance CD let the circle EFG be described; {Post. 3} let the straight line EG be bisected at H, {I. 10} and let the straight lines CG, CH, CE be joined. {Post. 1} I say that CH has been drawn perpendicular to the given infinite straight line AB from the given point C which is not on it.

For, since GH is equal to HE, and HC is common, the two sides GH, HC are equal to the two sides EH, HC respectively; and the base CG is equal to the base CE; therefore the angle CHG is equal to the angle EHC. {I. 8} And they are adjacent angles.

But, when a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands. {Def. 10} Therefore CH has been drawn perpendicular to the given infinite straight line AB from the given point C which is not on it.

Q. E. F.

Proposition 13. If a straight line set up on a straight line make angles, it will make either two right angles or angles equal to two right angles.

For let any straight line AB set up on the straight line CD make the angles CBA, ABD; I say that the angles CBA, ABD are either two right angles or equal to two right angles.

Now, if the angle CBA is equal to the angle ABD, they are two right angles. {Def. 10} But, if not, let BE be drawn from the point B at right angles to CD; {I. 11} therefore the angles CBE, EBD are two right angles.

Then, since the angle CBE is equal to the two angles CBA, ABE, let the angle EBD be added to each; therefore the angles CBE, EBD are equal to the three angles CBA, ABE, EBD. {C. N. 2} Again, since the angle DBA is equal to the two angles DBE, EBA, let the angle ABC be added to each; therefore the angles DBA, ABC are equal to the three angles DBE, EBA, ABC. {C. N. 2} But the angles CBE, EBD were also proved equal to the same three angles; and things which are equal to the same thing are also equal to one another; {C. N. 1} therefore the angles CBE, EBD are also equal to the angles DBA, ABC.

But the angles CBE, EBD are two right angles; therefore the angles DBA, ABC are also equal to two right angles.

Therefore etc.

Q. E. D.

Proposition 14. If with any straight line, and at a point on it, two straight lines not lying on the same side make the adjacent angles equal to two right angles, the two straight lines will be in a straight line with one another.

For with any straight line AB, and at the point B on it, let the two straight lines BC, BD not lying on the same side make the adjacent angles ABC, ABD equal to two right angles; I say that BD is in a straight line with CB.

For, if BD is not in a straight line with BC, let BE be in a straight line with CB.

Then, since the straight line AB stands on the straight line CBE, the angles ABC, ABE are equal to two right angles. {I. 13} But the angles ABC, ABD are also equal to two right angles; therefore the angles CBA, ABE are equal to the angles CBA, ABD. {Post. 4 and C.N. 1} Let the angle CBA be subtracted from each; therefore the remaining angle ABE is equal to the remaining angle ABD, {C.N. 3} the less to the greater: which is impossible. Therefore BE is not in a straight line with CB.

Similarly we can prove that neither is any other straight line except BD.

Therefore CB is in a straight line with BD.

Therefore etc.

Q. E. D.

Proposition 15. If two straight lines cut one another, they make the vertical angles equal to one another.

For let the straight lines AB, CD cut one another at the point E; I say that the angle AEC is equal to the angle DEB, and the angle CEB to the angle AED.

For, since the straight line AE stands on the straight line CD, making the angles CEA, AED, the angles CEA, AED are equal to two right angles {I. 13} Again, since the straight line DE stands on the straight line AB, making the angles AED, DEB, the angles AED, DEB are equal to two right angles. {I. 13} But the angles CEA, AED were also proved equal to two right angles; therefore the angles CEA, AED are equal to the angles AED DEB. {Post. 4 and C. N. 1} Let the angle AED be subtracted from each; therefore the remaining angle CEA is equal to the remaining angle BED. {C. N. 3} Similarly it can be proved that the angles CEB, DEA are also equal.

Therefore etc. Q. E. D.

Porism.

{From this it is manifest that, if two straight lines cut one another, they will make the angles at the point of section equal to four right angles.

Proposition 16. In any triangle, if one of the sides be produced, the exterior angle is greater than either of the interior and opposite angles.

Let ABC be a triangle, and let one side of it BC be produced to D; I say that the exterior angle ACD is greater than either of the interior and opposite angles CBA, BAC.

Let AC be bisected at E {I. 10}, and let BE be joined and produced in a straight line to F; let EF be made equal to BE {I. 3}, let FC be joined {Post. 1}, and let AC be drawn through to G {Post. 2}.

Then, since AE is equal to EC, and BE to EF, the two sides AE, EB are equal to the two sides CE, EF respectively; and the angle AEB is equal to the angle FEC, for they are vertical angles. {I. 15} Therefore the base AB is equal to the base FC, and the triangle ABE is equal to the triangle CFE, and the remaining angles are equal to the remaining angles respectively, namely those which the equal sides subtend; {I. 4} therefore the angle BAE is equal to the angle ECF.

But the angle ECD is greater than the angle ECF; {C. N. 5} therefore the angle ACD is greater than the angle BAE.

Similarly also, if BC be bisected, the angle BCG, that is, the angle ACD {I. 15}, can be proved greater than the angle ABC as well.

Therefore etc.

Q. E. D.

Proposition 17. In any triangle two angles taken together in any manner are less than two right angles.

Let ABC be a triangle; I say that two angles of the triangle ABC taken together in any manner are less than two right angles.

For let BC be produced to D . {Post. 2} Then, since the angle ACD is an exterior angle of the triangle ABC , it is greater than the interior and opposite angle ABC . {I. 16} Let the angle ACB be added to each; therefore the angles ACD , ACB are greater than the angles ABC , BCA . But the angles ACD , ACB are equal to two right angles. {I. 13} Therefore the angles ABC , BCA are less than two right angles.

Similarly we can prove that the angles BAC , ACB are also less than two right angles, and so are the angles CAB , ABC as well.

Therefore etc.

Q. E. D.

Proposition 18. In any triangle the greater side subtends the greater angle.

For let ABC be a triangle having the side AC greater than AB ; I say that the angle ABC is also greater than the angle BCA .

For, since AC is greater than AB , let AD be made equal to AB {I. 3}, and let BD be joined.

Then, since the angle ADB is an exterior angle of the triangle BCD , it is greater than the interior and opposite angle DCB . {I. 16} But the angle ADB is equal to the angle ABD , since the side AB is equal to AD ; therefore the angle ABD is also greater than the angle ACB ; therefore the angle ABC is much greater than the angle ACB .

Therefore etc.

Q. E. D.

Proposition 19. In any triangle the greater angle is subtended by the greater side.

Let ABC be a triangle having the angle ABC greater than the angle BCA; I say that the side AC is also greater than the side AB.

For, if not, AC is either equal to AB or less.

Now AC is not equal to AB; for then the angle ABC would also have been equal to the angle ACB; {I. 5} but it is not; therefore AC is not equal to AB.

Neither is AC less than AB, for then the angle ABC would also have been less than the angle ACB; {I. 18} but it is not; therefore AC is not less than AB.

And it was proved that it is not equal either.

Therefore AC is greater than AB.

Therefore etc.

Q. E. D.

Proposition 20. In any triangle two sides taken together in any manner are greater than the remaining one.

For let ABC be a triangle; I say that in the triangle ABC two sides taken together in any manner are greater than the remaining one, namely BA, AC greater than BC, AB, BC greater than AC, BC, CA greater than AB.

For let BA be drawn through to the point D, let DA be made equal to CA, and let DC be joined.

Then, since DA is equal to AC, the angle ADC is also equal to the angle ACD; {I. 5} therefore the angle BCD is greater than the angle ADC. {C.N. 5} And, since DCB is a triangle having the angle BCD greater than the angle BDC, and the greater angle is subtended by the greater side, {I. 19} therefore DB is greater than BC.

But DA is equal to AC; therefore BA, AC are greater than BC.

Similarly we can prove that AB, BC are also greater than CA , and BC, CA than AB .

Therefore etc.

Q. E. D.

Proposition 21. If on one of the sides of a triangle, from its extremities, there be constructed two straight lines meeting within the triangle, the straight lines so constructed will be less than the remaining two sides of the triangle, but will contain a greater angle.

On BC , one of the sides of the triangle ABC , from its extremities B, C , let the two straight lines BD, DC be constructed meeting within the triangle; I say that BD, DC are less than the remaining two sides of the triangle BA, AC , but contain an angle BDC greater than the angle BAC .

For let BD be drawn through to E .

Then, since in any triangle two sides are greater than the remaining one, {I. 20} therefore, in the triangle ABE , the two sides AB, AE are greater than BE .

Let EC be added to each; therefore BA, AC are greater than BE, EC .

Again, since, in the triangle CED , the two sides CE, ED are greater than CD , let DB be added to each; therefore CE, EB are greater than CD, DB .

But BA, AC were proved greater than BE, EC ; therefore BA, AC are much greater than BD, DC .

Again, since in any triangle the exterior angle is greater than the interior and opposite angle, {I. 16} therefore, in the triangle CDE , the exterior angle BDC is greater than the angle CED .

For the same reason, moreover, in the triangle ABE also, the exterior angle CEB is greater than the angle BAC.

But the angle BDC was proved greater than the angle CEB; therefore the angle BDC is much greater than the angle BAC.

Therefore etc.

Q. E. D.

Proposition 22. Out of three straight lines, which are equal to three given straight lines, to construct a triangle: thus it is necessary that two of the straight lines taken together in any manner should be greater than the remaining one. {I. 20}

Let the three given straight lines be A, B, C, and of these let two taken together in any manner be greater than the remaining one, namely A, B greater than C, A, C greater than B, and B, C greater than A; thus it is required to construct a triangle out of straight lines equal to A, B, C.

Let there be set out a straight line DE, terminated at D but of infinite length in the direction of E, and let DF be made equal to A, FG equal to B, and GH equal to C. {I. 3} With centre F and distance FD let the circle DKL be described; again, with centre G and distance GH let the circle KLH be described; and let KF, KG be joined; I say that the triangle KFG has been constructed out of three straight lines equal to A, B, C.

For, since the point F is the centre of the circle DKL, FD is equal to FK.

But FD is equal to A; therefore KF is also equal to A.

Again, since the point G is the centre of the circle LKH, GH is equal to GK.

But GH is equal to C; therefore GK is also equal to C. And FG is also equal to B; therefore the three straight lines KF, FG, GK are equal to the three straight lines A, B, C.

Therefore out of the three straight lines KF, FG, GK, which are equal to the three given straight lines A, B, C, the triangle KFG has been constructed.

Q. E. F.

Proposition 23. On a given straight line and at a point on it to construct a rectilinear angle equal to a given rectilinear angle.

Let AB be the given straight line, A the point on it, and the angle DCE the given rectilinear angle; thus it is required to construct on the given straight line AB, and at the point A on it, a rectilinear angle equal to the given rectilinear angle DCE.

On the straight lines CD, CE respectively let the points D, E be taken at random; let DE be joined, and out of three straight lines which are equal to the three straight lines CD, DE, CE let the triangle AFG be constructed in such a way that CD is equal to AF, CE to AG, and further DE to FG.

Then, since the two sides DC, CE are equal to the two sides FA, AG respectively, and the base DE is equal to the base FG, the angle DCE is equal to the angle FAG. {I. 8} Therefore on the given straight line AB, and at the point A on it, the rectilinear angle FAG has been constructed equal to the given rectilinear angle DCE.

Q. E. F.

Proposition 24. If two triangles have the two sides equal to two sides respectively, but have the one of the angles contained by the equal straight lines greater than the other, they will also have the base greater than the base.

Let ABC, DEF be two triangles having the two sides AB, AC equal to the two sides DE, DF respectively, namely AB to DE, and AC to DF, and let the angle at A be greater than the angle at D; I say that the base BC is also greater than the base EF.

For, since the angle BAC is greater than the angle EDF, let there be constructed, on the straight line DE, and at the point D on it, the angle EDG equal to the angle BAC; {I. 23} let DG be made equal to either of the two straight lines AC, DF, and let EG, FG be joined.

Then, since AB is equal to DE, and AC to DG, the two sides BA, AC are equal to the two sides ED, DG, respectively; and the angle BAC is equal to the angle EDG; therefore the base BC is equal to the base EG. {I. 4} Again, since DF is equal to DG, the angle DGF is also equal to the angle DFG; {I. 5} therefore the angle DFG is greater than the angle EGF.

Therefore the angle EFG is much greater than the angle EGF.

And, since EFG is a triangle having the angle EFG greater than the angle EGF, and the greater angle is subtended by the greater side, {I. 19} the side EG is also greater than EF.

But EG is equal to BC.

Therefore BC is also greater than EF.

Therefore etc.

Q. E. D.

Proposition 25. If two triangles have the two sides equal to two sides respectively, but have the base greater than the base, they will also have the one of the angles contained by the equal straight lines greater than the other.

Let ABC, DEF be two triangles having the two sides AB, AC equal to the two sides DE, DF respectively, namely AB to DE, and AC to DF; and let the base BC be greater than the base EF; I say that the angle BAC is also greater than the angle EDF.

For, if not, it is either equal to it or less.

Now the angle BAC is not equal to the angle EDF; for then the base BC would also have been equal to the base EF, {I. 4} but it is not; therefore the angle BAC is not equal to the angle EDF.

Neither again is the angle BAC less than the angle EDF; for then the base BC would also have been less than the base EF, {I. 24} but it is not; therefore the angle BAC is not less than the angle EDF.

But it was proved that it is not equal either; therefore the angle BAC is greater than the angle EDF.

Therefore etc.

Q. E. D.

Proposition 26. If two triangles have the two angles equal to two angles respectively, and one side equal to one side, namely, either the side adjoining the equal angles, or that subtending one of the equal angles, they will also have the remaining sides equal to the remaining sides and the remaining angle to the remaining angle.

Let ABC, DEF be two triangles having the two angles ABC, BCA equal to the two angles DEF, EFD respectively, namely the angle ABC to the angle DEF, and the angle BCA to the angle EFD; and let them also have one side equal to one side, first that adjoining the equal angles, namely BC to EF; I say that they will also have the remaining sides equal to the remaining sides respectively, namely AB to DE and AC to DF, and the remaining angle to the remaining angle, namely the angle BAC to the angle EDF.

For, if AB is unequal to DE, one of them is greater.

Let AB be greater, and let BG be made equal to DE; and let GC be joined.

Then, since BG is equal to DE, and BC to EF, the two sides GB, BC are equal to the two sides DE, EF respectively; and the angle GBC

is equal to the angle DEF; therefore the base GC is equal to the base DF, and the triangle GBC is equal to the triangle DEF, and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend; {I. 4} therefore the angle GCB is equal to the angle DFE.

But the angle DFE is by hypothesis equal to the angle BCA; therefore the angle BCG is equal to the angle BCA, the less to the greater: which is impossible. Therefore AB is not unequal to DE, and is therefore equal to it.

But BC is also equal to EF; therefore the two sides AB, BC are equal to the two sides DE, EF respectively, and the angle ABC is equal to the angle DEF; therefore the base AC is equal to the base DF, and the remaining angle BAC is equal to the remaining angle EDF. {I. 4} Again, let sides subtending equal angles be equal, as AB to DE; I say again that the remaining sides will be equal to the remaining sides, namely AC to DF and BC to EF, and further the remaining angle BAC is equal to the remaining angle EDF.

For, if BC is unequal to EF, one of them is greater.

Let BC be greater, if possible, and let BH be made equal to EF; let AH be joined.

Then, since BH is equal to EF, and AB to DE, the two sides AB, BH are equal to the two sides DE, EF respectively, and they contain equal angles; therefore the base AH is equal to the base DF, and the triangle ABH is equal to the triangle DEF, and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend; {I. 4} therefore the angle BHA is equal to the angle EFD.

But the angle EFD is equal to the angle BCA; therefore, in the triangle AHC, the exterior angle BHA is equal to the interior and opposite angle BCA: which is impossible. {I. 16} Therefore BC is

not unequal to EF, and is therefore equal to it.

But AB is also equal to DE; therefore the two sides AB, BC are equal to the two sides DE, EF respectively, and they contain equal angles; therefore the base AC is equal to the base DF, the triangle ABC equal to the triangle DEF, and the remaining angle BAC equal to the remaining angle EDF. {I. 4} Therefore etc.

Q. E. D.

Proposition 27. If a straight line falling on two straight lines make the alternate angles equal to one another, the straight lines will be parallel to one another.

For let the straight line EF falling on the two straight lines AB, CD make the alternate angles AEF, EFD equal to one another; I say that AB is parallel to CD.

For, if not, AB, CD when produced will meet either in the direction of B, D or towards A, C.

Let them be produced and meet, in the direction of B, D, at G.

Then, in the triangle GEF, the exterior angle AEF is equal to the interior and opposite angle EFG: which is impossible. {I. 16} Therefore AB, CD when produced will not meet in the direction of B, D.

Similarly it can be proved that neither will they meet towards A, C.

But straight lines which do not meet in either direction are parallel; {Def. 23} therefore AB is parallel to CD.

Therefore etc.

Q. E. D.

Proposition 28. If a straight line falling on two straight lines make the exterior angle equal to the interior and opposite angle on the same side, or the interior angles on the same side equal

to two right angles, the straight lines will be parallel to one another.

For let the straight line EF falling on the two straight lines AB, CD make the exterior angle EGB equal to the interior and opposite angle GHD, or the interior angles on the same side, namely BGH, GHD, equal to two right angles; I say that AB is parallel to CD.

For, since the angle EGB is equal to the angle GHD, while the angle EGB is equal to the angle AGH, {I. 15} the angle AGH is also equal to the angle GHD; and they are alternate; therefore AB is parallel to CD. {I. 27} Again, since the angles BGH, GHD are equal to two right angles, and the angles AGH, BGH are also equal to two right angles, {I. 13} the angles AGH, BGH are equal to the angles BGH, GHD.

Let the angle BGH be subtracted from each; therefore the remaining angle AGH is equal to the remaining angle GHD; and they are alternate; therefore AB is parallel to CD. {I. 27} Therefore etc.

Q. E. D.

Proposition 29. A straight line falling on parallel straight lines makes the alternate angles equal to one another, the exterior angle equal to the interior and opposite angle, and the interior angles on the same side equal to two right angles.

For let the straight line EF fall on the parallel straight lines AB, CD; I say that it makes the alternate angles AGH, GHD equal, the exterior angle EGB equal to the interior and opposite angle GHD, and the interior angles on the same side, namely BGH, GHD, equal to two right angles.

For, if the angle AGH is unequal to the angle GHD, one of them is greater.

Let the angle AGH be greater.

Let the angle BGH be added to each; therefore the angles AGH, BGH are greater than the angles BGH, GHD.

But the angles AGH, BGH are equal to two right angles; {I. 13} therefore the angles BGH, GHD are less than two right angles.

But straight lines produced indefinitely from angles less than two right angles meet; {Post. 5} therefore AB, CD, if produced indefinitely, will meet; but they do not meet, because they are by hypothesis parallel.

Therefore the angle AGH is not unequal to the angle GHD, and is therefore equal to it.

Again, the angle AGH is equal to the angle EGB; {I. 15} therefore the angle EGB is also equal to the angle GHD. {C.N. 1} Let the angle BGH be added to each; therefore the angles EGB, BGH are equal to the angles BGH, GHD. {C.N. 2} But the angles EGB, BGH are equal to two right angles; {I. 13} therefore the angles BGH, GHD are also equal to two right angles.

Therefore etc.

Q. E. D.

Proposition 30. Straight lines parallel to the same straight line are also parallel to one another.

Let each of the straight lines AB, CD be parallel to EF; I say that AB is also parallel to CD.

For let the straight line GK fall upon them; Then, since the straight line GK has fallen on the parallel straight lines AB, EF, the angle AGK is equal to the angle GHF. {I. 29} Again, since the straight line GK has fallen on the parallel straight lines EF, CD, the angle GHF is equal to the angle GKD. {I. 29} But the angle AGK was also proved equal to the angle GHF; therefore the angle AGK is also equal to the angle GKD; {C.N. 1} and they are alternate.

Therefore AB is parallel to CD.

Q. E. D.

Proposition 31. Through a given point to draw a straight line parallel to a given straight line.

Let A be the given point, and BC the given straight line; thus it is required to draw through the point A a straight line parallel to the straight line BC.

Let a point D be taken at random on BC, and let AD be joined; on the straight line DA, and at the point A on it, let the angle DAE be constructed equal to the angle ADC {I. 23}; and let the straight line AF be produced in a straight line with EA.

Then, since the straight line AD falling on the two straight lines BC, EF has made the alternate angles EAD, ADC equal to one another, therefore EAF is parallel to BC. {I. 27} Therefore through the given point A the straight line EAF has been drawn parallel to the given straight line BC.

Q. E. F.

Proposition 32. In any triangle, if one of the sides be produced, the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.

Let ABC be a triangle, and let one side of it BC be produced to D; I say that the exterior angle ACD is equal to the two interior and opposite angles CAB, ABC, and the three interior angles of the triangle ABC, BCA, CAB are equal to two right angles.

For let CE be drawn through the point C parallel to the straight line AB. {I. 31} Then, since AB is parallel to CE, and AC has fallen upon them, the alternate angles BAC, ACE are equal to one another. {I. 29} Again, since AB is parallel to CE, and the straight line BD has fallen upon them, the exterior angle ECD is equal to

the interior and opposite angle ABC. {I. 29} But the angle ACE was also proved equal to the angle BAC; therefore the whole angle ACD is equal to the two interior and opposite angles BAC, ABC.

Let the angle ACB be added to each; therefore the angles ACD, ACB are equal to the three angles ABC, BCA, CAB.

But the angles ACD, ACB are equal to two right angles; {I. 13} therefore the angles ABC, BCA, CAB are also equal to two right angles.

Therefore etc.

Q. E. D.

Proposition 33. The straight lines joining equal and parallel straight lines (at the extremities which are) in the same directions (respectively) are themselves also equal and parallel.

Let AB, CD be equal and parallel, and let the straight lines AC, BD join them (at the extremities which are) in the same directions (respectively); I say that AC, BD are also equal and parallel.

Let BC be joined.

Then, since AB is parallel to CD, and BC has fallen upon them, the alternate angles ABC, BCD are equal to one another. {I. 29} And, since AB is equal to CD, and BC is common, the two sides AB, BC are equal to the two sides DC, CB; and the angle ABC is equal to the angle BCD; therefore the base AC is equal to the base BD, and the triangle ABC is equal to the triangle DCB, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend; {I. 4} therefore the angle ACB is equal to the angle CBD.

And, since the straight line BC falling on the two straight lines AC, BD has made the alternate angles equal to one another, AC is parallel to BD. {I. 27} And it was also proved equal to it.

Therefore etc.

Q. E. D.

Proposition 34. In parallelogrammic areas the opposite sides and angles are equal to one another, and the diameter bisects the areas.

Let ACDB be a parallelogrammic area, and BC its diameter; I say that the opposite sides and angles of the parallelogram ACDB are equal to one another, and the diameter BC bisects it.

For, since AB is parallel to CD, and the straight line BC has fallen upon them, the alternate angles ABC, BCD are equal to one another. {I. 29} Again, since AC is parallel to BD, and BC has fallen upon them, the alternate angles ACB, CBD are equal to one another. {I. 29} Therefore ABC, DCB are two triangles having the two angles ABC, BCA equal to the two angles DCB, CBD respectively, and one side equal to one side, namely that adjoining the equal angles and common to both of them, BC; therefore they will also have the remaining sides equal to the remaining sides respectively, and the remaining angle to the remaining angle; {I. 26} therefore the side AB is equal to CD, and AC to BD, and further the angle BAC is equal to the angle CDB.

And, since the angle ABC is equal to the angle BCD, and the angle CBD to the angle ACB, the whole angle ABD is equal to the whole angle ACD. {C.N. 2} And the angle BAC was also proved equal to the angle CDB.

Therefore in parallelogrammic areas the opposite sides and angles are equal to one another.

I say, next, that the diameter also bisects the areas.

For, since AB is equal to CD, and BC is common, the two sides AB, BC are equal to the two sides DC, CB respectively; and the angle ABC is equal to the angle BCD; therefore the base AC is also equal to DB, and the triangle ABC is equal to the triangle

DCB. {I. 4} Therefore the diameter BC bisects the parallelogram ACDB.

Q. E. D.

Proposition 35. Parallelograms which are on the same base and in the same parallels are equal to one another.

Let ABCD, EBCF be parallelograms on the same base BC and in the same parallels AF, BC; I say that ABCD is equal to the parallelogram EBCF.

For, since ABCD is a parallelogram, AD is equal to BC. {I. 34} For the same reason also EF is equal to BC, so that AD is also equal to EF; {C.N. 1} and DE is common; therefore the whole AE is equal to the whole DF. {C.N. 2} But AB is also equal to DC; {I. 34} therefore the two sides EA, AB are equal to the two sides FD, DC respectively, and the angle FDC is equal to the angle EAB, the exterior to the interior; {I. 29} therefore the base EB is equal to the base FC, and the triangle EAB will be equal to the triangle FDC. {I. 4} Let DGE be subtracted from each; therefore the trapezium ABGD which remains is equal to the trapezium EGCF which remains. {C.N. 3} Let the triangle GBC be added to each; therefore the whole parallelogram ABCD is equal to the whole parallelogram EBCF. {C.N. 2} Therefore etc.

Q. E. D.

Proposition 36. Parallelograms which are on equal bases and in the same parallels are equal to one another.

Let ABCD, EFGH be parallelograms which are on equal bases BC, FG and in the same parallels AH, BG; I say that the parallelogram ABCD is equal to EFGH.

For let BE, CH be joined.

Then, since BC is equal to FG while FG is equal to EH, BC is also equal to EH. {C.N. 1} But they are also parallel.

And EB, HC join them; but straight lines joining equal and parallel straight lines (at the extremities which are) in the same directions (respectively) are equal and parallel. {I. 33} Therefore EBCH is a parallelogram. {I. 34} And it is equal to ABCD; for it has the same base BC with it, and is in the same parallels BC, AH with it. {I. 35} For the same reason also EFGH is equal to the same EBCH; {I. 35} so that the parallelogram ABCD is also equal to EFGH. {C.N. 1} Therefore etc. Q. E. D.

Proposition 37. Triangles which are on the same base and in the same parallels are equal to one another.

Let ABC, DBC be triangles on the same base BC and in the same parallels AD, BC; I say that the triangle ABC is equal to the triangle DBC.

Let AD be produced in both directions to E, F; through B let BE be drawn parallel to CA, {I. 31} and through C let CF be drawn parallel to BD. {I. 31} Then each of the figures EBCA, DBCF is a parallelogram; and they are equal, for they are on the same base BC and in the same parallels BC, EF. {I. 35} Moreover the triangle ABC is half of the parallelogram EBCA; for the diameter AB bisects it. {I. 34} And the triangle DBC is half of the parallelogram DBCF; for the diameter DC bisects it. {I. 34} {But the halves of equal things are equal to one another.} Therefore the triangle ABC is equal to the triangle DBC.

Therefore etc.

Q. E. D.

Proposition 38. Triangles which are on equal bases and in the same parallels are equal to one another.

Let ABC, DEF be triangles on equal bases BC, EF and in the same parallels BF, AD; I say that the triangle ABC is equal to the triangle DEF.

For let AD be produced in both directions to G, H; through B let BG be drawn parallel to CA, {I. 31} and through F let FH be drawn parallel to DE.

Then each of the figures GBCA, DEFH is a parallelogram; and GBCA is equal to DEFH; for they are on equal bases BC, EF and in the same parallels BF, GH. {I. 36} Moreover the triangle ABC is half of the parallelogram GBCA; for the diameter AB bisects it. {I. 34} And the triangle FED is half of the parallelogram DEFH; for the diameter DF bisects it. {I. 34} {But the halves of equal things are equal to one another.} Therefore the triangle ABC is equal to the triangle DEF.

Therefore etc.

Q. E. D.

Proposition 39. Equal triangles which are on the same base and on the same side are also in the same parallels.

Let ABC, DBC be equal triangles which are on the same base BC and on the same side of it; {I say that they are also in the same parallels.} And {For} let AD be joined; I say that AD is parallel to BC.

For, if not, let AE be drawn through the point A parallel to the straight line BC, {I. 31} and let EC be joined.

Therefore the triangle ABC is equal to the triangle EBC; for it is on the same base BC with it and in the same parallels. {I. 37} But ABC is equal to DBC; therefore DBC is also equal to EBC, {C.N. 1} the greater to the less: which is impossible.

Therefore AE is not parallel to BC.

Similarly we can prove that neither is any other straight line except AD; therefore AD is parallel to BC.

Therefore etc.

Q. E. D.

{Proposition 40.

Equal triangles which are on equal bases and on the same side are also in the same parallels.

Let ABC, CDE be equal triangles on equal bases BC, CE and on the same side.

I say that they are also in the same parallels.

For let AD be joined; I say that AD is parallel to BE.

For, if not, let AF be drawn through A parallel to BE {I. 31}, and let FE be joined.

Therefore the triangle ABC is equal to the triangle FCE; for they are on equal bases BC, CE and in the same parallels BE, AF. {I. 38} But the triangle ABC is equal to the triangle DCE; therefore the triangle DCE is also equal to the triangle FCE, {C.N. 1} the greater to the less: which is impossible. Therefore AF is not parallel to BE.

Similarly we can prove that neither is any other straight line except AD; therefore AD is parallel to BE.

Therefore etc. Q. E. D.}

Proposition 41. If a parallelogram have the same base with a triangle and be in the same parallels, the parallelogram is double of the triangle.

For let the parallelogram ABCD have the same base BC with the triangle EBC, and let it be in the same parallels BC, AE; I say that the parallelogram ABCD is double of the triangle BEC.

For let AC be joined.

Then the triangle ABC is equal to the triangle EBC; for it is on the same base BC with it and in the same parallels BC, AE. {I. 37} But the parallelogram ABCD is double of the triangle ABC; for

the diameter AC bisects it; {I. 34} so that the parallelogram ABCD is also double of the triangle EBC.

Therefore etc.

Q. E. D.

Proposition 42. To construct, in a given rectilineal angle, a parallelogram equal to a given triangle.

Let ABC be the given triangle, and D the given rectilineal angle; thus it is required to construct in the rectilineal angle D a parallelogram equal to the triangle ABC.

Let BC be bisected at E, and let AE be joined; on the straight line EC, and at the point E on it, let the angle CEF be constructed equal to the angle D; {I. 23} through A let AG be drawn parallel to EC, and {I. 31} through C let CG be drawn parallel to EF.

Then FECG is a parallelogram.

And, since BE is equal to EC, the triangle ABE is also equal to the triangle AEC, for they are on equal bases BE, EC and in the same parallels BC, AG; {I. 38} therefore the triangle ABC is double of the triangle AEC.

But the parallelogram FECG is also double of the triangle AEC, for it has the same base with it and is in the same parallels with it; {I. 41} therefore the parallelogram FECG is equal to the triangle ABC.

And it has the angle CEF equal to the given angle D.

Therefore the parallelogram FECG has been constructed equal to the given triangle ABC, in the angle CEF which is equal to D.
Q. E. F.

Proposition 43. In any parallelogram the complements of the parallelograms about the diameter are equal to one another.

Let ABCD be a parallelogram, and AC its diameter; and about AC let EH, FG be parallelograms, and BK, KD the so-called complements; I say that the complement BK is equal to the complement KD.

For, since ABCD is a parallelogram, and AC its diameter, the triangle ABC is equal to the triangle ACD. {I. 34} Again, since EH is a parallelogram, and AK is its diameter, the triangle AEK is equal to the triangle AHK.

For the same reason the triangle KFC is also equal to KGC.

Now, since the triangle AEK is equal to the triangle AHK, and KFC to KGC, the triangle AEK together with KGC is equal to the triangle AHK together with KFC. {C.N. 2} And the whole triangle ABC is also equal to the whole ADC; therefore the complement BK which remains is equal to the complement KD which remains. {C.N. 3} Therefore etc.

Q. E. D.

Proposition 44. To a given straight line to apply, in a given rectilineal angle, a parallelogram equal to a given triangle.

Let AB be the given straight line, C the given triangle and D the given rectilineal angle; thus it is required to apply to the given straight line AB, in an angle equal to the angle D, a parallelogram equal to the given triangle C.

Let the parallelogram BEFG be constructed equal to the triangle C, in the angle EBG which is equal to D {I. 42}; let it be placed so that BE is in a straight line with AB; let FG be drawn through to H, and let AH be drawn through A parallel to either BG or EF. {I. 31} Let HB be joined.

Then, since the straight line HF falls upon the parallels AH, EF, the angles AHF, HFE are equal to two right angles. {I. 29} Therefore the angles BHG, GFE are less than two right angles; and straight lines produced indefinitely from angles less than

two right angles meet; {Post. 5} therefore HB, FE, when produced, will meet.

Let them be produced and meet at K; through the point K let KL be drawn parallel to either EA or FH, {I. 31} and let HA, GB be produced to the points L, M.

Then HLKF is a parallelogram, HK is its diameter, and AG, ME are parallelograms. and LB, BF the so-called complements, about HK; therefore LB is equal to BF. {I. 43} But BF is equal to the triangle C; therefore LB is also equal to C. {C.N. 1} And, since the angle GBE is equal to the angle ABM, {I. 15} while the angle GBE is equal to D, the angle ABM is also equal to the angle D.

Therefore the parallelogram LB equal to the given triangle C has been applied to the given straight line AB, in the angle ABM which is equal to D.

Q. E. F.

Proposition 45. To construct, in a given rectilinear angle, a parallelogram equal to a given rectilinear figure.

Let ABCD be the given rectilinear figure and E the given rectilinear angle; thus it is required to construct, in the given angle E, a parallelogram equal to the rectilinear figure ABCD.

Let DB be joined, and let the parallelogram FH be constructed equal to the triangle ABD, in the angle HKF which is equal to E; {I. 42} let the parallelogram GM equal to the triangle DBC be applied to the straight line GH, in the angle GHM which is equal to E. {I. 44} Then, since the angle E is equal to each of the angles HKF, GHM, the angle HKF is also equal to the angle GHM. {C.N. 1} Let the angle KHG be added to each; therefore the angles FKH, KHG are equal to the angles KHG, GHM.

But the angles FKH, KHG are equal to two right angles; {I. 29} therefore the angles KHG, GHM are also equal to two right angles.

Thus, with a straight line GH, and at the point H on it, two straight lines KH, HM not lying on the same side make the adjacent angles equal to two right angles; therefore KH is in a straight line with HM. {I. 14} And, since the straight line HG falls upon the parallels KM, FG, the alternate angles MHG, HGF are equal to one another. {I. 29} Let the angle HGL be added to each; therefore the angles MHG, HGL are equal to the angles HGF, HGL. {C.N. 2} But the angles MHG, HGL are equal to two right angles; {I. 29} therefore the angles HGF, HGL are also equal to two right angles. {C.N. 1} Therefore FG is in a straight line with GL. {I. 14} And, since FK is equal and parallel to HG, {I. 34} and HG to ML also, KF is also equal and parallel to ML; {C.N. 1; I. 30} and the straight lines KM, FL join them (at their extremities); therefore KM, FL are also equal and parallel. {I. 33} Therefore KFLM is a parallelogram.

And, since the triangle ABD is equal to the parallelogram FH, and DBC to GM, the whole rectilineal figure ABCD is equal to the whole parallelogram KFLM.

Therefore the parallelogram KFLM has been constructed equal to the given rectilineal figure ABCD, in the angle FKM which is equal to the given angle E.

Q. E. F.

Proposition 46. On a given straight line to describe a square.

Let AB be the given straight line; thus it is required to describe a square on the straight line AB.

Let AC be drawn at right angles to the straight line AB from the point A on it {I. 11}, and let AD be made equal to AB; through the point D let DE be drawn parallel to AB, and through the point B let BE be drawn parallel to AD. {I. 31} Therefore ADEB is a parallelogram; therefore AB is equal to DE, and AD to BE. {I. 34} But AB is equal to AD; therefore the four straight lines BA, AD,

DE, EB are equal to one another; therefore the parallelogram ADEB is equilateral.

I say next that it is also right-angled.

For, since the straight line AD falls upon the parallels AB, DE, the angles BAD, ADE are equal to two right angles. {I. 29} But the angle BAD is right; therefore the angle ADE is also right.

And in parallelogrammic areas the opposite sides and angles are equal to one another; {I. 34} therefore each of the opposite angles ABE, BED is also right. Therefore ADEB is right-angled.

And it was also proved equilateral.

Therefore it is a square; and it is described on the straight line AB.

Q. E. F.

Proposition 47. In right-angled triangles the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.

Let ABC be a right-angled triangle having the angle BAC right; I say that the square on BC is equal to the squares on BA, AC.

For let there be described on BC the square BDEC, and on BA, AC the squares GB, HC; {I. 46} through A let AL be drawn parallel to either BD or CE, and let AD, FC be joined.

Then, since each of the angles BAC, BAG is right, it follows that with a straight line BA, and at the point A on it, the two straight lines AC, AG not lying on the same side make the adjacent angles equal to two right angles; therefore CA is in a straight line with AG. {I. 14} For the same reason BA is also in a straight line with AH.

And, since the angle DBC is equal to the angle FBA: for each is right: let the angle ABC be added to each; therefore the whole

angle DBA is equal to the whole angle FBC. {C.N. 2} And, since DB is equal to BC, and FB to BA, the two sides AB, BD are equal to the two sides FB, BC respectively, and the angle ABD is equal to the angle FBC; therefore the base AD is equal to the base FC, and the triangle ABD is equal to the triangle FBC. {I. 4} Now the parallelogram BL is double of the triangle ABD, for they have the same base BD and are in the same parallels BD, AL. {I. 41} And the square GB is double of the triangle FBC, for they again have the same base FB and are in the same parallels FB, GC. {I. 41} {But the doubles of equals are equal to one another.} Therefore the parallelogram BL is also equal to the square GB.

Similarly, if AE, BK be joined, the parallelogram CL can also be proved equal to the square HC; therefore the whole square BDEC is equal to the two squares GB, HC. {C.N. 2} And the square BDEC is described on BC, and the squares GB, HC on BA, AC.

Therefore the square on the side BC is equal to the squares on the sides BA, AC.

Therefore etc.

Q. E. D.

Proposition 48. If in a triangle the square on one of the sides be equal to the squares on the remaining two sides of the triangle, the angle contained by the remaining two sides of the triangle is right.

For in the triangle ABC let the square on one side BC be equal to the squares on the sides BA, AC; I say that the angle BAC is right.

For let AD be drawn from the point A at right angles to the straight line AC, let AD be made equal to BA, and let DC be joined.

Since DA is equal to AB, the square on DA is also equal to the square on AB.

Let the square on AC be added to each; therefore the squares on DA, AC are equal to the squares on BA, AC.

But the square on DC is equal to the squares on DA, AC, for the angle DAC is right; {I. 47} and the square on BC is equal to the squares on BA, AC, for this is the hypothesis; therefore the square on DC is equal to the square on BC, so that the side DC is also equal to BC.

And, since DA is equal to AB, and AC is common, the two sides DA, AC are equal to the two sides BA, AC; and the base DC is equal to the base BC; therefore the angle DAC is equal to the angle BAC. {I. 8} But the angle DAC is right; therefore the angle BAC is also right.

Therefore etc.

Q. E. D.

ENDNOTES.

The Greek usage differs from ours in that the definite article is employed in such a phrase as this where we have the indefinite. ἐπὶ τῆς δοθείσης εὐθείας πεπερασμένης, “on the given finite straight line,” i.e. the finite straight line which we choose to take.

To be strictly literal we should have to translate in the reverse order “let the given finite straight line be the (straight line) AB” ; but this order is inconvenient in other cases where there is more than one datum, e.g. in the setting-out of I. 2, “let the given point be A, and the given straight line BC,” the awkwardness arising from the omission of the verb in the second clause. Hence I have, for clearness’ sake, adopted the other order throughout the book.

Two things are here to be noted, (1) the elegant and practically universal use of the perfect passive imperative in constructions, γεγράφθω meaning of course “let it have been described” or

“suppose it described,” (2) the impossibility of expressing shortly in a translation the force of the words in their original order. κύκλος γεγράφθω ὁ ΒΓΔ means literally “let a circle have been described, the (circle, namely, which I denote by) BCD.” Similarly we have lower down “let straight lines, (namely) the (straight lines) CA, CB, be joined,” ἐπεζεύχθωσαν εὐθεῖαι αἱ ΓΑ, ΓΒ. There seems to be no practicable alternative, in English, but to translate as I have done in the text.

Euclid is careful to adhere to the phraseology of Postulate 1 except that he speaks of “joining” (ἐπεζεύχθωσαν) instead of “drawing” (γράφειν). He does not allow himself to use the shortened expression “let the straight line FC be joined” (without mention of the points F, C) until I. 5.

, ἐκατέρᾳ τῶν ΓΑ, ΓΒ and 24. the three straight lines CA, AB, BC, αἱ τρεῖς αἱ ΓΑ, ΑΒ, ΒΓ. I have, here and in all similar expressions, inserted the words “straight lines” which are not in the Greek. The possession of the inflected definite article enables the Greek to omit the words, but this is not possible in English, and it would scarcely be English to write “each of CA, CB” or “the three CA, AB, BC.”

I have inserted these words because “to place a straight line at a given point” (πρὸς τῷ δοθέντι σημείῳ) is not quite clear enough, at least in English.

It will be observed that in this first application of Postulate 2, and again in I. 5, Euclid speaks of the continuation of the straight line as that which is produced in such cases, ἐκβεβλήσθωσαν and προσεκβεβλήσθωσαν meaning little more than drawing straight lines “in a straight line with” the given straight lines. The first place in which Euclid uses phraseology exactly corresponding to ours when speaking of a straight line being produced is in I. 16: “let one side of it, BC, be produced to D” (προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἢ ΒΓ ἐπὶ τὸ Δ).

The Greek expressions are λοιπὴ ἡ ΑΛ and λοιπὴ τῇ ΒΗ, and the literal translation would be “AL (or BG) remaining,” but the shade of meaning conveyed by the position of the definite article can hardly be expressed in English.

It is a fact that Euclid’s enunciations not infrequently leave something to be desired in point of clearness and precision. Here he speaks of the triangles having “the angle equal to the angle, namely the angle contained by the equal straight lines” (τὴν γωνίαν τῇ γωνίᾳ ἴσην ἔχῃ τὴν ὑπὸ τῶν ἴσων εὐθειῶν περιεχομένην), only one of the two angles being described in the latter expression (in the accusative), and a similar expression in the dative being left to be understood of the other angle. It is curious too that, after mentioning two “sides,” he speaks of the angles contained by the equal “straight lines,” not “sides.” It may be that he wished to adhere scrupulously, at the outset, to the phraseology of the definitions, where the angle is the inclination to one another of two lines or straight lines. Similarly in the enunciation of I. 5 he speaks of producing the equal “straight lines” as if to keep strictly to the wording of Postulate 2.

I agree with Mr H. M. Taylor (Euclid, p. ix) that it is best to abandon the traditional translation of “each to each,” which would naturally seem to imply that all the four magnitudes are equal rather than (as the Greek ἐκατέρᾳ ἐκατέρᾳ does) that one is equal to one and the other to the other.

Here we have the word base used for the first time in the Elements. Proclus explains it (p. 236, 12-15) as meaning (1), when no side of a triangle has been mentioned before, the side “which is on a level with the sight” (τὴν πρὸς τῇ ὄψει κειμένην), and (2), when two sides have already been mentioned, the third side. Proclus thus avoids the mistake made by some modern editors who explain the term exclusively with reference to the case where two sides have been mentioned before. That this is

an error is proved (1) by the occurrence of the term in the enunciations of I. 37 etc. about triangles on the same base and equal bases, (2) by the application of the same term to the bases of parallelograms in I. 35 etc. The truth is that the use of the term must have been suggested by the practice of drawing the particular side horizontally, as it were, and the rest of the figure above it. The base of a figure was therefore spoken of, primarily, in the same sense as the base of anything else, e.g. of a pedestal or column; but when, as in I. 5, two triangles were compared occupying other than the normal positions which gave rise to the name, and when two sides had been previously mentioned, the base was, as Proclus says, necessarily the third side.

ὑποτείνειν ὑπό, “to stretch under,” with accusative.

The full Greek expression would be ἡ ὑπὸ τῶν ΒΑ, ΑΓ περιεχομένη γωνία, “the angle contained by the (straight lines) ΒΑ, ΑΓ.” But it was a common practice of Greek geometers, e.g. of Archimedes and Apollonius (and Euclid too in Books X. — XIII.), to use the abbreviation αἱ ΒΑΓ for αἱ ΒΑ, ΑΓ, “the (straight lines) ΒΑ, ΑΓ.” Thus, on περιεχομένη being dropped, the expression would become first ἡ ὑπὸ τῶν ΒΑΓ γωνία, then ἡ ὑπὸ ΒΑΓ γωνία, and finally ἡ ὑπὸ ΒΑΓ, without γωνία, as we regularly find it in Euclid.

The difference between the technical use of the passive ἐφαρμόζεσθαι “to be applied (to),” and of the active ἐφαρμόζειν “to coincide (with)” has been noticed above (note on Common Notion 4, pp. 224-5).

Heiberg (*Paralipomena su Euklid in Hermes*, XXXVIII., 1903, p. 56) has pointed out, as a conclusive reason for regarding these words as an early interpolation, that the text of an-Nairīzī (Codex Leidensis 399, 1, ed. Besthorn-Heiberg, p. 55) does not give the words in this place but after the conclusion Q.E.D., which shows that they constitute a scholium only. They were

doubtless added by some commentator who thought it necessary to explain the immediate inference that, since B coincides with E and C with F, the straight line BC coincides with the straight line EF, an inference which really follows from the definition of a straight line and Post. 1; and no doubt the Postulate that “Two straight lines cannot enclose a space” (afterwards placed among the Common Notions) was interpolated at the same time.

Where (as here) Euclid’s conclusion merely repeats the enunciation word for word, I shall avoid the repetition and write “Therefore etc.” simply.

(meaning the equal sides). Cf. note on the similar expression in Prop. 4, lines 2, 3.

, εἰλήφθω ἐπὶ τῆς ΒΔ τυχὸν σημεῖον τὸ Ζ, where τυχὸν σημεῖον means “a chance point.”

δύο αἱ ΖΑ, ΑΓ δυὸς ταῖς ΗΑ, ΑΒ ἴσαι εἰσὶν ἑκατέρᾳ ἑκατέρᾳ. Here, and in numberless later passages, I have inserted the word “sides” for the reason given in the note on I. 1, line 20. It would have been permissible to supply either “straight lines” or “sides” ; but on the whole “sides” seems to be more in accordance with the phraseology of I. 4.

i.e., apparently, common to the angles, as the αὐτῶν in βάσις αὐτῶν κοινὴ can only refer to γωνία and γωνία preceding. Simson wrote “and the base BC is common to the two triangles BFC, CGB” ; Todhunter left out these words as being of no use and tending to perplex a beginner. But Euclid evidently chose to quote the conclusion of I. 4 exactly; the first phrase of that conclusion is that the bases (of the two triangles) are equal, and, as the equal bases are here the same base, Euclid naturally substitutes the word “common” for “equal.”

As “(Being) what it was required to prove” (or “do”) is somewhat long, I shall henceforth write the time-honoured “Q. E. D.” and “Q. E. F.” for ὅπερ ἔδει δεῖξαι and ὅπερ ἔδει ποιῆσαι.

In an English translation of the enunciation of this proposition it is absolutely necessary, in order to make it intelligible, to insert some words which are not in the Greek. The reason is partly that the Greek enunciation is itself very elliptical, and partly that some words used in it conveyed more meaning than the corresponding words in English do. Particularly is this the case with οὐ συσταθήσονται ἐπὶ “there shall not be constructed upon,” since συνίστασθαι is the regular word for constructing a triangle in particular. Thus a Greek would easily understand συσταθήσονται ἐπὶ as meaning the construction of two lines forming a triangle on a given straight line as base; whereas to “construct two straight lines on a straight line” is not in English sufficiently definite unless we explain that they are drawn from the ends of the straight line to meet at a point. I have had the less hesitation in putting in the words “from its extremities” because they are actually used by Euclid in the somewhat similar enunciation of I. 21.

How impossible a literal translation into English is, if it is to convey the meaning of the enunciation intelligibly, will be clear from the following attempt to render literally: “On the same straight line there shall not be constructed two other straight lines equal, each to each, to the same two straight lines, (terminating) at different points on the same side, having the same extremities as the original straight lines” (ἐπὶ τῆς αὐτῆς εὐθείας δύο ταῖς αὐταῖς εὐθείαις ἄλλαι δύο εὐθεῖαι ἴσαι ἑκατέρα ἑκατέρα οὐ συσταθήσονται πρὸς ἄλλω καὶ ἄλλω σημείῳ ἐπὶ τὰ αὐτὰ μέρη τὰ αὐτὰ πέρατα ἔχουσαι ταῖς ἐξ ἀρχῆς εὐθείαις).

The reason why Euclid allowed himself to use, in this enunciation, language apparently so obscure is no doubt that the phraseology was traditional and therefore, vague as it was,

had a conventional meaning which the contemporary geometer well understood. This is proved, I think, by the occurrence in Aristotle (*Meteorologica* III. 5, 376 a 2 sqq.) of the very same, evidently technical, expressions. Aristotle is there alluding to the theorem given by Eutocius from Apollonius' *Plane Loci* to the effect that, if H, K be two fixed points and M such a variable point that the ratio of MH to MK is a given ratio (not one of equality), the locus of M is a circle. (For an account of this theorem see note on VI. 3 below.) Now Aristotle says "The lines drawn up from H, K in this ratio cannot be constructed to two different points of the semicircle A" (αἱ οὖν ἀπὸ τῶν HK ἀναγόμεναι γραμμαὶ ἐν τούτῳ τῷ λόγῳ οὐ συσταθήσονται τοῦ ἐφ' ᾧ A ἡμικυκλίου πρὸς ἄλλο καὶ ἄλλο σημεῖον).

If a paraphrase is allowed instead of a translation adhering as closely as possible to the original, Simson's is the best that could be found, since the fact that the straight lines form triangles on the same base is really conveyed in the Greek. Simson's enunciation is, Upon the same base, and on the same side of it, there cannot be two triangles that have their sides which are terminated in one extremity of the base equal to one another, and likewise those which are terminated at the other extremity. Th. Taylor (the translator of Proclus) attacks Simson's alteration as "indiscreet" and as detracting from the beauty and accuracy of Euclid's enunciation which are enlarged upon by Proclus in his commentary. Yet, when Taylor says "Whatever difficulty learners may find in conceiving this proposition abstractedly is easily removed by its exposition in the figure," he really gives his case away. The fact is that Taylor, always enthusiastic over his author, was nettled by Simson's slighting remarks on Proclus' comments on the proposition. Simson had said, with reference to Proclus' explanation of the bearing of the second part of I. 5 on I. 7, that it was not "worth while to relate his trifles at full length," to which Taylor retorts "But Mr Simson was no philosopher; and therefore the greatest part of these

Commentaries must be considered by him as trifles, from the want of a philosophic genius to comprehend their meaning, and a taste superior to that of a mere mathematician, to discover their beauty and elegance.”

It would be natural to insert here the step “but the angle ACD is greater than the angle BCD. {C.N. 5}.”

much greater, literally “greater by much” (πολλῷ μείζων). Simson and those who follow him translate: “much more then is the angle BDC greater than the angle BCD,” but the Greek for this would have to be πολλῷ (or πολὺ{ρπαρ } μᾶλλον ἐστὶ...μείζων. πολλῷ μᾶλλον, however, though used by Apollonius, is not, apparently, found in Euclid or Archimedes.

The text has here “BA, CA.”

fall beside them. The Greek has the future, παραλλάξουσι. παραλλάττω means “to pass by without touching,” “to miss” or “to deviate.”

The verb is κείσθω which, as well as the other parts of κείμαι, is constantly used for the passive of τίθημι “to place” ; and the latter word is constantly used in the sense of making, e.g., one straight line equal to another straight line.

, κάθετον εὐθεϊαν γραμμὴν. This is the full expression for a perpendicular, κάθετος meaning let fall or let down, so that the expression corresponds to our plumb-line. ἡ κάθετος is however constantly used alone for a perpendicular, γραμμὴ being understood.

, literally “towards the other parts of the straight line AB,” ἐπὶ τὰ ἕτερα μέρη τῆς AB. Cf. “on the same side” (ἐπὶ τὰ αὐτὰ μέρη) in Post. 5 and “in both directions” (ἐφ’ ἑκάτερα τὰ μέρη) in Def. 23.

literally “let the angle EBD be added (so as to be) common,” κοινὴ προσκείσθω ἢ ὑπὸ EBA. Similarly κοινὴ ἀφηρήσθω is used

of subtracting a straight line or angle from each of two others. “Let the common angle EBD be added” is clearly an inaccurate translation, for the angle is not common before it is added, i.e. the κοινὴ is proleptic. “Let the common angle be subtracted” as a translation of κοινὴ ἀφηρήσθω would be less unsatisfactory, it is true, but, as it is desirable to use corresponding words when translating the two expressions, it seems hopeless to attempt to keep the word “common,” and I have therefore said “to each” and “from each” simply.

There is no greater difficulty in translating the works of the Greek geometers than that of accurately giving the force of prepositions. πρὸς, for instance, is used in all sorts of expressions with various shades of meaning. The present enunciation begins ἐὰν πρὸς τινι εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείω, and it is really necessary in this one sentence to translate πρὸς by three different words, with, at, and on. The first πρὸς must be translated by with because two straight lines “make” an angle with one another. On the other hand, where the similar expression πρὸς τῇ δοθείσῃ εὐθείᾳ occurs in I. 23, but it is a question of “constructing” an angle (συστήσασθαι), we have to say “to construct on a given straight line.” Against would perhaps be the English word coming nearest to expressing all these meanings of πρὸς, but it would be intolerable as a translation.

Todhunter points out that for the inference in this line Post. 4, that all right angles are equal, is necessary as well as the Common Notion that things which are equal to the same thing (or rather, here, to equal things) are equal. A similar remark applies to steps in the proofs of I. 15 and I. 28.

The Greek expresses this by the future of the verb, δείξομεν, “we shall prove,” which however would perhaps be misleading in English.

The difference between adjacent angles (αἱ ἐφεξῆς γωνίαι) and vertical angles (αἱ κατὰ κορυφὴν γωνίαι) is thus explained by Proclus (p. 298, 14-24). The first term describes the angles made by two straight lines when one only is divided by the other, i.e. when one straight line meets another at a point which is not either of its extremities, but is not itself produced beyond the point of meeting. When the first straight line is produced, so that the lines cross at the point, they make two pairs of vertical angles (which are more clearly described as vertically opposite angles), and which are so called because their convergence is from opposite directions to one point (the intersection of the lines) as vertex (κορυφή).

literally “at the section,” πρὸς τῇ τομῇ.

literally “the outside angle,” ἡ ἐκτὸς γωνία.

τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν.

The word is διήχθω, a variation on the more usual ἐκβεβλήσθω, “let it be produced.”

in the text “FEC.”

πάντη μεταλαμβάνομεναι, i.e. any pair added together.

In the enunciation of this number we have ὑποτείνειν (“subtend”) used with the simple accusative instead of the more usual ὑπό with accusative. The latter construction is used in the enunciation of I. 19, which otherwise only differs from that of I. 18 in the order of the words. The point to remember in order to distinguish the two is that the datum comes first and the quaesitum second, the datum being in this proposition the greater side and in the next the greater angle. Thus the enunciations are (I. 18) παντὸς τριγώνου ἡ μείζων πλευρὰ τὴν μείζονα γωνίαν ὑποτείνει and (I. 19) παντὸς τριγώνου ὑπὸ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει. In order to keep the proper order in English we must use the passive of the verb in I.

19. Aristotle quotes the result of I. 19, using the exact wording, ὑπὸ γὰρ τὴν μείζω γωνίαν ὑποτείνει (Meteorologica III. 5, 376 a 12).

It was the habit of the Epicureans, says Proclus (p. 322), to ridicule this theorem as being evident even to an ass and requiring no proof, and their allegation that the theorem was “known” (γνώριμον) even to an ass was based on the fact that, if fodder is placed at one angular point and the ass at another, he does not, in order to get to his food, traverse the two sides of the triangle but only the one side separating them (an argument which makes Savile exclaim that its authors were “digni ipsi, qui cum Asino foenum essent,” p. 78). Proclus replies truly that a mere perception of the truth of the theorem is a different thing from a scientific proof of it and a knowledge of the reason why it is true. Moreover, as Simson says, the number of axioms should not be increased without necessity.

The word “meeting” is not in the Greek, where the words are ἐντὸς συσταθῶσιν. συνίστασθαι is the word used of constructing two straight lines to a point (cf. I. 7) or so as to form a triangle; but it is necessary in English to indicate that they meet.

Observe the elegant brevity of the Greek αἱ συσταθεῖσαι.

This is the first case in the Elements of a διορισμός to a problem in the sense of a statement of the conditions or limits of the possibility of a solution. The criterion is of course supplied by the preceding proposition.

This is usually translated (e.g. by Williamson and Simson) “But it is necessary,” which is however inaccurate, since the Greek is not δεῖ δέ but δεῖ δῆ. The words are the same as those used to introduce the διορισμός in the other sense of the “definition” or “particular statement” of a construction to be effected. Hence, as in the latter case we say “thus it is required” (e.g. to bisect the finite straight line AB, I. 10), we should here translate “thus

it is necessary.”

To this enunciation all the MSS. and Boethius add, after the διορισμός, the words “because in any triangle two sides taken together in any manner are greater than the remaining one.” But this explanation has the appearance of a gloss, and it is omitted by Proclus and Campanus. Moreover there is no corresponding addition to the διορισμός of VI. 28.

I have naturally left out the well-known words added by Simson in order to avoid the necessity of considering three cases: “Of the two sides DE, DF let DE be the side which is not greater than the other.” I doubt whether Euclid could have been induced to insert the words himself, even if it had been represented to him that their omission meant leaving two possible cases out of consideration. His habit and that of the great Greek geometers was, not to set out all possible cases, but to give as a rule one case, generally the most difficult, as here, and to leave the others to the reader to work out for himself. We have already seen one instance in I. 7.

πλευρὰν τὴν πρὸς ταῖς ἴσαις γωνίαις.

ὑπόκειται ἴση, according to the elegant Greek idiom. ὑπόκειμαι is used for the passive of ὑποτίθημι, as κέῖμαι is used for the passive of τίθημι, and so with the other compounds. Cf. προσκεῖσθαι, “to be added.”

εἰς δύο εὐθείας ἐμπίπτουσα, the phrase being the same as that used in Post. 5, meaning a transversal.

αἱ ἐναλλάξ γωνίαι. Proclus (p. 357, 9) explains that Euclid uses the word alternate (or, more exactly, alternately, ἐναλλάξ) in two connexions, (1) of a certain transformation of a proportion, as in Book V. and the arithmetical Books, (2) as here, of certain of the angles formed by parallels with a straight line crossing them. Alternate angles are, according to Euclid as interpreted

by Proclus, those which are not on the same side of the transversal, and are not adjacent, but are separated by the transversal, both being within the parallels but one “above” and the other “below.” The meaning is natural enough if we imagine the four internal angles to be taken in cyclic order and alternate angles to be any two of them not successive but separated by one angle of the four.

literally “towards the parts B, D or towards A, C,” ἐπὶ τὰ B, Δ μέρη ἢ ἐπὶ τὰ A Γ.

αἱ δὲ ἀπ: ἐλασσόνων ἢ δύο ὀρθῶν ἐκβαλλόμεναι εἰς ἄπειρον συμπίπτουσιν, a variation from the more explicit language of Postulate 5. A good deal is left to be understood, namely that the straight lines begin from points at which they meet a transversal, and make with it internal angles on the same side the sum of which is less than two right angles.

literally “because they are supposed parallel,” διὰ τὸ παραλλήλους αὐτὰς ὑποκεῖσθαι.

The usual conclusion in general terms (“Therefore etc.”) repeating the enunciation is, curiously enough, wanting at the end of this proposition.

I have for clearness’ sake inserted the words in brackets though they are not in the original Greek, which has “joining...in the same directions” or “on the same sides,” ἐπὶ τὰ αὐτὰ μέρη ἐπιζευγνύουσαι. The expression “towards the same parts,” though usage has sanctioned it, is perhaps not quite satisfactory.

and 18. DCB. The Greek has “BC, CD” and “BCD” in these places respectively. Euclid is not always careful to write in corresponding order the letters denoting corresponding points in congruent figures. On the contrary, he evidently prefers the alphabetical order, and seems to disdain to alter it for the sake

of beginners or others who might be confused by it. In the case of angles alteration is perhaps unnecessary; but in the case of triangles and pairs of corresponding sides I have ventured to alter the order to that which the mathematician of to-day expects.

It is to be observed that, when parallelograms have to be mentioned for the first time, Euclid calls them “parallelogrammic areas” or, more exactly, “parallelogram” areas (παράλληλόγραμμα χωρία). The meaning is simply areas bounded by parallel straight lines with the further limitation placed upon the term by Euclid that only four-sided figures are so called, although of course there are certain regular polygons which have opposite sides parallel, and which therefore might be said to be areas bounded by parallel straight lines. We gather from Proclus (p. 393) that the word “parallelogram” was first introduced by Euclid, that its use was suggested by I. 33, and that the formation of the word παράλληλόγραμμος (parallel-lined) was analogous to that of εὐθύγραμμος (straight-lined or rectilinear).

and 36. DC, CB. The Greek has in these places “BCD” and “CD, BC” respectively. Cf. note on I. 33, lines 15, 18.

The text has “DFC.”

Euclid speaks of the triangle DGE without any explanation that, in the case which he takes (where AD, EF have no point in common), BE, CD must meet at a point G between the two parallels. He allows this to appear from the figure simply.

Here and in the next proposition Heiberg brackets the words “But the halves of equal things are equal to one another” on the ground that, since the Common Notion which asserted this fact was interpolated at a very early date (before the time of Theon), it is probable that the words here were interpolated at the same time. Cf. note above (p. 224) on the interpolated Common

Notion.

{I say that they are also in the same parallels.} Heiberg has proved (Hermes, XXXVIII., 1903, p. 50) from a recently discovered papyrus-fragment (Fayūm towns and their papyri, p. 96, No. IX.) that these words are an interpolation by some one who did not observe that the words “And let AD be joined” are part of the setting-out (ἐκθεσις), but took them as belonging to the construction (κατασκευή) and consequently thought that a διορισμός or “definition” (of the thing to be proved) should precede. The interpolator then altered “And” into “For” in the next sentence.

complements, παραπληρώματα, the figures put in to fill up (interstices).

Euclid’s phraseology here and in the next proposition implies that the complements as well as the other parallelograms are “about” the diagonal. The words are here περὶ δὲ τὴν ΑΓ παραλληλόγραμμα μὲν ἔστω τὰ ΕΘ, ΖΗ, τὰ δὲ λεγόμενα παραπληρώματα τὰ ΒΚ, ΚΔ. The expression “the so-called complements” indicates that this technical use of παραπληρώματα was not new, though it might not be universally known.

The verb is in the aorist (ἐνέπεσεν) here and in similar expressions in the following propositions.

“rectilineal” simply, without “figure,” εὐθύγραμμον being here used as a substantive, like the similarly formed παραλληλόγραμμον.

Proclus (p. 423, 18 sqq.) notes the difference between the word construct (συστήσασθαι) applied by Euclid to the construction of a triangle (and, he might have added, of an angle) and the words describe on (ἀναγράφειν ἀπό) used of drawing a square on a given straight line as one side. The triangle (or angle) is, so

to say, pieced together, while the describing of a square on a given straight line is the making of a figure “from” one side, and corresponds to the multiplication of the number representing the side by itself.

, τὸ ἀπὸ...τετράγωνον, the word ἀναγραφέν or ἀναγεγραμμένον being understood.

subtending the right angle. Here ὑποτείνουσας, “subtending,” is used with the simple accusative (τὴν ὀρθὴν γωνίαν) instead of being followed by ὑπὸ and the accusative, which seems to be the original and more orthodox construction. Cf. I. 18, note.

Euclid actually writes “DB, BA,” and therefore the equal sides in the two triangles are not mentioned in corresponding order, though he adheres to the words ἐκάτερα ἐκάτερα “respectively.” Here DB is equal to BC and BA to FB.

{But the doubles of equals are equal to one another.} Heiberg brackets these words as an interpolation, since it quotes a Common Notion which is itself interpolated. Cf. notes on I. 37, p. 332, and on interpolated Common Notions, pp. 223-4.

BOOK II. DEFINITIONS.

1. ANY RECTANGULAR PARALLELOGRAM is said to be contained by the two straight lines containing the right angle.
2. And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

PROPOSITIONS.

Proposition 1. IF THERE BE two straight lines, and one of them be cut into any number of segments whatever, the rectangle contained by the two straight lines is equal to the rectangles

contained by the uncut straight line and each of the segments.

Let A, BC be two straight lines, and let BC be cut at random at the points D, E; I say that the rectangle contained by A, BC is equal to the rectangle contained by A, BD, that contained by A, DE and that contained by A, EC.

For let BF be drawn from B at right angles to BC; {I. 11} let BG be made equal to A, {I. 3} through G let GH be drawn parallel to BC, {I. 31} and through D, E, C let DK, EL, CH be drawn parallel to BG.

Then BH is equal to BK, DL, EH.

Now BH is the rectangle A, BC, for it is contained by GB, BC, and BG is equal to A; BK is the rectangle A, BD, for it is contained by GB, BD, and BG is equal to A; and DL is the rectangle A, DE, for DK, that is BG {I. 34}, is equal to A.

Similarly also EH is the rectangle A, EC.

Therefore the rectangle A, BC is equal to the rectangle A, BD, the rectangle A, DE and the rectangle A, EC.

Therefore etc. Q. E. D.

Proposition 2. If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

For let the straight line AB be cut at random at the point C; I say that the rectangle contained by AB, BC together with the rectangle contained by BA, AC is equal to the square on AB.

For let the square ADEB be described on AB {I. 46}, and let CF be drawn through C parallel to either AD or BE. {I. 31} Then AE is equal to AF, CE.

Now AE is the square on AB; AF is the rectangle contained by BA, AC, for it is contained by DA, AC, and AD is equal to AB; and CE is the rectangle AB, BC, for BE is equal to AB.

Therefore the rectangle BA, AC together with the rectangle AB, BC is equal to the square on AB.

Therefore etc. Q. E. D.

Proposition 3. If a straight line be cut at random, the rectangle contained by the whole and one of the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

For let the straight line AB be cut at random at C; I say that the rectangle contained by AB, BC is equal to the rectangle contained by AC, CB together with the square on BC.

For let the square CDEB be described on CB; {I. 46} let ED be drawn through to F, and through A let AF be drawn parallel to either CD or BE. {I. 31} Then AE is equal to AD, CE.

Now AE is the rectangle contained by AB, BC, for it is contained by AB, BE, and BE is equal to BC; AD is the rectangle AC, CB, for DC is equal to CB; and DB is the square on CB. Therefore the rectangle contained by AB, BC is equal to the rectangle contained by AC, CB together with the square on BC.

Therefore etc. Q. E. D.

Proposition 4. If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

For let the straight line AB be cut at random at C; I say that the square on AB is equal to the squares on AC, CB and twice the rectangle contained by AC, CB.

For let the square ADEB be described on AB, {I. 46} let BD be joined; through C let CF be drawn parallel to either AD or EB, and through G let HK be drawn parallel to either AB or DE. {I. 31} Then, since CF is parallel to AD, and BD has fallen on them, the exterior angle CGB is equal to the interior and opposite angle

ADB. {I. 29} But the angle ADB is equal to the angle ABD, since the side BA is also equal to AD; {I. 5} therefore the angle CGB is also equal to the angle GBC, so that the side BC is also equal to the side CG. {I. 6} But CB is equal to GK, and CG to KB; {I. 34} therefore GK is also equal to KB; therefore CGKB is equilateral.

I say next that it is also right-angled.

For, since CG is parallel to BK, the angles KBC, GCB are equal to two right angles. {I. 29} But the angle KBC is right; therefore the angle BCG is also right, so that the opposite angles CGK, GKB are also right. {I. 34} Therefore CGKB is right-angled; and it was also proved equilateral; therefore it is a square; and it is described on CB.

For the same reason HF is also a square; and it is described on HG, that is AC. {I. 34} Therefore the squares HF, KC are the squares on AC, CB.

Now, since AG is equal to GE, and AG is the rectangle AC, CB, for GC is equal to CB, therefore GE is also equal to the rectangle AC, CB.

Therefore AG, GE are equal to twice the rectangle AC, CB.

But the squares HF, CK are also the squares on AC, CB; therefore the four areas HF, CK, AG, GE are equal to the squares on AC, CB and twice the rectangle contained by AC, CB.

But HF, CK, AG, GE are the whole ADEB, which is the square on AB.

Therefore the square on AB is equal to the squares on AC, CB and twice the rectangle contained by AC, CB.

Therefore etc. Q. E. D.

Proposition 5. If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between

the points of section is equal to the square on the half.

For let a straight line AB be cut into equal segments at C and into unequal segments at D; I say that the rectangle contained by AD, DB together with the square on CD is equal to the square on CB.

For let the square CEFB be described on CB, {I. 46} and let BE be joined; through D let DG be drawn parallel to either CE or BF, through H again let KM be drawn parallel to either AB or EF, and again through A let AK be drawn parallel to either CL or BM. {I. 31} Then, since the complement CH is equal to the complement HF, {I. 43} let DM be added to each; therefore the whole CM is equal to the whole DF.

But CM is equal to AL, since AC is also equal to CB; {I. 36} therefore AL is also equal to DF.

Let CH be added to each; therefore the whole AH is equal to the gnomon NOP.

But AH is the rectangle AD, DB, for DH is equal to DB, therefore the gnomon NOP is also equal to the rectangle AD, DB.

Let LG, which is equal to the square on CD, be added to each; therefore the gnomon NOP and LG are equal to the rectangle contained by AD, DB and the square on CD.

But the gnomon NOP and LG are the whole square CEFB, which is described on CB; therefore the rectangle contained by AD, DB together with the square on CD is equal to the square on CB.

Therefore etc. Q. E. D.

Proposition 6. If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.

For let a straight line AB be bisected at the point C , and let a straight line BD be added to it in a straight line; I say that the rectangle contained by AD , DB together with the square on CB is equal to the square on CD .

For let the square $CEFD$ be described on CD , {I. 46} and let DE be joined; through the point B let BG be drawn parallel to either EC or DF , through the point H let KM be drawn parallel to either AB or EF , and further through A let AK be drawn parallel to either CL or DM . {I. 31} Then, since AC is equal to CB , AL is also equal to CH . {I. 36} But CH is equal to HF . {I. 43} Therefore AL is also equal to HF .

Let CM be added to each; therefore the whole AM is equal to the gnomon NOP .

But AM is the rectangle AD , DB , for DM is equal to DB ; therefore the gnomon NOP is also equal to the rectangle AD , DB .

Let LG , which is equal to the square on BC , be added to each; therefore the rectangle contained by AD , DB together with the square on CB is equal to the gnomon NOP and LG .

But the gnomon NOP and LG are the whole square $CEFD$, which is described on CD ; therefore the rectangle contained by AD , DB together with the square on CB is equal to the square on CD .

Therefore etc. $Q. E. D.$

Proposition 7. If a straight line be cut at random, the square on the whole and that on one of the segments both together are equal to twice the rectangle contained by the whole and the said segment and the square on the remaining segment.

For let a straight line AB be cut at random at the point C ; I say that the squares on AB , BC are equal to twice the rectangle contained by AB , BC and the square on CA .

For let the square ADEB be described on AB, {I. 46} and let the figure be drawn.

Then, since AG is equal to GE, {I. 43} let CF be added to each; therefore the whole AF is equal to the whole CE.

Therefore AF, CE are double of AG.

But AF, CE are the gnomon KLM and the square CF; therefore the gnomon KLM and the square CF are double of AG.

But twice the rectangle AB, BC is also double of AG; for BG is equal to BC; therefore the gnomon KLM and the square CF are equal to twice the rectangle AB, BC.

Let DG, which is the square on AC, be added to each; therefore the gnomon KLM and the squares BG, GD are equal to twice the rectangle contained by AB, BC and the square on AC.

But the gnomon KLM and the squares BG, GD are the whole ADEB and CF, which are squares described on AB, BC; therefore the squares on AB, BC are equal to twice the rectangle contained by AB, BC together with the square on AC.

Therefore etc. Q. E. D.

Proposition 8. If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and the aforesaid segment as on one straight line.

For let a straight line AB be cut at random at the point C; I say that four times the rectangle contained by AB, BC together with the square on AC is equal to the square described on AB, BC as on one straight line.

For let {the straight line} BD be produced in a straight line {with AB}, and let BD be made equal to CB; let the square AEFD be described on AD, and let the figure be drawn double.

Then, since CB is equal to BD, while CB is equal to GK, and BD to KN, therefore GK is also equal to KN.

For the same reason QR is also equal to RP.

And, since BC is equal to BD, and GK to KN, therefore CK is also equal to KD, and GR to RN. {I. 36} But CK is equal to RN, for they are complements of the parallelogram CP; {I. 43} therefore KD is also equal to GR; therefore the four areas DK, CK, GR, RN are equal to one another.

Therefore the four are quadruple of CK.

Again, since CB is equal to BD, while BD is equal to BK, that is CG, and CB is equal to GK, that is GQ, therefore CG is also equal to GQ.

And, since CG is equal to GQ, and QR to RP, AG is also equal to MQ, and QL to RF. {I. 36} But MQ is equal to QL, for they are complements of the parallelogram ML; {I. 43} therefore AG is also equal to RF; therefore the four areas AG, MQ, QL, RF are equal to one another.

Therefore the four are quadruple of AG.

But the four areas CK, KD, GR, RN were proved to be quadruple of CK; therefore the eight areas, which contain the gnomon STU, are quadruple of AK.

Now, since AK is the rectangle AB, BD, for BK is equal to BD, therefore four times the rectangle AB, BD is quadruple of AK.

But the gnomon STU was also proved to be quadruple of AK; therefore four times the rectangle AB, BD is equal to the gnomon STU.

Let OH, which is equal to the square on AC, be added to each; therefore four times the rectangle AB, BD together with the square on AC is equal to the gnomon STU and OH.

But the gnomon STU and OH are the whole square AEFD, which is described on AD.

therefore four times the rectangle AB, BD together with the square on AC is equal to the square on AD But BD is equal to BC; therefore four times the rectangle contained by AB, BC together with the square on AC is equal to the square on AD, that is to the square described on AB and BC as on one straight line.

Therefore etc. Q. E. D.

Proposition 9. If a straight line be cut into equal and unequal segments, the squares on the unequal segments of the whole are double of the square on the half and of the square on the straight line between the points of section.

For let a straight line AB be cut into equal segments at C, and into unequal segments at D; I say that the squares on AD, DB are double of the squares on AC, CD.

For let CE be drawn from C at right angles to AB, and let it be made equal to either AC or CB; let EA, EB be joined, let DF be drawn through D parallel to EC, and FG through F parallel to AB, and let AF be joined.

Then, since AC is equal to CE, the angle EAC is also equal to the angle AEC.

And, since the angle at C is right, the remaining angles EAC, AEC are equal to one right angle. {I. 32} And they are equal; therefore each of the angles CEA, CAE is half a right angle.

For the same reason each of the angles CEB, EBC is also half a right angle; therefore the whole angle AEB is right.

And, since the angle GEF is half a right angle, and the angle EGF is right, for it is equal to the interior and opposite angle ECB, {I. 29} the remaining angle EFG is half a right angle; {I. 32} therefore the angle GEF is equal to the angle EFG, so that the

side EG is also equal to GF. {I. 6} Again, since the angle at B is half a right angle, and the angle FDB is right, for it is again equal to the interior and opposite angle ECB, {I. 29} the remaining angle BFD is half a right angle; {I. 32} therefore the angle at B is equal to the angle DFB, so that the side FD is also equal to the side DB. {I. 6} Now, since AC is equal to CE, the square on AC is also equal to the square on CE; therefore the squares on AC, CE are double of the square on AC.

But the square on EA is equal to the squares on AC, CE, for the angle ACE is right; {I. 47} therefore the square on EA is double of the square on AC.

Again, since EG is equal to GF, the square on EG is also equal to the square on GF; therefore the squares on EG, GF are double of the square on GF.

But the square on EF is equal to the squares on EG, GF; therefore the square on EF is double of the square on GF.

But GF is equal to CD; {I. 34} therefore the square on EF is double of the square on CD.

But the square on EA is also double of the square on AC; therefore the squares on AE, EF are double of the squares on AC, CD.

And the square on AF is equal to the squares on AE, EF, for the angle AEF is right; {I. 47} therefore the square on AF is double of the squares on AC, CD.

But the squares on AD, DF are equal to the square on AF, for the angle at D is right; {I. 47} therefore the squares on AD, DF are double of the squares on AC, CD.

And DF is equal to DB; therefore the squares on AD, DB are double of the squares on AC, CD.

Therefore etc. Q. E. D.

Proposition 10. If a straight line be bisected, and a straight line be added to it in a straight line, the square on the whole with the added straight line and the square on the added straight line both together are double of the square on the half and of the square described on the straight line made up of the half and the added straight line as on one straight line.

For let a straight line AB be bisected at C , and let a straight line BD be added to it in a straight line; I say that the squares on AD , DB are double of the squares on AC , CD .

For let CE be drawn from the point C at right angles to AB {I. 11}, and let it be made equal to either AC or CB {I. 3}; let EA , EB be joined; through E let EF be drawn parallel to AD , and through D let FD be drawn parallel to CE . {I. 31} Then, since a straight line EF falls on the parallel straight lines EC , FD , the angles CEF , EFD are equal to two right angles; {I. 29} therefore the angles FEB , EFD are less than two right angles.

But straight lines produced from angles less than two right angles meet; {I. Post. 5} therefore EB , FD , if produced in the direction B , D , will meet.

Let them be produced and meet at G , and let AG be joined.

Then, since AC is equal to CE , the angle EAC is also equal to the angle AEC ; {I. 5} and the angle at C is right; therefore each of the angles EAC , AEC is half a right angle. {I. 32} For the same reason each of the angles CEB , EBC is also half a right angle; therefore the angle AEB is right.

And, since the angle EBC is half a right angle, the angle DBG is also half a right angle. {I. 15} But the angle BDG is also right, for it is equal to the angle DCE , they being alternate; {I. 29} therefore the remaining angle DGB is half a right angle; {I. 32} therefore the angle DGB is equal to the angle DBG , so that the side BD is also equal to the side GD . {I. 6} Again, since the angle

EGF is half a right angle, and the angle at F is right, for it is equal to the opposite angle, the angle at C, {I. 34} the remaining angle FEG is half a right angle; {I. 32} therefore the angle EGF is equal to the angle FEG, so that the side GF is also equal to the side EF. {I. 6} Now, since the square on EC is equal to the square on CA, the squares on EC, CA are double of the square on CA.

But the square on EA is equal to the squares on EC, CA; {I. 47} therefore the square on EA is double of the square on AC. {C. N. 1} Again, since FG is equal to EF, the square on FG is also equal to the square on FE; therefore the squares on GF, FE are double of the square on EF.

But the square on EG is equal to the squares on GF, FE; {I. 47} therefore the square on EG is double of the square on EF.

And EF is equal to CD; {I. 34} therefore the square on EG is double of the square on CD.

But the square on EA was also proved double of the square on AC; therefore the squares on AE, EG are double of the squares on AC, CD.

And the square on AG is equal to the squares on AE, EG; {I. 47} therefore the square on AG is double of the squares on AC, CD. But the squares on AD, DG are equal to the square on AG; {I. 47} therefore the squares on AD, DG are double of the squares on AC, CD.

And DG is equal to DB; therefore the squares on AD, DB are double of the squares on AC, CD.

Therefore etc. Q. E. D.

Proposition 11. To cut a given straight line so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

Let AB be the given straight line; thus it is required to cut AB so that the rectangle contained by the whole and one of the segments is equal to the square on the remaining segment.

For let the square ABDC be described on AB; {I. 46} let AC be bisected at the point E, and let BE be joined; let CA be drawn through to F, and let EF be made equal to BE; let the square FH be described on AF, and let GH be drawn through to K.

I say that AB has been cut at H so as to make the rectangle contained by AB, BH equal to the square on AH.

For, since the straight line AC has been bisected at E, and FA is added to it, the rectangle contained by CF, FA together with the square on AE is equal to the square on EF. {II. 6} But EF is equal to EB; therefore the rectangle CF, FA together with the square on AE is equal to the square on EB.

But the squares on BA, AE are equal to the square on EB, for the angle at A is right; {I. 47} therefore the rectangle CF, FA together with the square on AE is equal to the squares on BA, AE.

Let the square on AE be subtracted from each; therefore the rectangle CF, FA which remains is equal to the square on AB.

Now the rectangle CF, FA is FK, for AF is equal to FG; and the square on AB is AD; therefore FK is equal to AD.

Let AK be subtracted from each; therefore FH which remains is equal to HD.

And HD is the rectangle AB, BH, for AB is equal to BD; and FH is the square on AH; therefore the rectangle contained by AB, BH is equal to the square on HA. therefore the given straight line AB has been cut at H so as to make the rectangle contained by AB, BH equal to the square on HA. Q. E. F.

Proposition 12. In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on

the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.

Let ABC be an obtuse-angled triangle having the angle BAC obtuse, and let BD be drawn from the point B perpendicular to CA produced; I say that the square on BC is greater than the squares on BA, AC by twice the rectangle contained by CA, AD.

For, since the straight line CD has been cut at random at the point A, the square on DC is equal to the squares on CA, AD and twice the rectangle contained by CA, AD. {II. 4} Let the square on DB be added to each; therefore the squares on CD, DB are equal to the squares on CA, AD, DB and twice the rectangle CA, AD.

But the square on CB is equal to the squares on CD, DB, for the angle at D is right; {I. 47} and the square on AB is equal to the squares on AD, DB; {I. 47} therefore the square on CB is equal to the squares on CA, AB and twice the rectangle contained by CA, AD; so that the square on CB is greater than the squares on CA, AB by twice the rectangle contained by CA, AD.

Therefore etc. Q. E. D.

Proposition 13. In acute-angled triangles the square on the side subtending the acute angle is less than the squares on the sides containing the acute angle by twice the rectangle contained by one of the sides about the acute angle, namely that on which the perpendicular falls, and the straight line cut off within by the perpendicular towards the acute angle.

Let ABC be an acute-angled triangle having the angle at B acute, and let AD be drawn from the point A perpendicular to BC; I say that the square on AC is less than the squares on CB, BA by twice the rectangle contained by CB, BD.

For, since the straight line CB has been cut at random at D, the squares on CB, BD are equal to twice the rectangle contained by CB, BD and the square on DC. {II. 7} Let the square on DA be added to each; therefore the squares on CB, BD, DA are equal to twice the rectangle contained by CB, BD and the squares on AD, DC.

But the square on AB is equal to the squares on BD, DA, for the angle at D is right; {I. 47} and the square on AC is equal to the squares on AD, DC; therefore the squares on CB, BA are equal to the square on AC and twice the rectangle CB, BD, so that the square on AC alone is less than the squares on CB, BA by twice the rectangle contained by CB, BD.

Therefore etc. Q. E. D.

Proposition 14. To construct a square equal to a given rectilineal figure.

Let A be the given rectilineal figure; thus it is required to construct a square equal to the rectilineal figure A.

For let there be constructed the rectangular parallelogram BD equal to the rectilineal figure A. {I. 45} Then, if BE is equal to ED, that which was enjoined will have been done; for a square BD has been constructed equal to the rectilineal figure A.

But, if not, one of the straight lines BE, ED is greater.

Let BE be greater, and let it be produced to F; let EF be made equal to ED, and let BF be bisected at G.

With centre G and distance one of the straight lines GB, GF let the semicircle BHF be described; let DE be produced to H, and let GH be joined.

Then, since the straight line BF has been cut into equal segments at G, and into unequal segments at E, the rectangle contained by BE, EF together with the square on EG is equal to

the square on GF. {II. 5} But GF is equal to GH; therefore the rectangle BE, EF together with the square on GE is equal to the square on GH.

But the squares on HE, EG are equal to the square on GH; {I. 47} therefore the rectangle BE, EF together with the square on GE is equal to the squares on HE, EG.

Let the square on GE be subtracted from each; therefore the rectangle contained by BE, EF which remains is equal to the square on EH.

But the rectangle BE, EF is BD, for EF is equal to ED; therefore the parallelogram BD is equal to the square on HE.

And BD is equal to the rectilinear figure A.

Therefore the rectilinear figure A is also equal to the square which can be described on EH.

Therefore a square, namely that which can be described on EH, has been constructed equal to the given rectilinear figure A. Q. E. F.

ENDNOTES.

the rectangle A, BC. From this point onward I shall translate thus in cases where Euclid leaves out the word contained (περιεχόμενον). Though the word “rectangle” is also omitted in the Greek (the neuter article being sufficient to show that the rectangle is meant), it cannot be dispensed with in English. De Morgan advises the use of the expression “the rectangle under two lines.” This does not seem to me a very good expression, and, if used in a translation from the Greek, it might suggest that ὑπό in τὸ ὑπό meant under, which it does not.

twice the rectangle contained by the segments. By a curious idiom this is in Greek “the rectangle twice contained by the segments.” Similarly “twice the rectangle contained by AC, CB”

is expressed as “the rectangle twice contained by AC, CB” (τὸ δις ὑπὸ τῶν ΑΓ, ΒΒ περιεχόμενον ὀρθογ<*>ώνιον).

described. 39, 45. the squares (before “on”). These words are not in the Greek, which simply says that the squares “are on” (εἶσιν ἀπὸ) their respective sides.

areas. It is necessary to supply some substantive (the Greek leaves it to be understood); and I prefer “areas” to “figures.”

between the points of section, literally “between the sections,” the word being the same (τομή) as that used of a conic section.

It will be observed that the gnomon is indicated in the figure by three separate letters and a dotted curve. This is no doubt a clearer way of showing what exactly the gnomon is than the method usual in our text-books. In this particular case the figure of the MSS. has two M’s in it, the gnomon being MNE. I have corrected the lettering to avoid confusion.

that which was enjoined will have been done, literally “would have been done,” γεγονὸς ἂν εἴη τὸ ἐπιταχθέν.

which can be described, expressed by the future passive participle, ἀναγραφησομένῳ, ἀναγραφησόμενον.

BOOK III. DEFINITIONS.

1. EQUAL CIRCLES ARE those the diameters of which are equal, or the radii of which are equal.
2. A straight line is said to touch a circle which, meeting the circle and being produced, does not cut the circle.
3. Circles are said to touch one another which, meeting one another, do not cut one another.
4. In a circle straight lines are said to be equally distant from the centre when the perpendiculars drawn to them from the centre are equal.

5. And that straight line is said to be at a greater distance on which the greater perpendicular falls.
6. A segment of a circle is the figure contained by a straight line and a circumference of a circle.
7. An angle of a segment is that contained by a straight line and a circumference of a circle.
8. An angle in a segment is the angle which, when a point is taken on the circumference of the segment and straight lines are joined from it to the extremities of the straight line which is the base of the segment, is contained by the straight lines so joined.
9. And, when the straight lines containing the angle cut off a circumference, the angle is said to stand upon that circumference.
10. A sector of a circle is the figure which, when an angle is constructed at the centre of the circle, is contained by the straight lines containing the angle and the circumference cut off by them.
11. Similar segments of circles are those which admit equal angles, or in which the angles are equal to one another.

PROPOSITIONS.

PROPOSITION 1. TO FIND THE centre of a given circle.

Let ABC be the given circle; thus it is required to find the centre of the circle ABC.

Let a straight line AB be drawn through it at random, and let it be bisected at the point D; from D let DC be drawn at right angles to AB and let it be drawn through to E; let CE be bisected at F; I say that F is the centre of the circle ABC.

For suppose it is not, but, if possible, let G be the centre, and let GA , GD , GB be joined.

Then, since AD is equal to DB , and DG is common, the two sides AD , DG are equal to the two sides BD , DG respectively; and the base GA is equal to the base GB , for they are radii; therefore the angle ADG is equal to the angle GDB . {I. 8} But, when a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is right; {I. Def. 10} therefore the angle GDB is right.

But the angle FDB is also right; therefore the angle FDB is equal to the angle GDB , the greater to the less: which is impossible.

Therefore G is not the centre of the circle ABC .

Similarly we can prove that neither is any other point except F .

Therefore the point F is the centre of the circle ABC .

PORISM.

From this it is manifest that, if in a circle a straight line cut a straight line into two equal parts and at right angles, the centre of the circle is on the cutting straight line. Q. E. F.

PROPOSITION 2. If on the circumference of a circle two points be taken at random, the straight line joining the points will fall within the circle.

Let ABC be a circle, and let two points A , B be taken at random on its circumference; I say that the straight line joined from A to B will fall within the circle.

For suppose it does not, but, if possible, let it fall outside, as AEB ; let the centre of the circle ABC be taken {III. 1}, and let it be D ; let DA , DB be joined, and let DFE be drawn through.

Then, since DA is equal to DB , the angle DAE is also equal to the angle DBE . {I. 5} And, since one side AEB of the triangle DAE is

produced, the angle DEB is greater than the angle DAE. {I. 16} But the angle DAE is equal to the angle DBE; therefore the angle DEB is greater than the angle DBE.

And the greater angle is subtended by the greater side; {I. 19} therefore DB is greater than DE. But DB is equal to DF; therefore DF is greater than DE, the less than the greater : which is impossible.

Therefore the straight line joined from A to B will not fall outside the circle.

Similarly we can prove that neither will it fall on the circumference itself; therefore it will fall within.

Therefore etc. Q. E. D.

PROPOSITION 3. If in a circle a straight line through the centre bisect a straight line not through the centre, it also cuts it at right angles; and if it cut it at right angles, it also bisects it.

Let ABC be a circle, and in it let a straight line CD through the centre bisect a straight line AB not through the centre at the point F; I say that it also cuts it at right angles.

For let the centre of the circle ABC be taken, and let it be E; let EA, EB be joined.

Then, since AF is equal to FB, and FE is common, two sides are equal to two sides; and the base EA is equal to the base EB; therefore the angle AFE is equal to the angle BFE. {I. 8} But, when a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is right; {I. Def. 10} therefore each of the angles AFE, BFE is right.

Therefore CD, which is through the centre, and bisects AB which is not through the centre, also cuts it at right angles.

Again, let CD cut AB at right angles; I say that it also bisects it. that is, that AF is equal to FB.

For, with the same construction, since EA is equal to EB, the angle EAF is also equal to the angle EBF. {I. 5} But the right angle AFE is equal to the right angle BFE, therefore EAF, EBF are two triangles having two angles equal to two angles and one side equal to one side, namely EF, which is common to them, and subtends one of the equal angles; therefore they will also have the remaining sides equal to the remaining sides; {I. 26} therefore AF is equal to FB.

Therefore etc. Q. E. D.

PROPOSITION 4. If in a circle two straight lines cut one another which are not through the centre, they do not bisect one another.

Let ABCD be a circle, and in it let the two straight lines AC, BD, which are not through the centre, cut one another at E; I say that they do not bisect one another.

For, if possible, let them bisect one another, so that AE is equal to EC, and BE to ED; let the centre of the circle ABCD be taken {III. 1}, and let it be F; let FE be joined.

Then, since a straight line FE through the centre bisects a straight line AC not through the centre, it also cuts it at right angles; {III. 3} therefore the angle FEA is right.

Again, since a straight line FE bisects a straight line BD, it also cuts it at right angles; {III. 3} therefore the angle FEB is right.

But the angle FEA was also proved right; therefore the angle FEA is equal to the angle FEB, the less to the greater: which is impossible.

Therefore AC, BD do not bisect one another.

Therefore etc. Q. E. D.

PROPOSITION 5. If two circles cut one another, they will not have the same centre.

For let the circles ABC, CDG cut one another at the points B, C; I say that they will not have the same centre.

For, if possible, let it be E; let EC be joined, and let EFG be drawn through at random.

Then, since the point E is the centre of the circle ABC, EC is equal to EF. {I. Def. 15} Again, since the point E is the centre of the circle CDG, EC is equal to EG.

But EC was proved equal to EF also; therefore EF is also equal to EG, the less to the greater : which is impossible.

Therefore the point E is not the centre of the circles ABC, CDG.

Therefore etc. Q. E. D.

PROPOSITION 6. If two circles touch one another, they will not have the same centre.

For let the two circles ABC, CDE touch one another at the point C; I say that they will not have the same centre.

For, if possible, let it be F; let FC be joined, and let FEB be drawn through at random.

Then, since the point F is the centre of the circle ABC, FC is equal to FB.

Again, since the point F is the centre of the circle CDE, FC is equal to FE.

But FC was proved equal to FB; therefore FE is also equal to FB, the less to the greater: which is impossible.

Therefore F is not the centre of the circles ABC, CDE.

Therefore etc. Q. E. D.

PROPOSITION 7. If on the diameter of a circle a point be taken which is not the centre of the circle, and from the point straight lines fall upon the circle, that will be greatest on which the

centre is, the remainder of the same diameter will be least, and of the rest the nearer to the straight line through the centre is always greater than the more remote, and only two equal straight lines will fall from the point on the circle, one on each side of the least straight line.

Let ABCD be a circle, and let AD be a diameter of it; on AD let a point F be taken which is not the centre of the circle, let E be the centre of the circle, and from F let straight lines FB, FC, FG fall upon the circle ABCD; I say that FA is greatest, FD is least, and of the rest FB is greater than FC, and FC than FG.

For let BE, CE, GE be joined.

Then, since in any triangle two sides are greater than the remaining one, {I. 20} Eb, ef are greater than BF.

But AE is equal to BE; therefore AF is greater than BF.

Again, since BE is equal to CE, and FE is common, the two sides BE, EF are equal to the two sides CE, EF.

But the angle BEF is also greater than the angle CEF; therefore the base BF is greater than the base CF. {I. 24} For the same reason CF is also greater than FG.

Again, since GF, FE are greater than EG, and EG is equal to ED, Gf, fe are greater than ED.

Let EF be subtracted from each; therefore the remainder GF is greater than the remainder FD.

Therefore FA is greatest, FD is least, and FB is greater than FC, and FC than FG.

I say also that from the point F only two equal straight lines will fall on the circle ABCD, one on each side of the least FD.

For on the straight line EF, and at the point E on it, let the angle FEH be constructed equal to the angle GEF {I. 23}, and let FH be

joined.

Then, since GE is equal to EH, and EF is common, the two sides GE, EF are equal to the two sides HE, EF; and the angle GEF is equal to the angle HEF; therefore the base FG is equal to the base FH. {I. 4} I say again that another straight line equal to FG will not fall on the circle from the point F.

For, if possible, let FK so fall.

Then, since FK is equal to FG, and FH to FG, FK is also equal to FH, the nearer to the straight line through the centre being thus equal to the more remote: which is impossible.

Therefore another straight line equal to GF will not fall from the point F upon the circle; therefore only one straight line will so fall.

Therefore etc. Q. E. D.

PROPOSITION 8. If a point be taken outside a circle and from the point straight lines be drawn through to the circle, one of which is through the centre and the others are drawn at random, then, of the straight lines which fall on the concave circumference, that through the centre is greatest, while of the rest the nearer to that through the centre is always greater than the more remote, but, of the straight lines falling on the convex circumference, that between the point and the diameter is least, while of the rest the nearer to the least is always less than the more remote, and only two equal straight lines will fall on the circle from the point, one on each side of the least.

Let ABC be a circle, and let a point D be taken outside ABC; let there be drawn through from it straight lines DA, DE, DF, DC, and let DA be through the centre; I say that, of the straight lines falling on the concave circumference AEFC, the straight line DA through the centre is greatest, while DE is greater than DF and DF than DC.; but, of the straight lines falling on the convex

circumference HLKG, the straight line DG between the point and the diameter AG is least; and the nearer to the least DG is always less than the more remote, namely DK than DL, and DL than DH.

For let the centre of the circle ABC be taken {III. 1}, and let it be M; let ME, MF, MC, MK, ML, MH be joined.

Then, since AM is equal to EM, let MD be added to each; therefore AD is equal to EM, MD.

But EM, MD are greater than ED; {I. 20} therefore AD is also greater than ED.

Again, since ME is equal to MF, and MD is common, therefore EM, MD are equal to FM, MD; and the angle EMD is greater than the angle FMD; therefore the base ED is greater than the base FD. {I. 24} Similarly we can prove that FD is greater than CD; therefore DA is greatest, while DE is greater than DF, and DF than DC.

Next, since MK, KD are greater than MD, {I. 20} and MG is equal to MK, therefore the remainder KD is greater than the remainder GD, so that GD is less than KD.

And, since on MD, one of the sides of the triangle MLD, two straight lines MK, KD were constructed meeting within the triangle, therefore MK, KD are less than ML, LD; {I. 21} and MK is equal to ML; therefore the remainder DK is less than the remainder DL.

Similarly we can prove that DL is also less than DH; therefore DG is least, while DK is less than DL, and DL than DH.

I say also that only two equal straight lines will fall from the point D on the circle, one on each side of the least DG.

On the straight line MD, and at the point M on it, let the angle DMB be constructed equal to the angle KMD, and let DB be joined.

Then, since MK is equal to MB, and MD is common, the two sides KM, MD are equal to the two sides BM, MD respectively; and the angle KMD is equal to the angle BMD; therefore the base DK is equal to the base DB. {I. 4} I say that no other straight line equal to the straight line DK will fall on the circle from the point D.

For, if possible, let a straight line so fall, and let it be DN.

Then, since DK is equal to DN, while DK is equal to DB, DB is also equal to DN, that is, the nearer to the least DG equal to the more remote: which was proved impossible.

Therefore no more than two equal straight lines will fall on the circle ABC from the point D, one on each side of DG the least.

Therefore etc.

PROPOSITION 9. If a point be taken within a circle, and more than two equal straight lines fall from the point on the circle, the point taken is the centre of the circle.

Let ABC be a circle and D a point within it, and from D let more than two equal straight lines, namely DA, DB, DC, fall on the circle ABC; I say that the point D is the centre of the circle ABC.

For let AB, BC be joined and bisected at the points E, F, and let ED, FD be joined and drawn through to the points G, K, H, L.

Then, since AE is equal to EB, and ED is common, the two sides AE, ED are equal to the two sides BE, ED; and the base DA is equal to the base DB; therefore the angle AED is equal to the angle BED. {I. 8} Therefore each of the angles AED, BED is right; {I. Def. 10} therefore GK cuts AB into two equal parts and at right angles.

And since, if in a circle a straight line cut a straight line into two equal parts and at right angles, the centre of the circle is on the cutting straight line, {III. 1, Por.} the centre of the circle is on GK.

For the same reason the centre of the circle ABC is also on HL.

And the straight lines GK, HL have no other point common but the point D; therefore the point D is the centre of the circle ABC.

Therefore etc. Q. E. D.

PROPOSITION 10. A circle does not cut a circle at more points than two.

For, if possible, let the circle ABC cut the circle DEF at more points than two, namely B, C, F, H; let BH, BG be joined and bisected at the points K, L, and from K, L let KC, LM be drawn at right angles to BH, BG and carried through to the points A, E.

Then, since in the circle ABC a straight line AC cuts a straight line BH into two equal parts and at right angles, the centre of the circle ABC is on AC. {III. 1, Por.} Again, since in the same circle ABC a straight line NO cuts a straight line BG into two equal parts and at right angles, the centre of the circle ABC is on NO.

But it was also proved to be on AC, and the straight lines AC, NO meet at no point except at P; therefore the point P is the centre of the circle ABC.

Similarly we can prove that P is also the centre of the circle DEF; therefore the two circles ABC, DEF which cut one another have the same centre P: which is impossible. {III. 5} Therefore etc. Q. E. D.

PROPOSITION 11. If two circles touch one another internally, and their centres be taken, the straight line joining their centres, if it be also produced, will fall on the point of contact of the circles.

For let the two circles ABC, ADE touch one another internally at the point A, and let the centre F of the circle ABC, and the centre G of ADE, be taken; I say that the straight line joined from

G to F and produced will fall on A.

For suppose it does not, but, if possible, let it fall as FGH, and let AF, AG be joined.

Then, since AG, GF are greater than FA, that is, than FH, let FG be subtracted from each; therefore the remainder AG is greater than the remainder GH.

But AG is equal to GD; therefore GD is also greater than GH, the less than the greater: which is impossible.

Therefore the straight line joined from F to G will not fall outside; therefore it will fall at A on the point of contact.

Therefore etc. Q. E. D.

PROPOSITION 12. If two circles touch one another externally, the straight line joining their centres will pass through the point of contact.

For let the two circles ABC, ADE touch one another externally at the point A, and let the centre F of ABC, and the centre G of ADE, be taken; I say that the straight line joined from F to G will pass through the point of contact at A.

For suppose it does not, but, if possible, let it pass as FCDG, and let AF, AG be joined.

Then, since the point F is the centre of the circle ABC, FA is equal to FC.

Again, since the point G is the centre of the circle ADE, GA is equal to GD.

But FA was also proved equal to FC; therefore FA, AG are equal to FC, GD, so that the whole FG is greater than FA, AG; but it is also less {I. 20}: which is impossible.

Therefore the straight line joined from F to G will not fail to pass through the point of contact at A; therefore it will pass

through it.

Therefore etc. Q. E. D.} **PROPOSITION 13.** A circle does not touch a circle at more points than one, whether it touch it internally or externally.

For, if possible, let the circle ABDC touch the circle EBFD, first internally, at more points than one, namely D, B.

Let the centre G of the circle ABDC, and the centre H of EBFD, be taken.

Therefore the straight line joined from G to H will fall on B, D. {III. 11} Let it so fall, as BGHD.

Then, since the point G is the centre of the circle ABCD, BG is equal to GD; therefore BG is greater than HD; therefore BH is much greater than HD.

Again, since the point H is the centre of the circle EBFD, BH is equal to HD; but it was also proved much greater than it: which is impossible.

Therefore a circle does not touch a circle internally at more points than one.

I say further that neither does it so touch it externally.

For, if possible, let the circle ACK touch the circle ABDC at more points than one, namely A, C, and let AC be joined.

Then, since on the circumference of each of the circles ABDC, ACK two points A, C have been taken at random, the straight line joining the points will fall within each circle; {III. 2} but it fell within the circle ABCD and outside ACK {III. Def. 3}: which is absurd.

Therefore a circle does not touch a circle externally at more points than one.

And it was proved that neither does it so touch it internally.

Therefore etc. Q. E. D.

PROPOSITION 14. In a circle equal straight lines are equally distant from the centre, and those which are equally distant from the centre are equal to one another.

Let ABDC be a circle, and let AB, CD be equal straight lines in it; I say that AB, CD are equally distant from the centre.

For let the centre of the circle ABDC be taken {III. 1}, and let it be E; from E let EF, EG be drawn perpendicular to AB, CD, and let AE, EC be joined.

Then, since a straight line EF through the centre cuts a straight line AB not through the centre at right angles, it also bisects it. {III. 3} Therefore AF is equal to FB; therefore AB is double of AF.

For the same reason CD is also double of CG; and AB is equal to CD; therefore AF is also equal to CG.

And, since AE is equal to EC, the square on AE is also equal to the square on EC.

But the squares on AF, EF are equal to the square on AE, for the angle at F is right; and the squares on EG, GC are equal to the square on EC, for the angle at G is right; {I. 47} therefore the squares on AF, FE are equal to the squares on CG, GE, of which the square on AF is equal to the square on CG, for AF is equal to CG; therefore the square on FE which remains is equal to the square on EG, therefore EF is equal to EG.

But in a circle straight lines are said to be equally distant from the centre when the perpendiculars drawn to them from the centre are equal; {III. Def. 4} therefore AB, CD are equally distant from the centre.

Next, let the straight lines AB, CD be equally distant from the centre; that is, let EF be equal to EG.

I say that AB is also equal to CD.

For, with the same construction, we can prove, similarly, that AB is double of AF, and CD of CG.

And, since AE is equal to CE, the square on AE is equal to the square on CE.

But the squares on EF, FA are equal to the square on AE, and the squares on EG, GC equal to the square on CE. {I. 47} Therefore the squares on EF, FA are equal to the squares on EG, GC, of which the square on EF is equal to the square on EG, for EF is equal to EG; therefore the square on AF which remains is equal to the square on CG; therefore AF is equal to CG.

And AB is double of AF, and CD double of CG; therefore AB is equal to CD.

Therefore etc. Q. E. D.

PROPOSITION 15. Of straight lines in a circle the diameter is greatest, and of the rest the nearer to the centre is always greater than the more remote.

Let ABCD be a circle, let AD be its diameter and E the centre; and let BC be nearer to the diameter AD, and FG more remote; I say that AD is greatest and BC greater than FG.

For from the centre E let EH, EK be drawn perpendicular to BC, FG.

Then, since BC is nearer to the centre and FG more remote, EK is greater than EH. {III. Def. 5} Let EL be made equal to EH, through L let LM be drawn at right angles to EK and carried through to N, and let ME, EN, FE, EG be joined.

Then, since EH is equal to EL, BC is also equal to MN. {III. 14} Again, since AE is equal to EM, and ED to EN, AD is equal to ME, EN.

But ME, EN are greater than MN, {I. 20} and MN is equal to BC; therefore AD is greater than BC.

And, since the two sides ME, EN are equal to the two sides FE, EG, and the angle MEN greater than the angle FEG, therefore the base MN is greater than the base FG. {I. 24} But MN was proved equal to BC.

Therefore the diameter AD is greatest and BC greater than FG.

Therefore etc. Q. E. D.

PROPOSITION 16. The straight line drawn at right angles to the diameter of a circle from its extremity will fall outside the circle, and into the space between the straight line and the circumference another straight line cannot be interposed; further the angle of the semicircle is greater, and the remaining angle less, than any acute rectilineal angle.

Let ABC be a circle about D as centre and AB as diameter; I say that the straight line drawn from A at right angles to AB from its extremity will fall outside the circle.

For suppose it does not, but, if possible, let it fall within as CA, and let DC be joined.

Since DA is equal to DC, the angle DAC is also equal to the angle ACD. {I. 5} But the angle DAC is right; therefore the angle ACD is also right: thus, in the triangle ACD, the two angles DAC, ACD are equal to two right angles: which is impossible. {I. 17} Therefore the straight line drawn from the point A at right angles to BA will not fall within the circle.

Similarly we can prove that neither will it fall on the circumference; therefore it will fall outside.

Let it fall as AE; I say next that into the space between the straight line AE and the circumference CHA another straight line cannot be interposed.

For, if possible, let another straight line be so interposed, as FA, and let DG be drawn from the point D perpendicular to FA.

Then, since the angle AGD is right, and the angle DAG is less than a right angle, AD is greater than DG. {I. 19} But DA is equal to DH; therefore DH is greater than DG, the less than the greater: which is impossible.

Therefore another straight line cannot be interposed into the space between the straight line and the circumference.

I say further that the angle of the semicircle contained by the straight line BA and the circumference CHA is greater than any acute rectilineal angle, and the remaining angle contained by the circumference CHA and the straight line AE is less than any acute rectilineal angle.

For, if there is any rectilineal angle greater than the angle contained by the straight line BA and the circumference CHA, and any rectilineal angle less than the angle contained by the circumference CHA and the straight line AE, then into the space between the circumference and the straight line AE a straight line will be interposed such as will make an angle contained by straight lines which is greater than the angle contained by the straight line BA and the circumference CHA, and another angle contained by straight lines which is less than the angle contained by the circumference CHA and the straight line AE.

But such a straight line cannot be interposed; therefore there will not be any acute angle contained by straight lines which is greater than the angle contained by the straight line BA and the circumference CHA, nor yet any acute angle contained by straight lines which is less than the angle contained by the circumference CHA and the straight line AE. — PORISM.

From this it is manifest that the straight line drawn at right angles to the diameter of a circle from its extremity touches the circle. Q. E. D.

PROPOSITION 17. From a given point to draw a straight line touching a given circle.

Let A be the given point, and BCD the given circle; thus it is required to draw from the point A a straight line touching the circle BCD.

For let the centre E of the circle be taken; {III. 1} let AE be joined, and with centre E and distance EA let the circle AFG be described; from D let DF be drawn at right angles to EA, and let EF, AB be joined; I say that AB has been drawn from the point A touching the circle BCD.

For, since E is the centre of the circles BCD, AFG, EA is equal to EF, and ED to EB; therefore the two sides AE, EB are equal to the two sides FE, ED: and they contain a common angle, the angle at E; therefore the base DF is equal to the base AB, and the triangle DEF is equal to the triangle BEA, and the remaining angles to the remaining angles; {I. 4} therefore the angle EDF is equal to the angle EBA.

But the angle EDF is right; therefore the angle EBA is also right.

Now EB is a radius; and the straight line drawn at right angles to the diameter of a circle, from its extremity, touches the circle; {III. 16, Por.} therefore AB touches the circle BCD.

Therefore from the given point A the straight line AB has been drawn touching the circle BCD.

PROPOSITION 18. If a straight line touch a circle, and a straight line be joined from the centre to the point of contact, the straight line so joined will be perpendicular to the tangent.

For let a straight line DE touch the circle ABC at the point C, let the centre F of the circle ABC be taken, and let FC be joined from F to C; I say that FC is perpendicular to DE.

For, if not, let FG be drawn from F perpendicular to DE.

Then, since the angle FGC is right, the angle FCG is acute; {I. 17} and the greater angle is subtended by the greater side; {I. 19} therefore FC is greater than FG.

But FC is equal to FB; therefore FB is also greater than FG, the less than the greater: which is impossible.

Therefore FG is not perpendicular to DE.

Similarly we can prove that neither is any other straight line except FC; therefore FC is perpendicular to DE.

Therefore etc. Q. E. D.

PROPOSITION 19. If a straight line touch a circle, and from the point of contact a straight line be drawn at right angles to the tangent, the centre of the circle will be on the straight line so drawn.

For let a straight line DE touch the circle ABC at the point C, and from C let CA be drawn at right angles to DE; I say that the centre of the circle is on AC.

For suppose it is not, but, if possible, let F be the centre, and let CF be joined.

Since a straight line DE touches the circle ABC, and FC has been joined from the centre to the point of contact, FC is perpendicular to DE; {III. 18} therefore the angle FCE is right.

But the angle ACE is also right; therefore the angle FCE is equal to the angle ACE, the less to the greater: which is impossible.

Therefore F is not the centre of the circle ABC.

Similarly we can prove that neither is any other point except a point on AC.

Therefore etc. Q. E. D.

PROPOSITION 20. In a circle the angle at the centre is double of the angle at the circumference, when the angles have the same

circumference as base.

Let ABC be a circle, let the angle BEC be an angle at its centre, and the angle BAC an angle at the circumference, and let them have the same circumference BC as base; I say that the angle BEC is double of the angle BAC.

For let AE be joined and drawn through to F.

Then, since EA is equal to EB, the angle EAB is also equal to the angle EBA; {I. 5} therefore the angles EAB, EBA are double of the angle EAB.

But the angle BEF is equal to the angles EAB, EBA; {I. 32} therefore the angle BEF is also double of the angle EAB.

For the same reason the angle FEC is also double of the angle EAC.

Therefore the whole angle BEC is double of the whole angle BAC.

Again let another straight line be inflected, and let there be another angle BDC; let DE be joined and produced to G.

Similarly then we can prove that the angle GEC is double of the angle EDC, of which the angle GEB is double of the angle EDB; therefore the angle BEC which remains is double of the angle BDC.

Therefore etc. Q. E. D.

PROPOSITION 21. In a circle the angles in the same segment are equal to one another.

Let ABCD be a circle, and let the angles BAD, BED be angles in the same segment BAED; I say that the angles BAD, BED are equal to one another.

For let the centre of the circle ABCD be taken, and let it be F; let BF, FD be joined.

Now, since the angle BFD is at the centre, and the angle BAD at the circumference, and they have the same circumference BCD as base, therefore the angle BFD is double of the angle BAD. {III. 20} For the same reason the angle BFD is also double of the angle BED; therefore the angle BAD is equal to the angle BED.

Therefore etc. Q. E. D.

PROPOSITION 22. The opposite angles of quadrilaterals in circles are equal to two right angles.

Let ABCD be a circle, and let ABCD be a quadrilateral in it; I say that the opposite angles are equal to two right angles.

Let AC, BD be joined.

Then, since in any triangle the three angles are equal to two right angles, {I. 32} the three angles CAB, ABC, BCA of the triangle ABC are equal to two right angles.

But the angle CAB is equal to the angle BDC, for they are in the same segment BADC; {III. 21} and the angle ACB is equal to the angle ADB, for they are in the same segment ADCB; therefore the whole angle ADC is equal to the angles BAC, ACB.

Let the angle ABC be added to each; therefore the angles ABC, BAC, ACB are equal to the angles ABC, ADC. But the angles ABC, BAC, ACB are equal to two right angles; therefore the angles ABC, ADC are also equal to two right angles.

Similarly we can prove that the angles BAD, DCB are also equal to two right angles.

Therefore etc. Q. E. D.

PROPOSITION 23. On the same straight line there cannot be constructed two similar and unequal segments of circles on the same side.

For, if possible, on the same straight line AB let two similar and unequal segments of circles ACB, ADB be constructed on the same side; let ACD be drawn through, and let CB, DB be joined.

Then, since the segment ACB is similar to the segment ADB, and similar segments of circles are those which admit equal angles, {III. Def. 11} the angle ACB is equal to the angle ADB, the exterior to the interior: which is impossible. {I. 16} Therefore etc. Q. E. D.

PROPOSITION 24. Similar segments of circles on equal straight lines are equal to one another.

For let AEB, CFD be similar segments of circles on equal straight lines AB, CD; I say that the segment AEB is equal to the segment CFD.

For, if the segment AEB be applied to CFD, and if the point A be placed on C and the straight line AB on CD, the point B will also coincide with the point D, because AB is equal to CD; and, AB coinciding with CD, the segment AEB will also coincide with CFD.

For, if the straight line AB coincide with CD but the segment AEB do not coincide with CFD, it will either fall with it, or outside it; or it will fall awry, as CGD, and a circle cuts a circle at more points than two : which is impossible. {III. 10} Therefore, if the straight line AB be applied to CD, the segment AEB will not fail to coincide with CFD also; therefore it will coincide with it and will be equal to it.

Therefore etc. Q. E. D.

PROPOSITION 25. Given a segment of a circle, to describe the complete circle of which it is a segment.

Let ABC be the given segment of a circle; thus it is required to describe the complete circle belonging to the segment ABC, that is, of which it is a segment.

For let AC be bisected at D, let DB be drawn from the point D at right angles to AC, and let AB. be joined; the angle ABD is then greater than, equal to, or less than the angle BAD.

First let it be greater; and on the straight line BA, and at the point A on it, let the angle BAE be constructed equal to the angle ABD; let DB be drawn through to E, and let EC be joined.

Then, since the angle ABE is equal to the angle BAE, the straight line EB is also equal to EA. {I. 6} And, since AD is equal to DC, and DE is common, the two sides AD, DE are equal to the two sides CD, DE respectively; and the angle ADE is equal to the angle CDE, for each is right; therefore the base AE is equal to the base CE.

But AE was proved equal to BE; therefore BE is also equal to CE; therefore the three straight lines AE, EB, EC are equal to one another.

Therefore the circle drawn with centre E and distance one of the straight lines AE, EB, EC will also pass through the remaining points and will have been completed. {III. 9} Therefore, given a segment of a circle, the complete circle has been described.

And it is manifest that the segment ABC is less than a semicircle, because the centre E happens to be outside it.

Similarly, even if the angle ABD be equal to the angle BAD, AD being equal to each of the two BD, DC, the three straight lines DA, DB, DC will be equal to one another, D will be the centre of the completed circle, and ABC will clearly be a semicircle.

But, if the angle ABD be less than the angle BAD, and if we construct, on the straight line BA and at the point A on it, an angle equal to the angle ABD, the centre will fall on DB within the segment ABC, and the segment ABC will clearly be greater than a semicircle.

Therefore, given a segment of a circle, the complete circle has been described. Q. E. F.

PROPOSITION 26. In equal circles equal angles stand on equal circumferences, whether they stand at the centres or at the circumferences.

Let ABC, DEF be equal circles, and in them let there be equal angles, namely at the centres the angles BGC, EHF, and at the circumferences the angles BAC, EDF; I say that the circumference BKC is equal to the circumference ELF.

For let BC, EF be joined.

Now, since the circles ABC, DEF are equal, the radii are equal.

Thus the two straight lines BG, GC are equal to the two straight lines EH, HF; and the angle at G is equal to the angle at H; therefore the base BC is equal to the base EF. {I. 4} And, since the angle at A is equal to the angle at D, the segment BAC is similar to the segment EDF; {III. Def. 11} and they are upon equal straight lines.

But similar segments of circles on equal straight lines are equal to one another; {III. 24} therefore the segment BAC is equal to EDF.

But the whole circle ABC is also equal to the whole circle DEF; therefore the circumference BKC which remains is equal to the circumference ELF.

Therefore etc. Q. E. D.

PROPOSITION 27. In equal circles angles standing on equal circumferences are equal to one another, whether they stand at the centres or at the circumferences.

For in equal circles ABC, DEF, on equal circumferences BC, EF, let the angles BGC, EHF stand at the centres G, H, and the angles BAC, EDF at the circumferences; I say that the angle BGC is equal

to the angle EHF, and the angle BAC is equal to the angle EDF.

For, if the angle BGC is unequal to the angle EHF, one of them is greater.

Let the angle BGC be greater : and on the straight line BG, and at the point G on it, let the angle BGK be constructed equal to the angle EHF. {I. 23} Now equal angles stand on equal circumferences, when they are at the centres; {III. 26} therefore the circumference BK is equal to the circumference EF.

But EF is equal to BC; therefore BK is also equal to BC, the less to the greater : which is impossible.

Therefore the angle BGC is not unequal to the angle EHF; therefore it is equal to it.

And the angle at A is half of the angle BGC, and the angle at D half of the angle EHF; {III. 20} therefore the angle at A is also equal to the angle at D.

Therefore etc. Q. E. D.

PROPOSITION 28. In equal circles equal straight lines cut off equal circumferences, the greater equal to the greater and the less to the less.

Let ABC, DEF be equal circles, and in the circles let AB, DE be equal straight lines cutting off ACB, DFE as greater circumferences and AGB, DHE as lesser; I say that the greater circumference ACB is equal to the greater circumference DFE, and the less circumference AGB to DHE.

For let the centres K, L of the circles be taken, and let AK, KB, DL, LE be joined.

Now, since the circles are equal, the radii are also equal; therefore the two sides AK, KB are equal to the two sides DL, LE; and the base AB is equal to the base DE; therefore the angle AKB is equal to the angle DLE. {I. 8} But equal angles stand on equal

circumferences, when they are at the centres; {III. 26} therefore the circumference AGB is equal to DHE.

And the whole circle ABC is also equal to the whole circle DEF; therefore the circumference ACB which remains is also equal to the circumference DFE which remains.

Therefore etc. Q. E. D.

PROPOSITION 29. In equal circles equal circumferences are subtended by equal straight lines.

Let ABC, DEF be equal circles, and in them let equal circumferences BGC, EHF be cut off; and let the straight lines BC, EF be joined; I say that BC is equal to EF.

For let the centres of the circles be taken, and let them be K, L; let BK, KC, EL, LF be joined.

Now, since the circumference BGC is equal to the circumference EHF, the angle BKC is also equal to the angle ELF. {III. 27} And, since the circles ABC, DEF are equal, the radii are also equal; therefore the two sides BK, KC are equal to the two sides EL, LF; and they contain equal angles; therefore the base BC is equal to the base EF. {I. 4} Therefore etc.

PROPOSITION 30. To bisect a given circumference.

Let ADB be the given circumference; thus it is required to bisect the circumference ADB.

Let AB be joined and bisected at C; from the point C let CD be drawn at right angles to the straight line AB, and let AD, DB be joined.

Then, since AC is equal to CB, and CD is common, the two sides AC, CD are equal to the two sides BC, CD; and the angle ACD is equal to the angle BCD, for each is right; therefore the base AD is equal to the base DB. {I. 4} But equal straight lines cut off equal circumferences, the greater equal to the greater, and the less to

the less; {III. 28} and each of the circumferences AD, DB is less than a semicircle; therefore the circumference AD is equal to the circumference DB.

Therefore the given circumference has been bisected at the point D. Q. E. F.

PROPOSITION 31. In a circle the angle in the semicircle is right, that in a greater segment less than a right angle, and that in a less segment greater than a right angle; and further the angle of the greater segment is greater than a right angle, and the angle of the less segment less than a right angle.

Let ABCD be a circle, let BC be its diameter, and E its centre, and let BA, AC, AD, DC be joined; I say that the angle BAC in the semicircle BAC is right, the angle ABC in the segment ABC greater than the semicircle is less than a right angle, and the angle ADC in the segment ADC less than the semicircle is greater than a right angle.

Let AE be joined, and let BA be carried through to F.

Then, since BE is equal to EA, the angle ABE is also equal to the angle BAE. {I. 5} Again, since CE is equal to EA, the angle ACE is also equal to the angle CAE. {I. 5} Therefore the whole angle BAC is equal to the two angles ABC, ACB.

But the angle FAC exterior to the triangle ABC is also equal to the two angles ABC, ACB; {I. 32} therefore the angle BAC is also equal to the angle FAC; therefore each is right; {I. Def. 10} therefore the angle BAC in the semicircle BAC is right.

Next, since in the triangle ABC the two angles ABC, BAC are less than two right angles, {I. 17} and the angle BAC is a right angle, the angle ABC is less than a right angle; and it is the angle in the segment ABC greater than the semicircle.

Next, since ABCD is a quadrilateral in a circle, and the opposite angles of quadrilaterals in circles are equal to two right angles,

{III. 22} while the angle ABC is less than a right angle, therefore the angle ADC which remains is greater than a right angle; and it is the angle in the segment ADC less than the semicircle.

I say further that the angle of the greater segment, namely that contained by the circumference ABC and the straight line AC, is greater than a right angle; and the angle of the less segment, namely that contained by the circumference ADC and the straight line AC, is less than a right angle.

This is at once manifest. For, since the angle contained by the straight lines BA, AC is right, the angle contained by the circumference ABC and the straight line AC is greater than a right angle.

Again, since the angle contained by the straight lines AC, AF is right, the angle contained by the straight line CA and the circumference ADC is less than a right angle.

Therefore etc. Q. E. D.

PROPOSITION 32. If a straight line touch a circle, and from the point of contact there be drawn across, in the circle, a straight line cutting the circle, the angles which it makes with the tangent will be equal to the angles in the alternate segments of the circle.

For let a straight line EF touch the circle ABCD at the point B, and from the point B let there be drawn across, in the circle ABCD, a straight line BD cutting it; I say that the angles which BD makes with the tangent EF will be equal to the angles in the alternate segments of the circle, that is, that the angle FBD is equal to the angle constructed in the segment BAD, and the angle EBD is equal to the angle constructed in the segment DCB.

For let BA be drawn from B at right angles to EF, let a point C be taken at random on the circumference BD, and let AD, DC, CB be joined.

Then, since a straight line EF touches the circle ABCD at B, and BA has been drawn from the point of contact at right angles to the tangent, the centre of the circle ABCD is on BA. {III. 19} Therefore BA is a diameter of the circle ABCD; therefore the angle ADB, being an angle in a semicircle, is right. {III. 31} Therefore the remaining angles BAD, ABD are equal to one right angle. {I. 32} But the angle ABF is also right; therefore the angle ABF is equal to the angles BAD, ABD.

Let the angle ABD be subtracted from each; therefore the angle DBF which remains is equal to the angle BAD in the alternate segment of the circle.

Next, since ABCD is a quadrilateral in a circle, its opposite angles are equal to two right angles. {III. 22} But the angles DBF, DBE are also equal to two right angles; therefore the angles DBF, DBE are equal to the angles BAD, BCD, of which the angle BAD was proved equal to the angle DBF; therefore the angle DBE which remains is equal to the angle DCB in the alternate segment DCB of the circle.

Therefore etc. Q. E. D.

PROPOSITION 33. On a given straight line to describe a segment of a circle admitting an angle equal to a given rectilineal angle.

Let AB be the given straight line, and the angle at C the given rectilineal angle; thus it is required to describe on the given straight line AB a segment of a circle admitting an angle equal to the angle at C.

The angle at C is then acute, or right, or obtuse.

First let it be acute, and, as in the first figure, on the straight line AB, and at the point A, let the angle BAD be constructed equal to the angle at C; therefore the angle BAD is also acute.

Let AE be drawn at right angles to DA, let AB be bisected at F, let FG be drawn from the point F at right angles to AB, and let GB be joined.

Then, since AF is equal to FB, and FG is common, the two sides AF, FG are equal to the two sides BF, FG; and the angle AFG is equal to the angle BFG; therefore the base AG is equal to the base BG. {I. 4} Therefore the circle described with centre G and distance GA will pass through B also.

Let it be drawn, and let it be ABE; let EB be joined.

Now, since AD is drawn from A, the extremity of the diameter AE, at right angles to AE, therefore AD touches the circle ABE. {III. 16, Por.} Since then a straight line AD touches the circle ABE, and from the point of contact at A a straight line AB is drawn across in the circle ABE, the angle DAB is equal to the angle AEB in the alternate segment of the circle. {III. 32} But the angle DAB is equal to the angle at C; therefore the angle at C is also equal to the angle AEB.

Therefore on the given straight line AB the segment AEB of a circle has been described admitting the angle AEB equal to the given angle, the angle at C.

Next let the angle at C be right; and let it be again required to describe on AB a segment of a circle admitting an angle equal to the right angle at C.

Let the angle BAD be constructed equal to the right angle at C, as is the case in the second figure; let AB be bisected at F, and with centre F and distance either FA or FB let the circle AEB be described.

Therefore the straight line AD touches the circle ABE, because the angle at A is right. {III. 16, Por.} And the angle BAD is equal to the angle in the segment AEB, for the latter too is itself a right angle, being an angle in a semicircle. {III. 31} But the angle

BAD is also equal to the angle at C.

Therefore the angle AEB is also equal to the angle at C.

Therefore again the segment AEB of a circle has been described on AB admitting an angle equal to the angle at C.

Next, let the angle at C be obtuse; and on the straight line AB, and at the point A, let the angle BAD be constructed equal to it, as is the case in the third figure; let AE be drawn at right angles to AD, let AB be again bisected at F, let FG be drawn at right angles to AB, and let GB be joined.

Then, since AF is again equal to FB, and FG is common, the two sides AF, FG are equal to the two sides BF, FG; and the angle AFG is equal to the angle BFG; therefore the base AG is equal to the base BG. {I. 4} Therefore the circle described with centre G and distance GA will pass through B also; let it so pass, as AEB.

Now, since AD is drawn at right angles to the diameter AE from its extremity, AD touches the circle AEB. {III. 16, Por.} And AB has been drawn across from the point of contact at A; therefore the angle BAD is equal to the angle constructed in the alternate segment AHB of the circle. {III. 32} But the angle BAD is equal to the angle at C.

Therefore the angle in the segment AHB is also equal to the angle at C.

Therefore on the given straight line AB the segment AHB of a circle has been described admitting an angle equal to the angle at C. Q. E. F.

PROPOSITION 34. From a given circle to cut off a segment admitting an angle equal to a given rectilineal angle.

Let ABC be the given circle, and the angle at D the given rectilineal angle; thus it is required to cut off from the circle ABC a segment admitting an angle equal to the given rectilineal

angle, the angle at D.

Let EF be drawn touching ABC at the point B, and on the straight line FB, and at the point B on it, let the angle FBC be constructed equal to the angle at D. {I. 23} Then, since a straight line EF touches the circle ABC, and BC has been drawn across from the point of contact at B, the angle FBC is equal to the angle constructed in the alternate segment BAC. {III. 32} But the angle FBC is equal to the angle at D; therefore the angle in the segment BAC is equal to the angle at D.

Therefore from the given circle ABC the segment BAC. has been cut off admitting an angle equal to the given rectilineal angle, the angle at D. Q. E. F.

PROPOSITION 35. If in a circle two straight lines cut one another, the rectangle contained by the segments of the one is equal to the rectangle contained by the segments of the other.

For in the circle ABCD let the two straight lines AC, BD cut one another at the point E; I say that the rectangle contained by AE, EC is equal to the rectangle contained by DE, EB.

If now AC, BD are through the centre, so that E is the centre of the circle ABCD, it is manifest that, AE, EC, DE, EB being equal, the rectangle contained by AE, EC is also equal to the rectangle contained by DE, EB.

Next let AC, DB not be through the centre; let the centre of ABCD be taken, and let it be F; from F let FG, FH be drawn perpendicular to the straight lines AC, DB, and let FB, FC, FE be joined.

Then, since a straight line GF through the centre cuts a straight line AC not through the centre at right angles, it also bisects it; {III. 3} therefore AG is equal to GC.

Since, then, the straight line AC has been cut into equal parts at G and into unequal parts at E, the rectangle contained by AE, EC

together with the square on EG is equal to the square on GC; {II. 5} Let the square on GF be added; therefore the rectangle AE, EC together with the squares on GE, GF is equal to the squares on CG, GF.

But the square on FE is equal to the squares on EG, GF, and the square on FC is equal to the squares on CG, GF; {I. 47} therefore the rectangle AE, EC together with the square on FE is equal to the square on FC.

And FC is equal to FB; therefore the rectangle AE, EC together with the square on FE is equal to the square on FB.

For the same reason, also, the rectangle DE, EB together with the square on FE is equal to the square on FB.

But the rectangle AE, EC together with the square on FE was also proved equal to the square on FB; therefore the rectangle AE, EC together with the square on FE is equal to the rectangle DE, EB together with the square on FE.

Let the square on FE be subtracted from each; therefore the rectangle contained by AE, EC which remains is equal to the rectangle contained by DE, EB.

Therefore etc.

PROPOSITION 36. If a point be taken outside a circle and from it there fall on the circle two straight lines, and if one of them cut the circle and the other touch it, the rectangle contained by the whole of the straight line which cuts the circle and the straight line intercepted on it outside between the point and the convex circumference will be equal to the square on the tangent.

For let a point D be taken outside the circle ABC, and from D let the two straight lines DCA, DB fall on the circle ABC; let DCA cut the circle ABC and let DB touch it; I say that the rectangle contained by AD, DC is equal to the square on DB.

Then DCA is either through the centre or not through the centre.

First let it be through the centre, and let F be the centre of the circle ABC; let FB be joined; therefore the angle FBD is right. {III. 18} And, since AC has been bisected at F, and CD is added to it, the rectangle AD, DC together with the square on FC is equal to the square on FD. {II. 6} But FC is equal to FB; therefore the rectangle AD, DC together with the square on FB is equal to the square on FD.

And the squares on FB, BD are equal to the square on FD; {I. 47} therefore the rectangle AD, DC together with the square on FB is equal to the squares on FB, BD.

Let the square on FB be subtracted from each; therefore the rectangle AD, DC which remains is equal to the square on the tangent DB.

Again, let DCA not be through the centre of the circle ABC; let the centre E be taken, and from E let EF be drawn perpendicular to AC; let EB, EC, ED be joined.

Then the angle EBD is right. {III. 18} And, since a straight line EF through the centre cuts a straight line AC not through the centre at right angles, it also bisects it; {III. 3} therefore AF is equal to FC.

Now, since the straight line AC has been bisected at the point F, and CD is added to it, the rectangle contained by AD, DC together with the square on FC is equal to the square on FD. {II. 6} Let the square on FE be added to each; therefore the rectangle AD, DC together with the squares on CF, FE is equal to the squares on FD, FE.

But the square on EC is equal to the squares on CF, FE, for the angle EFC is right; {I. 47} and the square on ED is equal to the squares on DF, FE; therefore the rectangle AD, DC together with

the square on EC is equal to the square on ED.

And EC is equal to EB; therefore the rectangle AD, DC together with the square on EB is equal to the square on ED.

But the squares on EB, BD are equal to the square on ED, for the angle EBD is right; {I. 47} therefore the rectangle AD, DC together with the square on EB is equal to the squares on EB, BD.

Let the square on EB be subtracted from each; therefore the rectangle AD, DC which remains is equal to the square on DB.

Therefore etc. Q. E. D.

PROPOSITION 37. If a point be taken outside a circle and from the point there fall on the circle two straight lines, if one of them cut the circle, and the other fall on it, and if further the rectangle contained by the whole of the straight line which cuts the circle and the straight line intercepted on it outside between the point and the convex circumference be equal to the square on the straight line which falls on the circle, the straight line which falls on it will touch the circle.

For let a point D be taken outside the circle ABC; from D let the two straight lines DCA, DB fall on the circle ACB; let DCA cut the circle and DB fall on it; and let the rectangle AD, DC be equal to the square on DB.

I say that DB touches the circle ABC.

For let DE be drawn touching ABC; let the centre of the circle ABC be taken, and let it be F; let FE, FB, FD be joined.

Thus the angle FED is right. {III. 18} Now, since DE touches the circle ABC, and DCA cuts it, the rectangle AD, DC is equal to the square on DE. {III. 36} But the rectangle AD, DC was also equal to the square on DB; therefore the square on DE is equal to the square on DB; therefore DE is equal to DB.

And FE is equal to FB; therefore the two sides DE, EF are equal to the two sides DB, BF; and FD is the common base of the triangles; therefore the angle DEF is equal to the angle DBF. {I. 8} But the angle DEF is right; therefore the angle DBF is also right.

And FB produced is a diameter; and the straight line drawn at right angles to the diameter of a circle, from its extremity, touches the circle; {III. 16, Por.} therefore DB touches the circle.

Similarly this can be proved to be the case even if the centre be on AC.

Therefore etc. Q. E. D.

ENDNOTES.

12. For suppose it is not. This is expressed in the Greek by the two words *Μὴ γάρ*, but such an elliptical phrase is impossible in English.

the two sides AD, DG are equal to the two sides BD, DG respectively. As before observed, Euclid is not always careful to put the equals in corresponding order. The text here has “GD, DB.”

with the same construction, τῶν αὐτῶν κατασκευασθέντων.

of the same diameter. I have inserted these words for clearness’ sake. The text has simply ἐλάχιστη δὲ ἡ λοιπή, “and the remaining (straight line) least.”

one on each side. The word “one” is not in the Greek, but is necessary to give the force of ἐφ’ ἑκάτερα τῆς ἐλάχιστης, literally “on both sides,” or “on each of the two sides, of the least.”

The word circle (κύκλος) is here employed in the unusual sense of the circumference (περιφέρεια) of a circle. Cf. note on I. Def. 15.

the straight line joining their centres, literally “the straight line joined to their centres” (ἡ ἐπὶ τὰ κέντρα αὐτῶν ἐπιζευγνυμένη εὐθεῖα).

point of contact is here συναφή, and in the enunciation of the next proposition ἐπαφή.

will not fail to pass. The Greek has the double negative, οὐκ ἄρα ἡ...εὐθεῖα... οὐκ ἐλεύσεται, literally “the straight line...will not not-pass....”

ABDC. Euclid writes ABCD (here and in the next proposition), notwithstanding the order in which the points are placed in the figure.

does it so touch it. It is necessary to supply these words which the Greek (ὅτι οὐδὲ ἐκτός and ὅτι οὐδὲ ἐντός) leaves to be understood.

Of straight lines. The Greek leaves these words to be understood.

Nearer to the diameter AD. As BC, FG are not in general parallel to AD, Euclid should have said “nearer to the centre.”

cannot be interposed, literally “will not fall in between” (οὐ παρεμπεσῆται).

the tangent, ἡ ἐφαπτομένη.

let another straight line be inflected, κεκλάσθω δὴ πάλιν (without εὐθεῖα). The verb κλάω (to break off) was the regular technical term for drawing from a point a (broken) straight line which first meets another straight line or curve and is then bent back from it to another point, or (in other words) for drawing straight lines from two points meeting at a point on a curve or another straight line. κεκλάσθαι is one of the geometrical terms the definition of which must according to Aristotle be assumed (Anal. Post. I. 10, 76 b 9).

cannot be constructed, οὐ συσταθήσεται, the same phrase as in 1. 7.

fall awry, παραλλάξει, the same word as used in the like case in I. 8. The word implies that the applied figure will partly fall short of, and partly overlap, the figure to which it is applied.

to describe the complete circle, προσαναγράψαι τὸν κύκλον, literally “to describe the circle on to it.”

BOOK IV. DEFINITIONS.

1. A RECTILINEAL FIGURE is said to be inscribed in a rectilineal figure when the respective angles of the inscribed figure lie on the respective sides of that in which it is inscribed.

2. Similarly a figure is said to be circumscribed about a figure when the respective sides of the circumscribed figure pass through the respective angles of that about which it is circumscribed.

3. A rectilineal figure is said to be inscribed in a circle when each angle of the inscribed figure lies on the circumference of the circle.

4. A rectilineal figure is said to be circumscribed about a circle, when each side of the circumscribed figure touches the circumference of the circle.

5. Similarly a circle is said to be inscribed in a figure when the circumference of the circle touches each side of the figure in which it is inscribed.

6. A circle is said to be circumscribed about a figure when the circumference of the circle passes through each angle of the figure about which it is circumscribed.

7. A straight line is said to be fitted into a circle when its extremities are on the circumference of the circle.

PROPOSITIONS

PROPOSITION 1. INTO A GIVEN circle to fit a straight line equal to a given straight line which is not greater than the diameter of the circle.

Let ABC be the given circle, and D the given straight line not greater than the diameter of the circle; thus it is required to fit into the circle ABC a straight line equal to the straight line D.

Let a diameter BC of the circle ABC be drawn.

Then, if BC is equal to D, that which was enjoined will have been done; for BC has been fitted into the circle ABC equal to the straight line D.

But, if BC is greater than D, let CE be made equal to D, and with centre C and distance CE let the circle EAF be described; let CA be joined.

Then, since the point C is the centre of the circle EAF, CA is equal to CE.

But CE is equal to D; therefore D is also equal to CA.

Therefore into the given circle ABC there has been fitted CA equal to the given straight line D.

PROPOSITION 2. In a given circle to inscribe a triangle equiangular with a given triangle.

Let ABC be the given circle, and DEF the given triangle; thus it is required to inscribe in the circle ABC a triangle equiangular with the triangle DEF.

Let GH be drawn touching the circle ABC at A {III. 16, Por.}; on the straight line AH, and at the point A on it, let the angle HAC be constructed equal to the angle DEF, and on the straight line AG, and at the point A on it, let the angle GAB be constructed equal to the angle DFE; {I. 23} let BC be joined.

Then, since a straight line AH touches the circle ABC, and from the point of contact at A the straight line AC is drawn across in the circle, therefore the angle HAC is equal to the angle ABC in the alternate segment of the circle. {III. 32} But the angle HAC is equal to the angle DEF; therefore the angle ABC is also equal to the angle DEF.

For the same reason the angle ACB is also equal to the angle DFE; therefore the remaining angle BAC is also equal to the remaining angle EDF. {I. 32} Therefore in the given circle there has been inscribed a triangle equiangular with the given triangle. Q. E. F.

PROPOSITION 3. About a given circle to circumscribe a triangle equiangular with a given triangle.

Let ABC be the given circle, and DEF the given triangle; thus it is required to circumscribe about the circle ABC a triangle equiangular with the triangle DEF.

Let EF be produced in both directions to the points G, H, let the centre K of the circle ABC be taken {III. 1}, and let the straight line KB be drawn across at random; on the straight line KB, and at the point K on it, let the angle BKA be constructed equal to the angle DEG, and the angle BKC equal to the angle DFH; {I. 23} and through the points A, B, C let LAM, MBN, NCL be drawn touching the circle ABC. {III. 16, Por.} Now, since LM, MN, NL touch the circle ABC at the points A, B, C, and KA, KB, KC have been joined from the centre K to the points A, B, C, therefore the angles at the points A, B, C are right. {III. 18} And, since the four angles of the quadrilateral AMBK are equal to four right angles, inasmuch as AMBK is in fact divisible into two triangles, and the angles KAM, KBM are right, therefore the remaining angles AKB, AMB are equal to two right angles.

But the angles DEG, DEF are also equal to two right angles; {I. 13} therefore the angles AKB, AMB are equal to the angles DEG, DEF,

of which the angle AKB is equal to the angle DEG; therefore the angle AMB which remains is equal to the angle DEF which remains.

Similarly it can be proved that the angle LNB is also equal to the angle DFE; therefore the remaining angle MLN is equal to the angle EDF. {I. 32} Therefore the triangle LMN is equiangular with the triangle DEF; and it has been circumscribed about the circle ABC.

Therefore about a given circle there has been circumscribed a triangle equiangular with the given triangle. Q. E. F.

PROPOSITION 4. In a given triangle to inscribe a circle.

Let ABC be the given triangle; thus it is required to inscribe a circle in the triangle ABC.

Let the angles ABC, ACB be bisected by the straight lines BD, CD {I. 9}, and let these meet one another at the point D; from D let DE, DF, DG be drawn perpendicular to the straight lines AB, BC, CA.

Now, since the angle ABD is equal to the angle CBD, and the right angle BED is also equal to the right angle BFD, EBD, FBD are two triangles having two angles equal to two angles and one side equal to one side, namely that subtending one of the equal angles, which is BD common to the triangles; therefore they will also have the remaining sides equal to the remaining sides; {I. 26} therefore DE is equal to DF.

For the same reason DG is also equal to DF.

Therefore the three straight lines DE, DF, DG are equal to one another; therefore the circle described with centre D and distance one of the straight lines DE, DF, DG will pass also through the remaining points, and will touch the straight lines AB, BC, CA, because the angles at the points E, F, G are right.

For, if it cuts them, the straight line drawn at right angles to the diameter of the circle from its extremity will be found to fall within the circle : which was proved absurd; {III. 16} therefore the circle described with centre D and distance one of the straight lines DE, DF, DG will not cut the straight lines AB, BC, CA; therefore it will touch them, and will be the circle inscribed in the triangle ABC. {IV. Def. 5} Let it be inscribed, as FGE.

Therefore in the given triangle ABC the circle EFG has been inscribed. Q. E. F.

PROPOSITION 5. About a given triangle to circumscribe a circle.

Let ABC be the given triangle; thus it is required to circumscribe a circle about the given triangle ABC.

Let the straight lines AB, AC be bisected at the points D, E {I. 10}, and from the points D, E let DF, EF be drawn at right angles to AB, AC; they will then meet within the triangle ABC, or on the straight line BC, or outside BC.

First let them meet within at F, and let FB, FC, FA be joined.

Then, since AD is equal to DB, and DF is common and at right angles, therefore the base AF is equal to the base FB. {I. 4} Similarly we can prove that CF is also equal to AF; so that FB is also equal to FC; therefore the three straight lines FA, FB, FC are equal to one another.

Therefore the circle described with centre F and distance one of the straight lines FA, FB, FC will pass also through the remaining points, and the circle will have been circumscribed about the triangle ABC.

Let it be circumscribed, as ABC.

Next, let DF, EF meet on the straight line BC at F, as is the case in the second figure; and let AF be joined.

Then, similarly, we shall prove that the point F is the centre of the circle circumscribed about the triangle ABC.

Again, let DF, EF meet outside the triangle ABC at F, as is the case in the third figure, and let AF, BF, CF be joined.

Then again, since AD is equal to DB, and DF is common and at right angles, therefore the base AF is equal to the base BF. {I. 4} Similarly we can prove that CF is also equal to AF; so that BF is also equal to FC; therefore the circle described with centre F and distance one of the straight lines FA, FB, FC will pass also through the remaining points, and will have been circumscribed about the triangle ABC.

Therefore about the given triangle a circle has been circumscribed. Q. E. F.

And it is manifest that, when the centre of the circle falls within the triangle, the angle BAC, being in a segment greater than the semicircle, is less than a right angle; when the centre falls on the straight line BC, the angle BAC, being in a semicircle, is right; and when the centre of the circle falls outside the triangle, the angle BAC, being in a segment less than the semicircle, is greater than a right angle. {III. 31} **PROPOSITION 6.** In a given circle to inscribe a square.

Let ABCD be the given circle; thus it is required to inscribe a square in the circle ABCD.

Let two diameters AC, BD of the circle ABCD be drawn at right angles to one another, and let AB, BC, CD, DA be joined.

Then, since BE is equal to ED, for E is the centre, and EA is common and at right angles, therefore the base AB is equal to the base AD. {I. 4} For the same reason each of the straight lines BC, CD is also equal to each of the straight lines AB, AD; therefore the quadrilateral ABCD is equilateral.

I say next that it is also right-angled.

For, since the straight line BD is a diameter of the circle ABCD, therefore BAD is a semicircle; therefore the angle BAD is right. {III. 31} For the same reason each of the angles ABC, BCD, CDA is also right; therefore the quadrilateral ABCD is right-angled.

But it was also proved equilateral; therefore it is a square; {I. Def. 22} and it has been inscribed in the circle ABCD.

Therefore in the given circle the square ABCD has been inscribed. Q. E. F.

PROPOSITION 7. About a given circle to circumscribe a square.

Let ABCD be the given circle; thus it is required to circumscribe a square about the circle ABCD.

Let two diameters AC, BD of the circle ABCD be drawn at right angles to one another, and through the points A, B, C, D let FG, GH, HK, KF be drawn touching the circle ABCD. {III. 16, Por.} Then, since FG touches the circle ABCD, and EA has been joined from the centre E to the point of contact at A, therefore the angles at A are right. {III. 18} For the same reason the angles at the points B, C, D are also right.

Now, since the angle AEB is right, and the angle EBG is also right, therefore GH is parallel to AC. {I. 28} For the same reason AC is also parallel to FK, so that GH is also parallel to FK. {I. 30} Similarly we can prove that each of the straight lines GF, HK is parallel to BED.

Therefore GK, GC, AK, FB, BK are parallelograms; therefore GF is equal to HK, and GH to FK. {I. 34} And, since AC is equal to BD, and AC is also equal to each of the straight lines GH, FK, while BD is equal to each of the straight lines GF, HK, {I. 34} therefore the quadrilateral FGHK is equilateral.

I say next that it is also right-angled.

For, since GBEA is a parallelogram, and the angle AEB is right, therefore the angle AGB is also right. {I. 34} Similarly we can prove that the angles at H, K, F are also right.

Therefore FGHK is right-angled.

But it was also proved equilateral; therefore it is a square; and it has been circumscribed about the circle ABCD.

Therefore about the given circle a square has been circumscribed. Q. E. F.

PROPOSITION 8. In a given square to inscribe a circle.

Let ABCD be the given square; thus it is required to inscribe a circle in the given square ABCD.

Let the straight lines AD, AB be bisected at the points E, F respectively {I. 10}, through E let EH be drawn parallel to either AB or CD, and through F let FK be drawn parallel to either AD or BC; {I. 31} therefore each of the figures AK, KB, AH, HD, AG, GC, BG, GD is a parallelogram, and their opposite sides are evidently equal. {I. 34} Now, since AD is equal to AB, and AE is half of AD, and AF half of AB, therefore AE is equal to AF, so that the opposite sides are also equal; therefore FG is equal to GE.

Similarly we can prove that each of the straight lines GH, GK is equal to each of the straight lines FG, GE; therefore the four straight lines GE, GF, GH, GK are equal to one another.

Therefore the circle described with centre G and distance one of the straight lines GE, GF, GH, GK will pass also through the remaining points.

And it will touch the straight lines AB, BC, CD, DA, because the angles at E, F, H, K are right.

For, if the circle cuts AB, BC, CD, DA, the straight line drawn at right angles to the diameter of the circle from its extremity will fall within the circle : which was proved absurd; {III. 16}

therefore the circle described with centre G and distance one of the straight lines GE, GF, GH, GK will not cut the straight lines AB, BC, CD, DA.

Therefore it will touch them, and will have been inscribed in the square ABCD.

Therefore in the given square a circle has been inscribed. Q. E. F.

PROPOSITION 9. About a given square to circumscribe a circle.

Let ABCD be the given square; thus it is required to circumscribe a circle about the square ABCD.

For let AC, BD be joined, and let them cut one another at E.

Then, since DA is equal to AB, and AC is common, therefore the two sides DA, AC are equal to the two sides BA, AC; and the base DC is equal to the base BC; therefore the angle DAC is equal to the angle BAC. {I. 8} Therefore the angle DAB is bisected by AC.

Similarly we can prove that each of the angles ABC, BCD, CDA is bisected by the straight lines AC, DB.

Now, since the angle DAB is equal to the angle ABC, and the angle EAB is half the angle DAB, and the angle EBA half the angle ABC, therefore the angle EAB is also equal to the angle EBA; so that the side EA is also equal to EB. {I. 6} Similarly we can prove that each of the straight lines EA, EB is equal to each of the straight lines EC, ED.

Therefore the four straight lines EA, EB, EC, ED are equal to one another.

Therefore the circle described with centre E and distance one of the straight lines EA, EB, EC, ED will pass also through the remaining points; and it will have been circumscribed about the square ABCD.

Let it be circumscribed, as ABCD.

Therefore about the given square a circle has been circumscribed. Q. E. F.

PROPOSITION 10. To construct an isosceles triangle having each of the angles at the base double of the remaining one.

Let any straight line AB be set out, and let it be cut at the point C so that the rectangle contained by AB, BC is equal to the square on CA; {II. 11} with centre A and distance AB let the circle BDE be described, and let there be fitted in the circle BDE the straight line BD equal to the straight line AC which is not greater than the diameter of the circle BDE. {IV. 1} Let AD, DC be joined, and let the circle ACD be circumscribed about the triangle ACD. {IV. 5} Then, since the rectangle AB, BC is equal to the square on AC, and AC is equal to BD, therefore the rectangle AB, BC is equal to the square on BD.

And, since a point B has been taken outside the circle ACD, and from B the two straight lines BA, BD have fallen on the circle ACD, and one of them cuts it, while the other falls on it, and the rectangle AB, BC is equal to the square on BD, therefore BD touches the circle ACD. {III. 37} Since, then, BD touches it, and DC is drawn across from the point of contact at D, therefore the angle BDC is equal to the angle DAC in the alternate segment of the circle. {III. 32} Since, then, the angle BDC is equal to the angle DAC, let the angle CDA be added to each; therefore the whole angle BDA is equal to the two angles CDA, DAC.

But the exterior angle BCD is equal to the angles CDA, DAC; {I. 32} therefore the angle BDA is also equal to the angle BCD.

But the angle BDA is equal to the angle CBD, since the side AD is also equal to AB; {I. 5} so that the angle DBA is also equal to the angle BCD.

Therefore the three angles BDA, DBA, BCD are equal to one another.

And, since the angle DBC is equal to the angle BCD, the side BD is also equal to the side DC. {I. 6} But BD is by hypothesis equal to CA; therefore CA is also equal to CD, so that the angle CDA is also equal to the angle DAC; {I. 5} therefore the angles CDA, DAC are double of the angle DAC.

But the angle BCD is equal to the angles CDA, DAC; therefore the angle BCD is also double of the angle CAD.

But the angle BCD is equal to each of the angles BDA, DBA; therefore each of the angles BDA, DBA is also double of the angle DAB.

Therefore the isosceles triangle ABD has been constructed having each of the angles at the base DB double of the remaining one. Q. E. F.

PROPOSITION 11. In a given circle to inscribe an equilateral and equiangular pentagon.

Let ABCDE be the given circle; thus it is required to inscribe in the circle ABCDE an equilateral and equiangular pentagon.

Let the isosceles triangle FGH be set out having each of the angles at G, H double of the angle at F; {IV. 10} let there be inscribed in the circle ABCDE the triangle ACD equiangular with the triangle FGH, so that the angle CAD is equal to the angle at F and the angles at G, H respectively equal to the angles ACD, CDA; {IV. 2} therefore each of the angles ACD, CDA is also double of the angle CAD.

Now let the angles ACD, CDA be bisected respectively by the straight lines CE, DB {I. 9}, and let AB, BC, DE, EA be joined.

Then, since each of the angles ACD, CDA is double of the angle CAD, and they have been bisected by the straight lines CE, DB, therefore the five angles DAC, ACE, ECD, CDB, BDA are equal to one another.

But equal angles stand on equal circumferences; {III. 26} therefore the five circumferences AB, BC, CD, DE, EA are equal to one another.

But equal circumferences are subtended by equal straight lines; {III. 29} therefore the five straight lines AB, BC, CD, DE, EA are equal to one another; therefore the pentagon ABCDE is equilateral.

I say next that it is also equiangular.

For, since the circumference AB is equal to the circumference DE, let BCD be added to each; therefore the whole circumference ABCD is equal to the whole circumference EDCB.

And the angle AED stands on the circumference ABCD, and the angle BAE on the circumference EDCB; therefore the angle BAE is also equal to the angle AED. {III. 27} For the same reason each of the angles ABC, BCD, CDE is also equal to each of the angles BAE, AED; therefore the pentagon ABCDE is equiangular.

But it was also proved equilateral; therefore in the given circle an equilateral and equiangular pentagon has been inscribed. Q. E. F.

PROPOSITION 12. About a given circle to circumscribe an equilateral and equiangular pentagon.

Let ABCDE be the given circle; thus it is required to circumscribe an equilateral and equiangular pentagon about the circle ABCDE.

Let A, B, C, D, E be conceived to be the angular points of the inscribed pentagon, so that the circumferences AB, BC, CD, DE, EA are equal; {IV. 11} through A, B, C, D, E let GH, HK, KL, LM, MG be drawn touching the circle; {III. 16, Por.} let the centre F of the circle ABCDE be taken {III. 1}, and let FB, FK, FC, FL, FD be joined.

Then, since the straight line KL touches the circle ABCDE at C, and FC has been joined from the centre F to the point of contact at C, therefore FC is perpendicular to KL; {III. 18} therefore each of the angles at C is right.

For the same reason the angles at the points B, D are also right.

And, since the angle FCK is right, therefore the square on FK is equal to the squares on FC, CK.

For the same reason {I. 47} the square on FK is also equal to the squares on FB, BK; so that the squares on FC, CK are equal to the squares on FB, BK, of which the square on FC is equal to the square on FB; therefore the square on CK which remains is equal to the square on BK.

Therefore BK is equal to CK.

And, since FB is equal to FC, and FK common, the two sides BF, FK are equal to the two sides CF, FK; and the base BK equal to the base CK; therefore the angle BFK is equal to the angle KFC, {I. 8} and the angle BKF to the angle FKC.

Therefore the angle BFC is double of the angle KFC, and the angle BKC of the angle FKC.

For the same reason the angle CFD is also double of the angle CFL, and the angle DLC of the angle FLC.

Now, since the circumference BC is equal to CD, the angle BFC is also equal to the angle CFD. {III. 27} And the angle BFC is double of the angle KFC, and the angle DFC of the angle LFC; therefore the angle KFC is also equal to the angle LFC.

But the angle FCK is also equal to the angle FCL; therefore FKC, FLC are two triangles having two angles equal to two angles and one side equal to one side, namely FC which is common to them; therefore they will also have the remaining sides equal to the remaining sides, and the remaining angle to the remaining

angle; {I. 26} therefore the straight line KC is equal to CL, and the angle FKC to the angle FLC.

And, since KC is equal to CL, therefore KL is double of KC.

For the same reason it can be proved that HK is also double of BK.

And BK is equal to KC; therefore HK is also equal to KL.

Similarly each of the straight lines HG, GM, ML can also be proved equal to each of the straight lines HK, KL; therefore the pentagon GHKLM is equilateral.

I say next that it is also equiangular.

For, since the angle FKC is equal to the angle FLC, and the angle HKL was proved double of the angle FKC, and the angle KLM double of the angle FLC, therefore the angle HKL is also equal to the angle KLM.

Similarly each of the angles KHG, HGM, GML can also be proved equal to each of the angles HKL, KLM; therefore the five angles GHK, HKL, KLM, LMG, MGH are equal to one another.

Therefore the pentagon GHKLM is equiangular.

And it was also proved equilateral; and it has been circumscribed about the circle ABCDE. Q. E. F.

PROPOSITION 13. In a given pentagon, which is equilateral and equiangular, to inscribe a circle.

Let ABCDE be the given equilateral and equiangular pentagon; thus it is required to inscribe a circle in the pentagon ABCDE.

For let the angles BCD, CDE be bisected by the straight lines CF, DF respectively; and from the point F, at which the straight lines CF, DF meet one another, let the straight lines FB, FA, FE be joined.

Then, since BC is equal to CD , and CF common, the two sides BC , CF are equal to the two sides DC , CF ; and the angle BCF is equal to the angle DCF ; therefore the base BF is equal to the base DF , and the triangle BCF is equal to the triangle DCF , and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend. {I. 4} Therefore the angle CBF is equal to the angle CDF .

And, since the angle CDE is double of the angle CDF , and the angle CDE is equal to the angle ABC , while the angle CDF is equal to the angle CBF ; therefore the angle CBA is also double of the angle CBF ; therefore the angle ABF is equal to the angle FBC ; therefore the angle ABC has been bisected by the straight line BF .

Similarly it can be proved that the angles BAE , AED have also been bisected by the straight lines FA , FE respectively.

Now let FG , FH , FK , FL , FM be drawn from the point F perpendicular to the straight lines AB , BC , CD , DE , EA .

Then, since the angle HCF is equal to the angle KCF , and the right angle FHC is also equal to the angle FKC , FHC , FKC are two triangles having two angles equal to two angles and one side equal to one side, namely FC which is common to them and subtends one of the equal angles; therefore they will also have the remaining sides equal to the remaining sides; {I. 26} therefore the perpendicular FH is equal to the perpendicular FK .

Similarly it can be proved that each of the straight lines FL , FM , FG is also equal to each of the straight lines FH , FK ; therefore the five straight lines FG , FH , FK , FL , FM are equal to one another.

Therefore the circle described with centre F and distance one of the straight lines FG , FH , FK , FL , FM will pass also through the remaining points; and it will touch the straight lines AB , BC , CD ,

DE, EA, because the angles at the points G, H, K, L, M are right.

For, if it does not touch them. but cuts them, it will result that the straight line drawn at right angles to the diameter of the circle from its extremity falls within the circle: which was proved absurd. {III. 16} Therefore the circle described with centre F and distance one of the straight lines FG, FH, FK, FL, FM will not cut the straight lines AB, BC, CD, DE, EA; therefore it will touch them.

Let it be described, as GHKLM.

Therefore in the given pentagon, which is equilateral and equiangular, a circle has been inscribed. Q. E. F.

PROPOSITION 14. About a given pentagon, which is equilateral and equiangular, to circumscribe a circle.

Let ABCDE be the given pentagon, which is equilateral and equiangular; thus it is required to circumscribe a circle about the pentagon ABCDE.

Let the angles BCD, CDE be bisected by the straight lines CF, DF respectively, and from the point F, at which the straight lines meet, let the straight lines FB, FA, FE be joined to the points B, A, E.

Then in manner similar to the preceding it can be proved that the angles CBA, BAE, AED have also been bisected by the straight lines FB, FA, FE respectively.

Now, since the angle BCD is equal to the angle CDE, and the angle FCD is half of the angle BCD, and the angle CDF half of the angle CDE, therefore the angle FCD is also equal to the angle CDF, so that the side FC is also equal to the side FD. {I. 6} Similarly it can be proved that each of the straight lines FB, FA, FE is also equal to each of the straight lines FC, FD; therefore the five straight lines FA, FB, FC, FD, FE are equal to one another.

Therefore the circle described with centre F and distance one of the straight lines FA, FB, FC, FD, FE will pass also through the remaining points, and will have been circumscribed.

Let it be circumscribed, and let it be ABCDE.

Therefore about the given pentagon, which is equilateral and equiangular, a circle has been circumscribed. Q. E. F.

PROPOSITION 15. In a given circle to inscribe an equilateral and equiangular hexagon.

Let ABCDEF be the given circle; thus it is required to inscribe an equilateral and equiangular hexagon in the circle ABCDEF.

Let the diameter AD of the circle ABCDEF be drawn; let the centre G of the circle be taken, and with centre D and distance DG let the circle EGCH be described; let EG, CG be joined and carried through to the points B, F, and let AB, BC, CD, DE, EF, FA be joined.

I say that the hexagon ABCDEF is equilateral and equiangular.

For, since the point G is the centre of the circle ABCDEF, GE is equal to GD.

Again, since the point D is the centre of the circle GCH, DE is equal to DG.

But GE was proved equal to GD; therefore GE is also equal to ED; therefore the triangle EGD is equilateral; and therefore its three angles EGD, GDE, DEG are equal to one another, inasmuch as, in isosceles triangles, the angles at the base are equal to one another. {I. 5} And the three angles of the triangle are equal to two right angles; {I. 32} therefore the angle EGD is one-third of two right angles.

Similarly, the angle DGC can also be proved to be onethird of two right angles.

And, since the straight line CG standing on EB makes the adjacent angles EGC, CGB equal to two right angles, therefore the remaining angle CGB is also one-third of two right angles.

Therefore the angles EGD, DGC, CGB are equal to one another; so that the angles vertical to them, the angles BGA, AGF, FGE are equal. {I. 15} Therefore the six angles EGD, DGC, CGB, BGA, AGF, FGE are equal to one another.

But equal angles stand on equal circumferences; {III. 26} therefore the six circumferences AB, BC, CD, DE, EF, FA are equal to one another.

And equal circumferences are subtended by equal straight lines; {III. 29} therefore the six straight lines are equal to one another; therefore the hexagon ABCDEF is equilateral.

I say next that it is also equiangular.

For, since the circumference FA is equal to the circumference ED, let the circumference ABCD be added to each; therefore the whole FABCD is equal to the whole EDCBA; and the angle FED stands on the circumference FABCD, and the angle AFE on the circumference EDCBA; therefore the angle AFE is equal to the angle DEF. {III. 27} Similarly it can be proved that the remaining angles of the hexagon ABCDEF are also severally equal to each of the angles AFE, FED; therefore the hexagon ABCDEF is equiangular.

But it was also proved equilateral; and it has been inscribed in the circle ABCDEF.

Therefore in the given circle an equilateral and equiangular hexagon has been inscribed. Q. E. F.

PORISM.

From this it is manifest that the side of the hexagon is equal to the radius of the circle.

And, in like manner as in the case of the pentagon, if through the points of division on the circle we draw tangents to the circle, there will be circumscribed about the circle an equilateral and equiangular hexagon in conformity with what was explained in the case of the pentagon.

And further by means similar to those explained in the case of the pentagon we can both inscribe a circle in a given hexagon and circumscribe one about it. Q. E. F.

PROPOSITION 16. In a given circle to inscribe a fifteen-angled figure which shall be both equilateral and equiangular.

Let ABCD be the given circle; thus it is required to inscribe in the circle ABCD a fifteen-angled figure which shall be both equilateral and equiangular.

In the circle ABCD let there be inscribed a side AC of the equilateral triangle inscribed in it, and a side AB of an equilateral pentagon; therefore, of the equal segments of which there are fifteen in the circle ABCD, there will be five in the circumference ABC which is one-third of the circle, and there will be three in the circumference AB which is one-fifth of the circle; therefore in the remainder BC there will be two of the equal segments.

Let BC be bisected at E; {III. 30} therefore each of the circumferences BE, EC is a fifteenth of the circle ABCD.

If therefore we join BE, EC and fit into the circle ABCD straight lines equal to them and in contiguity, a fifteen-angled figure which is both equilateral and equiangular will have been inscribed in it. Q. E. F.

And, in like manner as in the case of the pentagon, if through the points of division on the circle we draw tangents to the circle, there will be circumscribed about the circle a fifteen-angled figure which is equilateral and equiangular.

And further, by proofs similar to those in the case of the pentagon, we can both inscribe a circle in the given fifteenangled figure and circumscribe one about it. Q. E. F.

ENDNOTES.

at random, literally “as it may chance,” ὡς ἔτυχεν. The same expression is used in III. 1 and commonly.

is in fact divisible, καὶ διαίρεῖται, literally “is actually divided.”

and distance one of the (straight lines D)E, (D)F, (D)G. The words and letters here shown in brackets are put in to fill out the rather careless language of the Greek. Here and in several other places in Book IV. Euclid says literally “and with distance one of the (points) E, F, G” (καὶ διαστήματι ἐνὶ τῶν E, Z, H) and the like. In one case (IV. 13) he actually has “with distance one of the points G, H, K, L, M” (διαστήματι ἐνὶ τῶν H, Θ, K, Λ, M σημείων). Heiberg notes “Graecam locutionem satis miram et negligentem,” but, in view of its frequent occurrence in good MSS., does not venture to correct it.

BOOK V. DEFINITIONS.

1. A MAGNITUDE IS a part of a magnitude, the less of the greater, when it measures the greater.
2. The greater is a multiple of the less when it is measured by the less.
3. A ratio is a sort of relation in respect of size between two magnitudes of the same kind.
4. Magnitudes are said to have a ratio to one another which are capable, when multiplied, of exceeding one another.
5. Magnitudes are said to be in the same ratio, the first to the second and the third to the fourth, when, if any equimultiples whatever be taken of the first and third, and any equimultiples

whatever of the second and fourth, the former equimultiples alike exceed, are alike equal to, or alike fall short of, the latter equimultiples respectively taken in corresponding order.

6. Let magnitudes which have the same ratio be called proportional.

7. When, of the equimultiples, the multiple of the first magnitude exceeds the multiple of the second, but the multiple of the third does not exceed the multiple of the fourth, then the first is said to have a greater ratio to the second than the third has to the fourth.

8. A proportion in three terms is the least possible.

9. When three magnitudes are proportional, the first is said to have to the third the duplicate ratio of that which it has to the second.

10. When four magnitudes are <continuously> proportional, the first is said to have to the fourth the triplicate ratio of that which it has to the second, and so on continually, whatever be the proportion.

11. The term corresponding magnitudes is used of antecedents in relation to antecedents, and of consequents in relation to consequents.

12. Alternate ratio means taking the antecedent in relation to the antecedent and the consequent in relation to the consequent.

13. Inverse ratio means taking the consequent as antecedent in relation to the antecedent as consequent.

14. Composition of a ratio means taking the antecedent together with the consequent as one in relation to the consequent by itself.

15. Separation of a ratio means taking the excess by which the antecedent exceeds the consequent in relation to the consequent by itself.

16. Conversion of a ratio means taking the antecedent in relation to the excess by which the antecedent exceeds the consequent.

17. A ratio *ex aequali* arises when, there being several magnitudes and another set equal to them in multitude which taken two and two are in the same proportion, as the first is to the last among the first magnitudes, so is the first to the last among the second magnitudes; Or, in other words, it means taking the extreme terms by virtue of the removal of the intermediate terms.

18. A perturbed proportion arises when, there being three magnitudes and another set equal to them in multitude, as antecedent is to consequent among the first magnitudes, so is antecedent to consequent among the second magnitudes, while, as the consequent is to a third among the first magnitudes, so is a third to the antecedent among the second magnitudes.

PROPOSITIONS.

PROPOSITION I. IF THERE BE any number of magnitudes whatever which are, respectively, equimultiples of any magnitudes equal in multitude, then, whatever multiple one of the magnitudes is of one, that multiple also will all be of all.

Let any number of magnitudes whatever AB, CD be respectively equimultiples of any magnitudes E, F equal in multitude; I say that, whatever multiple AB is of E, that multiple will AB, CD also be of E, F.

For, since AB is the same multiple of E that CD is of F, as many magnitudes as there are in AB equal to E, so many also are there

in CD equal to F.

Let AB be divided into the magnitudes AG, GB equal to E, and CD into CH, HD equal to F; then the multitude of the magnitudes AG, GB will be equal to the multitude of the magnitudes CH, HD.

Now, since AG is equal to E, and CH to F, therefore AG is equal to E, and AG, CH to E, F.

For the same reason GB is equal to E, and GB, HD to E, F; therefore, as many magnitudes as there are in AB equal to E, so many also are there in AB, CD equal to E, F; therefore, whatever multiple AB is of E, that multiple will AB, CD also be of E, F.

Therefore etc. Q. E. D.

PROPOSITION 2. If a first magnitude be the same multiple of a second that a third is of a fourth, and a fifth also be the same multiple of the second that a sixth is of the fourth, the sum of the first and fifth will also be the same multiple of the second that the sum of the third and sixth is of the fourth.

Let a first magnitude, AB, be the same multiple of a second, C, that a third, DE, is of a fourth, F, and let a fifth, BG, also be the same multiple of the second, C, that a sixth, EH, is of the fourth F; I say that the sum of the first and fifth, AG, will be the same multiple of the second, C, that the sum of the third and sixth, DH, is of the fourth, F.

For, since AB is the same multiple of C that DE is of F, therefore, as many magnitudes as there are in AB equal to C, so many also are there in DE equal to F.

For the same reason also, as many as there are in BG equal to C, so many are there also in EH equal to F; therefore, as many as there are in the whole AG equal to C, so many also are there in the whole DH equal to F.

Therefore, whatever multiple AG is of C, that multiple also is DH of F.

Therefore the sum of the first and fifth, AG, is the same multiple of the second, C, that the sum of the third and sixth, DH, is of the fourth, F.

Therefore etc. Q.E.D.

PROPOSITION 3. If a first magnitude be the same multiple of a second that a third is of a fourth, and if equimultiples be taken of the first and third, then also ex aequali the magnitudes taken will be equimultiples respectively, the one of the second and the other of the fourth.

Let a first magnitude A be the same multiple of a second B that a third C is of a fourth D, and let equimultiples EF, GH be taken of A, C; I say that EF is the same multiple of B that GH is of D.

For, since EF is the same multiple of A that GH is of C, therefore, as many magnitudes as there are in EF equal to A, so many also are there in GH equal to C.

Let EF be divided into the magnitudes EK, KF equal to A, and GH into the magnitudes GL, LH equal to C; then the multitude of the magnitudes EK, KF will be equal to the multitude of the magnitudes GL, LH.

And, since A is the same multiple of B that C is of D, while EK is equal to A, and GL to C, therefore EK is the same multiple of B that GL is of D.

For the same reason KF is the same multiple of B that LH is of D.

Since, then, a first magnitude EK is the same multiple of a second B that a third GL is of a fourth D, and a fifth KF is also the same multiple of the second B that a sixth LH is of the fourth D, therefore the sum of the first and fifth, EF, is also the same multiple of the second B that the sum of the third and sixth, GH,

is of the fourth D. {V. 2} Therefore etc. Q. E. D.

PROPOSITION 4. If a first magnitude have to a second the same ratio as a third to a fourth, any equimultiples whatever of the first and third will also have the same ratio to any equimultiples whatever of the second and fourth respectively, taken in corresponding order.

For let a first magnitude A have to a second B the same ratio as a third C to a fourth D; and let equimultiples E, F be taken of A, C, and G, H other, chance, equimultiples of B, D; I say that, as E is to G, so is F to H.

For let equimultiples K, L be taken of E, F, and other, chance, equimultiples M, N of G, H.

Since E is the same multiple of A that F is of C, and equimultiples K, L of E, F have been taken, therefore K is the same multiple of A that L is of C. {V. 3} For the same reason M is the same multiple of B that N is of D.

And, since, as A is to B, so is C to D, and of A, C equimultiples K, L have been taken, and of B, D other, chance, equimultiples M, N, therefore, if K is in excess of M, L also is in excess of N, if it is equal, equal, and if less, less. {V. Def. 5} And K, L are equimultiples of E, F, and M, N other, chance, equimultiples of G, H; therefore, as E is to G, so is F to H. {V. Def. 5} Therefore etc. Q. E. D.

PROPOSITION 5. If a magnitude be the same multiple of a magnitude that a part subtracted is of a part subtracted, the remainder will also be the same multiple of the remainder that the whole is of the whole.

For let the magnitude AB be the same multiple of the magnitude CD that the part AE subtracted is of the part CF subtracted; I say that the remainder EB is also the same multiple of the remainder FD that the whole AB is of the whole CD.

For, whatever multiple AE is of CF, let EB be made that multiple of CG.

Then, since AE is the same multiple of CF that EB is of GC, therefore AE is the same multiple of CF that AB is of GF. {V. 1}

But, by the assumption, AE is the same multiple of CF that AB is of CD.

Therefore AB is the same multiple of each of the magnitudes GF, CD; therefore GF is equal to CD.

Let CF be subtracted from each; therefore the remainder GC is equal to the remainder FD.

And, since AE is the same multiple of CF that EB is of GC, and GC is equal to DF, therefore AE is the same multiple of CF that EB is of FD.

But, by hypothesis, AE is the same multiple of CF that AB is of CD; therefore EB is the same multiple of FD that AB is of CD.

That is, the remainder EB will be the same multiple of the remainder FD that the whole AB is of the whole CD.

Therefore etc. Q. E. D.

PROPOSITION 6. If two magnitudes be equimultiples of two magnitudes, and any magnitudes subtracted from them be equimultiples of the same, the remainders also are either equal to the same or equimultiples of them.

For let two magnitudes AB, CD be equimultiples of two magnitudes E, F, and let AG, CH subtracted from them be equimultiples of the same two E, F; I say that the remainders also, GB, HD, are either equal to E, F or equimultiples of them.

For, first, let GB be equal to E; I say that HD is also equal to F.

For let CK be made equal to F.

Since AG is the same multiple of E that CH is of F, while GB is equal to E and KC to F, therefore AB is the same multiple of E that KH is of F. {V. 2} But, by hypothesis, AB is the same multiple of E that CD is of F; therefore KH is the same multiple of F that CD is of F.

Since then each of the magnitudes KH, CD is the same multiple of F, therefore KH is equal to CD.

Let CH be subtracted from each; therefore the remainder KC is equal to the remainder HD.

But F is equal to KC; therefore HD is also equal to F.

Hence, if GB is equal to E, HD is also equal to F.

Similarly we can prove that, even if GB be a multiple of E, HD is also the same multiple of F.

Therefore etc. Q. E. D.

PROPOSITION 7. Equal magnitudes have to the same the same ratio, as also has the same to equal magnitudes.

Let A, B be equal magnitudes and C any other, chance, magnitude; I say that each of the magnitudes A, B has the same ratio to C, and C has the same ratio to each of the magnitudes A, B.

For let equimultiples D, E of A, B be taken, and of C another, chance, multiple F.

Then, since D is the same multiple of A that E is of B, while A is equal to B, therefore D is equal to E.

But F is another, chance, magnitude.

If therefore D is in excess of F, E is also in excess of F, if equal to it, equal; and, if less, less.

And D, E are equimultiples of A, B, while F is another, chance, multiple of C; therefore, as A is to C, so is B to C. {V. Def. 5} I say

next that C also has the same ratio to each of the magnitudes A, B.

For, with the same construction, we can prove similarly that D is equal to E; and F is some other magnitude.

If therefore F is in excess of D, it is also in excess of E, if equal, equal; and, if less, less.

And F is a multiple of C, while D, E are other, chance, equimultiples of A, B; therefore, as C is to A, so is C to B. {V. Def. 5} Therefore etc.

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From this it is manifest that, if any magnitudes are proportional, they will also be proportional inversely. Q. E. D.

PROPOSITION 8. Of unequal magnitudes, the greater has to the same a greater ratio than the less has; and the same has to the less a greater ratio than it has to the greater.

Let AB, C be unequal magnitudes, and let AB be greater; let D be another, chance, magnitude; I say that AB has to D a greater ratio than C has to D, and D has to C a greater ratio than it has to AB.

For, since AB is greater than C, let BE be made equal to C; then the less of the magnitudes AE, EB, if multiplied, will sometime be greater than D. {V. Def. 4} {Case I.} First, let AE be less than EB; let AE be multiplied, and let FG be a multiple of it which is greater than D; then, whatever multiple FG is of AE, let GH be made the same multiple of EB and K of C; and let L be taken double of D, M triple of it, and successive multiples increasing by one, until what is taken is a multiple of D and the first that is greater than K. Let it be taken, and let it be N which is quadruple of D and the first multiple of it that is greater than K.

Then, since K is less than N first, therefore K is not less than M.

And, since FG is the same multiple of AE that GH is of EB, therefore FG is the same multiple of AE that FH is of AB. {V. 1} But FG is the same multiple of AE that K is of C; therefore FH is the same multiple of AB that K is of C; therefore FH, K are equimultiples of AB, C.

Again, since GH is the same multiple of EB that K is of C, and EB is equal to C, therefore GH is equal to K.

But K is not less than M; therefore neither is GH less than M.

And FG is greater than D; therefore the whole FH is greater than D, M together.

But D, M together are equal to N, inasmuch as M is triple of D, and M, D together are quadruple of D, while N is also quadruple of D; whence M, D together are equal to N.

But FH is greater than M, D; therefore FH is in excess of N, while K is not in excess of N.

And FH, K are equimultiples of AB, C, while N is another, chance, multiple of D; therefore AB has to D a greater ratio than C has to D. {V. Def. 7} I say next, that D also has to C a greater ratio than D has to AB.

For, with the same construction, we can prove similarly that N is in excess of K, while N is not in excess of FH.

And N is a multiple of D, while FH, K are other, chance, equimultiples of AB, C; therefore D has to C a greater ratio than D has to AB. {V. Def. 7} {Case 2.} Again, let AE be greater than EB.

Then the less, EB, if multiplied, will sometime be greater than D. {V. Def. 4} Let it be multiplied, and let GH be a multiple of EB and greater than D; and, whatever multiple GH is of EB, let FG be made the same multiple of AE, and K of C.

Then we can prove similarly that FH, K are equimultiples of AB, C; and, similarly, let N be taken a multiple of D but the first that

is greater than FG, so that FG is again not less than M.

But GH is greater than D; therefore the whole FH is in excess of D, M, that is, of N.

Now K is not in excess of N, inasmuch as FG also, which is greater than GH, that is, than K, is not in excess of N.

And in the same manner, by following the above argument, we complete the demonstration.

Therefore etc. Q. E. D.

PROPOSITION 9. Magnitudes which have the same ratio to the same are equal to one another; and magnitudes to which the same has the same ratio are equal.

For let each of the magnitudes A, B have the same ratio to C; I say that A is equal to B.

For, otherwise, each of the magnitudes A, B would not have had the same ratio to C; {V. 8} but it has; therefore A is equal to B.

Again, let C have the same ratio to each of the magnitudes A, B; I say that A is equal to B.

For, otherwise, C would not have had the same ratio to each of the magnitudes A, B; {V. 8} but it has; therefore A is equal to B.

Therefore etc. Q. E. D.

PROPOSITION 10. Of magnitudes which have a ratio to the same, that which has a greater ratio is greater; and that to which the same has a greater ratio is less.

For let A have to C a greater ratio than B has to C; I say that A is greater than B.

For, if not, A is either equal to B or less.

Now A is not equal to B; for in that case each of the magnitudes A, B would have had the same ratio to C; {V. 7} but they have

not; therefore A is not equal to B.

Nor again is A less than B; for in that case A would have had to C a less ratio than B has to C; {V. 8} but it has not; therefore A is not less than B.

But it was proved not to be equal either; therefore A is greater than B.

Again, let C have to B a greater ratio than C has to A; I say that B is less than A.

For, if not, it is either equal or greater.

Now B is not equal to A; for in that case C would have had the same ratio to each of the magnitudes A, B; {V. 7} but it has not; therefore A is not equal to B.

Nor again is B greater than A; for in that case C would have had to B a less ratio than it has to A; {V. 8} but it has not; therefore B is not greater than A.

But it was proved that it is not equal either; therefore B is less than A.

Therefore etc. Q. E. D.

PROPOSITION 11. Ratios which are the same with the same ratio are also the same with one another.

For, as A is to B, so let C be to D, and, as C is to D, so let E be to F; I say that, as A is to B, so is E to F.

For of A, C, E let equimultiples G, H, K be taken, and of B, D, F other, chance, equimultiples L, M, N.

Then since, as A is to B, so is C to D, and of A, C equimultiples G, H have been taken, and of B, D other, chance, equimultiples L, M, therefore, if G is in excess of L, H is also in excess of M, if equal, equal, and if less, less.

Again, since, as C is to D, so is E to F, and of C, E equimultiples H, K have been taken, and of D, F other, chance, equimultiples M, N, therefore, if H is in excess of M, K is also in excess of N, if equal, equal, and if less, less.

But we saw that, if H was in excess of M, G was also in excess of L; if equal, equal; and if less, less; so that, in addition, if G is in excess of L, K is also in excess of N, if equal, equal, and if less, less.

And G, K are equimultiples of A, E, while L, N are other, chance, equimultiples of B, F; therefore, as A is to B, so is E to F.

Therefore etc. Q. E. D.

PROPOSITION 12. If any number of magnitudes be proportional, as one of the antecedents is to one of the consequents, so will all the antecedents be to all the consequents.

Let any number of magnitudes A, B, C, D, E, F be proportional, so that, as A is to B, so is C to D and E to F; I say that, as A is to B, so are A, C, E to B, D, F.

For of A, C, E let equimultiples G, H, K be taken, and of B, D, F other, chance, equimultiples L, M, N.

Then since, as A is to B, so is C to D, and E to F, and of A, C, E equimultiples G, H, K have been taken, and of B, D, F other, chance, equimultiples L, M, N, therefore, if G is in excess of L, H is also in excess of M, and K of N, if equal, equal, and if less, less; so that, in addition, if G is in excess of L, then G, H, K are in excess of L, M, N, if equal, equal, and if less, less.

Now G and G, H, K are equimultiples of A and A, C, E, since, if any number of magnitudes whatever are respectively equimultiples of any magnitudes equal in multitude, whatever multiple one of the magnitudes is of one, that multiple also will all be of all. {V. 1} For the same reason L and L, M, N are also equimultiples of B

and B, D, F; therefore, as A is to B, so are A, C, E to B, D, F. {V. Def. 5} Therefore etc. Q. E. D.

PROPOSITION 13. If a first magnitude have to a second the same ratio as a third to a fourth, and the third have to the fourth a greater ratio than a fifth has to a sixth, the first will also have to the second a greater ratio than the fifth to the sixth.

For let a first magnitude A have to a second B the same ratio as a third C has to a fourth D, and let the third C have to the fourth D a greater ratio than a fifth E has to a sixth F; I say that the first A will also have to the second B a greater ratio than the fifth E to the sixth F.

For, since there are some equimultiples of C, E, and of D, F other, chance, equimultiples, such that the multiple of C is in excess of the multiple of D, while the multiple of E is not in excess of the multiple of F, {V. Def. 7} let them be taken, and let G, H be equimultiples of C, E, and K, L other, chance, equimultiples of D, F, so that G is in excess of K, but H is not in excess of L; and, whatever multiple G is of C, let M be also that multiple of A, and, whatever multiple K is of D, let N be also that multiple of B.

Now, since, as A is to B, so is C to D, and of A, C equimultiples M, G have been taken, and of B, D other, chance, equimultiples N, K, therefore, if M is in excess of N, G is also in excess of K, if equal, equal, and if less, less. {V. Def. 5} But G is in excess of K; therefore M is also in excess of N.

But H is not in excess of L; and M, H are equimultiples of A, E, and N, L other, chance, equimultiples of B, F; therefore A has to B a greater ratio than E has to F. {V. Def. 7} Therefore etc. Q. E. D.

PROPOSITION 14. If a first magnitude have to a second the same ratio as a third has to a fourth, and the first be greater than the third, the second will also be greater than the fourth; if

equal, equal; and if less, less.

For let a first magnitude A have the same ratio to a second B as a third C has to a fourth D; and let A be greater than C; I say that B is also greater than D.

For, since A is greater than C, and B is another, chance, magnitude, therefore A has to B a greater ratio than C has to B. {V. 8} But, as A is to B, so is C to D; therefore C has also to D a greater ratio than C has to B. {V. 13} But that to which the same has a greater ratio is less; {V. 10} therefore D is less than B; so that B is greater than D.

Similarly we can prove that, if A be equal to C, B will also be equal to D; and, if A be less than C, B will also be less than D.

Therefore etc. Q. E. D.

PROPOSITION 15. Parts have the same ratio as the same multiples of them taken in corresponding order.

For let AB be the same multiple of C that DE is of F; I say that, as C is to F, so is AB to DE.

For, since AB is the same multiple of C that DE is of F, as many magnitudes as there are in AB equal to C, so many are there also in DE equal to F.

Let AB be divided into the magnitudes AG, GH, HB equal to C, and DE into the magnitudes DK, KL, LE equal to F; then the multitude of the magnitudes AG, GH, HB will be equal to the multitude of the magnitudes DK, KL, LE.

And, since AG, GH, HB are equal to one another, and DK, KL, LE are also equal to one another, therefore, as AG is to DK, so is GH to KL, and HB to LE. {V. 7} Therefore, as one of the antecedents is to one of the consequents, so will all the antecedents be to all the consequents; {V. 12} therefore, as AG is to DK, so is AB to DE.

But AG is equal to C and DK to F; therefore, as C is to F, so is AB to DE.

Therefore etc. Q. E. D.

PROPOSITION 16. If four magnitudes be proportional, they will also be proportional alternately.

Let A, B, C, D be four proportional magnitudes, so that, as A is to B, so is C to D; I say that they will also be so alternately, that is, as A is to C, so is B to D.

For of A, B let equimultiples E, F be taken, and of C, D other, chance, equimultiples G, H.

Then, since E is the same multiple of A that F is of B, and parts have the same ratio as the same multiples of them, {V. 15} therefore, as A is to B, so is E to F.

But as A is to B, so is C to D; therefore also, as C is to D, so is E to F. {V. 11} Again, since G, H are equimultiples of C, D, therefore, as C is to D, so is G to H. {V. 15} But, as C is to D, so is E to F; therefore also, as E is to F, so is G to H. {V. 11} But, if four magnitudes be proportional, and the first be greater than the third, the second will also be greater than the fourth; if equal, equal; and if less, less. {V. 14} Therefore, if E is in excess of G, F is also in excess of H, if equal, equal, and if less, less.

Now E, F are equimultiples of A, B, and G, H other, chance, equimultiples of C, D; therefore, as A is to C, so is B to D. {V. Def. 5} Therefore etc. Q. E. D.

PROPOSITION 17. If magnitudes be proportional componendo, they will also be proportional separando.

Let AB, BE, CD, DF be magnitudes proportional componendo, so that, as AB is to BE, so is CD to DF; I say that they will also be proportional separando, that is, as AE is to EB, so is CF to DF.

For of AE, EB, CF, FD let equimultiples GH, HK, LM, MN be taken, and of EB, FD other, chance, equimultiples, KO, NP.

Then, since GH is the same multiple of AE that HK is of EB, therefore GH is the same multiple of AE that GK is of AB. {V. 1} But GH is the same multiple of AE that LM is of CF; therefore GK is the same multiple of AB that LM is of CF.

Again, since LM is the same multiple of CF that MN is of FD, therefore LM is the same multiple of CF that LN is of CD. {V. 1} But LM was the same multiple of CF that GK is of AB; therefore GK is the same multiple of AB that LN is of CD.

Therefore GK, LN are equimultiples of AB, CD.

Again, since HK is the same multiple of EB that MN is of FD, and KO is also the same multiple of EB that NP is of FD, therefore the sum HO is also the same multiple of EB that MP is of FD. {V. 2} And, since, as AB is to BE, so is CD to DF, and of AB, CD equimultiples GK, LN have been taken, and of EB, FD equimultiples HO, MP, therefore, if GK is in excess of HO, LN is also in excess of MP, if equal, equal, and if less, less.

Let GK be in excess of HO; then, if HK be subtracted from each, GH is also in excess of KO.

But we saw that, if GK was in excess of HO, LN was also in excess of MP; therefore LN is also in excess of MP, and, if MN be subtracted from each, LM is also in excess of NP; so that, if GH is in excess of KO, LM is also in excess of NP.

Similarly we can prove that, if GH be equal to KO, LM will also be equal to NP, and if less, less.

And GH, LM are equimultiples of AE, CF, while KO, NP are other, chance, equimultiples of EB, FD; therefore, as AE is to EB, so is CF to FD.

Therefore etc. Q. E. D.

PROPOSITION 18. If magnitudes be proportional separando, they will also be proportional componendo.

Let AE, EB, CF, FD be magnitudes proportional separando, so that, as AE is to EB, so is CF to FD; I say that they will also be proportional componendo, that is, as AB is to BE, so is CD to FD.

For, if CD be not to DF as AB to BE, then, as AB is to BE, so will CD be either to some magnitude less than DF or to a greater.

First, let it be in that ratio to a less magnitude DG.

Then, since, as AB is to BE, so is CD to DG, they are magnitudes proportional componendo; so that they will also be proportional separando. {V. 17} Therefore, as AE is to EB, so is CG to GD.

But also, by hypothesis, as AE is to EB, so is CF to FD.

Therefore also, as CG is to GD, so is CF to FD. {V. 11} But the first CG is greater than the third CF; therefore the second GD is also greater than the fourth FD. {V. 14} But it is also less: which is impossible.

Therefore, as AB is to BE, so is not CD to a less magnitude than FD.

Similarly we can prove that neither is it in that ratio to a greater; it is therefore in that ratio to FD itself.

Therefore etc. Q. E. D.

PROPOSITION 19. If, as a whole is to a whole, so is a part subtracted to a part subtracted, the remainder will also be to the remainder as whole to whole.

For, as the whole AB is to the whole CD, so let the part AE subtracted be to the part CF subtracted; I say that the remainder EB will also be to the remainder FD as the whole AB to the whole CD.

For since, as AB is to CD, so is AE to CF, alternately also, as BA is to AE, so is DC to CF. {V. 16} And, since the magnitudes are proportional componendo, they will also be proportional separando, {V. 17} that is, as BE is to EA, so is DF to CF, and, alternately, as BE is to DF, so is EA to FC. {V. 16} But, as AE is to CF, so by hypothesis is the whole AB to the whole CD.

Therefore also the remainder EB will be to the remainder FD as the whole AB is to the whole CD. {V. 11} Therefore etc. {PORISM.

From this it is manifest that, if magnitudes be proportional componendo, they will also be proportional convertendo.

} Q. E. D.

PROPOSITION 20. If there be three magnitudes, and others equal to them in multitude, which taken two and two are in the same ratio, and if ex aequali the first be greater than the third, the fourth will also be greater than the sixth; if equal, equal; and, if less, less.

Let there be three magnitudes A, B, C, and others D, E, F equal to them in multitude, which taken two and two are in the same ratio, so that, as A is to B, so is D to E, and as B is to C, so is E to F; and let A be greater than C ex aequali; I say that D will also be greater than F; if A is equal to C, equal; and, if less, less.

For, since A is greater than C, and B is some other magnitude, and the greater has to the same a greater ratio than the less has, {V. 8} therefore A has to B a greater ratio than C has to B.

But, as A is to B, so is D to E, and, as C is to B, inversely, so is F to E; therefore D has also to E a greater ratio than F has to E. {V. 13} But, of magnitudes which have a ratio to the same, that which has a greater ratio is greater; {V. 10} therefore D is greater than F.

Similarly we can prove that, if A be equal to C, D will also be equal to F; and if less, less.

Therefore etc. Q. E. D.

PROPOSITION 21. If there be three magnitudes, and others equal to them in multitude, which taken two and two together are in the same ratio, and the proportion of them be perturbed, then, if ex aequali the first magnitude is greater than the third, the fourth will also be greater than the sixth; if equal, equal; and if less, less.

Let there be three magnitudes A, B, C, and others D, E, F equal to them in multitude, which taken two and two are in the same ratio, and let the proportion of them be perturbed, so that, as A is to B, so is E to F, and, as B is to C, so is D to E, and let A be greater than C ex aequali; I say that D will also be greater than F; if A is equal to C, equal; and if less, less.

For, since A is greater than C, and B is some other magnitude, therefore A has to B a greater ratio than C has to B. {V. 8} But, as A is to B, so is E to F, and, as C is to B, inversely, so is E to D. Therefore also E has to F a greater ratio than E has to D. {V. 13} But that to which the same has a greater ratio is less; {V. 10} therefore F is less than D; therefore D is greater than F.

Similarly we can prove that, if A be equal to C, D will also be equal to F; and if less, less.

Therefore etc. Q. E. D.

PROPOSITION 22. If there be any number of magnitudes whatever, and others equal to them in multitude, which taken two and two together are in the same ratio, they will also be in the same ratio ex aequali.

Let there be any number of magnitudes A, B, C, and others D, E, F equal to them in multitude, which taken two and two together are in the same ratio, so that, as A is to B, so is D to E, and, as B is

to C, so is E to F; I say that they will also be in the same ratio ex aequali, <that is, as A is to C, so is D to F>.

For of A, D let equimultiples G, H be taken, and of B, E other, chance, equimultiples K, L; and, further, of C, F other, chance, equimultiples M, N.

Then, since, as A is to B, so is D to E, and of A, D equimultiples G, H have been taken, and of B, E other, chance, equimultiples K, L, therefore, as G is to K, so is H to L. {V. 4} For the same reason also, as K is to M, so is L to N.

Since, then, there are three magnitudes G, K, M, and others H, L, N equal to them in multitude, which taken two and two together are in the same ratio, therefore, ex aequali, if G is in excess of M, H is also in excess of N; if equal, equal; and if less, less. {V. 20} And G, H are equimultiples of A, D, and M, N other, chance, equimultiples of C, F.

Therefore, as A is to C, so is D to F. {V. Def. 5} Therefore etc. Q. E. D.

PROPOSITION 23. If there be three magnitudes, and others equal to them in multitude, which taken two and two together are in the same ratio, and the proportion of them be perturbed, they will also be in the same ratio ex aequali.

Let there be three magnitudes A, B, C, and others equal to them in multitude, which, taken two and two together, are in the same proportion, namely D, E, F; and let the proportion of them be perturbed, so that, as A is to B, so is E to F, and, as B is to C, so is D to E; I say that, as A is to C, so is D to F.

Of A, B, D let equimultiples G, H, K be taken, and of C, E, F other, chance, equimultiples L, M, N.

Then, since G, H are equimultiples of A, B, and parts have the same ratio as the same multiples of them, {V. 15} therefore, as A is to B, so is G to H.

For the same reason also, as E is to F, so is M to N.

And, as A is to B, so is E to F; therefore also, as G is to H, so is M to N. {V. 11} Next, since, as B is to C, so is D to E, alternately, also, as B is to D, so is C to E. {V. 16} And, since H, K are equimultiples of B, D, and parts have the same ratio as their equimultiples, therefore, as B is to D, so is H to K. {V. 15} But, as B is to D, so is C to E; therefore also, as H is to K, so is C to E. {V. 11} Again, since L, M are equimultiples of C, E, therefore, as C is to E, so is L to M. {V. 15} But, as C is to E, so is H to K; therefore also, as H is to K, so is L to M, {V. 11} and, alternately, as H is to L, so is K to M. {V. 16} But it was also proved that, as G is to H, so is M to N.

Since, then, there are three magnitudes G, H, L, and others equal to them in multitude K, M, N, which taken two and two together are in the same ratio, and the proportion of them is perturbed, therefore, ex aequali, if G is in excess of L, K is also in excess of N; if equal, equal; and if less, less. {V. 21} And G, K are equimultiples of A, D, and L, N of C, F.

Therefore, as A is to C, so is D to F.

Therefore etc. Q. E. D.

PROPOSITION 24. If a first magnitude have to a second the same ratio as a third has to a fourth, and also a fifth have to the second the same ratio as a sixth to the fourth, the first and fifth added together will have to the second the same ratio as the third and sixth have to the fourth.

Let a first magnitude AB have to a second C the same ratio as a third DE has to a fourth F; and let also a fifth BG have to the second C the same ratio as a sixth EH has to the fourth F; I say that the first and fifth added together, AG, will have to the second C the same ratio as the third and sixth, DH, has to the fourth F.

For since, as BG is to C, so is EH to F, inversely, as C is to BG, so is F to EH.

Since, then, as AB is to C, so is DE to F, and, as C is to BG, so is F to EH, therefore, ex aequali, as AB is to BG, so is DE to EH. {V. 22} And, since the magnitudes are proportional separando, they will also be proportional componendo; {V. 18} therefore, as AG is to GB, so is DH to HE.

But also, as BG is to C, so is EH to F; therefore, ex aequali, as AG is to C, so is DH to F. {V. 22} Therefore etc. Q. E. D.

PROPOSITION 25. If four magnitudes be proportional, the greatest and the least are greater than the remaining two.

Let the four magnitudes AB, CD, E, F be proportional so that, as AB is to CD, so is E to F, and let AB be the greatest of them and F the least; I say that AB, F are greater than CD, E.

For let AG be made equal to E, and CH equal to F.

Since, as AB is to CD, so is E to F, and E is equal to AG, and F to CH, therefore, as AB is to CD, so is AG to CH.

And since, as the whole AB is to the whole CD, so is the part AG subtracted to the part CH subtracted, the remainder GB will also be to the remainder HD as the whole AB is to the whole CD. {V. 19} But AB is greater than CD; therefore GB is also greater than HD.

And, since AG is equal to E, and CH to F, therefore AG, F are equal to CH, E.

And if, GB, HD being unequal, and GB greater, AG, F be added to GB and CH, E be added to HD, it follows that AB, F are greater than CD, E.

Therefore etc. Q. E. D.

ENDNOTES.

let EB be made that multiple of CG, τοσαυταπλάσιον γεγονέντω καὶ τὸ EB τοῦ ΓΗ. From this way of stating the construction one might suppose that CG was given and EB had to be found equal to a certain multiple of it. But in fact EB is what is given and CG has to be found, i.e. CG has to be constructed as a certain submultiple of EB.

“Let A, B, C, D be four proportional magnitudes, so that, as A is to B, so is C to D.” In a number of expressions like this it is absolutely necessary, when translating into English, to interpolate words which are not in the Greek. Thus the Greek here is: “Ἐστω τέσσαρα μεγέθη ἀνάλογον τὰ A, B, Γ, Δ, ὥς τὸ A πρὸς τὸ B, οὕτως τὸ Γ πρὸς τὸ Δ, literally “Let A, B, C, D be four proportional magnitudes, as A to B, so C to D.” The same remark applies to the corresponding expressions in the next propositions, V. 17, 18, and to other forms of expression in V. 20-23 and later propositions: e.g. in V. 20 we have a phrase meaning literally “Let there be magnitudes...which taken two and two are in the same ratio, as A to B, so D to E,” etc.: in V. 21 “(magnitudes)...which taken two and two are in the same ratio, and let the proportion of them be perturbed, as A to B, so E to F,” etc. In all such cases (where the Greek is so terse as to be almost ungrammatical) I shall insert the words necessary in English, without further remark.

BOOK VI. DEFINITIONS.

1. SIMILAR RECTILINEAL FIGURES are such as have their angles severally equal and the sides about the equal angles proportional.
2. {Reciprocally related figures. See note.} 3. A straight line is said to have been cut in extreme and mean ratio when, as the whole line is to the greater segment, so is the greater to the less.

4. The height of any figure is the perpendicular drawn from the vertex to the base.

5

PROPOSITIONS.

PROPOSITION 1. TRIANGLES AND PARALLELOGRAMS which are under the same height are to one another as their bases.

Let ABC, ACD be triangles and EC, CF parallelograms under the same height; I say that, as the base BC is to the base CD, so is the triangle ABC to the triangle ACD, and the parallelogram EC to the parallelogram CF.

For let BD be produced in both directions to the points H, L and let {any number of straight lines} BG, GH be made equal to the base BC, and any number of straight lines DK, KL equal to the base CD; let AG, AH, AK, AL be joined.

Then, since CB, BG, GH are equal to one another, the triangles ABC, AGB, AHG are also equal to one another. {I. 38} Therefore, whatever multiple the base HC is of the base BC, that multiple also is the triangle AHC of the triangle ABC.

For the same reason, whatever multiple the base LC is of the base CD, that multiple also is the triangle ALC of the triangle ACD; and, if the base HC is equal to the base CL, the triangle AHC is also equal to the triangle ACL, {I. 38} if the base HC is in excess of the base CL, the triangle AHC is also in excess of the triangle ACL, and, if less, less.

Thus, there being four magnitudes, two bases BC, CD and two triangles ABC, ACD, equimultiples have been taken of the base BC and the triangle ABC, namely the base HC and the triangle AHC, and of the base CD and the triangle ADC other, chance, equimultiples, namely the base LC and the triangle ALC; and it has been proved that, if the base HC is in excess of the base CL,

the triangle AHC is also in excess of the triangle ALC; if equal, equal; and, if less, less.

Therefore, as the base BC is to the base CD, so is the triangle ABC to the triangle ACD. {V. Def. 5} Next, since the parallelogram EC is double of the triangle ABC, {I. 41} and the parallelogram FC is double of the triangle ACD, while parts have the same ratio as the same multiples of them, {V. 15} therefore, as the triangle ABC is to the triangle ACD, so is the parallelogram EC to the parallelogram FC.

Since, then, it was proved that, as the base BC is to CD, so is the triangle ABC to the triangle ACD, and, as the triangle ABC is to the triangle ACD, so is the parallelogram EC to the parallelogram CF, therefore also, as the base BC is to the base CD, so is the parallelogram EC to the parallelogram FC. {V. 11} Therefore etc. Q. E. D.

PROPOSITION 2. If a straight line be drawn parallel to one of the sides of a triangle, it will cut the sides of the triangle proportionally; and, if the sides of the triangle be cut proportionally, the line joining the points of section will be parallel to the remaining side of the triangle.

For let DE be drawn parallel to BC, one of the sides of the triangle ABC; I say that, as BD is to DA, so is CE to EA.

For let BE, CD be joined.

Therefore the triangle BDE is equal to the triangle CDE; for they are on the same base DE and in the same parallels DE, BC. {I. 38} And the triangle ADE is another area.

But equals have the same ratio to the same; {V. 7} therefore, as the triangle BDE is to the triangle ADE, so is the triangle CDE to the triangle ADE.

But, as the triangle BDE is to ADE, so is BD to DA; for, being under the same height, the perpendicular drawn from E to AB,

they are to one another as their bases. {VI. 1} For the same reason also, as the triangle CDE is to ADE, so is CE to EA.

Therefore also, as BD is to DA, so is CE to EA. {V. 11} Again, let the sides AB, AC of the triangle ABC be cut proportionally, so that, as BD is to DA, so is CE to EA; and let DE be joined.

I say that DE is parallel to BC.

For, with the same construction, since, as BD is to DA, so is CE to EA, but, as BD is to DA, so is the triangle BDE to the triangle ADE, and, as CE is to EA, so is the triangle CDE to the triangle ADE, {VI. 1} therefore also, as the triangle BDE is to the triangle ADE, so is the triangle CDE to the triangle ADE. {V. 11} Therefore each of the triangles BDE, CDE has the same ratio to ADE.

Therefore the triangle BDE is equal to the triangle CDE; {V. 9} and they are on the same base DE.

But equal triangles which are on the same base are also in the same parallels. {I. 39} Therefore DE is parallel to BC.

Therefore etc. Q. E. D.

PROPOSITION 3. If an angle of a triangle be bisected and the straight line cutting the angle cut the base also, the segments of the base will have the same ratio as the remaining sides of the triangle; and, if the segments of the base have the same ratio as the remaining sides of the triangle, the straight line joined from the vertex to the point of section will bisect the angle of the triangle.

Let ABC be a triangle, and let the angle BAC be bisected by the straight line AD; I say that, as BD is to CD, so is BA to AC.

For let CE be drawn through C parallel to DA, and let BA be carried through and meet it at E.

Then, since the straight line AC falls upon the parallels AD, EC, the angle ACE is equal to the angle CAD. {I. 29} But the angle

CAD is by hypothesis equal to the angle BAD; therefore the angle BAD is also equal to the angle ACE.

Again, since the straight line BAE falls upon the parallels AD, EC, the exterior angle BAD is equal to the interior angle AEC. {I. 29} But the angle ACE was also proved equal to the angle BAD; therefore the angle ACE is also equal to the angle AEC, so that the side AE is also equal to the side AC. {I. 6} And, since AD has been drawn parallel to EC, one of the sides of the triangle BCE, therefore, proportionally, as BD is to DC, so is BA to AE.

But AE is equal to AC; {VI. 2} therefore, as BD is to DC, so is BA to AC.

Again, let BA be to AC as BD to DC, and let AD be joined; I say that the angle BAC has been bisected by the straight line AD.

For, with the same construction, since, as BD is to DC, so is BA to AC, and also, as BD is to DC, so is BA to AE : for AD has been drawn parallel to EC, one of the sides of the triangle BCE: {VI. 2} therefore also, as BA is to AC, so is BA to AE. {V. 11} Therefore AC is equal to AE, {V. 9} so that the angle AEC is also equal to the angle ACE. {I. 5} But the angle AEC is equal to the exterior angle BAD, {I. 29} and the angle ACE is equal to the alternate angle CAD; {id.} therefore the angle BAD is also equal to the angle CAD.

Therefore the angle BAC has been bisected by the straight line AD.

Therefore etc. Q. E. D.

PROPOSITION 4. In equiangular triangles the sides about the equal angles are proportional, and those are corresponding sides which subtend the equal angles.

Let ABC, DCE be equiangular triangles having the angle ABC equal to the angle DCE, the angle BAC to the angle CDE, and further the angle ACB to the angle CED; I say that in the triangles ABC, DCE the sides about the equal angles are

proportional, and those are corresponding sides which subtend the equal angles.

For let BC be placed in a straight line with CE.

Then, since the angles ABC, ACB are less than two right angles, {I. 17} and the angle ACB is equal to the angle DEC, therefore the angles ABC, DEC are less than two right angles; therefore BA, ED, when produced, will meet. {I. Post. 5} Let them be produced and meet at F.

Now, since the angle DCE is equal to the angle ABC, BF is parallel to CD. {I. 28} Again, since the angle ACB is equal to the angle DEC, AC is parallel to FE. {I. 28} Therefore FACD is a parallelogram; therefore FA is equal to DC, and AC to FD. {I. 34} And, since AC has been drawn parallel to FE, one side of the triangle FBE, therefore, as BA is to AF, so is BC to CE. {VI. 2} But AF is equal to CD; therefore, as BA is to CD, so is BC to CE, and alternately, as AB is to BC, so is DC to CE. {V. 16} Again, since CD is parallel to BF, therefore, as BC is to CE, so is FD to DE. {VI. 2} But FD is equal to AC; therefore, as BC is to CE, so is AC to DE, and alternately, as BC is to CA, so is CE to ED. {V. 16} Since then it was proved that, as AB is to BC, so is DC to CE, and, as BC is to CA, so is CE to ED; therefore, ex aequali, as BA is to AC, so is CD to DE. {V. 22} Therefore etc. Q. E. D.

PROPOSITION 5. If two triangles have their sides proportional, the triangles will be equiangular and will have those angles equal which the corresponding sides subtend.

Let ABC, DEF be two triangles having their sides proportional, so that, as AB is to BC, so is DE to EF, as BC is to CA, so is EF to FD, and further, as BA is to AC, so is ED to DF; I say that the triangle ABC is equiangular with the triangle DEF, and they will have those angles equal which the corresponding sides subtend, namely the angle ABC to the angle DEF, the angle BCA to the angle EFD, and further the angle BAC to the angle EDF.

For on the straight line EF, and at the points E, F on it, let there be constructed the angle FEG equal to the angle ABC, and the angle EFG equal to the angle ACB; {I. 23} therefore the remaining angle at A is equal to the remaining angle at G. {I. 32} Therefore the triangle ABC is equiangular with the triangle GEF.

Therefore in the triangles ABC, GEF the sides about the equal angles are proportional, and those are corresponding sides which subtend the equal angles; {VI. 4} therefore, as AB is to BC, so is GE to EF.

But, as AB is to BC, so by hypothesis is DE to EF; therefore, as DE is to EF, so is GE to EF. {V. 11} Therefore each of the straight lines DE, GE has the same ratio to EF; therefore DE is equal to GE. {V. 9} For the same reason DF is also equal to GF.

Since then DE is equal to EG, and EF is common, the two sides DE, EF are equal to the two sides GE, EF; and the base DF is equal to the base FG; therefore the angle DEF is equal to the angle GEF, {I. 8} and the triangle DEF is equal to the triangle GEF, and the remaining angles are equal to the remaining angles, namely those which the equal sides subtend. {I. 4} Therefore the angle DFE is also equal to the angle GFE, and the angle EDF to the angle EGF.

And, since the angle FED is equal to the angle GEF, while the angle GEF is equal to the angle ABC, therefore the angle ABC is also equal to the angle DEF.

For the same reason the angle ACB is also equal to the angle DFE, and further, the angle at A to the angle at D; therefore the triangle ABC is equiangular with the triangle DEF.

Therefore etc. Q. E. D.

PROPOSITION 6. If two triangles have one angle equal to one angle and the sides about the equal angles proportional, the triangles will be equiangular and will have those angles equal

which the corresponding sides subtend.

Let ABC , DEF be two triangles having one angle BAC equal to one angle EDF and the sides about the equal angles proportional, so that, as BA is to AC , so is ED to DF ; I say that the triangle ABC is equiangular with the triangle DEF , and will have the angle ABC equal to the angle DEF , and the angle ACB to the angle DFE .

For on the straight line DF , and at the points D , F on it, let there be constructed the angle FDG equal to either of the angles BAC , EDF , and the angle DFG equal to the angle ACB ; {I. 23} therefore the remaining angle at B is equal to the remaining angle at G . {I. 32} Therefore the triangle ABC is equiangular with the triangle DGF .

Therefore, proportionally, as BA is to AC , so is GD to DF . {VI. 4} But, by hypothesis, as BA is to AC , so also is ED to DF ; therefore also, as ED is to DF , so is GD to DF . {V. 11} Therefore ED is equal to DG ; {V. 9} and DF is common; therefore the two sides ED , DF are equal to the two sides GD , DF ; and the angle EDF is equal to the angle GDF ; therefore the base EF is equal to the base GF , and the triangle DEF is equal to the triangle DGF , and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend. {I. 4} Therefore the angle DFG is equal to the angle DFE , and the angle DGF to the angle DEF .

But the angle DFG is equal to the angle ACB ; therefore the angle ACB is also equal to the angle DFE .

And, by hypothesis, the angle BAC is also equal to the angle EDF ; therefore the remaining angle at B is also equal to the remaining angle at E ; {I. 32} therefore the triangle ABC is equiangular with the triangle DEF .

Therefore etc. Q. E. D.

PROPOSITION 7. If two triangles have one angle equal to one angle, the sides about other angles proportional, and the

remaining angles either both less or both not less than a right angle, the triangles will be equiangular and will have those angles equal, the sides about which are proportional.

Let ABC, DEF be two triangles having one angle equal to one angle, the angle BAC to the angle EDF, the sides about other angles ABC, DEF proportional, so that, as AB is to BC, so is DE to EF, and, first, each of the remaining angles at C, F less than a right angle; I say that the triangle ABC is equiangular with the triangle DEF, the angle ABC will be equal to the angle DEF, and the remaining angle, namely the angle at C, equal to the remaining angle, the angle at F.

For, if the angle ABC is unequal to the angle DEF, one of them is greater.

Let the angle ABC be greater; and on the straight line AB, and at the point B on it, let the angle ABG be constructed equal to the angle DEF. {I. 23} Then, since the angle A is equal to D, and the angle ABG to the angle DEF, therefore the remaining angle AGB is equal to the remaining angle DFE. {I. 32} Therefore the triangle ABG is equiangular with the triangle DEF.

Therefore, as AB is to BG, so is DE to EF {VI. 4} But, as DE is to EF, so by hypothesis is AB to BC; therefore AB has the same ratio to each of the straight lines BC, BG; {V. 11} therefore BC is equal to BG, {V. 9} so that the angle at C is also equal to the angle BGC. {I. 5} But, by hypothesis, the angle at C is less than a right angle; therefore the angle BGC is also less than a right angle; so that the angle AGB adjacent to it is greater than a right angle. {I. 13} And it was proved equal to the angle at F; therefore the angle at F is also greater than a right angle.

But it is by hypothesis less than a right angle : which is absurd.

Therefore the angle ABC is not unequal to the angle DEF; therefore it is equal to it.

But the angle at A is also equal to the angle at D; therefore the remaining angle at C is equal to the remaining angle at F. {I. 32} Therefore the triangle ABC is equiangular with the triangle DEF.

But, again, let each of the angles at C, F be supposed not less than a right angle; I say again that, in this case too, the triangle ABC is equiangular with the triangle DEF.

For, with the same construction, we can prove similarly that BC is equal to BG; so that the angle at C is also equal to the angle BGC. {I. 5} But the angle at C is not less than a right angle; therefore neither is the angle BGC less than a right angle.

Thus in the triangle BGC the two angles are not less than two right angles: which is impossible. {I. 17} Therefore, once more, the angle ABC is not unequal to the angle DEF; therefore it is equal to it.

But the angle at A is also equal to the angle at D; therefore the remaining angle at C is equal to the remaining angle at F. {I. 32} Therefore the triangle ABC is equiangular with the triangle DEF.

Therefore etc. Q. E. D.

PROPOSITION 8. If in a right-angled triangle a perpendicular be drawn from the right angle to the base, the triangles adjoining the perpendicular are similar both to the whole and to one another.

Let ABC be a right-angled triangle having the angle BAC right, and let AD be drawn from A perpendicular to BC; I say that each of the triangles ABD, ADC is similar to the whole ABC and, further, they are similar to one another.

For, since the angle BAC is equal to the angle ADB, for each is right, and the angle at B is common to the two triangles ABC and ABD, therefore the remaining angle ACB is equal to the remaining angle BAD; {I. 32} therefore the triangle ABC is equiangular with the triangle ABD.

Therefore, as BC which subtends the right angle in the triangle ABC is to BA which subtends the right angle in the triangle ABD, so is AB itself which subtends the angle at C in the triangle ABC to BD which subtends the equal angle BAD in the triangle ABD, and so also is AC to AD which subtends the angle at B common to the two triangles. {VI. 4} Therefore the triangle ABC is both equiangular to the triangle ABD and has the sides about the equal angles proportional.

Therefore the triangle ABC is similar to the triangle ABD. {VI. Def. 1} Similarly we can prove that the triangle ABC is also similar to the triangle ADC; therefore each of the triangles ABD, ADC is similar to the whole ABC.

I say next that the triangles ABD, ADC are also similar to one another.

For, since the right angle BDA is equal to the right angle ADC, and moreover the angle BAD was also proved equal to the angle at C, therefore the remaining angle at B is also equal to the remaining angle DAC; {I. 32} therefore the triangle ABD is equiangular with the triangle ADC.

Therefore, as BD which subtends the angle BAD in the triangle ABD is to DA which subtends the angle at C in the triangle ADC equal to the angle BAD, so is AD itself which subtends the angle at B in the triangle ABD to DC which subtends the angle DAC in the triangle ADC equal to the angle at B, and so also is BA to AC, these sides subtending the right angles; {VI. 4} therefore the triangle ABD is similar to the triangle ADC. {VI. Def. 1} Therefore etc.

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From this it is clear that, if in a right-angled triangle a perpendicular be drawn from the right angle to the base, the straight line so drawn is a mean proportional between the

segments of the base. Q. E. D.

PROPOSITION 9. From a given straight line to cut off a prescribed part.

Let AB be the given straight line; thus it is required to cut off from AB a prescribed part.

Let the third part be that prescribed.

Let a straight line AC be drawn through from A containing with AB any angle; let a point D be taken at random on AC, and let DE, EC be made equal to AD. {I. 3} Let BC be joined, and through D let DF be drawn parallel to it. {I. 31} Then, since FD has been drawn parallel to BC, one of the sides of the triangle ABC, therefore, proportionally, as CD is to DA, so is BF to FA. {VI. 2} But CD is double of DA; therefore BF is also double of FA; therefore BA is triple of AF.

Therefore from the given straight line AB the prescribed third part AF has been cut off. Q. E. F.

PROPOSITION 10. To cut a given uncut straight line similarly to a given cut straight line.

Let AB be the given uncut straight line, and AC the straight line cut at the points D, E; and let them be so placed as to contain any angle; let CB be joined, and through D, E let DF, EG be drawn parallel to BC, and through D let DHK be drawn parallel to AB. {I. 31} Therefore each of the figures FH, HB is a parallelogram; therefore DH is equal to FG and HK to GB. {I. 34} Now, since the straight line HE has been drawn parallel to KC, one of the sides of the triangle DKC, therefore, proportionally, as CE is to ED, so is KH to HD. {VI. 2} But KH is equal to BG, and HD to GF; therefore, as CE is to ED, so is BG to GF.

Again, since FD has been drawn parallel to GE, one of the sides of the triangle AGE, therefore, proportionally, as ED is to DA, so is GF to FA. {VI. 2} But it was also proved that, as CE is to ED, so is

BG to GF; therefore, as CE is to ED, so is BG to GF, and, as ED is to DA, so is GF to FA.

Therefore the given uncut straight line AB has been cut similarly to the given cut straight line AC. Q. E. F.

PROPOSITION 11. To two given straight lines to find a third proportional.

Let BA, AC be the two given straight lines, and let them be placed so as to contain any angle; thus it is required to find a third proportional to BA, AC.

For let them be produced to the points D, E, and let BD be made equal to AC; {I. 3} let BC be joined, and through D let DE be drawn parallel to it. {I. 31} Since, then, BC has been drawn parallel to DE, one of the sides of the triangle ADE, proportionally, as AB is to BD, so is AC to CE. {VI. 2} But BD is equal to AC; therefore, as AB is to AC, so is AC to CE.

Therefore to two given straight lines AB, AC a third proportional to them, CE, has been found. Q. E. F.

PROPOSITION 12. To three given straight lines to find a fourth proportional.

Let A, B, C be the three given straight lines; thus it is required to find a fourth proportional to A, B, C.

Let two straight lines DE, DF be set out containing any angle EDF; let DG be made equal to A, GE equal to B, and further DH equal to C; let GH be joined, and let EF be drawn through E parallel to it. {I. 31} Since, then, GH has been drawn parallel to EF, one of the sides of the triangle DEF, therefore, as DG is to GE, so is DH to HF. {VI. 2} But DG is equal to A, GE to B, and DH to C; therefore, as A is to B, so is C to HF.

Therefore to the three given straight lines A, B, C a fourth proportional HF has been found. Q. E. F.

PROPOSITION 13. To two given straight lines to find a mean proportional.

Let AB, BC be the two given straight lines; thus it is required to find a mean proportional to AB, BC.

Let them be placed in a straight line, and let the semicircle ADC be described on AC; let BD be drawn from the point B at right angles to the straight line AC, and let AD, DC be joined.

Since the angle ADC is an angle in a semicircle, it is right. {III. 31} And, since, in the right-angled triangle ADC, DB has been drawn from the right angle perpendicular to the base, therefore DB is a mean proportional between the segments of the base, AB, BC. {VI. 8, Por.} Therefore to the two given straight lines AB, BC a mean proportional DB has been found. Q. E. F.

PROPOSITION 14. In equal and equiangular parallelograms the sides about the equal angles are reciprocally proportional; and equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal.

Let AB, BC be equal and equiangular parallelograms having the angles at B equal, and let DB, BE be placed in a straight line; therefore FB, BG are also in a straight line. {I. 14} I say that, in AB, BC, the sides about the equal angles are reciprocally proportional, that is to say, that, as DB is to BE, so is GB to BF.

For let the parallelogram FE be completed.

Since, then, the parallelogram AB is equal to the parallelogram BC, and FE is another area, therefore, as AB is to FE, so is BC to FE. {V. 7} But, as AB is to FE, so is DB to BE, {VI. 1} and, as BC is to FE, so is GB to BF. {id.} therefore also, as DB is to BE, so is GB to BF. {V. 11} Therefore in the parallelograms AB, BC the sides about the equal angles are reciprocally proportional.

Next, let GB be to BF as DB to BE; I say that the parallelogram AB is equal to the parallelogram BC.

For since, as DB is to BE, so is GB to BF, while, as DB is to BE, so is the parallelogram AB to the parallelogram FE, {VI. 1} and, as GB is to BF, so is the parallelogram BC to the parallelogram FE, {VI. 1} therefore also, as AB is to FE, so is BC to FE; {V. 11} therefore the parallelogram AB is equal to the parallelogram BC. {V. 9} Therefore etc. Q. E. D.

PROPOSITION 15. In equal triangles which have one angle equal to one angle the sides about the equal angles are reciprocally proportional; and those triangles which have one angle equal to one angle, and in which the sides about the equal angles are reciprocally proportional, are equal.

Let ABC, ADE be equal triangles having one angle equal to one angle, namely the angle BAC to the angle DAE; I say that in the triangles ABC, ADE the sides about the equal angles are reciprocally proportional, that is to say, that, as CA is to AD, so is EA to AB.

For let them be placed so that CA is in a straight line with AD; therefore EA is also in a straight line with AB. {I. 14} Let BD be joined.

Since then the triangle ABC is equal to the triangle ADE, and BAD is another area, therefore, as the triangle CAB is to the triangle BAD, so is the triangle EAD to the triangle BAD. {V. 7} But, as CAB is to BAD, so is CA to AD, {VI. 1} and, as EAD is to BAD, so is EA to AB. {id.} Therefore also, as CA is to AD, so is EA to AB. {V. 11} Therefore in the triangles ABC, ADE the sides about the equal angles are reciprocally proportional.

Next, let the sides of the triangles ABC, ADE be reciprocally proportional, that is to say, let EA be to AB as CA to AD; I say that the triangle ABC is equal to the triangle ADE.

For, if BD be again joined, since, as CA is to AD, so is EA to AB, while, as CA is to AD, so is the triangle ABC to the triangle BAD,

and, as EA is to AB, so is the triangle EAD to the triangle BAD, {VI. 1} therefore, as the triangle ABC is to the triangle BAD, so is the triangle EAD to the triangle BAD. {V. 11} Therefore each of the triangles ABC, EAD has the same ratio to BAD.

Therefore the triangle ABC is equal to the triangle EAD. {V. 9} Therefore etc. Q. E. D.

PROPOSITION 16. If four straight lines be proportional, the rectangle contained by the extremes is equal to the rectangle contained by the means; and, if the rectangle contained by the extremes be equal to the rectangle contained by the means, the four straight lines will be proportional.

Let the four straight lines AB, CD, E, F be proportional, so that, as AB is to CD, so is E to F; I say that the rectangle contained by AB, F is equal to the rectangle contained by CD, E.

Let AG, CH be drawn from the points A, C at right angles to the straight lines AB, CD, and let AG be made equal to F, and CH equal to E.

Let the parallelograms BG, DH be completed.

Then since, as AB is to CD, so is E to F, while E is equal to CH, and F to AG, therefore, as AB is to CD, so is CH to AG.

Therefore in the parallelograms BG, DH the sides about the equal angles are reciprocally proportional.

But those equiangular parallelograms in which the sides about the equal angles are reciprocally proportional are equal; {VI. 14} therefore the parallelogram BG is equal to the parallelogram DH.

And BG is the rectangle AB, F, for AG is equal to F; and DH is the rectangle CD, E, for E is equal to CH; therefore the rectangle contained by AB, F is equal to the rectangle contained by CD, E.

Next, let the rectangle contained by AB, F be equal to the rectangle contained by CD, E; I say that the four straight lines will be proportional, so that, as AB is to CD, so is E to F.

For, with the same construction, since the rectangle AB, F is equal to the rectangle CD, E, and the rectangle AB, F is BG, for AG is equal to F, and the rectangle CD, E is DH, for CH is equal to E, therefore BG is equal to DH.

And they are equiangular. But in equal and equiangular parallelograms the sides about the equal angles are reciprocally proportional. {VI. 14} Therefore, as AB is to CD, so is CH to AG.

But CH is equal to E, and AG to F; therefore, as AB is to CD, so is E to F.

Therefore etc. Q. E. D.

PROPOSITION 17. If three straight lines be proportional, the rectangle contained by the extremes is equal to the square on the mean; and, if the rectangle contained by the extremes be equal to the square on the mean, the three straight lines will be proportional.

Let the three straight lines A, B, C be proportional, so that, as A is to B, so is B to C; I say that the rectangle contained by A, C is equal to the square on B.

Let D be made equal to B.

Then, since, as A is to B, so is B to C, and B is equal to D, therefore, as A is to B, so is D to C.

But, if four straight lines be proportional, the rectangle contained by the extremes is equal to the rectangle contained by the means. {VI. 16} Therefore the rectangle A, C is equal to the rectangle B, D.

But the rectangle B, D is the square on B, for B is equal to D; therefore the rectangle contained by A, C is equal to the square

on B.

Next, let the rectangle A, C be equal to the square on B; I say that, as A is to B, so is B to C.

For, with the same construction, since the rectangle A, C is equal to the square on B, while the square on B is the rectangle B, D, for B is equal to D, therefore the rectangle A, C is equal to the rectangle B, D.

But, if the rectangle contained by the extremes be equal to that contained by the means, the four straight lines are proportional. {VI. 16} Therefore, as A is to B, so is D to C.

But B is equal to D; therefore, as A is to B, so is B to C.

Therefore etc. Q. E. D.

PROPOSITION 18. On a given straight line to describe a rectilinear figure similar and similarly situated to a given rectilinear figure.

Let AB be the given straight line and CE the given rectilinear figure; thus it is required to describe on the straight line AB a rectilinear figure similar and similarly situated to the rectilinear figure CE.

Let DF be joined, and on the straight line AB, and at the points A, B on it, let the angle GAB be constructed equal to the angle at C, and the angle ABG equal to the angle CDF. {I. 23} Therefore the remaining angle CFD is equal to the angle AGB; {I. 32} therefore the triangle FCD is equiangular with the triangle GAB.

Therefore, proportionally, as FD is to GB, so is FC to GA, and CD to AB.

Again, on the straight line BG, and at the points B, G on it, let the angle BGH be constructed equal to the angle DFE, and the angle GBH equal to the angle FDE. {I. 23} Therefore the remaining angle at E is equal to the remaining angle at H; {I. 32} therefore

the triangle FDE is equiangular with the triangle GBH; therefore, proportionally, as FD is to GB, so is FE to GH, and ED to HB. {VI. 4} But it was also proved that, as FD is to GB, so is FC to GA, and CD to AB; therefore also, as FC is to AG, so is CD to AB, and FE to GH, and further ED to HB.

And, since the angle CFD is equal to the angle AGB, and the angle DFE to the angle BGH, therefore the whole angle CFE is equal to the whole angle AGH.

For the same reason the angle CDE is also equal to the angle ABH.

And the angle at C is also equal to the angle at A, and the angle at E to the angle at H.

Therefore AH is equiangular with CE; and they have the sides about their equal angles proportional; therefore the rectilineal figure AH is similar to the rectilineal figure CE. {VI. Def. 1} Therefore on the given straight line AB the rectilineal figure AH has been described similar and similarly situated to the given rectilineal figure CE. Q. E. F.

PROPOSITION 19. Similar triangles are to one another in the duplicate ratio of the corresponding sides.

Let ABC, DEF be similar triangles having the angle at B equal to the angle at E, and such that, as AB is to BC, so is DE to EF, so that BC corresponds to EF; {V. Def. 11} I say that the triangle ABC has to the triangle DEF a ratio duplicate of that which BC has to EF.

For let a third proportional BG be taken to BC, EF, so that, as BC is to EF, so is EF to BG; {VI. 11} and let AG be joined.

Since then, as AB is to BC, so is DE to EF, therefore, alternately, as AB is to DE, so is BC to EF. {V. 16} But, as BC is to EF, so is EF to BG; therefore also, as AB is to DE, so is EF to BG. {V. 11} Therefore in the triangles ABG, DEF the sides about the equal

angles are reciprocally proportional.

But those triangles which have one angle equal to one angle, and in which the sides about the equal angles are reciprocally proportional, are equal; {VI. 15} therefore the triangle ABG is equal to the triangle DEF.

Now since, as BC is to EF, so is EF to BG, and, if three straight lines be proportional, the first has to the third a ratio duplicate of that which it has to the second, {V. Def. 9} therefore BC has to BG a ratio duplicate of that which CB has to EF.

But, as CB is to BG, so is the triangle ABC to the triangle ABG; {VI. 1} therefore the triangle ABC also has to the triangle ABG a ratio duplicate of that which BC has to EF.

But the triangle ABG is equal to the triangle DEF; therefore the triangle ABC also has to the triangle DEF a ratio duplicate of that which BC has to EF.

Therefore etc.

PORISM.

From this it is manifest that, if three straight lines be proportional, then, as the first is to the third, so is the figure described on the first to that which is similar and similarly described on the second. Q. E. D.

PROPOSITION 20. Similar polygons are divided into similar triangles, and into triangles equal in multitude and in the same ratio as the wholes, and the polygon has to the polygon a ratio duplicate of that which the corresponding side has to the corresponding side.

Let ABCDE, FGHKL be similar polygons, and let AB correspond to FG; I say that the polygons ABCDE, FGHKL are divided into similar triangles, and into triangles equal in multitude and in the same ratio as the wholes, and the polygon ABCDE has to the

polygon FGHKL a ratio duplicate of that which AB has to FG.

Let BE, EC, GL, LH be joined.

Now, since the polygon ABCDE is similar to the polygon FGHKL, the angle BAE is equal to the angle GFL; and, as BA is to AE, so is GF to FL. {VI. Def. 1} Since then ABE, FGL are two triangles having one angle equal to one angle and the sides about the equal angles proportional, therefore the triangle ABE is equiangular with the triangle FGL; {VI. 6} so that it is also similar; {VI. 4 and Def. 1} therefore the angle ABE is equal to the angle FGL.

But the whole angle ABC is also equal to the whole angle FGH because of the similarity of the polygons; therefore the remaining angle EBC is equal to the angle LGH.

And, since, because of the similarity of the triangles ABE, FGL, as EB is to BA, so is LG to GF, and moreover also, because of the similarity of the polygons, as AB is to BC, so is FG to GH, therefore, ex aequali, as EB is to BC, so is LG to GH; {V. 22} that is, the sides about the equal angles EBC, LGH are proportional; therefore the triangle EBC is equiangular with the triangle LGH, {VI. 6} so that the triangle EBC is also similar to the triangle LGH. {VI. 4 and Def. 1} For the same reason the triangle ECD is also similar to the triangle LHK.

Therefore the similar polygons ABCDE, FGHKL have been divided into similar triangles, and into triangles equal in multitude.

I say that they are also in the same ratio as the wholes, that is, in such manner that the triangles are proportional, and ABE, EBC, ECD are antecedents, while FGL, LGH, LHK are their consequents, and that the polygon ABCDE has to the polygon FGHKL a ratio duplicate of that which the corresponding side has to the corresponding side, that is AB to FG.

For let AC, FH be joined.

Then since, because of the similarity of the polygons, the angle ABC is equal to the angle FGH, and, as AB is to BC, so is FG to GH, the triangle ABC is equiangular with the triangle FGH; {VI. 6} therefore the angle BAC is equal to the angle GFH, and the angle BCA to the angle GHF.

And, since the angle BAM is equal to the angle GFN, and the angle ABM is also equal to the angle FGN, therefore the remaining angle AMB is also equal to the remaining angle FNG; {I. 32} therefore the triangle ABM is equiangular with the triangle FGN.

Similarly we can prove that the triangle BMC is also equiangular with the triangle GNH.

Therefore, proportionally, as AM is to MB, so is FN to NG, and, as BM is to MC, so is GN to NH; so that, in addition, ex aequali, as AM is to MC, so is FN to NH.

But, as AM is to MC, so is the triangle ABM to MBC, and AME to EMC; for they are to one another as their bases. {VI. 1} Therefore also, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents; {V. 12} therefore, as the triangle AMB is to BMC, so is ABE to CBE.

But, as AMB is to BMC, so is AM to MC; therefore also, as AM is to MC, so is the triangle ABE to the triangle EBC.

For the same reason also, as FN is to NH, so is the triangle FGL to the triangle GLH.

And, as AM is to MC, so is FN to NH; therefore also, as the triangle ABE is to the triangle BEC, so is the triangle FGL to the triangle GLH; and, alternately, as the triangle ABE is to the triangle FGL, so is the triangle BEC to the triangle GLH.

Similarly we can prove, if BD, GK be joined, that, as the triangle BEC is to the triangle LGH, so also is the triangle ECD to the triangle LHK.

And since, as the triangle ABE is to the triangle FGL, so is EBC to LGH, and further ECD to LHK, therefore also, as one of the antecedents is to one of the consequents so are all the antecedents to all the consequents; {V. 12} therefore, as the triangle ABE is to the triangle FGL, so is the polygon ABCDE to the polygon FGHL.

But the triangle ABE has to the triangle FGL a ratio duplicate of that which the corresponding side AB has to the corresponding side FG; for similar triangles are in the duplicate ratio of the corresponding sides. {VI. 19} Therefore the polygon ABCDE also has to the polygon FGHL a ratio duplicate of that which the corresponding side AB has to the corresponding side FG.

Therefore etc.

PORISM.

Similarly also it can be proved in the case of quadrilaterals that they are in the duplicate ratio of the corresponding sides. And it was also proved in the case of triangles; therefore also, generally, similar rectilineal figures are to one another in the duplicate ratio of the corresponding sides. Q. E. D.

PROPOSITION 21. Figures which are similar to the same rectilineal figure are also similar to one another.

For let each of the rectilineal figures A, B be similar to C; I say that A is also similar to B.

For, since A is similar to C, it is equiangular with it and has the sides about the equal angles proportional. {VI. Def. 1} Again, since B is similar to C, it is equiangular with it and has the sides about the equal angles proportional.

Therefore each of the figures A, B is equiangular with C and with C has the sides about the equal angles proportional; therefore A is similar to B. Q. E. D.

PROPOSITION 22. If four straight lines be proportional, the rectilineal figures similar and similarly described upon them will also be proportional; and, if the rectilineal figures similar and similarly described upon them be proportional, the straight lines will themselves also be proportional.

Let the four straight lines AB, CD, EF, GH be proportional, so that, as AB is to CD, so is EF to GH, and let there be described on AB, CD the similar and similarly situated rectilineal figures KAB, LCD, and on EF, GH the similar and similarly situated rectilineal figures MF, NH; I say that, as KAB is to LCD, so is MF to NH.

For let there be taken a third proportional O to AB, CD, and a third proportional P to EF, GH. {VI. 11} Then since, as AB is to CD, so is EF to GH, and, as CD is to O, so is GH to P, therefore, ex aequali, as AB is to O, so is EF to P. {V. 22} But, as AB is to O, so is KAB to LCD, {VI. 19, Por.} and, as EF is to P, so is MF to NH; therefore also, as KAB is to LCD, so is MF to NH. {V. 11} Next, let MF be to NH as KAB is to LCD; I say also that, as AB is to CD, so is EF to GH.

For, if EF is not to GH as AB to CD, let EF be to QR as AB to CD, {VI. 12} and on QR let the rectilineal figure SR be described similar and similarly situated to either of the two MF, NH. {VI. 18} Since then, as AB is to CD, so is EF to QR, and there have been described on AB, CD the similar and similarly situated figures KAB, LCD, and on EF, QR the similar and similarly situated figures MF, SR, therefore, as KAB is to LCD, so is MF to SR.

But also, by hypothesis, as KAB is to LCD, so is MF to NH; therefore also, as MF is to SR, so is MF to NH. {V. 11} Therefore MF has the same ratio to each of the figures NH, SR; therefore NH is equal to SR. {V. 9} But it is also similar and similarly

situated to it; therefore GH is equal to QR.

And, since, as AB is to CD, so is EF to QR, while QR is equal to GH, therefore, as AB is to CD, so is EF to GH.

Therefore etc. Q. E. D.

PROPOSITION 23. Equiangular parallelograms have to one another the ratio compounded of the ratios of their sides.

Let AC, CF be equiangular parallelograms having the angle BCD equal to the angle ECG; I say that the parallelogram AC has to the parallelogram CF the ratio compounded of the ratios of the sides.

For let them be placed so that BC is in a straight line with CG; therefore DC is also in a straight line with CE.

Let the parallelogram DG be completed; let a straight line K be set out, and let it be contrived that, as BC is to CG, so is K to L, and, as DC is to CE, so is L to M. {VI. 12} Then the ratios of K to L and of L to M are the same as the ratios of the sides, namely of BC to CG and of DC to CE.

But the ratio of K to M is compounded of the ratio of K to L and of that of L to M; so that K has also to M the ratio compounded of the ratios of the sides.

Now since, as BC is to CG, so is the parallelogram AC to the parallelogram CH, {VI. 1} while, as BC is to CG, so is K to L, therefore also, as K is to L, so is AC to CH. {V. 11} Again, since, as DC is to CE, so is the parallelogram CH to CF, {VI. 1} while, as DC is to CE, so is L to M, therefore also, as L is to M, so is the parallelogram CH to the parallelogram CF. {V. 11} Since then it was proved that, as K is to L, so is the parallelogram AC to the parallelogram CH, and, as L is to M, so is the parallelogram CH to the parallelogram CF, therefore, ex aequali, as K is to M, so is AC to the parallelogram CF.

But K has to M the ratio compounded of the ratios of the sides; therefore AC also has to CF the ratio compounded of the ratios of the sides.

Therefore etc. Q. E. D.

PROPOSITION 24. In any parallelogram the parallelograms about the diameter are similar both to the whole and to one another.

Let ABCD be a parallelogram, and AC its diameter, and let EG, HK be parallelograms about AC; I say that each of the parallelograms EG, HK is similar both to the whole ABCD and to the other.

For, since EF has been drawn parallel to BC, one of the sides of the triangle ABC, proportionally, as BE is to EA, so is CF to FA. {VI. 2} Again, since FG has been drawn parallel to CD, one of the sides of the triangle ACD, proportionally, as CF is to FA, so is DG to GA. {VI. 2} But it was proved that, as CF is to FA, so also is BE to EA; therefore also, as BE is to EA, so is DG to GA, and therefore, componendo, as BA is to AE, so is DA to AG, {V. 18} and, alternately, as BA is to AD, so is EA to AG. {V. 16} Therefore in the parallelograms ABCD, EG, the sides about the common angle BAD are proportional.

And, since GF is parallel to DC, the angle AFG is equal to the angle DCA; and the angle DAC is common to the two triangles ADC, AGF; therefore the triangle ADC is equiangular with the triangle AGF.

For the same reason the triangle ACB is also equiangular with the triangle AFE, and the whole parallelogram ABCD is equiangular with the parallelogram EG.

Therefore, proportionally, as AD is to DC, so is AG to GF, as DC is to CA, so is GF to FA, as AC is to CB, so is AF to FE, and further, as CB is to BA, so is FE to EA.

And, since it was proved that, as DC is to CA, so is GF to FA, and, as AC is to CB, so is AF to FE, therefore, ex aequali, as DC is to CB, so is GF to FE. {V. 22} Therefore in the parallelograms ABCD, EG the sides about the equal angles are proportional; therefore the parallelogram ABCD is similar to the parallelogram EG. {VI. Def. 1} For the same reason the parallelogram ABCD is also similar to the parallelogram KH; therefore each of the parallelograms EG, HK is similar to ABCD.

But figures similar to the same rectilineal figure are also similar to one another; {VI. 21} therefore the parallelogram EG is also similar to the parallelogram HK.

Therefore etc. Q. E. D.

PROPOSITION 25. To construct one and the same figure similar to a given rectilineal figure and equal to another given rectilineal figure.

Let ABC be the given rectilineal figure to which the figure to be constructed must be similar, and D that to which it must be equal; thus it is required to construct one and the same figure similar to ABC and equal to D.

Let there be applied to BC the parallelogram BE equal to the triangle ABC {I. 44}, and to CE the parallelogram CM equal to D in the angle FCE which is equal to the angle CBL. {I. 45} Therefore BC is in a straight line with CF, and LE with EM.

Now let GH be taken a mean proportional to BC, CF {VI. 13}, and on GH let KGH be described similar and similarly situated to ABC. {VI. 18} Then, since, as BC is to GH, so is GH to CF, and, if three straight lines be proportional, as the first is to the third, so is the figure on the first to the similar and similarly situated figure described on the second, {VI. 19, Por.} therefore, as BC is to CF, so is the triangle ABC to the triangle KGH.

But, as BC is to CF, so also is the parallelogram BE to the parallelogram EF. {VI. 1} Therefore also, as the triangle ABC is to the triangle KGH, so is the parallelogram BE to the parallelogram EF; therefore, alternately, as the triangle ABC is to the parallelogram BE, so is the triangle KGH to the parallelogram EF. {V. 16} But the triangle ABC is equal to the parallelogram BE; therefore the triangle KGH is also equal to the parallelogram EF.

But the parallelogram EF is equal to D; therefore KGH is also equal to D.

And KGH is also similar to ABC.

Therefore one and the same figure KGH has been constructed similar to the given rectilineal figure ABC and equal to the other given figure D. Q. E. D.

PROPOSITION 26. If from a parallelogram there be taken away a parallelogram similar and similarly situated to the whole and having a common angle with it, it is about the same diameter with the whole For from the parallelogram ABCD let there be taken away the parallelogram AF similar and similarly situated to ABCD, and having the angle DAB common with it; I say that ABCD is about the same diameter with AF.

For suppose it is not, but, if possible, let AHC be the diameter $\angle$ of ABCD, let GF be produced and carried through to H, and let HK be drawn through H parallel to either of the straight lines AD, BC. {I. 31} Since, then, ABCD is about the same diameter with KG, therefore, as DA is to AB, so is GA to AK. {VI. 24} But also, because of the similarity of ABCD, EG, as DA is to AB, so is GA to AE; therefore also, as GA is to AK, so is GA to AE. {V. 11} Therefore GA has the same ratio to each of the straight lines AK, AE.

Therefore AE is equal to AK {V. 9}, the less to the greater : which is impossible.

Therefore ABCD cannot but be about the same diameter with AF; therefore the parallelogram ABCD is about the same diameter with the parallelogram AF.

Therefore etc. Q. E. D.

PROPOSITION 27. Of all the parallelograms applied to the same straight line and deficient by parallelogrammic figures similar and similarly situated to that described on the half of the straight line, that parallelogram is greatest which is applied to the half of the straight line and is similar to the defect.

Let AB be a straight line and let it be bisected at C; let there be applied to the straight line AB the parallelogram AD deficient by the parallelogrammic figure DB described on the half of AB, that is, CB; I say that, of all the parallelograms applied to AB and deficient by parallelogrammic figures similar and similarly situated to DB, AD is greatest.

For let there be applied to the straight line AB the parallelogram AF deficient by the parallelogrammic figure FB similar and similarly situated to DB; I say that AD is greater than AF.

For, since the parallelogram DB is similar to the parallelogram FB, they are about the same diameter. {VI. 26} Let their diameter DB be drawn, and let the figure be described.

Then, since CF is equal to FE, {I. 43} and FB is common, therefore the whole CH is equal to the whole KE.

But CH is equal to CG, since AC is also equal to CB. {I. 36} Therefore GC is also equal to EK.

Let CF be added to each; therefore the whole AF is equal to the gnomon LMN; so that the parallelogram DB, that is, AD, is greater than the parallelogram AF.

Therefore etc.

PROPOSITION 28. To a given straight line to apply a parallelogram equal to a given rectilinear figure and deficient by a parallelogrammic figure similar to a given one : thus the given rectilinear figure must not be greater than the parallelogram described on the half of the straight line and similar to the defect.

Let AB be the given straight line, C the given rectilinear figure to which the figure to be applied to AB is required to be equal, not being greater than the parallelogram described on the half of AB and similar to the defect, and D the parallelogram to which the defect is required to be similar; thus it is required to apply to the given straight line AB a parallelogram equal to the given rectilinear figure C and deficient by a parallelogrammic figure which is similar to D.

Let AB be bisected at the point E, and on EB let ECFG be described similar and similarly situated to D; {VI. 18} let the parallelogram AG be completed.

If then AG is equal to C, that which was enjoined will have been done; for there has been applied to the given straight line AB the parallelogram AG equal to the given rectilinear figure C and deficient by a parallelogrammic figure GB which is similar to D.

But, if not, let HE be greater than C.

Now HE is equal to GB; therefore GB is also greater than C.

Let KLMN be constructed at once equal to the excess by which GB is greater than C and similar and similarly situated to D. {VI. 25} But D is similar to GB; therefore KM is also similar to GB. {VI. 21} Let, then, KL correspond to GE, and LM to GF.

Now, since GB is equal to C, KM, therefore GB is greater than KM; therefore also GE is greater than KL, and GF than LM.

Let GO be made equal to KL , and GP equal to LM ; and let the parallelogram $OGPQ$ be completed; therefore it is equal and similar to KM .

Therefore GQ is also similar to GB ; {VI. 21} therefore GQ is about the same diameter with GB . {VI. 26} Let GQB be their diameter, and let the figure be described.

Then, since BG is equal to C , KM , and in them GQ is equal to KM , therefore the remainder, the gnomon UWV , is equal to the remainder C .

And, since PR is equal to OS , let QB be added to each; therefore the whole PB is equal to the whole OB .

But OB is equal to TE , since the side AE is also equal to the side EB ; {I. 36} therefore TE is also equal to PB .

Let OS be added to each; therefore the whole TS is equal to the whole, the gnomon VWU .

But the gnomon VWU was proved equal to C ; therefore TS is also equal to C .

Therefore to the given straight line AB there has been applied the parallelogram ST equal to the given rectilineal figure C and deficient by a parallelogrammic figure QB which is similar to D . $Q. E. F.$

PROPOSITION 29. To a given straight line to apply a parallelogram equal to a given rectilineal figure and exceeding by a parallelogrammic figure similar to a given one.

Let AB be the given straight line, C the given rectilineal figure to which the figure to be applied to AB is required to be equal, and D that to which the excess is required to be similar; thus it is required to apply to the straight line AB a parallelogram equal to the rectilineal figure C and exceeding by a parallelogrammic figure similar to D .

Let AB be bisected at E; let there be described on EB the parallelogram BF similar and similarly situated to D; and let GH be constructed at once equal to the sum of BF, C and similar and similarly situated to D. {VI. 25} Let KH correspond to FL and KG to FE.

Now, since GH is greater than FB, therefore KH is also greater than FL, and KG than FE.

Let FL, FE be produced, let FLM be equal to KH, and FEN to KG, and let MN be completed; therefore MN is both equal and similar to GH.

But GH is similar to EL; therefore MN is also similar to EL; {VI. 21} therefore EL is about the same diameter with MN. {VI. 26} Let their diameter FO be drawn, and let the figure be described.

Since GH is equal to EL, C, while GH is equal to MN, therefore MN is also equal to EL, C.

Let EL be subtracted from each; therefore the remainder, the gnomon XWV, is equal to C.

Now, since AE is equal to EB, AN is also equal to NB {I. 36}, that is, to LP {I. 43}.

Let EO be added to each; therefore the whole AO is equal to the gnomon VWX.

But the gnomon VWX is equal to C; therefore AO is also equal to C.

Therefore to the given straight line AB there has been applied the parallelogram AO equal to the given rectilineal figure C and exceeding by a parallelogrammic figure QP which is similar to D, since PQ is also similar to EL {VI. 24}. Q. E. F.

PROPOSITION 30. To cut a given finite straight line in extreme and mean ratio.

Let AB be the given finite straight line; thus it is required to cut AB in extreme and mean ratio.

On AB let the square BC be described; and let there be applied to AC the parallelogram CD equal to BC and exceeding by the figure AD similar to BC. {VI. 29} Now BC is a square; therefore AD is also a square.

And, since BC is equal to CD, let CE be subtracted from each; therefore the remainder BF is equal to the remainder AD.

But it is also equiangular with it; therefore in BF, AD the sides about the equal angles are reciprocally proportional; {VI. 14} therefore, as FE is to ED, so is AE to EB.

But FE is equal to AB, and ED to AE.

Therefore, as BA is to AE, so is AE to EB.

And AB is greater than AE; therefore AE is also greater than EB.

Therefore the straight line AB has been cut in extreme and mean ratio at E, and the greater segment of it is AE. Q. E. F.

PROPOSITION 31. In right-angled triangles the figure on the side subtending the right angle is equal to the similar and similarly described figures on the sides containing the right angle.

Let ABC be a right-angled triangle having the angle BAC right; I say that the figure on BC is equal to the similar and similarly described figures on BA, AC.

Let AD be drawn perpendicular.

Then since, in the right-angled triangle ABC, AD has been drawn from the right angle at A perpendicular to the base BC, the triangles ABD, ADC adjoining the perpendicular are similar both to the whole ABC and to one another. {VI. 8} And, since ABC is similar to ABD, therefore, as CB is to BA, so is AB to BD. {VI. Def.

1} And, since three straight lines are proportional, as the first is to the third, so is the figure on the first to the similar and similarly described figure on the second. {VI. 19, Por.} Therefore, as CB is to BD, so is the figure on CB to the similar and similarly described figure on BA.

For the same reason also, as BC is to CD, so is the figure on BC to that on CA; so that, in addition, as BC is to BD, DC, so is the figure on BC to the similar and similarly described figures on BA, AC.

But BC is equal to BD, DC; therefore the figure on BC is also equal to the similar and similarly described figures on BA, AC.

Therefore etc. Q. E. D.

PROPOSITION 32. If two triangles having two sides proportional to two sides be placed together at one angle so that their corresponding sides are also parallel, the remaining sides of the triangles will be in a straight line.

Let ABC, DCE be two triangles having the two sides BA, AC proportional to the two sides DC, DE, so that, as AB is to AC, so is DC to DE, and AB parallel to DC, and AC to DE; I say that BC is in a straight line with CE.

For, since AB is parallel to DC, and the straight line AC has fallen upon them, the alternate angles BAC, ACD are equal to one another. {I. 29} For the same reason the angle CDE is also equal to the angle ACD; so that the angle BAC is equal to the angle CDE.

And, since ABC, DCE are two triangles having one angle, the angle at A, equal to one angle, the angle at D, and the sides about the equal angles proportional, so that, as BA is to AC, so is CD to DE, therefore the triangle ABC is equiangular with the triangle DCE; {VI. 6} therefore the angle ABC is equal to the angle DCE.

But the angle ACD was also proved equal to the angle BAC; therefore the whole angle ACE is equal to the two angles ABC, BAC.

Let the angle ACB be added to each; therefore the angles ACE, ACB are equal to the angles BAC, ACB, CBA.

But the angles BAC, ABC, ACB are equal to two right angles; {I. 32} therefore the angles ACE, ACB are also equal to two right angles.

Therefore with a straight line AC, and at the point C on it, the two straight lines BC, CE not lying on the same side make the adjacent angles ACE, ACB equal to two right angles; therefore BC is in a straight line with CE. {I. 14} Therefore etc. Q. E. D.

PROPOSITION 33. In equal circles angles have the same ratio as the circumferences on which they stand, whether they stand at the centres or at the circumferences.

Let ABC, DEF be equal circles, and let the angles BGC, EHF be angles at their centres G, H, and the angles BAC, EDF angles at the circumferences; I say that, as the circumference BC is to the circumference EF, so is the angle BGC to the angle EHF, and the angle BAC to the angle EDF.

For let any number of consecutive circumferences CK, KL be made equal to the circumference BC, and any number of consecutive circumferences FM, MN equal to the circumference EF; and let GK, GL, HM, HN be joined.

Then, since the circumferences BC, CK, KL are equal to one another, the angles BGC, CGK, KGL are also equal to one another; {III. 27} therefore, whatever multiple the circumference BL is of BC, that multiple also is the angle BGL of the angle BGC.

For the same reason also, whatever multiple the circumference NE is of EF, that multiple also is the angle NHE of the angle EHF.

If then the circumference BL is equal to the circumference EN, the angle BGL is also equal to the angle EHN; {III. 27} if the circumference BL is greater than the circumference EN, the angle BGL is also greater than the angle EHN; and, if less, less.

There being then four magnitudes, two circumferences BC, EF, and two angles BGC, EHF, there have been taken, of the circumference BC and the angle BGC equimultiples, namely the circumference BL and the angle BGL, and of the circumference EF and the angle EHF equimultiples, namely the circumference EN and the angle EHN.

And it has been proved that, if the circumference BL is in excess of the circumference EN, the angle BGL is also in excess of the angle EHN; if equal, equal; and if less, less.

Therefore, as the circumference BC is to EF, so is the angle BGC to the angle EHF. {V. Def. 5} But, as the angle BGC is to the angle EHF, so is the angle BAC to the angle EDF; for they are doubles respectively.

Therefore also, as the circumference BC is to the circumference EF, so is the angle BGC to the angle EHF, and the angle BAC to the angle EDF.

Therefore etc. Q. E. D.

ENDNOTES.

Under the same height. The Greek text has “under the same height AC,” with a figure in which the side AC common to the two triangles is perpendicular to the base and is therefore itself the “height.” But, even if the two triangles are placed contiguously so as to have a common side AC, it is quite gratuitous to require it to be perpendicular to the base. Theon, on this occasion making an improvement, altered to “which are (ὄντα) under the same height, (namely) the perpendicular drawn from A to BD.” I have ventured to alter so far as to omit

“AC” and to draw the figure in the usual way.

Abc, agb, Ahg. Euclid, indifferent to exact order, writes “AHG, AGB, ABC.”

Since then it was proved that, as the base BC is to CD, so is the triangle ABC to the triangle ACD. Here again words have to be supplied in translating the extremely terse Greek ἐπεὶ οὖν ἐδείχθη, ὥς μὲν ἡ βάσις ΒΓ πρὸς τὴν ΓΔ, οὕτως τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΑΓΔ τρίγωνον, literally “since was proved, as the base BC to CD, so the triangle ABC to the triangle ACD.” Cf. note on V. 16, p. 165.

any angle. The expression here and in the two following propositions is τυχοῦσα γωνία, corresponding exactly to τυχὸν σημεῖον which I have translated as “a point (taken) at random” ; but “an angle (taken) at random” would not be so appropriate where it is a question, not of taking any angle at all, but of drawing a straight line casually so as to make any angle with another straight line.

to find. The Greek word, here and in the next two propositions, is προσευρεῖν, literally “to find in addition.”

and such that, as AB is to BC, so is DE to EF, literally “(triangles) having the angle at B equal to the angle at E, and (having), as AB to BC, so DE to EF.”

in the same ratio as the wholes. The same word ὁμόλογος is used which I have generally translated by “corresponding.” But here it is followed by a dative, ὁμόλογα τοῖς ὅλοις “homologous with the wholes,” instead of being used absolutely. The meaning can therefore here be nothing else but “in the same ratio with” or “proportional to the wholes” ; and Euclid seems to recognise that he is making a special use of the word, because he explains it lower down (I. 46): “the triangles are homologous to the wholes, that is, in such manner that the triangles are

proportional, and ABE, EBC, ECD are antecedents, while FGL, LGH, LHK are their consequents.”

ἐπόμενα αὐτῶν, “their consequents,” is a little awkward, but may be supposed to indicate which triangles correspond to which as consequent to antecedent.

the ratio compounded of the ratios of the sides, λόγον τὸν συγκεείμενον ἐκ τῶν πλευρῶν which, meaning literally “the ratio compounded of the sides,” is negligently written here and commonly for λόγον τὸν συγκεείμενον ἐκ τῶν τῶν πλευρῶν (sc. λόγων).

let it be contrived that, as BC is to CG, so is K to L. The Greek phrase is of the usual terse kind, untranslatable literally : καὶ γεγονέτω ὥς μὲν ἡ ΒΓ πρὸς τὴν ΓΗ, οὕτως ἡ Κ πρὸς τὸ Λ, the words meaning “and let (there) be made, as BC to CG, so K to L,” where L is the straight line which has to be constructed.

to which the figure to be constructed must be similar, literally “to which it is required to construct (one) similar,” ὃ δεῖ ὅμοιον συστήσασθαι.

BOOK VII. DEFINITIONS.

1. AN UNIT IS that by virtue of which each of the things that exist is called one.
2. A number is a multitude composed of units.
3. A number is a part of a number, the less of the greater, when it measures the greater; 4. but parts when it does not measure it.
5. The greater number is a multiple of the less when it is measured by the less.
6. An even number is that which is divisible into two equal parts.

7. An odd number is that which is not divisible into two equal parts, or that which differs by an unit from an even number.
8. An even-times even number is that which is measured by an even number according to an even number.
9. An even-times odd number is that which is measured by an even number according to an odd number.
10. An odd-times odd number is that which is measured by an odd number according to an odd number.
11. A prime number is that which is measured by an unit alone.
12. Numbers prime to one another are those which are measured by an unit alone as a common measure.
13. A composite number is that which is measured by some number.
14. Numbers composite to one another are those which are measured by some number as a common measure.
15. A number is said to multiply a number when that which is multiplied is added to itself as many times as there are units in the other, and thus some number is produced.
16. And, when two numbers having multiplied one another make some number, the number so produced is called plane, and its sides are the numbers which have multiplied one another.
17. And, when three numbers having multiplied one another make some number, the number so produced is solid, and its sides are the numbers which have multiplied one another.
18. A square number is equal multiplied by equal, or a number which is contained by two equal numbers.
19. And a cube is equal multiplied by equal and again by equal, or a number which is contained by three equal numbers.

20. Numbers are proportional when the first is the same multiple, or the same part, or the same parts, of the second that the third is of the fourth.

21. Similar plane and solid numbers are those which have their sides proportional.

22. A perfect number is that which is equal to its own parts.

PROPOSITIONS.

PROPOSITION I. TWO UNEQUAL NUMBERS being set out, and the less being continually subtracted in turn from the greater, if the number which is left never measures the one before it until an unit is left, the original numbers will be prime to one another.

For, the less of two unequal numbers AB, CD being continually subtracted from the greater, let the number which is left never measure the one before it until an unit is left; I say that AB, CD are prime to one another, that is, that an unit alone measures AB, CD.

For, if AB, CD are not prime to one another, some number will measure them.

Let a number measure them, and let it be E; let CD, measuring BF, leave FA less than itself, let AF, measuring DG, leave GC less than itself, and let GC, measuring FH, leave an unit HA.

Since, then, E measures CD, and CD measures BF, therefore E also measures BF.

But it also measures the whole BA; therefore it will also measure the remainder AF.

But AF measures DG; therefore E also measures DG.

But it also measures the whole DC therefore it will also measure the remainder CG.

But CG measures FH; therefore E also measures FH.

But it also measures the whole FA; therefore it will also measure the remainder, the unit AH, though it is a number: which is impossible.

Therefore no number will measure the numbers AB, CD; therefore AB, CD are prime to one another. {VII. Def. 12} Q. E. D.

PROPOSITION 2. Given two numbers not prime to one another, to find their greatest common measure.

Let AB, CD be the two given numbers not prime to one another.

Thus it is required to find the greatest common measure of AB, CD.

If now CD measures AB — and it also measures itself — CD is a common measure of CD, AB.

And it is manifest that it is also the greatest; for no greater number than CD will measure CD.

But, if CD does not measure AB, then, the less of the numbers AB, CD being continually subtracted from the greater, some number will be left which will measure the one before it.

For an unit will not be left; otherwise AB, CD will be prime to one another {VII. 1}, which is contrary to the hypothesis.

Therefore some number will be left which will measure the one before it.

Now let CD, measuring BE, leave EA less than itself, let EA, measuring DF, leave FC less than itself, and let CF measure AE.

Since then, CF measures AE, and AE measures DF, therefore CF will also measure DF.

But it also measures itself; therefore it will also measure the whole CD.

But CD measures BE; therefore CF also measures BE.

But it also measures EA; therefore it will also measure the whole BA.

But it also measures CD; therefore CF measures AB, CD.

Therefore CF is a common measure of AB, CD.

I say next that it is also the greatest.

For, if CF is not the greatest common measure of AB, CD, some number which is greater than CF will measure the numbers AB, CD.

Let such a number measure them, and let it be G.

Now, since G measures CD, while CD measures BE, G also measures BE.

But it also measures the whole BA; therefore it will also measure the remainder AE.

But AE measures DF; therefore G will also measure DF.

But it also measures the whole DC; therefore it will also measure the remainder CF, that is, the greater will measure the less: which is impossible.

Therefore no number which is greater than CF will measure the numbers AB, CD; therefore CF is the greatest common measure of AB, CD.

PORISM.

From this it is manifest that, if a number measure two numbers, it will also measure their greatest common measure.

Q. E. D.

PROPOSITION 3. Given three numbers not prime to one another, to find their greatest common measure.

Let A, B, C be the three given numbers not prime to one another; thus it is required to find the greatest common measure of A, B, C.

For let the greatest common measure, D, of the two numbers A, B be taken; {VII. 2} then D either measures, or does not measure, C.

First, let it measure it.

But it measures A, B also; therefore D measures A, B, C; therefore D is a common measure of A, B, C.

I say that it is also the greatest.

For, if D is not the greatest common measure of A, B, C, some number which is greater than D will measure the numbers A, B, C.

Let such a number measure them, and let it be E.

Since then E measures A, B, C, it will also measure A, B; therefore it will also measure the greatest common measure of A, B. {VII. 2, Por.} But the greatest common measure of A, B is D; therefore E measures D, the greater the less: which is impossible.

Therefore no number which is greater than D will measure the numbers A, B, C; therefore D is the greatest common measure of A, B, C.

Next, let D not measure C; I say first that C, D are not prime to one another.

For, since A, B, C are not prime to one another, some number will measure them.

Now that which measures A, B, C will also measure A, B, and will measure D, the greatest common measure of A, B. {VII. 2, Por.} But it measures C also; therefore some number will measure the

numbers D, C; therefore D, C are not prime to one another.

Let then their greatest common measure E be taken. {VII. 2} Then, since E measures D, and D measures A, B, therefore E also measures A, B.

But it measures C also; therefore E measures A, B, C; therefore E is a common measure of A, B, C.

I say next that it is also the greatest.

For, if E is not the greatest common measure of A, B, C, some number which is greater than E will measure the numbers A, B, C.

Let such a number measure them, and let it be F.

Now, since F measures A, B, C, it also measures A, B; therefore it will also measure the greatest common measure of A, B. {VII. 2, Por.} But the greatest common measure of A, B is D; therefore F measures D.

And it measures C also; therefore F measures D, C; therefore it will also measure the greatest common measure of D, C. {VII. 2, Por.} But the greatest common measure of D, C is E; therefore F measures E, the greater the less: which is impossible.

Therefore no number which is greater than E will measure the numbers A, B, C; therefore E is the greatest common measure of A, B, C. Q. E. D.

PROPOSITION 4. Any number is either a part or parts of any number, the less of the greater.

Let A, BC be two numbers, and let BC be the less; I say that BC is either a part, or parts, of A.

For A, BC are either prime to one another or not.

First, let A, BC be prime to one another.

Then, if BC be divided into the units in it, each unit of those in BC will be some part of A; so that BC is parts of A.

Next let A, BC not be prime to one another; then BC either measures, or does not measure, A.

If now BC measures A, BC is a part of A.

But, if not, let the greatest common measure D of A, BC be taken; {VII. 2} and let BC be divided into the numbers equal to D, namely BE, EF, FC.

Now, since D measures A, D is a part of A.

But D is equal to each of the numbers BE, EF, FC; therefore each of the numbers BE, EF, FC is also a part of A; so that BC is parts of A.

Therefore etc. Q. E. D.

PROPOSITION 5. If a number be a part of a number, and another be the same part of another, the sum will also be the same part of the sum that the one is of the one.

For let the number A be a part of BC, and another, D, the same part of another EF that A is of BC; I say that the sum of A, D is also the same part of the sum of BC, EF that A is of BC.

For since, whatever part A is of BC, D is also the same part of EF, therefore, as many numbers as there are in BC equal to A, so many numbers are there also in EF equal to D.

Let BC be divided into the numbers equal to A, namely BG, GC, and EF into the numbers equal to D, namely EH, HF; then the multitude of BG, GC will be equal to the multitude of EH, HF.

And, since BG is equal to A, and EH to D, therefore BG, EH are also equal to A, D.

For the same reason GC, HF are also equal to A, D.

Therefore, as many numbers as there are in BC equal to A, so many are there also in BC, EF equal to A, D.

Therefore, whatever multiple BC is of A, the same multiple also is the sum of BC, EF of the sum of A, D.

Therefore, whatever part A is of BC, the same part also is the sum of A, D of the sum of BC, EF. Q. E. D.

PROPOSITION 6. If a number be parts of a number, and another be the same parts of another, the sum will also be the same parts of the sum that the one is of the one.

For let the number AB be parts of the number C, and another, DE, the same parts of another, F, that AB is of C; I say that the sum of AB, DE is also the same parts of the sum of C, F that AB is of C.

For since, whatever parts AB is of C, DE is also the same parts of F, therefore, as many parts of C as there are in AB, so many parts of F are there also in DE.

Let AB be divided into the parts of C, namely AG, GB, and DE into the parts of F, namely DH, HE; thus the multitude of AG, GB will be equal to the multitude of DH, HE.

And since, whatever part AG is of C, the same part is DH of F also, therefore, whatever part AG is of C, the same part also is the sum of AG, DH of the sum of C, F. {VII. 5} For the same reason, whatever part GB is of C, the same part also is the sum of GB, HE of the sum of C, F.

Therefore, whatever parts AB is of C, the same parts also is the sum of AB, DE of the sum of C, F. Q. E. D.

PROPOSITION 7. If a number be that part of a number, which a number subtracted is of a number subtracted, the remainder will also be the same part of the remainder that the whole is of the whole.

For let the number AB be that part of the number CD which AE subtracted is of CF subtracted; I say that the remainder EB is also the same part of the remainder FD that the whole AB is of the whole CD.

For, whatever part AE is of CF, the same part also let EB be of CG.

Now since, whatever part AE is of CF, the same part also is EB of CG, therefore, whatever part AE is of CF, the same part also is AB of GF. {VII. 5} But, whatever part AE is of CF, the same part also, by hypothesis, is AB of CD; therefore, whatever part AB is of GF, the same part is it of CD also; therefore GF is equal to CD.

Let CF be subtracted from each; therefore the remainder GC is equal to the remainder FD.

Now since, whatever part AE is of CF, the same part also is EB of GC, while GC is equal to FD, therefore, whatever part AE is of CF, the same part also is EB of FD.

But, whatever part AE is of CF, the same part also is AB of CD; therefore also the remainder EB is the same part of the remainder FD that the whole AB is of the whole CD. Q. E. D.

PROPOSITION 8. If a number be the same parts of a number that a number subtracted is of a number subtracted, the remainder will also be the same parts of the remainder that the whole is of the whole.

For let the number AB be the same parts of the number CD that AE subtracted is of CF subtracted; I say that the remainder EB is also the same parts of the remainder FD that the whole AB is of the whole CD.

For let GH be made equal to AB.

Therefore, whatever parts GH is of CD, the same parts also is AE of CF.

Let GH be divided into the parts of CD, namely GK, KH, and AE into the parts of CF, namely AL, LE; thus the multitude of GK, KH will be equal to the multitude of AL, LE.

Now since, whatever part GK is of CD, the same part also is AL of CF, while CD is greater than CF, therefore GK is also greater than AL.

Let GM be made equal to AL.

Therefore, whatever part GK is of CD, the same part also is GM of CF; therefore also the remainder MK is the same part of the remainder FD that the whole GK is of the whole CD. {VII. 7} Again, since, whatever part KH is of CD, the same part also is EL of CF, while CD is greater than CF, therefore HK is also greater than EL.

Let KN be made equal to EL.

Therefore, whatever part KH is of CD, the same part also is KN of CF; therefore also the remainder NH is the same part of the remainder FD that the whole KH is of the whole CD. {VII. 7} But the remainder MK was also proved to be the same part of the remainder FD that the whole GK is of the whole CD; therefore also the sum of MK, NH is the same parts of DF that the whole HG is of the whole CD.

But the sum of MK, NH is equal to EB, and HG is equal to BA; therefore the remainder EB is the same parts of the remainder FD that the whole AB is of the whole CD. Q. E. D.

PROPOSITION 9. If a number be a part of a number, and another be the same part of another, alternately also, whatever part or parts the first is of the third, the same part, or the same parts, will the second also be of the fourth.

For let the number A be a part of the number BC, and another, D, the same part of another, EF, that A is of BC; I say that, alternately also, whatever part or parts A is of D, the same part

or parts is BC of EF also.

For since, whatever part A is of BC, the same part also is D of EF, therefore, as many numbers as there are in BC equal to A, so many also are there in EF equal to D.

Let BC be divided into the numbers equal to A, namely BG, GC, and EF into those equal to D, namely EH, HF; thus the multitude of BG, GC will be equal to the multitude of EH, HF.

Now, since the numbers BG, GC are equal to one another, and the numbers EH, HF are also equal to one another, while the multitude of BG, GC is equal to the multitude of EH, HF, therefore, whatever part or parts BG is of EH, the same part or the same parts is GC of HF also; so that, in addition, whatever part or parts BG is of EH, the same part also, or the same parts, is the sum BC of the sum EF. {VII. 5, 6} But BG is equal to A, and EH to D; therefore, whatever part or parts A is of D, the same part or the same parts is BC of EF also. Q. E. D.

PROPOSITION 10. If a number be parts of a number, and another be the same parts of another, alternately also, whatever parts or part the first is of the third, the same parts or the same part will the second also be of the fourth.

For let the number AB be parts of the number C, and another, DE, the same parts of another, F; I say that, alternately also, whatever parts or part AB is of DE, the same parts or the same part is C of F also.

For since, whatever parts AB is of C, the same parts also is DE of F, therefore, as many parts of C as there are in AB, so many parts also of F are there in DE.

Let AB be divided into the parts of C, namely AG, GB, and DE into the parts of F, namely DH, HE; thus the multitude of AG, GB will be equal to the multitude of DH, HE.

Now since, whatever part AG is of C, the same part also is DH of F, alternately also, whatever part or parts AG is of DH, the same part or the same parts is C of F also. {VII. 9} For the same reason also, whatever part or parts GB is of HE, the same part or the same parts is C of F also; so that, in addition, whatever parts or part AB is of DE, the same parts also, or the same part, is C of F. {VII. 5, 6} Q. E. D.

PROPOSITION 11. If, as whole is to whole, so is a number subtracted to a number subtracted, the remainder will also be to the remainder as whole to whole.

As the whole AB is to the whole CD, so let AE subtracted be to CF subtracted; I say that the remainder EB is also to the remainder FD as the whole AB to the whole CD.

Since, as AB is to CD, so is AE to CF, whatever part or parts AB is of CD, the same part or the same parts is AE of CF also; {VII. Def. 20} Therefore also the remainder EB is the same part or parts of FD that AB is of CD. {VII. 7, 8} Therefore, as EB is to FD, so is AB to CD. {VII. Def. 20} Q. E. D.

PROPOSITION 12. If there be as many numbers as we please in proportion, then, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents.

Let A, B, C, D be as many numbers as we please in proportion, so that, as A is to B, so is C to D; I say that, as A is to B, so are A, C to B, D.

For since, as A is to B, so is C to D, whatever part or parts A is of B, the same part or parts is C of D also. {VII. Def. 20} Therefore also the sum of A, C is the same part or the same parts of the sum of B, D that A is of B. {VII. 5, 6} Therefore, as A is to B, so are A, C to B, D. {VII. Def. 20} **PROPOSITION 13.** If four numbers be proportional, they will also be proportional alternately.

Let the four numbers A, B, C, D be proportional, so that, as A is to B, so is C to D; I say that they will also be proportional alternately, so that, as A is to C, so will B be to D.

For since, as A is to B, so is C to D, therefore, whatever part or parts A is of B, the same part or the same parts is C of D also. {VII. Def. 20} Therefore, alternately, whatever part or parts A is of C, the same part or the same parts is B of D also. {VII. 10} Therefore, as A is to C, so is B to D. {VII. Def. 20} Q. E. D.

PROPOSITION 14. If there be as many numbers as we please, and others equal to them in multitude, which taken two and two are in the same ratio, they will also be in the same ratio ex aequali.

Let there be as many numbers as we please A, B, C, and others equal to them in multitude D, E, F, which taken two and two are in the same ratio, so that, as A is to B, so is D to E, and, as B is to C, so is E to F; I say that, ex aequali, as A is to C, so also is D to F.

For, since, as A is to B, so is D to E, therefore, alternately, as A is to D, so is B to E. {VII. 13} Again, since, as B is to C, so is E to F, therefore, alternately, as B is to E, so is C to F. {VII. 13} But, as B is to E, so is A to D; therefore also, as A is to D, so is C to F.

Therefore, alternately, as A is to C, so is D to F. {id.}

PROPOSITION 15. If an unit measure any number, and another number measure any other number the same number of times, alternately also, the unit will measure the third number the same number of times that the second measures the fourth.

For let the unit A measure any number BC, and let another number D measure any other number EF the same number of times; I say that, alternately also, the unit A measures the number D the same number of times that BC measures EF.

For, since the unit A measures the number BC the same number of times that D measures EF, therefore, as many units as there

are in BC, so many numbers equal to D are there in EF also.

Let BC be divided into the units in it, BG, GH, HC, and EF into the numbers EK, KL, LF equal to D.

Thus the multitude of BG, GH, HC will be equal to the multitude of EK, KL, LF.

And, since the units BG, GH, HC are equal to one another, and the numbers EK, KL, LF are also equal to one another, while the multitude of the units BG, GH, HC is equal to the multitude of the numbers EK, KL, LF, therefore, as the unit BG is to the number EK, so will the unit GH be to the number KL, and the unit HC to the number LF.

Therefore also, as one of the antecedents is to one of the consequents, so will all the antecedents be to all the consequents; {VII. 12} therefore, as the unit BG is to the number EK, so is BC to EF.

But the unit BG is equal to the unit A, and the number EK to the number D.

Therefore, as the unit A is to the number D, so is BC to EF.

Therefore the unit A measures the number D the same number of times that BC measures EF. Q. E. D.

PROPOSITION 16. If two numbers by multiplying one another make certain numbers, the numbers so produced will be equal to one another.

Let A, B be two numbers, and let A by multiplying B make C, and B by multiplying A make D; I say that C is equal to D.

For, since A by multiplying B has made C, therefore B measures C according to the units in A.

But the unit E also measures the number A according to the units in it; therefore the unit E measures A the same number of

times that B measures C.

Therefore, alternately, the unit E measures the number B the same number of times that A measures C. {VII. 15} Again, since B by multiplying A has made D, therefore A measures D according to the units in B.

But the unit E also measures B according to the units in it; therefore the unit E measures the number B the same number of times that A measures D.

But the unit E measured the number B the same number of times that A measures C; therefore A measures each of the numbers C, D the same number of times.

Therefore C is equal to D. Q. E. D.

PROPOSITION 17. If a number by multiplying two numbers make certain numbers, the numbers so produced will have the same ratio as the numbers multiplied.

For let the number A by multiplying the two numbers B, C make D, E; I say that, as B is to C, so is D to E.

For, since A by multiplying B has made D, therefore B measures D according to the units in A.

But the unit F also measures the number A according to the units in it; therefore the unit F measures the number A the same number of times that B measures D.

Therefore, as the unit F is to the number A, so is B to D. {VII. Def. 20} For the same reason, as the unit F is to the number A, so also is C to E; therefore also, as B is to D, so is C to E.

Therefore, alternately, as B is to C, so is D to E. {VII. 13} Q. E. D.

PROPOSITION 18. If two numbers by multiplying any number make certain numbers, the numbers so produced will have the same ratio as the multipliers.

For let two numbers A, B by multiplying any number C make D, E; I say that, as A is to B, so is D to E.

For, since A by multiplying C has made D, therefore also C by multiplying A has made D. {VII. 16} For the same reason also C by multiplying B has made E.

Therefore the number C by multiplying the two numbers A, B has made D, E.

Therefore, as A is to B, so is D to E. {VII. 17} **PROPOSITION 19.** If four numbers be proportional, the number produced from the first and fourth will be equal to the number produced from the second and third; and, if the number produced from the first and fourth be equal to that produced from the second and third, the four numbers will be proportional.

Let A, B, C, D be four numbers in proportion, so that, as A is to B, so is C to D; and let A by multiplying D make E, and let B by multiplying C make F; I say that E is equal to F.

For let A by multiplying C make G.

Since, then, A by multiplying C has made G, and by multiplying D has made E, the number A by multiplying the two numbers C, D has made G, E.

Therefore, as C is to D, so is G to E. {VII. 17} But, as C is to D, so is A to B; therefore also, as A is to B, so is G to E.

Again, since A by multiplying C has made G, but, further, B has also by multiplying C made F, the two numbers A, B by multiplying a certain number C have made G, F.

Therefore, as A is to B, so is G to F. {VII. 18} But further, as A is to B, so is G to E also; therefore also, as G is to E, so is G to F.

Therefore G has to each of the numbers E, F the same ratio; therefore E is equal to F. {cf. V. 9} Again, let E be equal to F; I say that, as A is to B, so is C to D.

For, with the same construction, since E is equal to F, therefore, as G is to E, so is G to F. {cf. V. 7} But, as G is to E, so is C to D, {VII. 17} and, as G is to F, so is A to B. {VII. 18} Therefore also, as A is to B, so is C to D. Q. E. D.

PROPOSITION 20. The least numbers of those which have the same ratio with them measure those which have the same ratio the same number of times, the greater the greater and the less the less.

For let CD, EF be the least numbers of those which have the same ratio with A, B; I say that CD measures A the same number of times that EF measures B.

Now CD is not parts of A.

For, if possible, let it be so; therefore EF is also the same parts of B that CD is of A. {VII. 13 and Def. 20} Therefore, as many parts of A as there are in CD, so many parts of B are there also in EF.

Let CD be divided into the parts of A, namely CG, GD, and EF into the parts of B, namely EH, HF; thus the multitude of CG, GD will be equal to the multitude of EH, HF.

Now, since the numbers CG, GD are equal to one another, and the numbers EH, HF are also equal to one another, while the multitude of CG, GD is equal to the multitude of EH, HF, therefore, as CG is to EH, so is GD to HF.

Therefore also, as one of the antecedents is to one of the consequents, so will all the antecedents be to all the consequents. {VII. 12} Therefore, as CG is to EH, so is CD to EF.

Therefore CG, EH are in the same ratio with CD, EF, being less than they: which is impossible, for by hypothesis CD, EF are the least numbers of those which have the same ratio with them.

Therefore CD is not parts of A; therefore it is a part of it. {VII. 4} And EF is the same part of B that CD is of A; {VII. 13 and Def. 20}

therefore CD measures A the same number of times that EF measures B. Q. E. D.

PROPOSITION 21. Numbers prime to one another are the least of those which have the same ratio with them.

Let A, B be numbers prime to one another; I say that A, B are the least of those which have the same ratio with them.

For, if not, there will be some numbers less than A, B which are in the same ratio with A, B.

Let them be C, D.

Since, then, the least numbers of those which have the same ratio measure those which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent, {VII. 20} therefore C measures A the same number of times that D measures B.

Now, as many times as C measures A, so many units let there be in E.

Therefore D also measures B according to the units in E.

And, since C measures A according to the units in E, therefore E also measures A according to the units in C. {VII. 16} For the same reason E also measures B according to the units in D. {VII. 16} Therefore E measures A, B which are prime to one another: which is impossible. {VII. Def. 12} Therefore there will be no numbers less than A, B which are in the same ratio with A, B.

Therefore A, B are the least of those which have the same ratio with them. Q. E. D.

PROPOSITION 22. The least numbers of those which have the same ratio with them are prime to one another.

Let A, B be the least numbers of those which have the same ratio with them; I say that A, B are prime to one another.

For, if they are not prime to one another, some number will measure them.

Let some number measure them, and let it be C .

And, as many times as C measures A , so many units let there be in D , and, as many times as C measures B , so many units let there be in E . Since C measures A according to the units in D , therefore C by multiplying D has made A . {VII. Def. 15} For the same reason also C by multiplying E has made B .

Thus the number C by multiplying the two numbers D, E has made A, B ; therefore, as D is to E , so is A to B ; {VII. 17} therefore D, E are in the same ratio with A, B , being less than they: which is impossible.

Therefore no number will measure the numbers A, B .

Therefore A, B are prime to one another. Q. E. D.

PROPOSITION 23. If two number be prime to one another, the number which measures the one of them will be prime to the remaining number.

Let A, B be two numbers prime to one another, and let any number C measure A ; I say that C, B are also prime to one another.

For, if C, B are not prime to one another, some number will measure C, B .

Let a number measure them, and let it be D .

Since D measures C , and C measures A , therefore D also measures A .

But it also measures B ; therefore D measures A, B which are prime to one another: which is impossible. {VII. Def. 12}

Therefore no number will measure the numbers C, B.

Therefore C, B are prime to one another. Q. E. D.

PROPOSITION 24. If two numbers be prime to any number, their product also will be prime to the same.

For let the two numbers A, B be prime to any number C, and let A by multiplying B make D; I say that C, D are prime to one another.

For, if C, D are not prime to one another, some number will measure C, D.

Let a number measure them, and let it be E.

Now, since C, A are prime to one another, and a certain number E measures C, therefore A, E are prime to one another. {VII. 23} As many times, then, as E measures D, so many units let there be in F; therefore F also measures D according to the units in E. {VII. 16} Therefore E by multiplying F has made D. {VII. Def. 15} But, further, A by multiplying B has also made D; therefore the product of E, F is equal to the product of A, B.

But, if the product of the extremes be equal to that of the means, the four numbers are proportional; {VII. 19} therefore, as E is to A, so is B to F.

But A, E are prime to one another, numbers which are prime to one another are also the least of those which have the same ratio, {VII. 21} and the least numbers of those which have the same ratio with them measure those which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent; {VII. 20} therefore E measures B.

But it also measures C; therefore E measures B, C which are prime to one another: which is impossible. {VII. Def. 12} Therefore no number will measure the numbers C, D.

Therefore C, D are prime to one another. Q. E. D.

PROPOSITION 25. If two numbers be prime to one another, the product of one of them into itself will be prime to the remaining one.

Let A, B be two numbers prime to one another, and let A by multiplying itself make C: I say that B, C are prime to one another.

For let D be made equal to A.

Since A, B are prime to one another, and A is equal to D, therefore D, B are also prime to one another.

Therefore each of the two numbers D, A is prime to B; therefore the product of D, A will also be prime to B. {VII. 24} But the number which is the product of D, A is C.

Therefore C, B are prime to one another. Q. E. D.

PROPOSITION 26. If two numbers be prime to two numbers, both to each, their products also will be prime to one another.

For let the two numbers A, B be prime to the two numbers C, D; both to each, and let A by multiplying B make E, and let C by multiplying D make F; I say that E, F are prime to one another.

For, since each of the numbers A, B is prime to C, therefore the product of A, B will also be prime to C. {VII. 24} But the product of A, B is E; therefore E, C are prime to one another.

For the same reason E, D are also prime to one another.

Therefore each of the numbers C, D is prime to E.

Therefore the product of C, D will also be prime to E. {VII. 24} But the product of C, D is F.

Therefore E, F are prime to one another. Q. E. D.

PROPOSITION 27. If two numbers be prime to one another, and each by multiplying itself make a certain number, the products will be prime to one another; and, if the original numbers by multiplying the products make certain numbers, the latter will also be prime to one another {and this is always the case with the extremes}.

Let A, B be two numbers prime to one another, let A by multiplying itself make C, and by multiplying C make D, and let B by multiplying itself make E, and by multiplying E make F; I say that both C, E and D, F are prime to one another.

For, since A, B are prime to one another, and A by multiplying itself has made C, therefore C, B are prime to one another. {VII. 25} Since then C, B are prime to one another, and B by multiplying itself has made E, therefore C, E are prime to one another. {id.} Again, since A, B are prime to one another, and B by multiplying itself has made E, therefore A, E are prime to one another. {id.} Since then the two numbers A, C are prime to the two numbers B, E, both to each, therefore also the product of A, C is prime to the product of B, E. {VII. 26} And the product of A, C is D, and the product of B, E is F.

Therefore D, F are prime to one another. Q. E. D.

PROPOSITION 28. If two numbers be prime to one another, the sum will also be prime to each of them; and, if the sum of two numbers be prime to any one of them, the original numbers will also be prime to one another.

For let two numbers AB, BC prime to one another be added; I say that the sum AC is also prime to each of the numbers AB, BC.

For, if CA, AB are not prime to one another, some number will measure CA, AB.

Let a number measure them, and let it be D.

Since then D measures CA, AB, therefore it will also measure the remainder BC.

But it also measures BA; therefore D measures AB, BC which are prime to one another: which is impossible. {VII. Def. 12} Therefore no number will measure the numbers CA, AB; therefore CA, AB are prime to one another.

For the same reason AC, CB are also prime to one another.

Therefore CA is prime to each of the numbers AB, BC.

Again, let CA, AB be prime to one another; I say that AB, BC are also prime to one another.

For, if AB, BC are not prime to one another, some number will measure AB, BC.

Let a number measure them, and let it be D.

Now, since D measures each of the numbers AB, BC, it will also measure the whole CA.

But it also measures AB; therefore D measures CA, AB which are prime to one another: which is impossible. {VII. Def. 12} Therefore no number will measure the numbers AB, BC.

Therefore AB, BC are prime to one another. Q. E. D.

PROPOSITION 29. Any prime number is prime to any number which it does not measure.

Let A be a prime number, and let it not measure B; I say that B, A are prime to one another.

For, if B, A are not prime to one another, some number will measure them.

Let C measure them.

Since C measures B, and A does not measure B, therefore C is not the same with A.

Now, since C measures B, A, therefore it also measures A which is prime, though it is not the same with it: which is impossible.

Therefore no number will measure B, A.

Therefore A, B are prime to one another. Q. E. D.

PROPOSITION 30. If two numbers by multiplying one another make some number, and any prime number measure the product, it will also measure one of the original numbers.

For let the two numbers A, B by multiplying one another make C, and let any prime number D measure C; I say that D measures one of the numbers A, B.

For let it not measure A.

Now D is prime; therefore A, D are prime to one another. {VII. 29} And, as many times as D measures C, so many units let there be in E.

Since then D measures C according to the units in E, therefore D by multiplying E has made C. {VII. Def. 15} Further, A by multiplying B has also made C; therefore the product of D, E is equal to the product of A, B.

Therefore, as D is to A, so is B to E. {VII. 19} But D, A are prime to one another, primes are also least, {VII. 21} and the least measure the numbers which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent; {VII. 20} therefore D measures B.

Similarly we can also show that, if D do not measure B, it will measure A.

Therefore D measures one of the numbers A, B. Q. E. D.

PROPOSITION 31. Any composite number is measured by some prime number.

Let A be a composite number; I say that A is measured by some prime number.

For, since A is composite, some number will measure it.

Let a number measure it, and let it be B.

Now, if B is prime, what was enjoined will have been done.

But if it is composite, some number will measure it.

Let a number measure it, and let it be C.

Then, since C measures B, and B measures A, therefore C also measures A.

And, if C is prime, what was enjoined will have been done.

But if it is composite, some number will measure it.

Thus, if the investigation be continued in this way, some prime number will be found which will measure the number before it, which will also measure A.

For, if it is not found, an infinite series of numbers will measure the number A, each of which is less than the other: which is impossible in numbers.

Therefore some prime number will be found which will measure the one before it, which will also measure A.

Therefore any composite number is measured by some prime number.

PROPOSITION 32. Any number either is prime or is measured by some prime number.

Let A be a number; I say that A either is prime or is measured by some prime number.

If now A is prime, that which was enjoined will have been done.

But if it is composite, some prime number will measure it. {VII. 31} Therefore any number either is prime or is measured by some prime number. Q. E. D.

PROPOSITION 33. Given as many numbers as we please, to find the least of those which have the same ratio with them.

Let A, B, C be the given numbers, as many as we please; thus it is required to find the least of those which have the same ratio with A, B, C.

A, b, C are either prime to one another or not.

Now, if A, B, C are prime to one another, they are the least of those which have the same ratio with them. {VII. 21} But, if not, let D the greatest common measure of A, B, C be taken, {VII. 3} and, as many times as D measures the numbers A, B, C respectively, so many units let there be in the numbers E, F, G respectively.

Therefore the numbers E, F, G measure the numbers A, B, C respectively according to the units in D. {VII. 16} Therefore E, F, G measure A, B, C the same number of times; therefore E, F, G are in the same ratio with A, B, C. {VII. Def. 20} I say next that they are the least that are in that ratio.

For, if E, F, G are not the least of those which have the same ratio with A, B, C, there will be numbers less than E, F, G which are in the same ratio with A, B, C.

Let them be H, K, L; therefore H measures A the same number of times that the numbers K, L measure the numbers B, C respectively.

Now, as many times as H measures A, so many units let there be in M; therefore the numbers K, L also measure the numbers B, C respectively according to the units in M.

And, since H measures A according to the units in M, therefore M also measures A according to the units in H. {VII. 16} For the same reason M also measures the numbers B, C according to the units in the numbers K, L respectively; Therefore M measures A, B, C.

Now, since H measures A according to the units in M, therefore H by multiplying M has made A. {VII. Def. 15} For the same reason also E by multiplying D has made A.

Therefore the product of E, D is equal to the product of H, M.

Therefore, as E is to H, so is M to D. {VII. 19} But E is greater than H; therefore M is also greater than D.

And it measures A, B, C: which is impossible, for by hypothesis D is the greatest common measure of A, B, C.

Therefore there cannot be any numbers less than E, F, G which are in the same ratio with A, B, C.

Therefore E, F, G are the least of those which have the same ratio with A, B, C. Q. E. D.

PROPOSITION 34. Given two numbers, to find the least number which they measure.

Let A, B be the two given numbers; thus it is required to find the least number which they measure.

Now A, B are either prime to one another or not.

First, let A, B be prime to one another, and let A by multiplying B make C; therefore also B by multiplying A has made C. {VII. 16} Therefore A, B measure C I say next that it is also the least number they measure.

For, if not, A, B will measure some number which is less than C.

Let them measure D.

Then, as many times as A measures D, so many units let there be in E, and, as many times as B measures D, so many units let there be in F; therefore A by multiplying E has made D, and B by multiplying F has made D; {VII. Def. 15} therefore the product of A, E is equal to the product of B, F.

Therefore, as A is to B, so is F E. {VII. 19} But A, B are prime, primes are also least, {VII. 21} and the least measure the numbers which have the same ratio the same number of times, the greater the greater and the less the less; {VII. 20} therefore B measures E, as consequent consequent.

And, since A by multiplying B, E has made C, D, therefore, as B is to E, so is C to D. {VII. 17} But B measures E; therefore C also measures D, the greater the less: which is impossible.

Therefore A, B do not measure any number less than C; therefore C is the least that is measured by A, B.

Next, let A, B not be prime to one another, and let F, E, the least numbers of those which have the same ratio with A, B, be taken; {VII. 33} therefore the product of A, E is equal to the product of B, F. {VII. 19} And let A by multiplying E make C; therefore also B by multiplying F has made C; therefore A, B measure C.

I say next that it is also the least number that they measure.

For, if not, A, B will measure some number which is less than C.

Let them measure D.

And, as many times as A measures D, so many units let there be in G, and, as many times as B measures D, so many units let there be in H.

Therefore A by multiplying G has made D, and B by multiplying H has made D.

Therefore the product of A, G is equal to the product of B, H; therefore, as A is to B, so is H to G. {VII. 19} But, as A is to B, so is

F to E.

Therefore also, as F is to E, so is H to G.

But F, E are least, and the least measure the numbers which have the same ratio the same number of times, the greater the greater and the less the less; {VII. 20} therefore E measures G.

And, since A by multiplying E, G has made C, D, therefore, as E is to G, so is C to D. {VII. 17} But E measures G; therefore C also measures D, the greater the less: which is impossible.

Therefore A, B will not measure any number which is less than C.

Therefore C is the least that is measured by A, B. Q. E. D.

PROPOSITION 35. If two numbers measure any number, the least number measured by them will also measure the same.

For let the two numbers A, B measure any number CD, and let E be the least that they measure; I say that E also measures CD.

For, if E does not measure CD, let E, measuring DF, leave CF less than itself.

Now, since A, B measure E, and E measures DF, therefore A, B will also measure DF.

But they also measure the whole CD; therefore they will also measure the remainder CF which is less than E: which is impossible.

Therefore E cannot fail to measure CD; therefore it measures it. Q. E. D.

PROPOSITION 36. Given three numbers, to find the least number which they measure.

Let A, B, C be the three given numbers; thus it is required to find the least number which they measure.

Let D, the least number measured by the two numbers A, B, be taken. {VII. 34} Then C either measures, or does not measure, D.

First, let it measure it.

But A, B also measure D; therefore A, B, C measure D.

I say next that it is also the least that they measure.

For, if not, A, B, C will measure some number which is less than D.

Let them measure E.

Since A, B, C measure E, therefore also A, B measure E.

Therefore the least number measured by A, B will also measure E. {VII. 35} But D is the least number measured by A, B; therefore D will measure E, the greater the less: which is impossible.

Therefore A, B, C will not measure any number which is less than D; therefore D is the least that A, B, C measure.

Again, let C not measure D, and let E, the least number measured by C, D, be taken. {VII. 34} Since A, B measure D, and D measures E, therefore also A, B measure E.

But C also measures E; therefore also A, B, C measure E.

I say next that it is also the least that they measure.

For, if not, A, B, C will measure some number which is less than E.

Let them measure F.

Since A, B, C measure F, therefore also A, B measure F; therefore the least number measured by A, B will also measure F. {VII. 35} But D is the least number measured by A, B; therefore D measures F.

But C also measures F; therefore D, C measure F, so that the least number measured by D, C will also measure F.

But E is the least number measured by C, D; therefore E measures F, the greater the less: which is impossible.

Therefore A, B, C will not measure any number which is less than E.

Therefore E is the least that is measured by A, B, C. Q. E. D.

PROPOSITION 37. If a number be measured by any number, the number which is measured will have a part called by the same name as the measuring number.

For let the number A be measured by any number B; I say that A has a part called by the same name as B.

For, as many times as B measures A, so many units let there be in C.

Since B measures A according to the units in C, and the unit D also measures the number C according to the units in it, therefore the unit D measures the number C the same number of times as B measures A.

Therefore, alternately, the unit D measures the number B the same number of times as C measures A; {VII. 15} therefore, whatever part the unit D is of the number B, the same part is C of A also.

But the unit D is a part of the number B called by the same name as it; therefore C is also a part of A called by the same name as B, so that A has a part C which is called by the same name as B. Q. E. D.

PROPOSITION 38. If a number have any part whatever, it will be measured by a number called by the same name as the part.

For let the number A have any part whatever, B, and let C be a number called by the same name as the part B; I say that C measures A.

For, since B is a part of A called by the same name as C, and the unit D is also a part of C called by the same name as it, therefore, whatever part the unit D is of the number C, the same part is B of A also; therefore the unit D measures the number C the same number of times that B measures A.

Therefore, alternately, the unit D measures the number B the same number of times that C measures A. {VII. 15} Therefore C measures A. Q. E. D.

PROPOSITION 39. To find the number which is the least that will have given parts.

Let A, B, C be the given parts; thus it is required to find the number which is the least that will have the parts A, B, C.

Let D, E, F be numbers called by the same name as the parts A, B, C, and let G, the least number measured by D, E, F, be taken. {VII. 36} Therefore G has parts called by the same name as D, E, F. {VII. 37} But A, B, C are parts called by the same name as D, E, F; therefore G has the parts A, B, C.

I say next that it is also the least number that has.

For, if not, there will be some number less than G which will have the parts A, B, C.

Let it be H.

Since H has the parts A, B, C, therefore H will be measured by numbers called by the same name as the parts A, B, C. {VII. 38} But D, E, F are numbers called by the same name as the parts A, B, C; therefore H is measured by D, E, F.

And it is less than G : which is impossible.

Therefore there will be no number less than G that will have the parts A, B, C. Q. E. D.

ENDNOTES.

The Greek has οἱ γινόμενοι ἐξ αὐτῶν, “the (numbers) produced from them.” By “from them” Euclid means “from the original numbers,” though this is not very clear even in the Greek. I think ambiguity is best avoided by leaving out the words.

ὁ ἐξ αὐτῶν γινόμενος, literally “the (number) produced from them,” will henceforth be translated as “their product.”

The Greek, ὁ ἐκ τοῦ ἑνὸς αὐτῶν γινόμενος, literally “the number produced from the one of them,” leaves “multiplied into itself” to be understood.

i.e. the implied problem of finding a prime number which measures A.

In the Greek the sentence stops here, but it is necessary to add the words “the number before it, which will also measure A,” which are found a few lines further down. It is possible that the words may have fallen out of P here by a simple mistake due to ὁμοιοτέλευτον (Heiberg).

literally (as usual) “each of the numbers E, F, G measures each of the numbers A, B, C.”

BOOK VIII. PROPOSITIONS

PROPOSITION 1. IF THERE BE as many numbers as we please in continued proportion, and the extremes of them be prime to one another, the numbers are the least of those which have the same ratio with them.

Let there be as many numbers as we please, A, B, C, D, in continued proportion, and let the extremes of them A, D be prime to one another; I say that A, B, C, D are the least of those which have the same ratio with them.

For, if not, let E, F, G, H be less than A, B, C, D, and in the same ratio with them.

Now, since A, B, C, D are in the same ratio with E, F, G, H, and the multitude of the numbers A, B, C, D is equal to the multitude of the numbers E, F, G, H, therefore, ex aequali, as A is to D, so is E to H. {VII. 14} But A, D are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent. {VII. 20} Therefore A measures E, the greater the less: which is impossible.

Therefore E, F, G, H which are less than A, B, C, D are not in the same ratio with them.

Therefore A, B, C, D are the least of those which have the same ratio with them. Q. E. D.

PROPOSITION 2. To find numbers in continued proportion, as many as may be prescribed, and the least that are in a given ratio.

Let the ratio of A to B be the given ratio in least numbers; thus it is required to find numbers in continued proportion, as many as may be prescribed, and the least that are in the ratio of A to B.

Let four be prescribed; let A by multiplying itself make C, and by multiplying B let it make D; let B by multiplying itself make E; further, let A by multiplying C, D, E make F, G, H, and let B by multiplying E make K.

Now, since A by multiplying itself has made C, and by multiplying B has made D, therefore, as A is to B, so is C to D. {VII. 17} Again, since A by multiplying B has made D, and B by multiplying itself has made E, therefore the numbers A, B by multiplying B have made the numbers D, E respectively.

Therefore, as A is to B, so is D to E. {VII. 18} But, as A is to B, so is C to D; therefore also, as C is to D, so is D to E.

And, since A by multiplying C, D has made F, G, therefore, as C is to D, so is F to G. {VII. 17} But, as C is to D, so was A to B; therefore also, as A is to B, so is F to G.

Again, since A by multiplying D, E has made G, H, therefore, as D is to E, so is G to H. {VII. 17} But, as D is to E, so is A to B.

Therefore also, as A is to B, so is G to H.

And, since A, B by multiplying E have made H, K, therefore, as A is to B, so is H to K. {VII. 18} But, as A is to B, so is F to G, and G to H.

Therefore also, as F is to G, so is G to H, and H to K; therefore C, D, E, and F, G, H, K are proportional in the ratio of A to B.

I say next that they are the least numbers that are so.

For, since A, B are the least of those which have the same ratio with them, and the least of those which have the same ratio are prime to one another, {VII. 22} therefore A, B are prime to one another.

And the numbers A, B by multiplying themselves respectively have made the numbers C, E, and by multiplying the numbers C, E respectively have made the numbers F, K; therefore C, E and F, K are prime to one another respectively. {VII. 27} But, if there be as many numbers as we please in continued proportion, and the extremes of them be prime to one another, they are the least of those which have the same ratio with them. {VIII. 1} Therefore C, D, E and F, G, H, K are the least of those which have the same ratio with A, B. Q. E. D.

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From this it is manifest that, if three numbers in continued proportion be the least of those which have the same ratio with them, the extremes of them are squares, and, if four numbers, cubes.

PROPOSITION 3. If as many numbers as we please in continued proportion be the least of those which have the same ratio with them, the extremes of them are prime to one another.

Let as many numbers as we please, A, B, C, D, in continued proportion be the least of those which have the same ratio with them; I say that the extremes of them A, D are prime to one another.

For let two numbers E, F, the least that are in the ratio of A, B, C, D, be taken, {VII. 33} then three others G, H, K with the same property; and others, more by one continually, {VIII. 2} until the multitude taken becomes equal to the multitude of the numbers A, B, C, D.

Let them be taken, and let them be L, M, N, O.

Now, since E, F are the least of those which have the same ratio with them, they are prime to one another. {VII. 22} And, since the numbers E, F by multiplying themselves respectively have made the numbers G, K, and by multiplying the numbers G, K respectively have made the numbers L, O, {VIII. 2, Por.} therefore both G, K and L, O are prime to one another. {VII. 27} And, since A, B, C, D are the least of those which have the same ratio with them, while L, M, N, O are the least that are in the same ratio with A, B, C, D, and the multitude of the numbers A, B, C, D is equal to the multitude of the numbers L, M, N, O, therefore the numbers A, B, C, D are equal to the numbers L, M, N, O respectively; therefore A is equal to L, and D to O.

And L, O are prime to one another.

Therefore A, D are also prime to one another. Q. E. D.

PROPOSITION 4. Given as many ratios as we please in least numbers, to find numbers in continued proportion which are the least in the given ratios.

Let the given ratios in least numbers be that of A to B, that of C to D, and that of E to F; thus it is required to find numbers in continued proportion which are the least that are in the ratio of A to B, in the ratio of C to D, and in the ratio of E to F.

Let G, the least number measured by B, C, be taken. {VII. 34} And, as many times as B measures G, so many times also let A measure H, and, as many times as C measures G, so many times also let D measure K.

Now E either measures or does not measure K.

First, let it measure it.

And, as many times as E measures K, so many times let F measure L also.

Now, since A measures H the same number of times that B measures G, therefore, as A is to B, so is H to G. {VII. Def. 20, VII. 13} For the same reason also, as C is to D, so is G to K, and further, as E is to F, so is K to L; therefore H, G, K, L are continuously proportional in the ratio of A to B, in the ratio of C to D, and in the ratio of E to F.

I say next that they are also the least that have this property.

For, if H, G, K, L are not the least numbers continuously proportional in the ratios of A to B, of C to D, and of E to F, let them be N, O, M, P.

Then since, as A is to B, so is N to O, while A, B are least, and the least numbers measure those which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent; therefore B measures O. {VII. 20} For the same reason C also measures O; therefore B, C measure O; therefore the least number measured by B, C will also measure O. {VII. 35} But G is the least number measured by B, C; therefore G measures O, the greater the less: which is impossible.

Therefore there will be no numbers less than H, G, K, L which are continuously in the ratio of A to B, of C to D, and of E to F.

Next, let E not measure K.

Let M, the least number measured by E, K, be taken.

And, as many times as K measures M, so many times let H, G measure N, O respectively, and, as many times as E measures M, so many times let F measure P also.

Since H measures N the same number of times that G measures O, therefore, as H is to G, so is N to O. {VII. 13 and Def. 20} But, as H is to G, so is A to B; therefore also, as A is to B, so is N to O.

For the same reason also, as C is to D, so is O to M.

Again, since E measures M the same number of times that F measures P, therefore, as E is to F, so is M to P; {VII. 13 and Def. 20} therefore N, O, M, P are continuously proportional in the ratios of A to B, of C to D, and of E to F.

I say next that they are also the least that are in the ratios A : B, C : D, E : F.

For, if not, there will be some numbers less than N, O, M, P continuously proportional in the ratios A : B, C : D, E : F.

Let them be Q, R, S, T.

Now since, as Q is to R, so is A to B, while A, B are least, and the least numbers measure those which have the same ratio with them the same number of times, the antecedent the antecedent and the consequent the consequent, {VII. 20} therefore B measures R.

For the same reason C also measures R; therefore B, C measure R.

Therefore the least number measured by B, C will also measure R. {VII. 35} But G is the least number measured by B, C; therefore

G measures R.

And, as G is to R, so is K to S: {VII. 13} therefore K also measures S.

But E also measures S; therefore E, K measure S.

Therefore the least number measured by E, K will also measure S. {VII. 35} But M is the least number measured by E, K; therefore M measures S, the greater the less: which is impossible.

Therefore there will not be any numbers less than N, O, M, P continuously proportional in the ratios of A to B, of C to D, and of E to F; therefore N, O, M, P are the least numbers continuously proportional in the ratios $A : B, C : D, E : F$. Q. E. D.

PROPOSITION 5. Plane numbers have to one another the ratio compounded of the ratios of their sides.

Let A, B be plane numbers, and let the numbers C, D be the sides of A, and E, F of B; I say that A has to B the ratio compounded of the ratios of the sides.

For, the ratios being given which C has to E and D to F, let the least numbers G, H, K that are continuously in the ratios $C : E, D : F$ be taken, so that, as C is to E, so is G to H, and, as D is to F, so is H to K. {VIII. 4} And let D by multiplying E make L.

Now, since D by multiplying C has made A, and by multiplying E has made L, therefore, as C is to E, so is A to L. {VII. 17} But, as C is to E, so is G to H; therefore also, as G is to H, so is A to L.

Again, since E by multiplying D has made L, and further by multiplying F has made B, therefore, as D is to F, so is L to B. {VII. 17} But, as D is to F, so is H to K; therefore also, as H is to K, so is L to B.

But it was also proved that, as G is to H, so is A to L; therefore, ex aequali, as G is to K, so is A to B. {VII. 14} But G has to K the ratio

compounded of the ratios of the sides; therefore A also has to B the ratio compounded of the ratios of the sides. Q. E. D.

PROPOSITION 6. If there be as many numbers as we please in continued proportion, and the first do not measure the second, neither will any other measure any other.

Let there be as many numbers as we please, A, B, C, D, E, in continued proportion, and let A not measure B; I say that neither will any other measure any other.

Now it is manifest that A, B, C, D, E do not measure one another in order; for A does not even measure B.

I say, then, that neither will any other measure any other.

For, if possible, let A measure C.

And, however many A, B, C are, let as many numbers F, G, H, the least of those which have the same ratio with A, B, C, be taken. {VII. 33} Now, since F, G, H are in the same ratio with A, B, C, and the multitude of the numbers A, B, C is equal to the multitude of the numbers F, G, H, therefore, *ex aequali*, as A is to C, so is F to H. {VII. 14} And since, as A is to B, so is F to G, while A does not measure B, therefore neither does F measure G; {VII. Def. 20} therefore F is not an unit, for the unit measures any number.

Now F, H are prime to one another. {VIII. 3} And, as F is to H, so is A to C; therefore neither does A measure C.

Similarly we can prove that neither will any other measure any other. Q. E. D.

PROPOSITION 7. If there be as many numbers as we please in continued proportion, and the first measure the last, it will measure the second also.

Let there be as many numbers as we please, A, B, C, D, in continued proportion; and let A measure D; I say that A also measures B.

For, if A does not measure B, neither will any other of the numbers measure any other. {VIII. 6} But A measures D.

Therefore A also measures B. Q. E. D.

PROPOSITION 8. If between two numbers there fall numbers in continued proportion with them, then, however many numbers fall between them in continued proportion, so many will also fall in continued proportion between the numbers which have the same ratio with the original numbers.

Let the numbers C, D fall between the two numbers A, B in continued proportion with them, and let E be made in the same ratio to F as A is to B; I say that, as many numbers as have fallen between A, B in continued proportion, so many will also fall between E, F in continued proportion.

For, as many as A, B, C, D are in multitude, let so many numbers G, H, K, L, the least of those which have the same ratio with A, C, D, B, be taken; {VII. 33} therefore the extremes of them G, L are prime to one another. {VIII. 3} Now, since A, C, D, B are in the same ratio with G, H, K, L, and the multitude of the numbers A, C, D, B is equal to the multitude of the numbers G, H, K, L, therefore, ex aequali, as A is to B, so is G to L. {VII. 14} But, as A is to B, so is E to F; therefore also, as G is to L, so is E to F.

But G, L are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent. {VII. 20} Therefore G measures E the same number of times as L measures F.

Next, as many times as G measures E, so many times let H, K also measure M, N respectively; therefore G, H, K, L measure E, M, N, F the same number of times.

Therefore G, H, K, L are in the same ratio with E, M, N, F. {VII. Def. 20} But G, H, K, L are in the same ratio with A, C, D, B; therefore A, C, D, B are also in the same ratio with E, M, N, F.

But A, C, D, B are in continued proportion; therefore E, M, N, F are also in continued proportion.

Therefore, as many numbers as have fallen between A, B in continued proportion with them, so many numbers have also fallen between E, F in continued proportion. Q. E. D.

PROPOSITION 9. If two numbers be prime to one another, and numbers fall between them in continued proportion, then, however many numbers fall between them in continued proportion, so many will also fall between each of them and an unit in continued proportion.

Let A, B be two numbers prime to one another, and let C, D fall between them in continued proportion, and let the unit E be set out; I say that, as many numbers as fall between A, B in continued proportion, so many will also fall between either of the numbers A, B and the unit in continued proportion.

For let two numbers F, G, the least that are in the ratio of A, C, D, B, be taken, three numbers H, K, L with the same property, and others more by one continually, until their multitude is equal to the multitude of A, C, D, B. {VIII. 2} Let them be taken, and let them be M, N, O, P.

It is now manifest that F by multiplying itself has made H and by multiplying H has made M, while G by multiplying itself has made L and by multiplying L has made P. {VIII. 2, Por.} And, since M, N, O, P are the least of those which have the same ratio with F, G, and A, C, D, B are also the least of those which have the same ratio with F, G, {VIII. 1} while the multitude of the numbers M, N, O, P is equal to the multitude of the numbers A, C, D, B, therefore M, N, O, P are equal to A, C, D, B respectively;

therefore M is equal to A, and P to B.

Now, since F by multiplying itself has made H, therefore F measures H according to the units in F.

But the unit E also measures F according to the units in it; therefore the unit E measures the number F the same number of times as F measures H.

Therefore, as the unit E is to the number F, so is F to H. {VII. Def. 20} Again, since F by multiplying H has made M, therefore H measures M according to the units in F.

But the unit E also measures the number F according to the units in it; therefore the unit E measures the number F the same number of times as H measures M.

Therefore, as the unit E is to the number F, so is H to M.

But it was also proved that, as the unit E is to the number F, so is F to H; therefore also, as the unit E is to the number F, so is F to H, and H to M.

But M is equal to A; therefore, as the unit E is to the number F, so is F to H, and H to A.

For the same reason also, as the unit E is to the number G, so is G to L and L to B.

Therefore, as many numbers as have fallen between A, B in continued proportion, so many numbers also have fallen between each of the numbers A, B and the unit E in continued proportion. Q. E. D.

PROPOSITION 10. If numbers fall between each of two numbers and an unit in continued proportion, however many numbers fall between each of them and an unit in continued proportion, so many also will fall between the numbers themselves in continued proportion.

For let the numbers D, E and F, G respectively fall between the two numbers A, B and the unit C in continued proportion; I say that, as many numbers as have fallen between each of the numbers A, B and the unit C in continued proportion, so many numbers will also fall between A, B in continued proportion.

For let D by multiplying F make H, and let the numbers D, F by multiplying H make K, L respectively.

Now, since, as the unit C is to the number D, so is D to E, therefore the unit C measures the number D the same number of times as D measures E. {VII. Def. 20} But the unit C measures the number D according to the units in D; therefore the number D also measures E according to the units in D; therefore D by multiplying itself has made E.

Again, since, as C is to the number D, so is E to A, therefore the unit C measures the number D the same number of times as E measures A.

But the unit C measures the number D according to the units in D; therefore E also measures A according to the units in D; therefore D by multiplying E has made A.

For the same reason also F by multiplying itself has made G, and by multiplying G has made B.

And, since D by multiplying itself has made E and by multiplying F has made H, therefore, as D is to F, so is E to H. {VII. 17} For the same reason also, as D is to F, so is H to G. {VII. 18} Therefore also, as E is to H, so is H to G.

Again, since D by multiplying the numbers E, H has made A, K respectively, therefore, as E is to H, so is A to K. {VII. 17} But, as E is to H, so is D to F; therefore also, as D is to F, so is A to K.

Again, since the numbers D, F by multiplying H have made K, L respectively, therefore, as D is to F, so is K to L. {VII. 18} But, as D is to F, so is A to K; therefore also, as A is to K, so is K to L.

Further, since F by multiplying the numbers H, G has made L, B respectively, therefore, as H is to G, so is L to B. {VII. 17} But, as H is to G, so is D to F; therefore also, as D is to F, so is L to B.

But it was also proved that, as D is to F, so is A to K and K to L; therefore also, as A is to K, so is K to L and L to B.

Therefore A, K, L, B are in continued proportion.

Therefore, as many numbers as fall between each of the numbers A, B and the unit C in continued proportion, so many also will fall between A, B in continued proportion. Q. E. D.

PROPOSITION 11. Between two square numbers there is one mean proportional number, and the square has to the square the ratio duplicate of that which the side has to the side.

Let A, B be square numbers, and let C be the side of A, and D of B; I say that between A, B there is one mean proportional number, and A has to B the ratio duplicate of that which C has to D.

For let C by multiplying D make E.

Now, since A is a square and C is its side, therefore C by multiplying itself has made A.

For the same reason also D by multiplying itself has made B.

Since then C by multiplying the numbers C, D has made A, E respectively, therefore, as C is to D, so is A to E. {VII. 17} For the same reason also, as C is to D, so is E to B. {VII. 18} Therefore also, as A is to E, so is E to B.

Therefore between A, B there is one mean proportional number.

I say next that A also has to B the ratio duplicate of that which C has to D.

For, since A, E, B are three numbers in proportion, therefore A has to B the ratio duplicate of that which A has to E. {V. Def. 9} But, as A is to E, so is C to D.

Therefore A has to B the ratio duplicate of that which the side C has to D. Q. E. D.

PROPOSITION 12. Between two cube numbers there are two mean proportional numbers, and the cube has to the cube the ratio triplicate of that which the side has to the side.

Let A, B be cube numbers, and let C be the side of A, and D of B; I say that between A, B there are two mean proportional numbers, and A has to B the ratio triplicate of that which C has to D.

For let C by multiplying itself make E, and by multiplying D let it make F; let D by multiplying itself make G, and let the numbers C, D by multiplying F make H, K respectively.

Now, since A is a cube, and C its side, and C by multiplying itself has made E, therefore C by multiplying itself has made E and by multiplying E has made A.

For the same reason also D by multiplying itself has made G and by multiplying G has made B.

And, since C by multiplying the numbers C, D has made E, F respectively, therefore, as C is to D, so is E to F. {VII. 17} For the same reason also, as C is to D, so is F to G. {VII. 18} Again, since C by multiplying the numbers E, F has made A, H respectively, therefore, as E is to F, so is A to H. {VII. 17} But, as E is to F, so is C to D.

Therefore also, as C is to D, so is A to H.

Again, since the numbers C, D by multiplying F have made H, K respectively, therefore, as C is to D, so is H to K. {VII. 18} Again, since D by multiplying each of the numbers F, G has made K, B respectively, therefore, as F is to G, so is K to B. {VII. 17} But, as F is to G, so is C to D; therefore also, as C is to D, so is A to H, H to K, and K to B.

Therefore H, K are two mean proportionals between A, B.

I say next that A also has to B the ratio triplicate of that which C has to D.

For, since A, H, K, B are four numbers in proportion, therefore A has to B the ratio triplicate of that which A has to H. {V. Def. 10} But, as A is to H, so is C to D; therefore A also has to B the ratio triplicate of that which C has to D. Q. E. D.

PROPOSITION 13. If there be as many numbers as we please in continued proportion, and each by multiplying itself make some number, the products will be proportional; and, if the original numbers by multiplying the products make certain numbers, the latter will also be proportional.

Let there be as many numbers as we please, A, B, C, in continued proportion, so that, as A is to B, so is B to C; let A, B, C by multiplying themselves make D, E, F, and by multiplying D, E, F let them make G, H, K; I say that D, E, F and G, H, K are in continued proportion.

For let A by multiplying B make L, and let the numbers A, B by multiplying L make M, N respectively.

And again let B by multiplying C make O, and let the numbers B, C by multiplying O make P, Q respectively.

Then, in manner similar to the foregoing, we can prove that D, L, E and G, M, N, H are continuously proportional in the ratio of A to B, and further E, O, F and H, P, Q, K are continuously proportional in the ratio of B to C.

Now, as A is to B, so is B to C; therefore D, L, E are also in the same ratio with E, O, F, and further G, M, N, H in the same ratio with H, P, Q, K.

And the multitude of D, L, E is equal to the multitude of E, O, F, and that of G, M, N, H to that of H, P, Q, K; therefore, ex aequali,

as D is to E, so is E to F, and, as G is to H, so is H to K. {VII. 14} Q. E. D.

PROPOSITION 14. If a square measure a square, the side will also measure the side; and, if the side measure the side, the square will also measure the square.

Let A, B be square numbers, let C, D be their sides, and let A measure B; I say that C also measures D.

For let C by multiplying D make E; therefore A, E, B are continuously proportional in the ratio of C to D. {VIII. 11} And, since A, E, B are continuously proportional, and A measures B, therefore A also measures E. {VIII. 7} And, as A is to E, so is C to D; therefore also C measures D. {VII. Def. 20} Again, let C measure D; I say that A also measures B.

For, with the same construction, we can in a similar manner prove that A, E, B are continuously proportional in the ratio of C to D.

And since, as C is to D, so is A to E, and C measures D, therefore A also measures E. {VII. Def. 20} And A, E, B are continuously proportional; therefore A also measures B.

Therefore etc. Q. E. D.

PROPOSITION 15. If a cube number measure a cube number, the side will also measure the side; and, if the side measure the side, the cube will also measure the cube.

For let the cube number A measure the cube B, and let C be the side of A and D of B; I say that C measures D.

For let C by multiplying itself make E, and let D by multiplying itself make G; further, let C by multiplying D make F, and let C, D by multiplying F make H, K respectively.

Now it is manifest that E, F, G and A, H, K, B are continuously proportional in the ratio of C to D. {VIII. 11, 12} And, since A, H,

K, B are continuously proportional, and A measures B, therefore it also measures H. {VIII. 7} And, as A is to H, so is C to D; therefore C also measures D. {VII. Def. 20} Next, let C measure D; I say that A will also measure B.

For, with the same construction, we can prove in a similar manner that A, H, K, B are continuously proportional in the ratio of C to D.

And, since C measures D, and, as C is to D, so is A to H, therefore A also measures H, {VII. Def. 20} so that A measures B also. Q. E. D.

PROPOSITION 16. If a square number do not measure a square number, neither will the side measure the side; and, if the side do not measure the side, neither will the square measure the square.

Let A, B be square numbers, and let C, D be their sides; and let A not measure B; I say that neither does C measure D.

For, if C measures D, A will also measure B. {VIII. 14} But A does not measure B; therefore neither will C measure D.

Again, let C not measure D; I say that neither will A measure B.

For, if A measures B, C will also measure D. {VIII. 14} But C does not measure D; therefore neither will A measure B. Q. E. D.

PROPOSITION 17. If a cube number do not measure a cube number, neither will the side measure the side; and, if the side do not measure the side, neither will the cube measure the cube.

For let the cube number A not measure the cube number B, and let C be the side of A, and D of B; I say that C will not measure D.

For if C measures D, A will also measure B. {VIII. 15} But A does not measure B; therefore neither does C measure D.

Again, let C not measure D; I say that neither will A measure B.

For, if A measures B, C will also measure D. {VIII. 15} But C does not measure D; therefore neither will A measure B. Q. E. D.

PROPOSITION 18. Between two similar plane numbers there is one mean proportional number; and the plane number has to the plane number the ratio duplicate of that which the corresponding side has to the corresponding side.

Let A, B be two similar plane numbers, and let the numbers C, D be the sides of A, and E, F of B.

Now, since similar plane numbers are those which have their sides proportional, {VII. Def. 21} therefore, as C is to D, so is E to F.

I say then that between A, B there is one mean proportional number, and A has to B the ratio duplicate of that which C has to E, or D to F, that is, of that which the corresponding side has to the corresponding side.

Now since, as C is to D, so is E to F, therefore, alternately, as C is to E, so is D to F. {VII. 13} And, since A is plane, and C, D are its sides, therefore D by multiplying C has made A.

For the same reason also E by multiplying F has made B.

Now let D by multiplying E make G.

Then, since D by multiplying C has made A, and by multiplying E has made G, therefore, as C is to E, so is A to G. {VII. 17} But, as C is to E, so is D to F; therefore also, as D is to F, so is A to G.

Again, since E by multiplying D has made G, and by multiplying F has made B, therefore, as D is to F, so is G to B. {VII. 17} But it was also proved that, as D is to F, so is A to G; therefore also, as A is to G, so is G to B.

Therefore A, G, B are in continued proportion.

Therefore between A, B there is one mean proportional number.

I say next that A also has to B the ratio duplicate of that which the corresponding side has to the corresponding side, that is, of that which C has to E or D to F.

For, since A, G, B are in continued proportion, A has to B the ratio duplicate of that which it has to G. {V. Def. 9} And, as A is to G, so is C to E, and so is D to F.

Therefore A also has to B the ratio duplicate of that which C has to E or D to F. Q. E. D.

PROPOSITION 19. Between two similar solid numbers there fall two mean proportional numbers; and the solid number has to the similar solid number the ratio triplicate of that which the corresponding side has to the corresponding side.

Let A, B be two similar solid numbers, and let C, D, E be the sides of A, and F, G, H of B.

Now, since similar solid numbers are those which have their sides proportional, {VII. Def. 21} therefore, as C is to D, so is F to G, and, as D is to E, so is G to H.

I say that between A, B there fall two mean proportional numbers, and A has to B the ratio triplicate of that which C has to F, D to G, and also E to H.

For let C by multiplying D make K, and let F by multiplying G make L.

Now, since C, D are in the same ratio with F, G, and K is the product of C, D, and L the product of F, G, K, L are similar plane numbers; {VII. Def. 21} therefore between K, L there is one mean proportional number. {VIII. 18} Let it be M Therefore M is the product of D, F, as was proved in the theorem preceding this. {VIII. 18} Now, since D by multiplying C has made K, and by multiplying F has made M, therefore, as C is to F, so is K to M.

{VII. 17} But, as K is to M, so is M to L.

Therefore K, M, L are continuously proportional in the ratio of C to F.

And since, as C is to D, so is F to G, alternately therefore, as C is to F, so is D to G. {VII. 13} For the same reason also, as D is to G, so is E to H.

Therefore K, M, L are continuously proportional in the ratio of C to F, in the ratio of D to G, and also in the ratio of E to H.

Next, let E, H by multiplying M make N, O respectively.

Now, since A is a solid number, and C, D, E are its sides, therefore E by multiplying the product of C, D has made A.

But the product of C, D is K; therefore E by multiplying K has made A.

For the same reason also H by multiplying L has made B.

Now, since E by multiplying K has made A, and further also by multiplying M has made N, therefore, as K is to M, so is A to N. {VII. 17} But, as K is to M, so is C to F, D to G, and also E to H; therefore also, as C is to F, D to G, and E to H, so is A to N.

Again, since E, H by multiplying M have made N, O respectively, therefore, as E is to H, so is N to O. {VII. 18} But, as E is to H, so is C to F and D to G; therefore also, as C is to F, D to G, and E to H, so is A to N and N to O.

Again, since H by multiplying M has made O, and further also by multiplying L has made B, therefore, as M is to L, so is O to B. {VII. 17} But, as M is to L, so is C to F, D to G, and E to H.

Therefore also, as C is to F, D to G, and E to H, so not only is O to B, but also A to N and N to O.

Therefore A, N, O, B are continuously proportional in the aforesaid ratios of the sides.

I say that A also has to B the ratio triplicate of that which the corresponding side has to the corresponding side, that is, of the ratio which the number C has to F, or D to G, and also E to H.

For, since A, N, O, B are four numbers in continued proportion, therefore A has to B the ratio triplicate of that which A has to N. {V. Def. 10} But, as A is to N, so it was proved that C is to F, D to G, and also E to H.

Therefore A also has to B the ratio triplicate of that which the corresponding side has to the corresponding side, that is, of the ratio which the number C has to F, D to G, and also E to H. Q. E. D.

PROPOSITION 20. If one mean proportional number fall between two numbers, the numbers will be similar plane numbers.

For let one mean proportional number C fall between the two numbers A, B; I say that A, B are similar plane numbers.

Let D, E, the least numbers of those which have the same ratio with A, C, be taken; {VII. 33} therefore D measures A the same number of times that E measures C. {VII. 20} Now, as many times as D measures A, so many units let there be in F; therefore F by multiplying D has made A, so that A is plane, and D, F are its sides.

Again, since D, E are the least of the numbers which have the same ratio with C, B, therefore D measures C the same number of times that E measures B. {VII. 20} As many times, then, as E measures B, so many units let there be in G; therefore E measures B according to the units in G; therefore G by multiplying E has made B.

Therefore B is plane, and E, G are its sides.

Therefore A, B are plane numbers.

I say next that they are also similar.

For, $\langle * \rangle$ since F by multiplying D has made A, and by multiplying E has made C, therefore, as D is to E, so is A to C, that is, C to B. {VII. 17} Again, $\langle * \rangle$ since E by multiplying F, G has made C, B respectively, therefore, as F is to G, so is C to B. {VII. 17} But, as C is to B, so is D to E; therefore also, as D is to E, so is F to G.

And alternately, as D is to F, so is E to G. {VII. 13} Therefore A, B are similar plane numbers; for their sides are proportional. Q. E. D.

PROPOSITION 21. If two mean proportional numbers fall between two numbers, the numbers are similar solid numbers.

For let two mean proportional numbers C, D fall between the two numbers A, B; I say that A, B are similar solid numbers.

For let three numbers E, F, G, the least of those which have the same ratio with A, C, D, be taken; {VII. 33 or VIII. 2} therefore the extremes of them E, G are prime to one another. {VIII. 3} Now, since one mean proportional number F has fallen between E, G, therefore E, G are similar plane numbers. {VIII. 20} Let, then, H, K be the sides of E, and L, M of G.

Therefore it is manifest from the theorem before this that E, F, G are continuously proportional in the ratio of H to L and that of K to M.

Now, since E, F, G are the least of the numbers which have the same ratio with A, C, D, and the multitude of the numbers E, F, G is equal to the multitude of the numbers A, C, D, therefore, ex aequali, as E is to G, so is A to D. {VII. 14} But E, G are prime, primes are also least, {VII. 21} and the least measure those which have the same ratio with them the same number of times, the greater the greater and the less the less, that is, the antecedent the antecedent and the consequent the consequent;

{VII. 20} therefore E measures A the same number of times that G measures D.

Now, as many times as E measures A, so many units let there be in N.

Therefore N by multiplying E has made A.

But E is the product of H, K; therefore N by multiplying the product of H, K has made A.

Therefore A is solid, and H, K, N are its sides.

Again, since E, F, G are the least of the numbers which have the same ratio as C, D, B, therefore E measures C the same number of times that G measures B.

Now, as many times as E measures C, so many units let there be in O.

Therefore G measures B according to the units in O; therefore O by multiplying G has made B.

But G is the product of L, M; therefore O by multiplying the product of L, M has made B.

Therefore B is solid, and L, M, O are its sides; therefore A, B are solid.

I say that they are also similar.

For since N, O by multiplying E have made A, C, therefore, as N is to O, so is A to C, that is, E to F. {VII. 18} But, as E is to F, so is H to L and K to M; therefore also, as H is to L, so is K to M and N to O.

And H, K, N are the sides of A, and O, L, M the sides of B.

Therefore A, B are similar solid numbers. Q. E. D.

PROPOSITION 22. If three numbers be in continued proportion, and the first be square, the third will also be square.

Let A, B, C be three numbers in continued proportion, and let A the first be square; I say that C the third is also square.

For, since between A, C there is one mean proportional number, B, therefore A, C are similar plane numbers. {VIII. 20} But A is square; therefore C is also square. Q. E. D.

PROPOSITION 23. If four numbers be in continued proportion, and the first be cube, the fourth will also be cube.

Let A, B, C, D be four numbers in continued proportion, and let A be cube; I say that D is also cube.

For, since between A, D there are two mean proportional numbers B, C, therefore A, D are similar solid numbers. {VIII. 21} But A is cube; therefore D is also cube. Q. E. D.

PROPOSITION 24. If two numbers have to one another the ratio which a square number has to a square number, and the first be square, the second will also be square.

For let the two numbers A, B have to one another the ratio which the square number C has to the square number D, and let A be square; I say that B is also square.

For, since C, D are square, C, D are similar plane numbers.

Therefore one mean proportional number falls between C, D. {VIII. 18} And, as C is to D, so is A to B; therefore one mean proportional number falls between A, B also. {VIII. 8} And A is square; therefore B is also square. {VIII. 22} Q. E. D.

PROPOSITION 25. If two numbers have to one another the ratio which a cube number has to a cube number, and the first be cube, the second will also be cube.

For let the two numbers A, B have to one another the ratio which the cube number C has to the cube number D, and let A be cube; I say that B is also cube.

For, since C, D are cube, C, D are similar solid numbers.

Therefore two mean proportional numbers fall between C, D. {VIII. 19} And, as many numbers as fall between C, D in continued proportion, so many will also fall between those which have the same ratio with them; {VIII. 8} so that two mean proportional numbers fall between A, B also.

Let E, F so fall.

Since, then, the four numbers A, E, F, B are in continued proportion, and A is cube, therefore B is also cube. {VIII. 23} Q. E. D.

PROPOSITION 26. Similar plane numbers have to one another the ratio which a square number has to a square number.

Let A, B be similar plane numbers; I say that A has to B the ratio which a square number has to a square number.

For, since A, B are similar plane numbers, therefore one mean proportional number falls between A, B. {VIII. 18} Let it so fall, and let it be C; and let D, E, F, the least numbers of those which have the same ratio with A, C, B, be taken; {VII. 33 or VIII. 2} therefore the extremes of them D, F are square. {VIII. 2, Por.} And since, as D is to F, so is A to B, and D, F are square, therefore A has to B the ratio which a square number has to a square number. Q. E. D.

PROPOSITION 27. Similar solid numbers have to one another the ratio which a cube number has to a cube number.

Let A, B be similar solid numbers; I say that A has to B the ratio which a cube number has to a cube number.

For, since A, B are similar solid numbers, therefore two mean proportional numbers fall between A, B. {VIII. 19} Let C, D so fall, and let E, F, G, H, the least numbers of those which have the same ratio with A, C, D, B, and equal with them in multitude, be

taken; {VII. 33 or VIII. 2} therefore the extremes of them E, H are cube. {VIII. 2, Por.} And, as E is to H, so is A to B; therefore A also has to B the ratio which a cube number has to a cube number. Q. E. D.

ENDNOTES.

69, 71, 99. the ratios $A : B, C : D, E : F$. This abbreviated expression is in the Greek ὁ $AB, \Gamma\Delta, EZ$ λόγοι.

1, 5, 29, 31. compounded of the ratios of their sides. As in VI. 23, the Greek has the less exact phrase, “compounded of their sides.”

1. fall. The Greek word is ἐμπίπτειν, “fall in” = “can be interpolated.”

25. For, since $F \dots 27. C$ to B . The text has clearly suffered corruption here. It is not necessary to infer from other facts that, as D is to E , so is A to C ; for this is part of the hypotheses (II. 6, 7). Again, there is no explanation of the statement (I. 25) that F by multiplying E has made C . It is the statement and explanation of this latter fact which are alone wanted; after which the proof proceeds as in I. 28. We might therefore substitute for II. 25-28 the following.

“For, since E measures C the same number of times that D measures A {I. 8}, that is, according to the units in F {I. 10}, therefore F by multiplying E has made C .

And, since E by multiplying F, G ,” etc. etc.

BOOK IX. PROPOSITIONS

PROPOSITION 1. IF TWO SIMILAR plane numbers by multiplying one another make some number, the product will be square.

Let A, B be two similar plane numbers, and let A by multiplying B make C; I say that C is square.

For let A by multiplying itself make D.

Therefore D is square.

Since then A by multiplying itself has made D, and by multiplying B has made C, therefore, as A is to B, so is D to C. {VII. 17} And, since A, B are similar plane numbers, therefore one mean proportional number falls between A, B. {VIII. 18} But, if numbers fall between two numbers in continued proportion, as many as fall between them, so many also fall between those which have the same ratio; {VIII. 8} so that one mean proportional number falls between D, C also.

And D is square; therefore C is also square. {VIII. 22} Q. E. D.

PROPOSITION 2. If two numbers by multiplying one another make a square number, they are similar plane numbers.

Let A, B be two numbers, and let A by multiplying B make the square number C; I say that A, B are similar plane numbers.

For let A by multiplying itself make D; therefore D is square.

Now, since A by multiplying itself has made D, and by multiplying B has made C, therefore, as A is to B, so is D to C. {VII. 17} And, since D is square, and C is so also, therefore D, C are similar plane numbers.

Therefore one mean proportional number falls between D, C. {VIII. 18} And, as D is to C, so is A to B; therefore one mean proportional number falls between A, B also. {VIII. 8} But, if one mean proportional number fall between two numbers, they are similar plane numbers; {VIII. 20} therefore A, B are similar plane numbers. Q. E. D.

PROPOSITION 3. If a cube number by multiplying itself make some number, the product will be cube.

For let the cube number A by multiplying itself make B; I say that B is cube.

For let C, the side of A, be taken, and let C by multiplying itself make D.

It is then manifest that C by multiplying D has made A.

Now, since C by multiplying itself has made D, therefore C measures D according to the units in itself.

But further the unit also measures C according to the units in it; therefore, as the unit is to C, so is C to D. {VII. Def. 20} Again, since C by multiplying D has made A, therefore D measures A according to the units in C.

But the unit also measures C according to the units in it; therefore, as the unit is to C, so is D to A.

But, as the unit is to C, so is C to D; therefore also, as the unit is to C, so is C to D, and D to A.

Therefore between the unit and the number A two mean proportional numbers C, D have fallen in continued proportion.

Again, since A by multiplying itself has made B, therefore A measures B according to the units in itself.

But the unit also measures A according to the units in it; therefore, as the unit is to A, so is A to B. {VII. Def. 20} But between the unit and A two mean proportional numbers have fallen; therefore two mean proportional numbers will also fall between A, B. {VIII. 8} But, if two mean proportional numbers fall between two numbers, and the first be cube, the second will also be cube. {VIII. 23} And A is cube; therefore B is also cube. Q. E. D.

PROPOSITION 4. If a cube number by multiplying a cube number make some number, the product will be cube.

For let the cube number A by multiplying the cube number B make C; I say that C is cube.

For let A by multiplying itself make D; therefore D is cube. {IX. 3} And, since A by multiplying itself has made D, and by multiplying B has made C therefore, as A is to B, so is D to C. {VII. 17} And, since A, B are cube numbers, A, B are similar solid numbers.

Therefore two mean proportional numbers fall between A, B; {VIII. 19} so that two mean proportional numbers will fall between D, C also. {VIII. 8} And D is cube; therefore C is also cube {VIII. 23} Q. E. D.

PROPOSITION 5. If a cube number by multiplying any number make a cube number, the multiplied number will also be cube.

For let the cube number A by multiplying any number B make the cube number C; I say that B is cube.

For let A by multiplying itself make D; therefore D is cube. {IX. 3} Now, since A by multiplying itself has made D, and by multiplying B has made C, therefore, as A is to B, so is D to C. {VII. 17} And since D, C are cube, they are similar solid numbers.

Therefore two mean proportional numbers fall between D, C. {VIII. 19} And, as D is to C, so is A to B; therefore two mean proportional numbers fall between A, B also. {VIII. 8} And A is cube; therefore B is also cube. {VIII. 23} **PROPOSITION 6.** If a number by multiplying itself make a cube number, it will itself also be cube.

For let the number A by multiplying itself make the cube number B; I say that A is also cube.

For let A by multiplying B make C.

Since, then, A by multiplying itself has made B, and by multiplying B has made C, therefore C is cube.

And, since A by multiplying itself has made B, therefore A measures B according to the units in itself.

But the unit also measures A according to the units in it.

Therefore, as the unit is to A, so is A to B. {VII. Def. 20} And, since A by multiplying B has made C, therefore B measures C according to the units in A.

But the unit also measures A according to the units in it.

Therefore, as the unit is to A, so is B to C. {VII. Def. 20} But, as the unit is to A, so is A to B; therefore also, as A is to B, so is B to C.

And, since B, C are cube, they are similar solid numbers.

Therefore there are two mean proportional numbers between B, C. {VIII. 19} And, as B is to C, so is A to B.

Therefore there are two mean proportional numbers between A, B also. {VIII. 8} And B is cube; therefore A is also cube. {cf. VIII. 23} Q. E. D.

PROPOSITION 7. If a composite number by multiplying any number make some number, the product will be solid.

For let the composite number A by multiplying any number B make C; I say that C is solid.

For, since A is composite, it will be measured by some number. {VII. Def. 13} Let it be measured by D; and, as many times as D measures A, so many units let there be in E.

Since then D measures A according to the units in E, therefore E by multiplying D has made A. {VII. Def. 15} And, since A by multiplying B has made C, and A is the product of D, E, therefore the product of D, E by multiplying B has made C.

Therefore C is solid, and D, E, B are its sides. Q. E. D.

PROPOSITION 8. If as many numbers as we please beginning from an unit be in continued proportion, the third from the unit will be square, as will also those which successively leave out one; the fourth will be cube, as will also all those which leave out two; and the seventh will be at once cube and square, as will also those which leave out five.

Let there be as many numbers as we please, A, B, C, D, E, F, beginning from an unit and in continued proportion; I say that B, the third from the unit, is square, as are also all those which leave out one; C, the fourth, is cube, as are also all those which leave out two; and F, the seventh, is at once cube and square, as are also all those which leave out five.

For since, as the unit is to A, so is A to B, therefore the unit measures the number A the same number of times that A measures B. {VII. Def. 20} But the unit measures the number A according to the units in it; therefore A also measures B according to the units in A.

Therefore A by multiplying itself has made B; therefore B is square.

And, since B, C, D are in continued proportion, and B is square, therefore D is also square. {VIII. 22} For the same reason F is also square.

Similarly we can prove that all those which leave out one are square.

I say next that C, the fourth from the unit, is cube, as are also all those which leave out two.

For since, as the unit is to A, so is B to C, therefore the unit measures the number A the same number of times that B measures C.

But the unit measures the number A according to the units in A; therefore B also measures C according to the units in A.

Therefore A by multiplying B has made C.

Since then A by multiplying itself has made B, and by multiplying B has made C, therefore C is cube.

And, since C, D, E, F are in continued proportion, and C is cube, therefore F is also cube. {VIII. 23} But it was also proved square; therefore the seventh from the unit is both cube and square.

Similarly we can prove that all the numbers which leave out five are also both cube and square. Q. E. D.

PROPOSITION 9. If as many numbers as we please beginning from an unit be in continued proportion, and the number after the unit be square, all the rest will also be square. And, if the number after the unit be cube, all the rest will also be cube.

Let there be as many numbers as we please, A, B, C, D, E, F, beginning from an unit and in continued proportion, and let A, the number after the unit, be square; I say that all the rest will also be square.

Now it has been proved that B, the third from the unit, is square, as are also all those which leave out one; {IX. 8} I say that all the rest are also square.

For, since A, B, C are in continued proportion, and A is square, therefore C is also square. {VIII. 22} Again, since B, C, D are in continued proportion, and B is square, D is also square. {VIII. 22} Similarly we can prove that all the rest are also square.

Next, let A be cube; I say that all the rest are also cube.

Now it has been proved that C, the fourth from the unit, is cube, as also are all those which leave out two; {IX. 8} I say that all the rest are also cube.

For, since, as the unit is to A, so is A to B, therefore the unit measures A the same number of times as A measures B.

But the unit measures A according to the units in it; therefore A also measures B according to the units in itself; therefore A by multiplying itself has made B.

And A is cube.

But, if a cube number by multiplying itself make some number, the product is cube. {IX. 3} Therefore B is also cube.

And, since the four numbers A, B, C, D are in continued proportion, and A is cube, D also is cube. {VIII. 23} For the same reason E is also cube, and similarly all the rest are cube. Q. E. D.

PROPOSITION 10. If as many numbers as we please beginning from an unit be in continued proportion, and the number after the unit be not square, neither will any other be square except the third from the unit and all those which leave out one. And, if the number after the unit be not cube, neither will any other be cube except the fourth from the unit and all those which leave out two.

Let there be as many numbers as we please, A, B, C, D, E, F, beginning from an unit and in continued proportion, and let A, the number after the unit, not be square; I say that neither will any other be square except the third from the unit <and those which leave out one>.

For, if possible, let C be square.

But B is also square; {IX. 8} {therefore B, C have to one another the ratio which a square number has to a square number}.

And, as B is to C, so is A to B; therefore A, B have to one another the ratio which a square number has to a square number; {so that A, B are similar plane numbers}. {VIII. 26, converse} And B is square; therefore A is also square: which is contrary to the hypothesis.

Therefore C is not square.

Similarly we can prove that neither is any other of the numbers square except the third from the unit and those which leave out one.

Next, let A not be cube.

I say that neither will any other be cube except the fourth from the unit and those which leave out two.

For, if possible, let D be cube.

Now C is also cube; for it is fourth from the unit. {IX. 8} And, as C is to D, so is B to C; therefore B also has to C the ratio which a cube has to a cube.

And C is cube; therefore B is also cube. {VIII. 25} And since, as the unit is to A, so is A to B, and the unit measures A according to the units in it, therefore A also measures B according to the units in itself; therefore A by multiplying itself has made the cube number B.

But, if a number by multiplying itself make a cube number, it is also itself cube. {IX. 6} Therefore A is also cube: which is contrary to the hypothesis.

Therefore D is not cube.

Similarly we can prove that neither is any other of the numbers cube except the fourth from the unit and those which leave out two. Q. E. D.

PROPOSITION II. If as many numbers as we please beginning from an unit be in continued proportion, the less measures the greater according to some one of the numbers which have place among the proportional numbers.

Let there be as many numbers as we please, B, C, D, E, beginning from the unit A and in continued proportion; I say that B, the least of the numbers B, C, D, E, measures E according to some one of the numbers C, D.

For since, as the unit A is to B, so is D to E, therefore the unit A measures the number B the same number of times as D measures E; therefore, alternately, the unit A measures D the same number of times as B measures E. {VII. 15} But the unit A measures D according to the units in it; therefore B also measures E according to the units in D; so that B the less measures E the greater according to some number of those which have place among the proportional numbers. — PORISM.

And it is manifest that, whatever place the measuring number has, reckoned from the unit, the same place also has the number according to which it measures, reckoned from the number measured, in the direction of the number before it. — Q. E. D.

PROPOSITION 12. If as many numbers as we please beginning from an unit be in continued proportion, by however many prime numbers the last is measured, the next to the unit will also be measured by the same.

Let there be as many numbers as we please, A, B, C, D, beginning from an unit, and in continued proportion; I say that, by however many prime numbers D is measured, A will also be measured by the same.

For let D be measured by any prime number E; I say that E measures A.

For suppose it does not; now E is prime, and any prime number is prime to any which it does not measure; {VII. 29} therefore E, A are prime to one another.

And, since E measures D, let it measure it according to F, therefore E by multiplying F has made D.

Again, since A measures D according to the units in C, {IX. 11 and Por.} therefore A by multiplying C has made D.

But, further, E has also by multiplying F made D; therefore the product of A, C is equal to the product of E, F.

Therefore, as A is to E, so is F to C. {VII. 19} But A, E are prime, primes are also least, {VII. 21} and the least measure those which have the same ratio the same number of times, the antecedent the antecedent and the consequent the consequent; {VII. 20} therefore E measures C.

Let it measure it according to G; therefore E by multiplying G has made C.

But, further, by the theorem before this, A has also by multiplying B made C. {IX. 11 and Por.} Therefore the product of A, B is equal to the product of E, G.

Therefore, as A is to E, so is G to B. {VII. 19} But A, E are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio with them the same number of times, the antecedent the antecedent and the consequent the consequent: {VII. 20} therefore E measures B.

Let it measure it according to H; therefore E by multiplying H has made B.

But further A has also by multiplying itself made B; {IX. 8} therefore the product of E, H is equal to the square on A.

Therefore, as E is to A, so is A to H. {VII. 19} But A, E are prime, primes are also least, {VII. 21} and the least measure those which have the same ratio the same number of times, the antecedent the antecedent and the consequent the consequent; {VII. 20} therefore E measures A, as antecedent antecedent.

But, again, it also does not measure it: which is impossible.

Therefore E, A are not prime to one another.

Therefore they are composite to one another.

But numbers composite to one another are measured by some number. {VII. Def. 14} And, since E is by hypothesis prime, and the prime is not measured by any number other than itself, therefore E measures A, E, so that E measures A.

{But it also measures D; therefore E measures A, D.} Similarly we can prove that, by however many prime numbers D is measured, A will also be measured by the same. Q. E. D.

PROPOSITION 13. If as many numbers as we please beginning from an unit be in continued proportion, and the number after the unit be prime, the greatest will not be measured by any except those which have a place among the proportional numbers.

Let there be as many numbers as we please, A, B, C, D, beginning from an unit and in continued proportion, and let A, the number after the unit, be prime; I say that D, the greatest of them, will not be measured by any other number except A, B, C.

For, if possible, let it be measured by E, and let E not be the same with any of the numbers A, B, C.

It is then manifest that E is not prime.

For, if E is prime and measures D, it will also measure A {IX. 12}, which is prime, though it is not the same with it: which is impossible.

Therefore E is not prime.

Therefore it is composite.

But any composite number is measured by some prime number; {VII. 31} therefore E is measured by some prime number.

I say next that it will not be measured by any other prime except A.

For, if E is measured by another, and E measures D, that other will also measure D; so that it will also measure A {IX. 12}, which is prime, though it is not the same with it: which is impossible.

Therefore A measures E.

And, since E measures D, let it measure it according to F.

I say that F is not the same with any of the numbers A, B, C.

For, if F is the same with one of the numbers A, B, C, and measures D according to E, therefore one of the numbers A, B, C also measures D according to E.

But one of the numbers A, B, C measures D according to some one of the numbers A, B, C; {IX. 11} therefore E is also the same with one of the numbers A, B, C: which is contrary to the hypothesis.

Therefore F is not the same as any one of the numbers A, B, C.

Similarly we can prove that F is measured by A, by proving again that F is not prime.

For, if it is, and measures D, it will also measure A {IX. 12}, which is prime, though it is not the same with it: which is impossible; therefore F is not prime.

Therefore it is composite.

But any composite number is measured by some prime number; {VII. 31} therefore F is measured by some prime number.

I say next that it will not be measured by any other prime except A.

For, if any other prime number measures F, and F measures D, that other will also measure D; so that it will also measure A {IX. 12}, which is prime, though it is not the same with it: which is impossible.

Therefore A measures F.

And, since E measures D according to F, therefore E by multiplying F has made D.

But, further, A has also by multiplying C made D; {IX. 11} therefore the product of A, C is equal to the product of E, F.

Therefore, proportionally, as A is to E, so is F to C. {VII. 19} But A measures E; therefore F also measures C.

Let it measure it according to G.

Similarly, then, we can prove that G is not the same with any of the numbers A, B, and that it is measured by A.

And, since F measures C according to G therefore F by multiplying G has made C.

But, further, A has also by multiplying B made C; {IX. 11} therefore the product of A, B is equal to the product of F, G.

Therefore, proportionally, as A is to F, so is G to B. {VII. 19} But A measures F; therefore G also measures B.

Let it measure it according to H.

Similarly then we can prove that H is not the same with A.

And, since G measures B according to H, therefore G by multiplying H has made B.

But further A has also by multiplying itself made B; {IX. 8} therefore the product of H, G is equal to the square on A.

Therefore, as H is to A, so is A to G. {VII. 19} But A measures G; therefore H also measures A, which is prime, though it is not the same with it: which is absurd.

Therefore D the greatest will not be measured by any other number except A, B, C. Q. E. D.

PROPOSITION 14. If a number be the least that is measured by prime numbers, it will not be measured by any other prime

number except those originally measuring it.

For let the number A be the least that is measured by the prime numbers B, C, D; I say that A will not be measured by any other prime number except B, C, D.

For, if possible, let it be measured by the prime number E, and let E not be the same with any one of the numbers B, C, D.

Now, since E measures A, let it measure it according to F; therefore E by multiplying F has made A.

And A is measured by the prime numbers B, C, D.

But, if two numbers by multiplying one another make some number, and any prime number measure the product, it will also measure one of the original numbers; {VII. 30} therefore B, C, D will measure one of the numbers E, F.

Now they will not measure E; for E is prime and not the same with any one of the numbers B, C, D.

Therefore they will measure F, which is less than A: which is impossible, for A is by hypothesis the least number measured by B, C, D.

Therefore no prime number will measure A except B, C, D. Q. E. D.

PROPOSITION 15. If three numbers in continued proportion be the least of those which have the same ratio with them, any two whatever added together will be prime to the remaining number.

Let A, B, C, three numbers in continued proportion, be the least of those which have the same ratio with them; I say that any two of the numbers A, B, C whatever added together are prime to the remaining number, namely A, B to C; B, C to A; and further A, C to B.

For let two numbers DE, EF, the least of those which have the same ratio with A, B, C, be taken. {VIII. 2} It is then manifest that DE by multiplying itself has made A, and by multiplying EF has made B, and, further, EF by multiplying itself has made C. {VIII. 2} Now, since DE, EF are least, they are prime to one another. {VII. 22} But, if two numbers be prime to one another, their sum is also prime to each; {VII. 28} therefore DF is also prime to each of the numbers DE, EF.

But further DE is also prime to EF; therefore DF, DE are prime to EF.

But, if two numbers be prime to any number, their product is also prime to the other; {VII. 24} so that the product of FD, DE is prime to EF; hence the product of FD, DE is also prime to the square on EF. {VII. 25} But the product of FD, DE is the square on DE together with the product of DE, EF; {II. 3} therefore the square on DE together with the product of DE, EF is prime to the square on EF.

And the square on DE is A, the product of DE, EF is B, and the square on EF is C; therefore A, B added together are prime to C.

Similarly we can prove that B, C added together are prime to A.

I say next that A, C added together are also prime to B.

For, since DF is prime to each of the numbers DE, EF, the square on DF is also prime to the product of DE, EF. {VII. 24, 25} But the squares on DE, EF together with twice the product of DE, EF are equal to the square on DF; {II. 4} therefore the squares on DE, EF together with twice the product of DE, EF are prime to the product of DE, EF.

Separando, the squares on DE, EF together with once the product of DE, EF are prime to the product of DE, EF.

Therefore, separando again, the squares on DE, EF are prime to the product of DE, EF.

And the square on DE is A, the product of DE, EF is B, and the square on EF is C.

Therefore A, C added together are prime to B. Q. E. D.

PROPOSITION 16. If two numbers be prime to one another, the second will not be to any other number as the first is to the second.

For let the two numbers A, B be prime to one another; I say that B is not to any other number as A is to B.

For, if possible, as A is to B, so let B be to C.

Now A, B are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio the same number of times, the antecedent the antecedent and the consequent the consequent; {VII. 20} therefore A measures B as antecedent antecedent.

But it also measures itself; therefore A measures A, B which are prime to one another: which is absurd.

Therefore B will not be to C, as A is to B. Q. E. D.

PROPOSITION 17. If there be as many numbers as we please in continued proportion, and the extremes of them be prime to one another, the last will not be to any other number as the first to the second.

For let there be as many numbers as we please, A, B, C, D, in continued proportion, and let the extremes of them, A, D, be prime to one another; I say that D is not to any other number as A is to B.

For, if possible, as A is to B, so let D be to E; therefore, alternately, as A is to D, so is B to E. {VII. 13} But A, D are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio the same number of times, the antecedent the antecedent and the consequent the consequent.

{VII. 20} Therefore A measures B.

And, as A is to B, so is B to C.

Therefore B also measures C; so that A also measures C.

And since, as B is to C, so is C to D, and B measures C, therefore C also measures D.

But A measured C; so that A also measures D.

But it also measures itself; therefore A measures A, D which are prime to one another : which is impossible.

Therefore D will not be to any other number as A is to B. Q. E. D.

PROPOSITION 18. Given two numbers, to investigate whether it is possible to find a third proportional to them.

Let A, B be the given two numbers, and let it be required to investigate whether it is possible to find a third proportional to them.

Now A, B are either prime to one another or not.

And, if they are prime to one another, it has been proved that it is impossible to find a third proportional to them. {IX. 16} Next, let A, B not be prime to one another, and let B by multiplying itself make C.

Then A either measures C or does not measure it.

First, let it measure it according to D; therefore A by multiplying D has made C.

But, further, B has also by multiplying itself made C; therefore the product of A, D is equal to the square on B.

Therefore, as A is to B, so is B to D; {VII. 19} therefore a third proportional number D has been found to A, B.

Next, let A not measure C; I say that it is impossible to find a third proportional number to A, B.

For, if possible, let D, such third proportional, have been found.

Therefore the product of A, D is equal to the square on B.

But the square on B is C; therefore the product of A, D is equal to C.

Hence A by multiplying D has made C; therefore A measures C according to D.

But, by hypothesis, it also does not measure it: which is absurd.

Therefore it is not possible to find a third proportional number to A, B when A does not measure C. Q. E. D.

PROPOSITION 19. Given three numbers, to investigate when it is possible to find a fourth proportional to them.

Let A, B, C be the given three numbers, and let it be required to investigate when it is possible to find a fourth proportional to them.

Now either they are not in continued proportion, and the extremes of them are prime to one another; or they are in continued proportion, and the extremes of them are not prime to one another; or they are not in continued proportion, nor are the extremes of them prime to one another; or they are in continued proportion, and the extremes of them are prime to one another.

If then A, B, C are in continued proportion, and the extremes of them A, C are prime to one another, it has been proved that it is impossible to find a fourth proportional number to them. {IX. 17} <*>Next, let A, B, C not be in continued proportion, the extremes being again prime to one another; I say that in this case also it is impossible to find a fourth proportional to them.

For, if possible, let D have been found, so that, as A is to B, so is C to D, and let it be contrived that, as B is to C, so is D to E.

Now, since, as A is to B, so is C to D, and, as B is to C, so is D to E, therefore, *ex aequali*, as A is to C, so is C to E. {VII. 14} But A, C are prime, primes are also least, {VII. 21} and the least numbers measure those which have the same ratio, the antecedent the antecedent and the consequent the consequent. {VII. 20} Therefore A measures C as antecedent antecedent.

But it also measures itself; therefore A measures A, C which are prime to one another: which is impossible.

Therefore it is not possible to find a fourth proportional to A, B, C.〈*〉 Next, let A, B, C be again in continued proportion, but let A, C not be prime to one another.

I say that it is possible to find a fourth proportional to them.

For let B by multiplying C make D; therefore A either measures D or does not measure it.

First, let it measure it according to E; therefore A by multiplying E has made D.

But, further, B has also by multiplying C made D; therefore the product of A, E is equal to the product of B, C; therefore, proportionally, as A is to B, so is C to E; {VII. 19} therefore E has been found a fourth proportional to A, B, C.

Next, let A not measure D; I say that it is impossible to find a fourth proportional number to A, B, C.

For, if possible, let E have been found; therefore the product of A, E is equal to the product of B, C. {VII. 19} But the product of B, C is D; therefore the product of A, E is also equal to D.

Therefore A by multiplying E has made D; therefore A measures D according to E, so that A measures D.

But it also does not measure it: which is absurd.

Therefore it is not possible to find a fourth proportional number to A, B, C when A does not measure D.

Next, let A, B, C not be in continued proportion, nor the extremes prime to one another.

And let B by multiplying C make D.

Similarly then it can be proved that, if A measures D, it is possible to find a fourth proportional to them, but, if it does not measure it, impossible. Q. E. D.

PROPOSITION 20. Prime numbers are more than any assigned multitude of prime numbers.

Let A, B, C be the assigned prime numbers; I say that there are more prime numbers than A, B, C.

For let the least number measured by A, B, C be taken, and let it be DE; let the unit DF be added to DE.

Then EF is either prime or not.

First, let it be prime; then the prime numbers A, B, C, EF have been found which are more than A, B, C.

Next, let EF not be prime; therefore it is measured by some prime number. {VII. 31} Let it be measured by the prime number G.

I say that G is not the same with any of the numbers A, B, C.

For, if possible, let it be so.

Now A, B, C measure DE; therefore G also will measure DE.

But it also measures EF.

Therefore G, being a number, will measure the remainder, the unit DF: which is absurd.

Therefore G is not the same with any one of the numbers A, B, C.

And by hypothesis it is prime.

Therefore the prime numbers A, B, C, G have been found which are more than the assigned multitude of A, B, C. Q. E. D.

PROPOSITION 21. If as many even numbers as we please be added together, the whole is even.

For let as many even numbers as we please, AB, BC, CD, DE, be added together; I say that the whole AE is even.

For, since each of the numbers AB, BC, CD, DE is even, it has a half part; {VII. Def. 6} so that the whole AE also has a half part.

But an even number is that which is divisible into two equal parts; {id.} therefore AE is even. Q. E. D.

PROPOSITION 22. If as many odd numbers as we please be added together, and their multitude be even, the whole will be even.

For let as many odd numbers as we please, AB, BC, CD, DE, even in multitude, be added together; I say that the whole AE is even.

For, since each of the numbers AB, BC, CD, DE is odd, if an unit be subtracted from each, each of the remainders will be even; {VII. Def. 7} so that the sum of them will be even. {IX. 21} But the multitude of the units is also even.

Therefore the whole AE is also even. {IX. 21} Q. E. D.

PROPOSITION 23. If as many odd numbers as we please be added together, and their multitude be odd, the whole will also be odd.

For let as many odd numbers as we please, AB, BC, CD, the multitude of which is odd, be added together; I say that the whole AD is also odd.

Let the unit DE be subtracted from CD; therefore the remainder CE is even. {VII. Def. 7} But CA is also even; {IX. 22} therefore the

whole AE is also even. {IX. 21} And DE is an unit.

Therefore AD is odd. {VII. Def. 7} Q. E. D.

PROPOSITION 24. If from an even number an even number be subtracted, the remainder will be even.

For from the even number AB let the even number BC be subtracted; I say that the remainder CA is even.

For, since AB is even, it has a half part. {VII. Def. 6} For the same reason BC also has a half part; so that the remainder {CA also has a half part, and} AC is therefore even. Q. E. D.

PROPOSITION 25. If from an even number an odd number be subtracted, the remainder will be odd.

For from the even number AB let the odd number BC be subtracted; I say that the remainder CA is odd.

For let the unit CD be subtracted from BC; therefore DB is even. {VII. Def. 7} But AB is also even; therefore the remainder AD is also even. {IX. 24} And CD is an unit; therefore CA is odd. {VII. Def. 7} Q. E. D.

PROPOSITION 26. If from an odd number an odd number be subtracted, the remainder will be even.

For from the odd number AB let the odd number BC be subtracted; I say that the remainder CA is even.

For, since AB is odd, let the unit BD be subtracted; therefore the remainder AD is even. {VII. Def. 7} For the same reason CD is also even; {VII. Def. 7} so that the remainder CA is also even. {IX. 24} Q. E. D.

PROPOSITION 27. If from an odd number an even number be subtracted, the remainder will be odd.

For from the odd number AB let the even number BC be subtracted; I say that the remainder CA is odd.

Let the unit AD be subtracted; therefore DB is even. {VII. Def. 7}
But BC is also even; therefore the remainder CD is even. {IX. 24}
Therefore CA is odd. {VII. Def. 7} Q. E. D.

PROPOSITION 28. If an odd number by multiplying an even number make some number, the product will be even.

For let the odd number A by multiplying the even number B make C; I say that C is even.

For, since A by multiplying B has made C, therefore C is made up of as many numbers equal to B as there are units in A. {VII. Def. 15} And B is even; therefore C is made up of even numbers.

But, if as many even numbers as we please be added together, the whole is even. {IX. 21} Therefore C is even. Q. E. D.

PROPOSITION 29. If an odd number by multiplying an odd number make some number, the product will be odd.

For let the odd number A by multiplying the odd number B make C; I say that C is odd.

For, since A by multiplying B has made C, therefore C is made up of as many numbers equal to B as there are units in A. {VII. Def. 15} And each of the numbers A, B is odd; therefore C is made up of odd numbers the multitude of which is odd.

Thus C is odd. {IX. 23} Q. E. D.

PROPOSITION 30. If an odd number measure an even number, it will also measure the half of it.

For let the odd number A measure the even number B; I say that it will also measure the half of it.

For, since A measures B, let it measure it according to C; I say that C is not odd.

For, if possible, let it be so.

Then, since A measures B according to C, therefore A by multiplying C has made B.

Therefore B is made up of odd numbers the multitude of which is odd.

Therefore B is odd: {IX. 23} which is absurd, for by hypothesis it is even.

Therefore C is not odd; therefore C is even.

Thus A measures B an even number of times.

For this reason then it also measures the half of it. Q. E. D.

PROPOSITION 31. If an odd number be prime to any number, it will also be prime to the double of it.

For let the odd number A be prime to any number B, and let C be double of B; I say that A is prime to C.

For, if they are not prime to one another, some number will measure them.

Let a number measure them, and let it be D.

Now A is odd; therefore D is also odd.

And since D which is odd measures C, and C is even, therefore {D} will measure the half of C also. {IX. 30} But B is half of C; therefore D measures B.

But it also measures A; therefore D measures A, B which are prime to one another: which is impossible.

Therefore A cannot but be prime to C.

Therefore A, C are prime to one another. Q. E. D.

PROPOSITION 32. Each of the numbers which are continually doubled beginning from a dyad is even-times even only.

For let as many numbers as we please, B, C, D, have been continually doubled beginning from the dyad A; I say that B, C, D are eventimes even only.

Now that each of the numbers B, C, D is even-times even is manifest; for it is doubled from a dyad.

I say that it is also even-times even only.

For let an unit be set out.

Since then as many numbers as we please beginning from an unit are in continued proportion, and the number A after the unit is prime, therefore D, the greatest of the numbers A, B, C, D, will not be measured by any other number except A, B, C. {IX. 13} And each of the numbers A, B, C is even; therefore D is even-times even only. {VII. Def. 8} Similarly we can prove that each of the numbers B, C is even-times even only. Q. E. D.

PROPOSITION 33. If a number have its half odd, it is even-times odd only.

For let the number A have its half odd; I say that A is even-times odd only.

Now that it is even-times odd is manifest; for the half of it, being odd, measures it an even number of times. {VII. Def. 9} I say next that it is also even-times odd only.

For, if A is even-times even also, it will be measured by an even number according to an even number; {VII. Def. 8} so that the half of it will also be measured by an even number though it is odd: which is absurd.

Therefore A is even-times odd only. Q. E. D.

PROPOSITION 34. If a number neither be one of those which are continually doubled from a dyad, nor have its half odd, it is both eventimes even and even-times odd.

For let the number A neither be one of those doubled from a dyad, nor have its half odd; I say that A is both even-times even and even-times odd.

Now that A is even-times even is manifest; for it has not its half odd. {VII. Def. 8} I say next that it is also even-times odd.

For, if we bisect A, then bisect its half, and do this continually, we shall come upon some odd number which will measure A according to an even number.

For, if not, we shall come upon a dyad, and A will be among those which are doubled from a dyad: which is contrary to the hypothesis.

Thus A is even-times odd.

But it was also proved even-times even.

Therefore A is both even-times even and even-times odd. Q. E. D.

PROPOSITION 35. If as many numbers as we please be in continued proportion, and there be subtracted from the second and the last numbers equal to the first, then, as the excess of the second is to the first, so will the excess of the last be to all those before it.

Let there be as many numbers as we please in continued proportion, A, BC, D, EF, beginning from A as least, and let there be subtracted from BC and EF the numbers BG, FH, each equal to A; I say that, as GC is to A, so is EH to A, BC, D.

For let FK be made equal to BC, and FL equal to D.

Then, since FK is equal to BC, and of these the part FH is equal to the part BG, therefore the remainder HK is equal to the remainder GC.

And since, as EF is to D, so is D to BC, and BC to A, while D is equal to FL, BC to FK, and A to FH, therefore, as EF is to FL, so is LF to FK, and FK to FH.

Separando, as EL is to LF, so is LK to FK, and KH to FH. {VII. 11, 13} Therefore also, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents; {VII. 12} therefore, as KH is to FH, so are EL, LK, KH to LF, FK, HF.

But KH is equal to CG, FH to A, and LF, FK, HF to D, BC, A; therefore, as CG is to A, so is EH to D, BC, A.

Therefore, as the excess of the second is to the first, so is the excess of the last to all those before it. Q. E. D.

PROPOSITION 36. If as many numbers as we please beginning from an unit be set out continuously in double proportion, until the sum of all becomes prime, and if the sum multiplied into the last make some number, the product will be perfect.

For let as many numbers as we please, A, B, C, D, beginning from an unit be set out in double proportion, until the sum of all becomes prime, let E be equal to the sum, and let E by multiplying D make FG; I say that FG is perfect.

For, however many A, B, C, D are in multitude, let so many E, HK, L, M be taken in double proportion beginning from E; therefore, ex aequali, as A is to D, so is E to M. {VII. 14} Therefore the product of E, D is equal to the product of A, M. {VII. 19} And the product of E, D is FG; therefore the product of A, M is also FG.

Therefore A by multiplying M has made FG; therefore M measures FG according to the units in A.

And A is a dyad; therefore FG is double of M.

But M, L, HK, E are continuously double of each other; therefore E, HK, L, M, FG are continuously proportional in double proportion.

Now let there be subtracted from the second HK and the last FG the numbers HN, FO, each equal to the first E; therefore, as the excess of the second is to the first, so is the excess of the last to all those before it. {IX. 35} Therefore, as NK is to E, so is OG to M, L, KH, E.

And NK is equal to E; therefore OG is also equal to M, L, HK, E.

But FO is also equal to E, and E is equal to A, B, C, D and the unit.

Therefore the whole FG is equal to E, HK, L, M and A, B, C, D and the unit; and it is measured by them.

I say also that FG will not be measured by any other number except A, B, C, D, E, HK, L, M and the unit.

For, if possible, let some number P measure FG, and let P not be the same with any of the numbers A, B, C, D, E, HK, L, M.

And, as many times as P measures FG, so many units let there be in Q; therefore Q by multiplying P has made FG.

But, further, E has also by multiplying D made FG; therefore, as E is to Q, so is P to D. {VII. 19} And, since A, B, C, D are continuously proportional beginning from an unit, therefore D will not be measured by any other number except A, B, C. {IX. 13} And, by hypothesis, P is not the same with any of the numbers A, B, C; therefore P will not measure D.

But, as P is to D, so is E to Q; therefore neither does E measure Q. {VII. Def. 20} And E is prime; and any prime number is prime to any number which it does not measure. {VII. 29} Therefore E, Q are prime to one another.

But primes are also least, {VII. 21} and the least numbers measure those which have the same ratio the same number of times, the antecedent the antecedent and the consequent the consequent; {VII. 20} and, as E is to Q, so is P to D; therefore E measures P the same number of times that Q measures D.

But D is not measured by any other number except A, B, C; therefore Q is the same with one of the numbers A, B, C.

Let it be the same with B.

And, however many B, C, D are in multitude, let so many E, HK, L be taken beginning from E.

Now E, HK, L are in the same ratio with B, C, D; therefore, ex aequali, as B is to D, so is E to L. {VII. 14} Therefore the product of B, L is equal to the product of D, E. {VII. 19} But the product of D, E is equal to the product of Q, P; therefore the product of Q, P is also equal to the product of B, L.

Therefore, as Q is to B, so is L to P. {VII. 19} And Q is the same with B; therefore L is also the same with P; which is impossible, for by hypothesis P is not the same with any of the numbers set out.

Therefore no number will measure FG except A, B, C, D, E, HK, L, M and the unit.

And FG was proved equal to A, B, C, D, E, HK, L, M and the unit; and a perfect number is that which is equal to its own parts; {VII. Def. 22} therefore FG is perfect. Q. E. D.

ENDNOTE.

3. Literally “let there be as many numbers as we please, of which let the multitude be odd.” This form, natural in Greek, is awkward in English.

BOOK X. DEFINITIONS I.

1. THOSE MAGNITUDES ARE said to be commensurable which are measured by the same measure, and those incommensurable which cannot have any common measure.

2. Straight lines are commensurable in square when the squares on them are measured by the same area, and incommensurable

in square when the squares on them cannot possibly have any area as a common measure.

3. With these hypotheses, it is proved that there exist straight lines infinite in multitude which are commensurable and incommensurable respectively, some in length only, and others in square also, with an assigned straight line. Let then the assigned straight line be called rational, and those straight lines which are commensurable with it, whether in length and in square or in square only, rational, but those which are incommensurable with it irrational.

4. And let the square on the assigned straight line be called rational and those areas which are commensurable with it rational, but those which are incommensurable with it irrational, and the straight lines which produce them irrational, that is, in case the areas are squares, the sides themselves, but in case they are any other rectilineal figures, the straight lines on which are described squares equal to them.

PROPOSITIONS 1-47.

PROPOSITION 1. TWO UNEQUAL MAGNITUDES being set out, if from the greater there be subtracted a magnitude greater than its half, and from that which is left a magnitude greater than its half, and if this process be repeated continually, there will be left some magnitude which will be less than the lesser magnitude set out.

Let AB, C be two unequal magnitudes of which AB is the greater: I say that, if from AB there be subtracted a magnitude greater than its half, and from that which is left a magnitude greater than its half, and if this process be repeated continually, there will be left some magnitude which will be less than the magnitude C.

For C if multiplied will sometime be greater than AB. {cf. v. Def. 4} Let it be multiplied, and let DE be a multiple of C, and greater than. AB; let DE be divided into the parts DF, FG, GE equal to C, from AB let there be subtracted BH greater than its half, and, from AH, HK greater than its half, and let this process be repeated continually until the divisions in AB are equal in multitude with the divisions in DE.

Let, then, AK, KH, HB be divisions which are equal in multitude with DF, FG, GE.

Now, since DE is greater than AB, and from DE there has been subtracted EG less than its half, and, from AB, BH greater than its half, therefore the remainder GD is greater than the remainder HA.

And, since GD is greater than HA, and there has been subtracted, from GD, the half GF, and, from HA, HK greater than its half, therefore the remainder DF is greater than the remainder AK.

But DF is equal to C; therefore C is also greater than AK.

Therefore AK is less than C.

Therefore there is left of the magnitude AB the magnitude AK which is less than the lesser magnitude set out, namely C. Q. E. D.

And the theorem can be similarly proved even if the parts subtracted be halves.

PROPOSITION 2. If, when the less of two unequal magnitudes is continually subtracted in turn from the greater, that which is left never measures the one before it, the magnitudes will be incommensurable.

For, there being two unequal magnitudes AB, CD, and AB being the less, when the less is continually subtracted in turn from the greater, let that which is left over never measure the one before

it; I say that the magnitudes AB, CD are incommensurable.

For, if they are commensurable, some magnitude will measure them.

Let a magnitude measure them, if possible, and let it be E; let AB, measuring FD, leave CF less than itself, let CF measuring BG, leave AG less than itself, and let this process be repeated continually, until there is left some magnitude which is less than E.

Suppose this done, and let there be left AG less than E.

Then, since E measures AB, while AB measures DF, therefore E will also measure FD.

But it measures the whole CD also; therefore it will also measure the remainder CF.

But CF measures BG; therefore E also measures BG.

But it measures the whole AB also; therefore it will also measure the remainder AG, the greater the less: which is impossible.

Therefore no magnitude will measure the magnitudes AB, CD; therefore the magnitudes AB, CD are incommensurable.

Therefore etc. {X. Def. 1} **PROPOSITION 3.** Given two commensurable magnitudes, to find their greatest common measure.

Let the two given commensurable magnitudes be AB, CD of which AB is the less; thus it is required to find the greatest common measure of AB, CD.

Now the magnitude AB either measures CD or it does not.

If then it measures it — and it measures itself also — AB is a common measure of AB, CD.

And it is manifest that it is also the greatest; for a greater magnitude than the magnitude AB will not measure AB.

Next, let AB not measure CD.

Then, if the less be continually subtracted in turn from the greater, that which is left over will sometime measure the one before it, because AB, CD are not incommensurable; {cf. X. 2} let AB, measuring ED, leave EC less than itself, let EC, measuring FB, leave AF less than itself, and let AF measure CE.

Since, then, AF measures CE, while CE measures FB, therefore AF will also measure FB.

But it measures itself also; therefore AF will also measure the whole AB.

But AB measures DE; therefore AF will also measure ED.

But it measures CE also; therefore it also measures the whole CD.

Therefore AF is a common measure of AB, CD.

I say next that it is also the greatest.

For, if not, there will be some magnitude greater than AF which will measure AB, CD.

Let it be G.

Since then G measures AB, while AB measures ED, therefore G will also measure ED.

But it measures the whole CD also; therefore G will also measure the remainder CE.

But CE measures FB; therefore G will also measure FB.

But it measures the whole AB also, and it will therefore measure the remainder AF, the greater the less: which is impossible.

Therefore no magnitude greater than AF will measure AB, CD; therefore AF is the greatest common measure of AB, CD.

Therefore the greatest common measure of the two given commensurable magnitudes AB, CD has been found. Q. E. D.

PORISM.

From this it is manifest that, if a magnitude measure two magnitudes, it will also measure their greatest common measure.

PROPOSITION 4. Given three commensurable magnitudes, to find their greatest common measure.

Let A, B, C be the three given commensurable magnitudes; thus it is required to find the greatest common measure of A, B, C.

Let the greatest common measure of the two magnitudes A, B be taken, and let it be D; {X. 3} then D either measures C, or does not measure it.

First, let it measure it.

Since then D measures C, while it also measures A, B, therefore D is a common measure of A, B, C.

And it is manifest that it is also the greatest; for a greater magnitude than the magnitude D does not measure A, B.

Next, let D not measure C.

I say first that C, D are commensurable.

For, since A, B, C are commensurable, some magnitude will measure them, and this will of course measure A, B also; so that it will also measure the greatest common measure of A, B, namely D. {X. 3, Por.} But it also measures C; so that the said magnitude will measure C, D; therefore C, D are commensurable.

Now let their greatest common measure be taken, and let it be E. {X. 3} Since then E measures D, while D measures A, B, therefore E will also measure A, B.

But it measures C also; therefore E measures A, B, C; therefore E is a common measure of A, B, C.

I say next that it is also the greatest.

For, if possible, let there be some magnitude F greater than E, and let it measure A, B, C.

Now, since F measures A, B, C, it will also measure A, B, and will measure the greatest common measure of A, B. {X. 3, Por.} But the greatest common measure of A, B is D; therefore F measures D.

But it measures C also; therefore F measures C, D; therefore F will also measure the greatest common measure of C, D. {X. 3, Por.} But that is E; therefore F will measure E, the greater the less: which is impossible.

Therefore no magnitude greater than the magnitude E will measure A, B, C; therefore E is the greatest common measure of A, B, C if D do not measure C, and, if it measure it, D is itself the greatest common measure.

Therefore the greatest common measure of the three given commensurable magnitudes has been found.

PORISM.

From this it is manifest that, if a magnitude measure three magnitudes, it will also measure their greatest common measure.

Similarly too, with more magnitudes, the greatest common measure can be found, and the porism can be extended. Q. E. D.

PROPOSITION 5. Commensurable magnitudes have to one another the ratio which a number has to a number.

Let A, B be commensurable magnitudes; I say that A has to B the ratio which a number has to a number.

For, since A, B are commensurable, some magnitude will measure them.

Let it measure them, and let it be C.

And, as many times as C measures A, so many units let there be in D; and, as many times as C measures B, so many units let there be in E.

Since then C measures A according to the units in D, while the unit also measures D according to the units in it, therefore the unit measures the number D the same number of times as the magnitude C measures A; therefore, as C is to A, so is the unit to D; {VII. Def. 20} therefore, inversely, as A is to C, so is D to the unit. {cf. V. 7, Por.} Again, since C measures B according to the units in E, while the unit also measures E according to the units in it, therefore the unit measures E the same number of times as C measures B; therefore, as C is to B, so is the unit to E.

But it was also proved that, as A is to C, so is D to the unit; therefore, ex aequali, as A is to B, so is the number D to E. {V. 22} Therefore the commensurable magnitudes A, B have to one another the ratio which the number D has to the number E. Q. E. D.

PROPOSITION 6. If two magnitudes have to one another the ratio which a number has to a number, the magnitudes will be commensurable.

For let the two magnitudes A, B have to one another the ratio which the number D has to the number E; I say that the magnitudes A, B are commensurable.

For let A be divided into as many equal parts as there are units in D, and let C be equal to one of them; and let F be made up of as many magnitudes equal to C as there are units in E.

Since then there are in A as many magnitudes equal to C as there are units in D, whatever part the unit is of D, the same

part is C of A also; therefore, as C is to A, so is the unit to D. {VII. Def. 20} But the unit measures the number D; therefore C also measures A.

And since, as C is to A, so is the unit to D, therefore, inversely, as A is to C, so is the number D to the unit. {cf. V. 7, Por.} Again, since there are in F as many magnitudes equal to C as there are units in E, therefore, as C is to F, so is the unit to E. {VII. Def. 20} But it was also proved that, as A is to C, so is D to the unit; therefore, ex aequali, as A is to F, so is D to E. {v. 22} But, as D is to E, so is A to B; therefore also, as A is to B, so is it to F also. {V. 11} Therefore A has the same ratio to each of the magnitudes B, F; therefore B is equal to F. {V. 9} But C measures F; therefore it measures B also.

Further it measures A also; therefore C measures A, B.

Therefore A is commensurable with B.

Therefore etc.

PORISM.

From this it is manifest that, if there be two numbers, as D, E, and a straight line, as A, it is possible to make a straight line {F} such that the given straight line is to it as the number D is to the number E.

And, if a mean proportional be also taken between A, F, as B, as A is to F, so will the square on A be to the square on B, that is, as the first is to the third, so is the figure on the first to that which is similar and similarly described on the second. {VI. 19, Por.} But, as A is to F, so is the number D to the number E; therefore it has been contrived that, as the number D is to the number E, so also is the figure on the straight line A to the figure on the straight line B. Q. E. D.

PROPOSITION 7. Incommensurable magnitudes have not to one another the ratio which a number has to a number.

Let A, B be incommensurable magnitudes; I say that A has not to B the ratio which a number has to a number.

For, if A has to B the ratio which a number has to a number, A will be commensurable with B. {X. 6} But it is not; therefore A has not to B the ratio which a number has to a number.

Therefore etc.

PROPOSITION 8. If two magnitudes have not to one another the ratio which a number has to a number, the magnitudes will be incommensurable.

For let the two magnitudes A, B not have to one another the ratio which a number has to a number; I say that the magnitudes A, B are incommensurable.

For, if they are commensurable, A will have to B the ratio which a number has to a number. {X. 5} But it has not; therefore the magnitudes A, B are incommensurable.

Therefore etc.

PROPOSITION 9. The squares on straight lines commensurable in length have to one another the ratio which a square number has to a square number; and squares which have to one another the ratio which a square number has to a square number will also have their sides commensurable in length. But the squares on straight lines incommensurable in length have not to one another the ratio which a square number has to a square number; and squares which have not to one another the ratio which a square number has to a square number will not have their sides commensurable in length either.

For let A, B be commensurable in length; I say that the square on A has to the square on B the ratio which a square number has to a square number.

For, since A is commensurable in length with B, therefore A has to B the ratio which a number has to a number. {X. 5} Let it have to it the ratio which C has to D.

Since then, as A is to B, so is C to D, while the ratio of the square on A to the square on B is duplicate of the ratio of A to B, for similar figures are in the duplicate ratio of their corresponding sides; {VI. 20, Por.} and the ratio of the square on C to the square on D is duplicate of the ratio of C to D, for between two square numbers there is one mean proportional number, and the square number has to the square number the ratio duplicate of that which the side has to the side; {VIII. 11} therefore also, as the square on A is to the square on B, so is the square on C to the square on D.

Next, as the square on A is to the square on B, so let the square on C be to the square on D; I say that A is commensurable in length with B.

For since, as the square on A is to the square on B, so is the square on C to the square on D, while the ratio of the square on A to the square on B is duplicate of the ratio of A to B, and the ratio of the square on C to the square on D is duplicate of the ratio of C to D, therefore also, as A is to B, so is C to D.

Therefore A has to B the ratio which the number C has to the number D; therefore A is commensurable in length with B. {X. 6} Next, let A be incommensurable in length with B; I say that the square on A has not to the square on B the ratio which a square number has to a square number.

For, if the square on A has to the square on B the ratio which a square number has to a square number, A will be commensurable with B.

But it is not; therefore the square on A has not to the square on B the ratio which a square number has to a square number.

Again, let the square on A not have to the square on B the ratio which a square number has to a square number; I say that A is incommensurable in length with B.

For, if A is commensurable with B, the square on A will have to the square on B the ratio which a square number has to a square number.

But it has not; therefore A is not commensurable in length with B.

Therefore etc.

PORISM.

And it is manifest from what has been proved that straight lines commensurable in length are always commensurable in square also, but those commensurable in square are not always commensurable in length also.

LEMMA.

{It has been proved in the arithmetical books that similar plane numbers have to one another the ratio which a square number has to a square number, {VIII. 26} and that, if two numbers have to one another the ratio which a square number has to a square number, they are similar plane numbers. {Converse of VIII. 26} And it is manifest from these propositions that numbers which are not similar plane numbers, that is, those which have not their sides proportional, have not to one another the ratio which a square number has to a square number.

For, if they have, they will be similar plane numbers: which is contrary to the hypothesis.

Therefore numbers which are not similar plane numbers have not to one another the ratio which a square number has to a square number.} {PROPOSITION 10.

To find two straight lines incommensurable, the one in length only, and the other in square also, with an assigned straight line.

Let A be the assigned straight line; thus it is required to find two straight lines incommensurable, the one in length only, and the other in square also, with A .

Let two numbers B, C be set out which have not to one another the ratio which a square number has to a square number, that is, which are not similar plane numbers; and let it be contrived that, as B is to C , so is the square on A to the square on D

— for we have learnt how to do this — {X. 6, Por.} therefore the square on A is commensurable with the square on D . {X. 6} And, since B has not to C the ratio which a square number has to a square number, therefore neither has the square on A to the square on D the ratio which a square number has to a square number; therefore A is incommensurable in length with D . {X. 9} Let E be taken a mean proportional between A, D ; therefore, as A is to D , so is the square on A to the square on E . {V. Def. 9} But A is incommensurable in length with D ; therefore the square on A is also incommensurable with the square on E ; {X. 11} therefore A is incommensurable in square with E .

Therefore two straight lines D, E have been found incommensurable, D in length only, and E in square and of course in length also, with the assigned straight line A .
PROPOSITION 11. If four magnitudes be proportional, and the first be commensurable with the second, the third will also be commensurable with the fourth; and, if the first be incommensurable with the second, the third will also be incommensurable with the fourth.

Let A, B, C, D be four magnitudes in proportion, so that, as A is to B , so is C to D , and let A be commensurable with B ; I say that C will also be commensurable with D .

For, since A is commensurable with B, therefore A has to B the ratio which a number has to a number. {X. 5} And, as A is to B, so is C to D; therefore C also has to D the ratio which a number has to a number; therefore C is commensurable with D. {X. 6} Next, let A be incommensurable with B; I say that C will also be incommensurable with D.

For, since A is incommensurable with B, therefore A has not to B the ratio which a number has to a number. {X. 7} And, as A is to B, so is C to D; therefore neither has C to D the ratio which a number has to a number; therefore C is incommensurable with D. {X. 8} Therefore etc.

PROPOSITION 12. Magnitudes commensurable with the same magnitude are commensurable with one another also.

For let each of the magnitudes A, B be commensurable with C; I say that A is also commensurable with B.

For, since A is commensurable with C, therefore A has to C the ratio which a number has to a number. {X. 5} Let it have the ratio which D has to E.

Again, since C is commensurable with B, therefore C has to B the ratio which a number has to a number. {X. 5} Let it have the ratio which F has to G.

And, given any number of ratios we please, namely the ratio which D has to E and that which F has to G, let the numbers H, K, L be taken continuously in the given ratios; {cf. VIII. 4} so that, as D is to E, so is H to K, and, as F is to G, so is K to L.

Since, then, as A is to C, so is D to E, while, as D is to E, so is H to K, therefore also, as A is to C, so is H to K. {V. 11} Again, since, as C is to B, so is F to G, while, as F is to G, so is K to L, therefore also, as C is to B, so is K to L. {V. 11} But also, as A is to C, so is H to K; therefore, ex aequali, as A is to B, so is H to L. {V. 22} Therefore A has to B the ratio which a number has to a number;

therefore A is commensurable with B. {X. 6} Therefore etc. Q. E. D.

PROPOSITION 13. If two magnitudes be commensurable, and the one of them be incommensurable with any magnitude, the remaining one will also be incommensurable with the same.

Let A, B be two commensurable magnitudes, and let one of them, A, be incommensurable with any other magnitude C; I say that the remaining one, B, will also be incommensurable with C.

For, if B is commensurable with C, while A is also commensurable with B, A is also commensurable with C. {X. 12} But it is also incommensurable with it: which is impossible.

Therefore B is not commensurable with C; therefore it is incommensurable with it.

Therefore etc.

LEMMA.

Given two unequal straight lines, to find by what square the square on the greater is greater than the square on the less.

Let AB, C be the given two unequal straight lines, and let AB be the greater of them; thus it is required to find by what square the square on AB is greater than the square on C.

Let the semicircle ADB be described on AB, and let AD be fitted into it equal to C; {IV. 1} let DB be joined.

It is then manifest that the angle ADB is right, {III. 31} and that the square on AB is greater than the square on AD, that is, C, by the square on DB. {I. 47} Similarly also, if two straight lines be given, the straight line the square on which is equal to the sum of the squares on them is found in this manner.

Let AD, DB be the given two straight lines, and let it be required to find the straight line the square on which is equal to the sum

of the squares on them.

Let them be placed so as to contain a right angle, that formed by AD, DB; and let AB be joined.

It is again manifest that the straight line the square on which is equal to the sum of the squares on AD, DB is AB. {I. 47} Q. E. D.

PROPOSITION 14. If four straight lines be proportional, and the square on the first be greater than the square on the second by the square on a straight line commensurable with the first, the square on the third will also be greater than the square on the fourth by the square on a straight line commensurable with the third.

And, if the square on the first be greater than the square on the second by the square on a straight line incommensurable with the first, the square on the third will also be greater than the square on the fourth by the square on a straight line incommensurable with the third.

Let A, B, C, D be four straight lines in proportion, so that, as A is to B, so is C to D; and let the square on A be greater than the square on B by the square on E, and let the square on C be greater than the square on D by the square on F; I say that, if A is commensurable with E, C is also commensurable with F, and, if A is incommensurable with E, C is also incommensurable with F.

For since, as A is to B, so is C to D, therefore also, as the square on A is to the square on B, so is the square on C to the square on D. {VI. 22} But the squares on E, B are equal to the square on A, and the squares on D, F are equal to the square on C.

Therefore, as the squares on E, B are to the square on B, so are the squares on D, F to the square on D; therefore, separando, as the square on E is to the square on B, so is the square on F to the square on D; {V. 17} therefore also, as E is to B, so is F to D; {VI. 22} therefore, inversely, as B is to E, so is D to F.

But, as A is to B, so also is C to D; therefore, *ex aequali*, as A is to E, so is C to F. {V. 22} Therefore, if A is commensurable with E, C is also commensurable with F, and, if A is incommensurable with E, C is also incommensurable with F. {X. 11} Therefore etc.

PROPOSITION 15. If two commensurable magnitudes be added together, the whole will also be commensurable with each of them; and, if the whole be commensurable with one of them, the original magnitudes will also be commensurable.

For let the two commensurable magnitudes AB, BC be added together; I say that the whole AC is also commensurable with each of the magnitudes AB, BC.

For, since AB, BC are commensurable, some magnitude will measure them.

Let it measure them, and let it be D.

Since then D measures AB, BC, it will also measure the whole AC.

But it measures AB, BC also; therefore D measures AB, BC, AC; therefore AC is commensurable with each of the magnitudes AB, BC. {X. Def. 1} Next, let AC be commensurable with AB; I say that AB, BC are also commensurable.

For, since AC, AB are commensurable, some magnitude will measure them.

Let it measure them, and let it be D.

Since then D measures CA, AB, it will also measure the remainder BC.

But it measures AB also; therefore D will measure AB, BC; therefore AB, BC are commensurable. {X. Def. 1} Therefore etc.

PROPOSITION 16. If two incommensurable magnitudes be added together, the whole will also be incommensurable with each of them; and, if the whole be incommensurable with one of

them, the original magnitudes will also be incommensurable.

For let the two incommensurable magnitudes AB, BC be added together; I say that the whole AC is also incommensurable with each of the magnitudes AB, BC.

For, if CA, AB are not incommensurable, some magnitude will measure them.

Let it measure them, if possible, and let it be D.

Since then D measures CA, AB, therefore it will also measure the remainder BC.

But it measures AB also; therefore D measures AB, BC.

Therefore AB, BC are commensurable; but they were also, by hypothesis, incommensurable: which is impossible.

Therefore no magnitude will measure CA, AB; therefore CA, AB are incommensurable. {X. Def. 1} Similarly we can prove that AC, CB are also incommensurable.

Therefore AC is incommensurable with each of the magnitudes AB, BC.

Next, let AC be incommensurable with one of the magnitudes AB, BC.

First, let it be incommensurable with AB; I say that AB, BC are also incommensurable.

For, if they are commensurable, some magnitude will measure them.

Let it measure them, and let it be D.

Since then D measures AB, BC. therefore it will also measure the whole AC.

But it measures AB also; therefore D measures CA, AB.

Therefore CA, AB are commensurable; but they were also, by hypothesis, incommensurable: which is impossible.

Therefore no magnitude will measure AB, BC; therefore AB, BC are incommensurable. {X. Def. 1} Therefore etc.

LEMMA.

If to any straight line there be applied a parallelogram deficient by a square figure, the applied parallelogram is equal to the rectangle contained by the segments of the straight line resulting from the application.

For let there be applied to the straight line AB the parallelogram AD deficient by the square figure DB; I say that AD is equal to the rectangle contained by AC, CB.

This is indeed at once manifest; for, since DB is a square, DC is equal to CB; and AD is the rectangle AC, CD, that is, the rectangle AC, CB.

Therefore etc.

PROPOSITION 17. If there be two unequal straight lines, and to the greater there be applied a parallelogram equal to the fourth part of the square on the less and deficient by a square figure, and if it divide it into parts which are commensurable in length, then the square on the greater will be greater than the square on the less by the square on a straight line commensurable with the greater.

And, if the square on the greater be greater than the square on the less by the square on a straight line commensurable with the greater, and if there be applied to the greater a parallelogram equal to the fourth part of the square on the less and deficient by a square figure, it will divide it into parts which are commensurable in length.

Let A , BC be two unequal straight lines, of which BC is the greater, and let there be applied to BC a parallelogram equal to the fourth part of the square on the less, A , that is, equal to the square on the half of A , and deficient by a square figure. Let this be the rectangle BD , DC , {cf. Lemma} and let BD be commensurable in length with DC ; I say that the square on BC is greater than the square on A by the square on a straight line commensurable with BC .

For let BC be bisected at the point E , and let EF be made equal to DE .

Therefore the remainder DC is equal to BF .

And, since the straight line BC has been cut into equal parts at E , and into unequal parts at D , therefore the rectangle contained by BD , DC , together with the square on ED , is equal to the square on EC ; {II. 5} And the same is true of their quadruples; therefore four times the rectangle BD , DC , together with four times the square on DE , is equal to four times the square on EC .

But the square on A is equal to four times the rectangle BD , DC ; and the square on DF is equal to four times the square on DE , for DF is double of DE .

And the square on BC is equal to four times the square on EC , for again BC is double of CE .

Therefore the squares on A , DF are equal to the square on BC , so that the square on BC is greater than the square on A by the square on DF .

It is to be proved that BC is also commensurable with DF .

Since BD is commensurable in length with DC , therefore BC is also commensurable in length with CD . {X. 15} But CD is commensurable in length with CD , BF , for CD is equal to BF . {X. 6} Therefore BC is also commensurable in length with BF , CD , {X. 12} so that BC is also commensurable in length with the

remainder FD; {X. 15} therefore the square on BC is greater than the square on A by the square on a straight line commensurable with BC.

Next, let the square on BC be greater than the square on A by the square on a straight line commensurable with BC, let a parallelogram be applied to BC equal to the fourth part of the square on A and deficient by a square figure, and let it be the rectangle BD, DC.

It is to be proved that BD is commensurable in length with DC.

With the same construction, we can prove similarly that the square on BC is greater than the square on A by the square on FD.

But the square on BC is greater than the square on A by the square on a straight line commensurable with BC.

Therefore BC is commensurable in length with FD, so that BC is also commensurable in length with the remainder, the sum of BF, DC. {X. 15} But the sum of BF, DC is commensurable with DC, {X. 6} so that BC is also commensurable in length with CD; {X. 12} and therefore, separando, BD is commensurable in length with DC. {X. 15} Therefore etc.

PROPOSITION 18. If there be two unequal straight lines, and to the greater there be applied a parallelogram equal to the fourth part of the square on the less and deficient by a square figure, and if it divide it into parts which are incommensurable, the square on the greater will be greater than the square on the less by the square on a straight line incommensurable with the greater.

And, if the square on the greater be greater than the square on the less by the square on a straight line incommensurable with the greater, and if there be applied to the greater a parallelogram equal to the fourth part of the square on the less

and deficient by a square figure, it divides it into parts which are incommensurable.

Let A , BC be two unequal straight lines, of which BC is the greater, and to BC let there be applied a parallelogram equal to the fourth part of the square on the less, A , and deficient by a square figure. Let this be the rectangle BD , DC , {cf. Lemma before X. 17} and let BD be incommensurable in length with DC ; I say that the square on BC is greater than the square on A by the square on a straight line incommensurable with BC .

For, with the same construction as before, we can prove similarly that the square on BC is greater than the square on A by the square on FD .

It is to be proved that BC is incommensurable in length with DF .

Since BD is incommensurable in length with DC , therefore BC is also incommensurable in length with CD . {X. 16} But DC is commensurable with the sum of BF , DC ; {X. 6} therefore BC is also incommensurable with the sum of BF , DC ; {X. 13} so that BC is also incommensurable in length with the remainder FD . {X. 16} And the square on BC is greater than the square on A by the square on FD ; therefore the square on BC is greater than the square on A by the square on a straight line incommensurable with BC .

Again, let the square on BC be greater than the square on A by the square on a straight line incommensurable with BC , and let there be applied to BC a parallelogram equal to the fourth part of the square on A and deficient by a square figure. Let this be the rectangle BD , DC .

It is to be proved that BD is incommensurable in length with DC .

For, with the same construction, we can prove similarly that the square on BC is greater than the square on A by the square on FD .

But the square on BC is greater than the square on A by the square on a straight line incommensurable with BC; therefore BC is incommensurable in length with FD. so that BC is also incommensurable with the remainder, the sum of BF, DC. {X. 16} But the sum of BF, DC is commensurable in length with DC; {X. 6} therefore BC is also incommensurable in length with DC, {X. 13} so that, separando, BD is also incommensurable in length with DC. {X. 16} Therefore etc.

LEMMA.

{Since it has been proved that straight lines commensurable in length are always commensurable in square also, while those commensurable in square are not always commensurable in length also, but can of course be either commensurable or incommensurable in length, it is manifest that, if any straight line be commensurable in length with a given rational straight line, it is called rational and commensurable with the other not only in length but in square also, since straight lines commensurable in length are always commensurable in square also.

But, if any straight line be commensurable in square with a given rational straight line, then, if it is also commensurable in length with it, it is called in this case also rational and commensurable with it both in length and in square; but, if again any straight line, being commensurable in square with a given rational straight line, be incommensurable in length with it, it is called in this case also rational but commensurable in square only.} **PROPOSITION 19.** The rectangle contained by rational straight lines commensurable in length is rational.

For let the rectangle AC be contained by the rational straight lines AB, BC commensurable in length; I say that AC is rational.

For on AB let the square AD be described; therefore AD is rational. {X. Def. 4} And, since AB is commensurable in length

with BC, while AB is equal to BD, therefore BD is commensurable in length with BC.

And, as BD is to BC, so is DA to AC. {VI. 1} Therefore DA is commensurable with AC. {X. 11} But DA is rational; therefore AC is also rational. {X. Def. 4} Therefore etc.

PROPOSITION 20. If a rational area be applied to a rational straight line, it produces as breadth a straight line rational and commensurable in length with the straight line to which it is applied.

For let the rational area AC be applied to AB, a straight line once more rational in any of the aforesaid ways, producing BC as breadth; I say that BC is rational and commensurable in length with BA. For on AB let the square AD be described; therefore AD is rational. {X. Def. 4} But AC is also rational; therefore DA is commensurable with AC.

And, as DA is to AC, so is DB to BC. {VI. 1} Therefore DB is also commensurable with BC; {X. 11} and DB is equal to BA; therefore AB is also commensurable with BC.

But AB is rational; therefore BC is also rational and commensurable in length with AB.

Therefore etc.

PROPOSITION 21. The rectangle contained by rational straight lines commensurable in square only is irrational, and the side of the square equal to it is irrational. Let the latter be called medial.

For let the rectangle AC be contained by the rational straight lines AB, BC commensurable in square only; I say that AC is irrational, and the side of the square equal to it is irrational; and let the latter be called medial.

For on AB let the square AD be described; therefore AD is rational. {X. Def. 4} And, since AB is incommensurable in length with BC, for by hypothesis they are commensurable in square only, while AB is equal to BD, therefore DB is also incommensurable in length with BC.

And, as DB is to BC, so is AD to AC; {VI. 1} therefore DA is incommensurable with AC. {X. 11} But DA is rational; therefore AC is irrational, so that the side of the square equal to AC is also irrational. {X. Def. 4} And let the latter be called medial. Q. E. D.

LEMMA.

If there be two straight lines, then, as the first is to the second, so is the square on the first to the rectangle contained by the two straight lines.

Let FE, EG be two straight lines.

I say that, as FE is to EG, so is the square on FE to the rectangle FE, EG.

For on FE let the square DF be described, and let GD be completed.

Since then, as FE is to EG, so is FD to DG, {VI. 1} and FD is the square on FE, and DG the rectangle DE, EG, that is, the rectangle FE, EG, therefore, as FE is to EG, so is the square on FE to the rectangle FE, EG.

Similarly also, as the rectangle GE, EF is to the square on EF, that is, as GD is to FD, so is GE to EF. Q. E. D.

PROPOSITION 22. The square on a medial straight line, if applied to a rational straight line, produces as breadth a straight line rational and incommensurable in length with that to which it is applied.

Let A be medial and CB rational, and let a rectangular area BD equal to the square on A be applied to BC, producing CD as

breadth; I say that CD is rational and incommensurable in length with CB.

For, since A is medial, the square on it is equal to a rectangular area contained by rational straight lines commensurable in square only. {X. 21} Let the square on it be equal to GF.

But the square on it is also equal to BD; therefore BD is equal to GF.

But it is also equiangular with it; and in equal and equiangular parallelograms the sides about the equal angles are reciprocally proportional; {VI. 14} therefore, proportionally, as BC is to EG, so is EF to CD.

Therefore also, as the square on BC is to the square on EG, so is the square on EF to the square on CD. {VI. 22} But the square on CB is commensurable with the square on EG, for each of these straight lines is rational; therefore the square on EF is also commensurable with the square on CD. {X. 11} But the square on EF is rational; therefore the square on CD is also rational; {X. Def. 4} therefore CD is rational.

And, since EF is incommensurable in length with EG, for they are commensurable in square only, and, as EF is to EG, so is the square on EF to the rectangle FE, EG, {Lemma} therefore the square on EF is incommensurable with the rectangle FE, EG. {X. 11} But the square on CD is commensurable with the square on EF, for the straight lines are rational in square; and the rectangle DC, CB is commensurable with the rectangle FE, EG, for they are equal to the square on A; therefore the square on CD is also incommensurable with the rectangle DC, CB. {X. 13} But, as the square on CD is to the rectangle DC, CB, so is DC to CB; {Lemma} therefore DC is incommensurable in length with CB. {X. 11} Therefore CD is rational and incommensurable in length with CB. Q. E. D.

PROPOSITION 23. A straight line commensurable with a medial straight line is medial.

Let A be medial, and let B be commensurable with A; I say that B is also medial.

For let a rational straight line CD be set out, and to CD let the rectangular area CE equal to the square on A be applied, producing ED as breadth; therefore ED is rational and incommensurable in length with CD. {X. 22} And let the rectangular area CF equal to the square on B be applied to CD, producing DF as breadth.

Since then A is commensurable with B, the square on A is also commensurable with the square on B.

But EC is equal to the square on A, and CF is equal to the square on B; therefore EC is commensurable with CF.

And, as EC is to CF, so is ED to DF; {VI. 1} therefore ED is commensurable in length with DF. {X. 11} But ED is rational and incommensurable in length with DC; therefore DF is also rational {X. Def. 3} and incommensurable in length with DC. {X. 13} Therefore CD, DF are rational and commensurable in square only.

But the straight line the square on which is equal to the rectangle contained by rational straight lines commensurable in square only is medial; {X. 21} therefore the side of the square equal to the rectangle CD, DF is medial.

And B is the side of the square equal to the rectangle CD, DF; therefore B is medial.

PORISM.

From this it is manifest that an area commensurable with a medial area is medial.

{And in the same way as was explained in the case of rationals {Lemma following X. 18} it follows, as regards medials, that a straight line commensurable in length with a medial straight line is called medial and commensurable with it not only in length but in square also, since, in general, straight lines commensurable in length are always commensurable in square also.

But, if any straight line be commensurable in square with a medial straight line, then, if it is also commensurable in length with it, the straight lines are called, in this case too, medial and commensurable in length and in square, but, if in square only, they are called medial straight lines commensurable in square only.} **PROPOSITION 24.** The rectangle contained by medial straight lines commensurable in length is medial.

For let the rectangle AC be contained by the medial straight lines AB, BC which are commensurable in length; I say that AC is medial.

For on AB let the square AD be described; therefore AD is medial.

And, since AB is commensurable in length with BC, while AB is equal to BD, therefore DB is also commensurable in length with BC; so that DA is also commensurable with AC. {VI. 1, X. 11} But DA is medial; therefore AC is also medial. {X. 23, Por.} Q. E. D.

PROPOSITION 25. The rectangle contained by medial straight lines commensurable in square only is either rational or medial.

For let the rectangle AC be contained by the medial straight lines AB, BC which are commensurable in square only; I say that AC is either rational or medial.

For on AB, BC let the squares AD, BE be described; therefore each of the squares AD, BE is medial.

Let a rational straight line FG be set out, to FG let there be applied the rectangular parallelogram GH equal to AD , producing FH as breadth, to HM let there be applied the rectangular parallelogram MK equal to AC , producing HK as breadth, and further to KN let there be similarly applied NL equal to BE , producing KL as breadth; therefore FH , HK , KL are in a straight line.

Since then each of the squares AD , BE is medial, and AD is equal to GH , and BE to NL , therefore each of the rectangles GH , NL is also medial.

And they are applied to the rational straight line FG ; therefore each of the straight lines FH , KL is rational and incommensurable in length with FG . {X. 22} And, since AD is commensurable with BE , therefore GH is also commensurable with NL .

And, as GH is to NL , so is FH to KL ; {VI. 1} therefore FH is commensurable in length with KL . {X. 11} Therefore FH , KL are rational straight lines commensurable in length; therefore the rectangle FH , KL is rational. {X. 19} And, since DB is equal to BA , and OB to BC , therefore, as DB is to BC , so is AB to BO .

But, as DB is to BC , so is DA to AC , {VI. 1} and, as AB is to BO , so is AC to CO ; {id.} therefore, as DA is to AC , so is AC to CO .

But AD is equal to GH , AC to MK and CO to NL ; therefore, as GH is to MK , so is MK to NL ; therefore also, as FH is to HK , so is HK to KL ; {VI. 1, V. 11} therefore the rectangle FH , KL is equal to the square on HK . {VI. 17} But the rectangle FH , KL is rational; therefore the square on HK is also rational.

Therefore HK is rational.

And, if it is commensurable in length with FG , HN is rational; {X. 19} but, if it is incommensurable in length with FG , KH , HM are rational straight lines commensurable in square only, and

therefore HN is medial. {X. 21} Therefore HN is either rational or medial.

But HN is equal to AC; therefore AC is either rational or medial.

Therefore etc.

PROPOSITION 26. 4 medial area does not exceed a medial area by a rational area.

For, if possible, let the medial area AB exceed the medial area AC by the rational area DB, and let a rational straight line EF be set out; to EF let there be applied the rectangular parallelogram FH equal to AB, producing EH as breadth, and let the rectangle FG equal to AC be subtracted; therefore the remainder BD is equal to the remainder KH.

But DB is rational; therefore KH is also rational.

Since, then, each of the rectangles AB, AC is medial, and AB is equal to FH, and AC to FG, therefore each of the rectangles FH, FG is also medial.

And they are applied to the rational straight line EF; therefore each of the straight lines HE, EG is rational and incommensurable in length with EF. {X. 22} And, since {DB is rational and is equal to KH, therefore} KH is {also} rational; and it is applied to the rational straight line EF; therefore GH is rational and commensurable in length with EF. {X. 20} But EG is also rational, and is incommensurable in length with EF; therefore EG is incommensurable in length with GH. {X. 13} And, as EG is to GH, so is the square on EG to the rectangle EG, GH; therefore the square on EG is incommensurable with the rectangle EG, GH. {X. 11} But the squares on EG, GH are commensurable with the square on EG, for both are rational; and twice the rectangle EG, GH is commensurable with the rectangle EG, GH, for it is double of it; {X. 6} therefore the squares on EG, GH are incommensurable with twice the

rectangle EG, GH; {X. 13} therefore also the sum of the squares on EG, GH and twice the rectangle EG, GH, that is, the square on EH {II. 4}, is incommensurable with the squares on EG, GH. {X. 16} But the squares on EG, GH are rational; therefore the square on EH is irrational. {X. Def. 4} Therefore EH is irrational.

But it is also rational: which is impossible.

Therefore etc. Q. E. D.

PROPOSITION 27. To find medial straight lines commensurable in square only which contain a rational rectangle.

Let two rational straight lines A, B commensurable in square only be set out; let C be taken a mean proportional between A, B, {VI. 13} and let it be contrived that, as A is to B, so is C to D. {VI. 12} Then, since A, B are rational and commensurable in square only, the rectangle A, B, that is, the square on C {VI.17}, is medial. {X. 21} Therefore C is medial. {X. 21} And since, as A is to B, so is C to D, and A, B are commensurable in square only, therefore C, D are also commensurable in square only. {X. 11} And C is medial; therefore D is also medial. {X. 23, addition} Therefore C, D are medial and commensurable in square only.

I say that they also contain a rational rectangle.

For since, as A is to B, so is C to D, therefore, alternately, as A is to C, so is B to D. {V. 16} But, as A is to C, so is C to B; therefore also, as C is to B, so is B to D; therefore the rectangle C, D is equal to the square on B.

But the square on B is rational; therefore the rectangle C, D is also rational.

Therefore medial straight lines commensurable in square only have been found which contain a rational rectangle. Q. E. D.

PROPOSITION 28. To find medial straight lines commensurable in square only which contain a medial rectangle.

Let the rational straight lines A, B, C commensurable in square only be set out; let D be taken a mean proportional between A, B, {VI. 13} and let it be contrived that, as B is to C, so is D to E. {VI. 12} Since A, B are rational straight lines commensurable in square only, therefore the rectangle A, B, that is, the square on D {VI. 17}, is medial. {X. 21} Therefore D is medial. {X. 21} And since B, C are commensurable in square only, and, as B is to C, so is D to E, therefore D, E are also commensurable in square only. {X. 11} But D is medial; therefore E is also medial. {X. 23, addition} Therefore D, E are medial straight lines commensurable in square only.

I say next that they also contain a medial rectangle.

For since, as B is to C, so is D to E, therefore, alternately, as B is to D, so is C to E. {V. 16} But, as B is to D, so is D to A; therefore also, as D is to A, so is C to E; therefore the rectangle A, C is equal to the rectangle D, E. {VI. 16} But the rectangle A, C is medial; {X. 21} therefore the rectangle D, E is also medial.

Therefore medial straight lines commensurable in square only have been found which contain a medial rectangle. Q. E. D.

LEMMA I. To find two square numbers such that their sum is also square.

Let two numbers AB, BC be set out, and let them be either both even or both odd.

Then since, whether an even number is subtracted from an even number, or an odd number from an odd number, the remainder is even, {IX. 24, 26} therefore the remainder AC is even.

Let AC be bisected at D.

Let AB, BC also be either similar plane numbers, or square numbers, which are themselves also similar plane numbers.

Now the product of AB, BC together with the square on CD is equal to the square on BD. {II. 6} And the product of AB, BC is square, inasmuch as it was proved that, if two similar plane numbers by multiplying one another make some number the product is square. {IX. 1} Therefore two square numbers, the product of AB, BC, and the square on CD, have been found which, when added together, make the square on BD.

And it is manifest that two square numbers, the square on BD and the square on CD, have again been found such that their difference, the product of AB, BC, is a square, whenever AB, BC are similar plane numbers.

But when they are not similar plane numbers, two square numbers, the square on BD and the square on DC, have been found such that their difference, the product of AB, BC, is not square. Q. E. D.

LEMMA 2. To find two square numbers such that their sum is not square.

For let the product of AB, BC, as we said, be square, and CA even, and let CA be bisected by D.

It is then manifest that the square product of AB, BC together with the square on CD is equal to the square on BD. {See Lemma 1} Let the unit DE be subtracted; therefore the product of AB, BC together with the square on CE is less than the square on BD.

I say then that the square product of AB, BC together with the square on CE will not be square.

For, if it is square, it is either equal to the square on BE, or less than the square on BE, but cannot any more be greater, lest the unit be divided.

First, if possible, let the product of AB, BC together with the square on CE be equal to the square on BE, and let GA be double of the unit DE.

Since then the whole AC is double of the whole CD, and in them AG is double of DE, therefore the remainder GC is also double of the remainder EC; therefore GC is bisected by E.

Therefore the product of GB, BC together with the square on CE is equal to the square on BE. {II. 6} But the product of AB, BC together with the square on CE is also, by hypothesis, equal to the square on BE; therefore the product of GB, BC together with the square on CE is equal to the product of AB, BC together with the square on CE.

And, if the common square on CE be subtracted, it follows that AB is equal to GB: which is absurd.

Therefore the product of AB, BC together with the square on CE is not equal to the square on BE.

I say next that neither is it less than the square on BE.

For, if possible, let it be equal to the square on BF, and let HA be double of DF.

Now it will again follow that HC is double of CF; so that CH has also been bisected at F, and for this reason the product of HB, BC together with the square on FC is equal to the square on BF. {II. 6} But, by hypothesis, the product of AB, BC together with the square on CE is also equal to the square on BF.

Thus the product of HB, BC together with the square on CF will also be equal to the product of AB, BC together with the square on CE: which is absurd.

Therefore the product of AB, BC together with the square on CE is not less than the square on BE.

And it was proved that neither is it equal to the square on BE.

Therefore the product of AB, BC together with the square on CE is not square. Q. E. D.

PROPOSITION 29. To find two rational straight lines commensurable in square only and such that the square on the greater is greater than the square on the less by the square on a straight line commensurable in length with the greater.

For let there be set out any rational straight line AB, and two square numbers CD, DE such that their difference CE is not square; {Lemma 1} let there be described on AB the semicircle AFB, and let it be contrived that, as DC is to CE, so is the square on BA to the square on AF. {X. 6, Por.} Let FB be joined.

Since, as the square on BA is to the square on AF, so is DC to CE, therefore the square on BA has to the square on AF the ratio which the number DC has to the number CE; therefore the square on BA is commensurable with the square on AF. {X. 6} But the square on AB is rational; {X. Def. 4} therefore the square on AF is also rational; {id.} therefore AF is also rational.

And, since DC has not to CE the ratio which a square number has to a square number, neither has the square on BA to the square on AF the ratio which a square number has to a square number; therefore AB is incommensurable in length with AF. {X. 9} Therefore BA, AF are rational straight lines commensurable in square only.

And since, as DC is to CE, so is the square on BA to the square on AF, therefore, convertendo, as CD is to DE, so is the square on AB to the square on BF. {V. 19, Por., III. 31, I. 47} But CD has to DE the ratio which a square number has to a square number: therefore also the square on AB has to the square on BF the ratio which a square number has to a square number; therefore AB is commensurable in length with BF. {X. 9} And the square on AB is equal to the squares on AF, FB; therefore the square on AB is greater than the square on AF by the square on BF commensurable with AB.

Therefore there have been found two rational straight lines BA, AF commensurable in square only and such that the square on the greater AB is greater than the square on the less AF by the square on BF commensurable in length with AB.

PROPOSITION 30. To find two rational straight lines commensurable in square only and such that the square on the greater is greater than the square on the less by the square on a straight line incommensurable in length with the greater.

Let there be set out a rational straight line AB, and two square numbers CE, ED such that their sum CD is not square; {Lemma 2} let there be described on AB the semicircle AFB, let it be contrived that, as DC is to CE, so is the square on BA to the square on AF, {X. 6, Por.} and let FB be joined.

Then, in a similar manner to the preceding, we can prove that BA, AF are rational straight lines commensurable in square only.

And since, as DC is to CE, so is the square on BA to the square on AF, therefore, convertendo, as CD is to DE, so is the square on AB to the square on BF. {V. 19, Por., III. 31, I. 47} But CD has not to DE the ratio which a square number has to a square number; therefore neither has the square on AB to the square on BF the ratio which a square number has to a square number; therefore AB is incommensurable in length with BF. {X. 9} And the square on AB is greater than the square on AF by the square on FB incommensurable with AB.

Therefore AB, AF are rational straight lines commensurable in square only, and the square on AB is greater than the square on AF by the square on FB incommensurable in length with AB. Q. E. D.

PROPOSITION 31. To find two medial straight lines commensurable in square only, containing a rational rectangle,

and such that the square on the greater is greater than the square on the less by the square on a straight line commensurable in length with the greater.

Let there be set out two rational straight lines A , B commensurable in square only and such that the square on A , being the greater, is greater than the square on B the less by the square on a straight line commensurable in length with A . {X. 29} And let the square on C be equal to the rectangle A , B .

Now the rectangle A , B is medial; {X. 21} therefore the square on C is also medial; therefore C is also medial. {X. 21} Let the rectangle C , D be equal to the square on B .

Now the square on B is rational; therefore the rectangle C , D is also rational.

And since, as A is to B , so is the rectangle A , B to the square on B , while the square on C is equal to the rectangle A , B , and the rectangle C , D is equal to the square on B , therefore, as A is to B , so is the square on C to the rectangle C , D .

But, as the square on C is to the rectangle C , D , so is C to D ; therefore also, as A is to B , so is C to D .

But A is commensurable with B in square only; therefore C is also commensurable with D in square only. {X. 11} And C is medial; therefore D is also medial. {X. 23, addition} And since, as A is to B , so is C to D , and the square on A is greater than the square on B by the square on a straight line commensurable with A , therefore also the square on C is greater than the square on D by the square on a straight line commensurable with C . {X. 14} Therefore two medial straight lines C , D , commensurable in square only and containing a rational rectangle, have been found, and the square on C is greater than the square on D by the square on a straight line commensurable in length with C .

Similarly also it can be proved that the square on C exceeds the square on D by the square on a straight line incommensurable with C, when the square on A is greater than the square on B by the square on a straight line incommensurable with A. {X. 30} **PROPOSITION 32.** To find two medial straight lines commensurable in square only, containing a medial rectangle, and such that the square on the greater is greater than the square on the less by the square on a straight line commensurable with the greater.

Let there be set out three rational straight lines A, B, C commensurable in square only, and such that the square on A is greater than the square on C by the square on a straight line commensurable with A, {X. 29} and let the square on D be equal to the rectangle A, B.

Therefore the square on D is medial; therefore D is also medial. {X. 21} Let the rectangle D, E be equal to the rectangle B, C.

Then since, as the rectangle A, B is to the rectangle B, C, so is A to C; while the square on D is equal to the rectangle A, B, and the rectangle D, E is equal to the rectangle B, C, therefore, as A is to C, so is the square on D to the rectangle D, E.

But, as the square on D is to the rectangle D, E, so is D to E; therefore also, as A is to C, so is D to E.

But A is commensurable with C in square only; therefore D is also commensurable with E in square only. {X. 11} But D is medial; therefore E is also medial. {X. 23, addition} And, since, as A is to C, so is D to E, while the square on A is greater than the square on C by the square on a straight line commensurable with A, therefore also the square on D will be greater than the square on E by the square on a straight line commensurable with D. {X. 14} I say next that the rectangle D, E is also medial.

For, since the rectangle B, C is equal to the rectangle D, E, while the rectangle B, C is medial, {X. 21} therefore the rectangle D, E is also medial.

Therefore two medial straight lines D, E, commensurable in square only, and containing a medial rectangle, have been found such that the square on the greater is greater than the square on the less by the square on a straight line commensurable with the greater.

Similarly again it can be proved that the square on D is greater than the square on E by the square on a straight line incommensurable with D, when the square on A is greater than the square on C by the square on a straight line incommensurable with A. {X. 30} LEMMA.

Let ABC be a right-angled triangle having the angle A right, and let the perpendicular AD be drawn; I say that the rectangle CB, BD is equal to the square on BA, the rectangle BC, CD equal to the square on CA, the rectangle BD, DC equal to the square on AD, and, further, the rectangle BC, AD equal to the rectangle BA, AC.

And first that the rectangle CB, BD is equal to the square on BA.

For, since in a right-angled triangle AD has been drawn from the right angle perpendicular to the base, therefore the triangles ABD, ADC are similar both to the whole ABC and to one another. {VI. 8} And since the triangle ABC is similar to the triangle ABD, therefore, as CB is to BA, so is BA to BD; {VI. 4} therefore the rectangle CB, BD is equal to the square on AB. {VI. 17} For the same reason the rectangle BC, CD is also equal to the square on AC.

And since, if in a right-angled triangle a perpendicular be drawn from the right angle to the base, the perpendicular so drawn is a mean proportional between the segments of the base, {VI. 8,

Por.} therefore, as BD is to DA, so is AD to DC; therefore the rectangle BD, DC is equal to the square on AD. {VI. 17} I say that the rectangle BC, AD is also equal to the rectangle BA, AC.

For since, as we said, ABC is similar to ABD, therefore, as BC is to CA, so is BA to AD. {VI. 4} Therefore the rectangle BC, AD is equal to the rectangle BA, AC. {VI. 16} Q. E. D.

PROPOSITION 33. To find two straight lines incommensurable in square which make the sum of the squares on them rational but the rectangle contained by them medial.

Let there be set out two rational straight lines AB, BC commensurable in square only and such that the square on the greater AB is greater than the square on the less BC by the square on a straight line incommensurable with AB, {X. 30} let BC be bisected at D, let there be applied to AB a parallelogram equal to the square on either of the straight lines BD, DC and deficient by a square figure, and let it be the rectangle AE, EB; {VI. 28} let the semicircle AFB be described on AB, let EF be drawn at right angles to AB, and let AF, FB be joined.

Then, since AB, BC are unequal straight lines, and the square on AB is greater than the square on BC by the square on a straight line incommensurable with AB, while there has been applied to AB a parallelogram equal to the fourth part of the square on BC, that is, to the square on half of it, and deficient by a square figure, making the rectangle AE, EB, therefore AE is incommensurable with EB. {X. 18} And, as AE is to EB, so is the rectangle BA, AE to the rectangle AB, BE, while the rectangle BA, AE is equal to the square on AF, and the rectangle AB, BE to the square on BF; therefore the square on AF is incommensurable with the square on BF; therefore AF, FB are incommensurable in square.

And, since AB is rational, therefore the square on AB is also rational; so that the sum of the squares on AF, FB is also

rational. {I. 47} And since, again, the rectangle AE, EB is equal to the square on EF, and, by hypothesis, the rectangle AE, EB is also equal to the square on BD, therefore FE is equal to BD; therefore BC is double of FE, so that the rectangle AB, BC is also commensurable with the rectangle AB, EF.

But the rectangle AB, BC is medial; {X. 21} therefore the rectangle AB, EF is also medial. {X. 23, Por.} But the rectangle AB, EF is equal to the rectangle AF, FB; {Lemma} therefore the rectangle AF, FB is also medial.

But it was also proved that the sum of the squares on these straight lines is rational.

Therefore two straight lines AF, FB incommensurable in square have been found which make the sum of the squares on them rational, but the rectangle contained by them medial. Q. E. D.

PROPOSITION 34. To find two straight lines incommensurable in square which make the sum of the squares on them medial but the rectangle contained by them rational.

Let there be set out two medial straight lines AB, BC, commensurable in square only, such that the rectangle which they contain is rational, and the square on AB is greater than the square on BC by the square on a straight line incommensurable with AB; {X. 31, ad fin.} let the semicircle ADB be described on AB, let BC be bisected at E, let there be applied to AB a parallelogram equal to the square on BE and deficient by a square figure, namely the rectangle AF, FB; {VI. 28} therefore AF is incommensurable in length with FB. {X. 18} Let FD be drawn from F at right angles to AB, and let AD, DB be joined.

Since AF is incommensurable in length with FB, therefore the rectangle BA, AF is also incommensurable with the rectangle AB, BF. {X. 11} But the rectangle BA, AF is equal to the square on AD, and the rectangle AB, BF to the square on DB; therefore the

square on AD is also incommensurable with the square on DB.

And, since the square on AB is medial, therefore the sum of the squares on AD, DB is also medial. {III. 31, I. 47} And, since BC is double of DF, therefore the rectangle AB, BC is also double of the rectangle AB, FD.

But the rectangle AB, BC is rational; therefore the rectangle AB, FD is also rational. {X. 6} But the rectangle AB, FD is equal to the rectangle AD, DB; {Lemma} so that the rectangle AD, DB is also rational.

Therefore two straight lines AD, DB incommensurable in square have been found which make the sum of the squares on them medial, but the rectangle contained by them rational. Q. E. D.

PROPOSITION 35. To find two straight lines incommensurable in square which make the sum of the squares on them medial and the rectangle contained by them medial and moreover incommensurable with the sum of the squares on them.

Let there be set out two medial straight lines AB, BC commensurable in square only, containing a medial rectangle, and such that the square on AB is greater than the square on BC by the square on a straight line incommensurable with AB; {X. 32, ad fin.} let the semicircle ADB be described on AB, and let the rest of the construction be as above.

Then, since AF is incommensurable in length with FB, {X. 18} AD is also incommensurable in square with DB. {X. 11} And, since the square on AB is medial, therefore the sum of the squares on AD, DB is also medial. {III. 31, I. 47} And, since the rectangle AF, FB is equal to the square on each of the straight lines BE, DF, therefore BE is equal to DF; therefore BC is double of FD, so that the rectangle AB, BC is also double of the rectangle AB, FD.

But the rectangle AB, BC is medial; therefore the rectangle AB, FD is also medial. {X. 32, Por.} And it is equal to the rectangle AD, DB; {Lemma after X. 32 } therefore the rectangle AD, DB is also medial.

And, since AB is incommensurable in length with BC, while CB is commensurable with BE, therefore AB is also incommensurable in length with BE, {X. 13 } so that the square on AB is also incommensurable with the rectangle AB, BE. {X. 11 } But the squares on AD, DB are equal to the square on AB, {I. 47 } and the rectangle AB, FD, that is, the rectangle AD, DB, is equal to the rectangle AB, BE; therefore the sum of the squares on AD, DB is incommensurable with the rectangle AD, DB.

Therefore two straight lines AD, DB incommensurable in square have been found which make the sum of the squares on them medial and the rectangle contained by them medial and moreover incommensurable with the sum of the squares on them. Q. E. D.

PROPOSITION 36. If two rational straight lines commensurable in square only be added together, the whole is irrational; and let it be called binomial.

For let two rational straight lines AB, BC commensurable in square only be added together; I say that the whole AC is irrational.

For, since AB is incommensurable in length with BC — for they are commensurable in square only — and, as AB is to BC, so is the rectangle AB, BC to the square on BC, therefore the rectangle AB, BC is incommensurable with the square on BC. {X. 11 } But twice the rectangle AB, BC is commensurable with the rectangle AB, BC {X. 6 }, and the squares on AB, BC are commensurable with the square on BC — for AB, BC are rational straight lines commensurable in square only — {X. 15 } therefore twice the rectangle AB, BC is incommensurable with

the squares on AB, BC. {X. 13 } And, componendo, twice the rectangle AB, BC together with the squares on AB, BC, that is, the square on AC {II. 4 }, is incommensurable with the sum of the squares on AB, BC. {X. 16 } But the sum of the squares on AB, BC is rational; therefore the square on AC is irrational, so that AC is also irrational. {X. Def. 4 } And let it be called binomial.

PROPOSITION 37. If two medial straight lines commensurable in square only and containing a rational rectangle be added together, the whole is irrational; and let it be called a first bimedial straight line.

For let two medial straight lines AB, BC commensurable in square only and containing a rational rectangle be added together; I say that the whole AC is irrational.

For, since AB is incommensurable in length with BC, therefore the squares on AB, BC are also incommensurable with twice the rectangle AB, BC; {cf. X. 36, ll. 9-20} and, componendo, the squares on AB, BC together with twice the rectangle AB, BC, that is, the square on AC {II. 4}, is incommensurable with the rectangle AB, BC. {X. 16 } But the rectangle AB, BC is rational, for, by hypothesis, AB, BC are straight lines containing a rational rectangle; therefore the square on AC is irrational; therefore AC is irrational. {X. Def. 4 } And let it be called a first bimedial straight line. Q. E. D.

PROPOSITION 38. If two medial straight lines commensurable in square only and containing a medial rectangle be added together, the whole is irrational; and let it be called a second bimedial straight line.

For let two medial straight lines AB, BC commensurable in square only and containing a medial rectangle be added together; I say that AC is irrational.

For let a rational straight line DE be set out, and let the parallelogram DF equal to the square on AC be applied to DE , producing DG as breadth. {I. 44 } Then, since the square on AC is equal to the squares on AB , BC and twice the rectangle AB , BC , {II. 4 } let EH , equal to the squares on AB , BC , be applied to DE ; therefore the remainder HF is equal to twice the rectangle AB , BC .

And, since each of the straight lines AB , BC is medial, therefore the squares on AB , BC are also medial.

But, by hypothesis, twice the rectangle AB , BC is also medial.

And EH is equal to the squares on AB , BC , while FH is equal to twice the rectangle AB , BC ; therefore each of the rectangle EH , HF is medial.

And they are applied to the rational straight line DE ; therefore each of the straight lines DH , HG is rational and incommensurable in length with DE . {X. 22 } Since then AB is incommensurable in length with BC , and, as AB is to BC , so is the square on AB to the rectangle AB , BC , therefore the square on AB is incommensurable with the rectangle AB , BC . {X. 11 } But the sum of the squares on AB , BC is commensurable with the square on AB , {X. 15 } and twice the rectangle AB , BC is commensurable with the rectangle AB , BC . {X. 6 } Therefore the sum of the squares on AB , BC is incommensurable with twice the rectangle AB , BC . {X. 13 } But EH is equal to the squares on AB , BC , and HF is equal to twice the rectangle AB , BC .

Therefore EH is incommensurable with HF , so that DH is also incommensurable in length with HG . {VI. 1 , X. 11 } Therefore DH , HG are rational straight lines commensurable in square only; so that DG is irrational. {X. 36 } But DE is rational; and the rectangle contained by an irrational and a rational straight line is irrational; {cf. X. 20 } therefore the area DF is irrational, and the side of the square equal to it is irrational. {X. Def. 4 } But AC

is the side of the square equal to DF; therefore AC is irrational.

And let it be called a second bimedial straight line. Q. E. D.

PROPOSITION 39. If two straight lines incommensurable in square which make the sum of the squares on them rational, but the rectangle contained by them medial, be added together, the whole straight line is irrational : and let it be called major.

For let two straight lines AB, BC incommensurable in square, and fulfilling the given conditions {X. 33 }, be added together; I say that AC is irrational.

For, since the rectangle AB, BC is medial, twice the rectangle AB, BC is also medial. {X. 6 and 23, Por.} But the sum of the squares on AB, BC is rational; therefore twice the rectangle AB, BC is incommensurable with the sum of the squares on AB, BC, so that the squares on AB, BC together with twice the rectangle AB, BC that is, the square on AC, is also incommensurable with the sum of the squares on AB, BC; {X. 16 } therefore the square on AC is irrational, so that AC is also irrational. {X. Def. 4 } And let it be called major. Q. E. D.

PROPOSITION 40. If two straight lines incommensurable in square which make the sum of the squares on them medial, but the rectangle contained by them rational, be added together, the whole straight line is irrational; and let it be called the side of a rational plus a medial area.

For let two straight lines AB, BC incommensurable in square, and fulfilling the given conditions {X. 34 }, be added together; I say that AC is irrational.

For, since the sum of the squares on AB, BC is medial, while twice the rectangle AB, BC is rational, therefore the sum of the squares on AB, BC is incommensurable with twice the rectangle AB, BC; so that the square on AC is also incommensurable with twice the rectangle AB, BC. {X. 16 } But twice the rectangle AB,

BC is rational; therefore the square on AC is irrational.

Therefore AC is irrational. {X. Def. 4 } And let it be called the side of a rational plus a medial area. Q. E. D.

PROPOSITION 41. If two straight lines incommensurable in square which make the sum of the squares on them medial, and the rectangle contained by them medial and also incommensurable with the sum of the squares on them, be added together, the whole straight line is irrational; and let it be called the side of the sum of two medial areas.

For let two straight lines AB, BC incommensurable in square and satisfying the given conditions {X. 35 } be added together; I say that AC is irrational.

Let a rational straight line DE be set out, and let there be applied to DE the rectangle DF equal to the squares on AB, BC, and the rectangle GH equal to twice the rectangle AB, BC; therefore the whole DH is equal to the square on AC. {II. 4 } Now, since the sum of the squares on AB, BC is medial, and is equal to DF, therefore DF is also medial.

And it is applied to the rational straight line DE; therefore DG is rational and incommensurable in length with DE. {X. 22 } For the same reason GK is also rational and incommensurable in length with GF, that is, DE.

And, since the squares on AB, BC are incommensurable with twice the rectangle AB, BC, DF is incommensurable with GH; so that DG is also incommensurable with GK. {VI. 1 , X. 11 } And they are rational; therefore DG, GK are rational straight lines commensurable in square only; therefore DK is irrational and what is called binomial. {X. 36 } But DE is rational; therefore DH is irrational, and the side of the square which is equal to it is irrational. {X. Def. 4 } But AC is the side of the square equal to HD; therefore AC is irrational.

And let it be called the side of the sum of two medial areas. Q. E. D.

LEMMA.

And that the aforesaid irrational straight lines are divided only in one way into the straight lines of which they are the sum and which produce the types in question, we will now prove after premising the following lemma.

Let the straight line AB be set out, let the whole be cut into unequal parts at each of the points C, D, and let AC be supposed greater than DB; I say that the squares on AC, CB are greater than the squares on AD, DB.

For let AB be bisected at E.

Then, since AC is greater than DB, let DC be subtracted from each; therefore the remainder AD is greater than the remainder CB.

But AE is equal to EB; therefore DE is less than EC; therefore the points C, D are not equidistant from the point of bisection.

And, since the rectangle AC, CB together with the square on EC is equal to the square on EB, {II. 5} and, further, the rectangle AD, DB together with the square on DE is equal to the square on EB, {id.} therefore the rectangle AC, CB together with the square on EC is equal to the rectangle AD, DB together with the square on DE.

And of these the square on DE is less than the square on EC; therefore the remainder, the rectangle AC, CB, is also less than the rectangle AD, DB, so that twice the rectangle AC, CB is also less than twice the rectangle AD, DB.

Therefore also the remainder, the sum of the squares on AC, CB, is greater than the sum of the squares on AD, DB. Q. E. D.

PROPOSITION 42. A binomial straight line is divided into its terms at one point only .

Let AB be a binomial straight line divided into its terms at C; therefore AC, CB are rational straight lines commensurable in square only. {X. 36 } I say that AB is not divided at another point into two rational straight lines commensurable in square only.

For, if possible, let it be divided at D also, so that AD, DB are also rational straight lines commensurable in square only.

It is then manifest that AC is not the same with DB.

For, if possible, let it be so.

Then AD will also be the same as CB, and, as AC is to CB, so will BD be to DA; thus AB will be divided at D also in the same way as by the division at C: which is contrary to the hypothesis.

Therefore AC is not the same with DB.

For this reason also the points C, D are not equidistant from the point of bisection.

Therefore that by which the squares on AC, CB differ from the squares on AD, DB is also that by which twice the rectangle AD, DB differs from twice the rectangle AC, CB, because both the squares on AC, CB together with twice the rectangle AC, CB, and the squares on AD, DB together with twice the rectangle AD, DB, are equal to the square on AB. {II. 4 } But the squares on AC, CB differ from the squares on AD, DB by a rational area, for both are rational; therefore twice the rectangle AD, DB also differs from twice the rectangle AC, CB by a rational area, though they are medial {X. 21 }; which is absurd, for a medial area does not exceed a medial by a rational area. {x. 26 } Therefore a binomial straight line is not divided at different points; therefore it is divided at one point only. Q. E. D.

PROPOSITION 43. A first bimedial straight line is divided at one point only.

Let AB be a first bimedial straight line divided at C, so that AC, CB are medial straight lines commensurable in square only and containing a rational rectangle; I say that AB is not so divided at another point.

For, if possible, let it be divided at D also, so that AD, DB are also medial straight lines commensurable in square only and containing a rational rectangle.

Since, then, that by which twice the rectangle AD, DB differs from twice the rectangle AC, CB is that by which the squares on AC, CB differ from the squares on AD, DB, while twice the rectangle AD, DB differs from twice the rectangle AC, CB by a rational area — for both are rational — therefore the squares on AC, CB also differ from the squares on AD, DB by a rational area, though they are medial: which is absurd. {x. 26 } Therefore a first bimedial straight line is not divided into its terms at different points; therefore it is so divided at one point only.

PROPOSITION 44. A second bimedial straight line is divided at one point only.

Let AB be a second bimedial straight line divided at C, so that AC, CB are medial straight lines commensurable in square only and containing a medial rectangle; {X. 38 } it is then manifest that C is not at the point of bisection, because the segments are not commensurable in length.

I say that AB is not so divided at another point.

For, if possible, let it be divided at D also, so that AC is not the same with DB, but AC is supposed greater; it is then clear that the squares on AD, DB are also, as we proved above {Lemma}, less than the squares on AC, CB; and suppose that AD, DB are medial straight lines commensurable in square only and

containing a medial rectangle.

Now let a rational straight line EF be set out, let there be applied to EF the rectangular parallelogram EK equal to the square on AB, and let EG equal to the squares on AC, CB be subtracted; therefore the remainder HK is equal to twice the rectangle AC, CB. {II. 4} Again, let there be subtracted EL, equal to the squares on AD, DB, which were proved less than the squares on AC, CB {Lemma}; therefore the remainder MK is also equal to twice the rectangle AD, DB.

Now, since the squares on AC, CB are medial, therefore EG is medial.

And it is applied to the rational straight line EF; therefore EH is rational and incommensurable in length with EF. {X. 22} For the same reason HN is also rational and incommensurable in length with EF.

And, since AC, CB are medial straight lines commensurable in square only, therefore AC is incommensurable in length with CB.

But, as AC is to CB, so is the square on AC to the rectangle AC, CB; therefore the square on AC is incommensurable with the rectangle AC, CB. {X. 11} But the squares on AC, CB are commensurable with the square on AC; for AC, CB are commensurable in square. {x. 15} And twice the rectangle AC, CB is commensurable with the rectangle AC, CB. {X. 6} Therefore the squares on AC, CB are also incommensurable with twice the rectangle AC, CB. {X. 13} But EG is equal to the squares on AC, CB, and HK is equal to twice the rectangle AC, CB; therefore EG is incommensurable with HK, so that EH is also incommensurable in length with HN. {VI. 1, X. 11} And they are rational; therefore EH, HN are rational straight lines commensurable in square only.

But, if two rational straight lines commensurable in square only be added together, the whole is the irrational which is called binomial. {X. 36 } Therefore EN is a binomial straight line divided at H.

In the same way EM, MN will also be proved to be rational straight lines commensurable in square only; and EN will be a binomial straight line divided at different points, H and M.

And EH is not the same with MN.

For the squares on AC, CB are greater than the squares on AD, DB.

But the squares on AD, DB are greater than twice the rectangle AD, DB; therefore also the squares on AC, CB, that is, EG, are much greater than twice the rectangle AD, DB, that is, MK, so that EH is also greater than MN.

Therefore EH is not the same with MN. Q. E. D.

PROPOSITION 45. A major straight line is divided at one and the same point only.

Let AB be a major straight line divided at C, so that AC, CB are incommensurable in square and make the sum of the squares on AC, CB rational, but the rectangle AC, CB medial; {X. 39 } I say that AB is not so divided at another point.

For, if possible, let it be divided at D also, so that AD, DB are also incommensurable in square and make the sum of the squares on AD, DB rational, but the rectangle contained by them medial.

Then, since that by which the squares on AC, CB differ from the squares on AD, DB is also that by which twice the rectangle AD, DB differs from twice the rectangle AC, CB, while the squares on AC, CB exceed the squares on AD, DB by a rational area — for both are rational — therefore twice the rectangle AD, DB also exceeds twice the rectangle AC, CB by a rational area, though

they are medial: which is impossible. {X. 26 } Therefore a major straight line is not divided at different points; therefore it is only divided at one and the same point. Q. E. D.

PROPOSITION 46. The side of a rational plus a medial area is divided at one point only.

Let AB be the side of a rational plus a medial area divided at C, so that AC, CB are incommensurable in square and make the sum of the squares on AC, CB medial, but twice the rectangle AC, CB rational; {X. 40 } I say that AB is not so divided at another point.

For, if possible, let it be divided at D also, so that AD, DB are also incommensurable in square and make the sum of the squares on AD, DB medial, but twice the rectangle AD, DB rational.

Since then that by which twice the rectangle AC, CB differs from twice the rectangle AD, DB is also that by which the squares on AD, DB differ from the squares on AC, CB, while twice the rectangle AC, CB exceeds twice the rectangle AD, DB by a rational area, therefore the squares on AD, DB also exceed the squares on AC, CB by a rational area, though they are medial: which is impossible. {X. 26 } Therefore the side of a rational plus a medial area is not divided at different points; therefore it is divided at one point only. Q. E. D.

PROPOSITION 47. The side of the sum of two medial areas is divided at one point only.

Let AB be divided at C, so that AC, CB are incommensurable in square and make the sum of the squares on AC, CB medial, and the rectangle AC, CB medial and also incommensurable with the sum of the squares on them; I say that AB is not divided at another point so as to fulfil the given conditions.

For, if possible, let it be divided at D, so that again AC is of course not the same as BD, but AC is supposed greater; let a

rational straight line EF be set out, and let there be applied to EF the rectangle EG equal to the squares on AC, CB, and the rectangle HK equal to twice the rectangle AC, CB; therefore the whole EK is equal to the square on AB. {II. 4} Again, let EL, equal to the squares on AD, DB, be applied to EF; therefore the remainder, twice the rectangle AD, DB, is equal to the remainder MK.

And since, by hypothesis, the sum of the squares on AC, CB is medial, therefore EG is also medial.

And it is applied to the rational straight line EF; therefore HE is rational and incommensurable in length with EF. {X. 22} For the same reason HN is also rational and incommensurable in length with EF.

And, since the sum of the squares on AC, CB is incommensurable with twice the rectangle AC, CB, therefore EG is also incommensurable with GN, so that EH is also incommensurable with HN. {VI. 1, X. 11} And they are rational; therefore EH, HN are rational straight lines commensurable in square only; therefore EN is a binomial straight line divided at H. {X. 36} Similarly we can prove that it is also divided at M.

And EH is not the same with MN; therefore a binomial has been divided at different points: which is absurd. {X. 42} Therefore a side of the sum of two medial areas is not divided at different points; therefore it is divided at one point only.

DEFINITIONS II.

1. GIVEN A RATIONAL straight line and a binomial, divided into its terms, such that the square on the greater term is greater than the square on the lesser by the square on a straight line commensurable in length with the greater, then, if the greater term be commensurable in length with the rational straight line set out, let the whole be called a first binomial straight line; 2.

but if the lesser term be commensurable in length with the rational straight line set out, let the whole be called a second binomial; **3.** and if neither of the terms be commensurable in length with the rational straight line set out, let the whole be called a third binomial.

4. Again, if the square on the greater term be greater than the square on the lesser by the square on a straight line incommensurable in length with the greater, then, if the greater term be commensurable in length with the rational straight line set out, let the whole be called a fourth binomial; **5.** if the lesser, a fifth binomial; **6.** and if neither, a sixth binomial.

PROPOSITIONS 48-84.

PROPOSITION 48. TO FIND THE first binomial straight line.

Let two numbers AC, CB be set out such that the sum of them AB has to BC the ratio which a square number has to a square number, but has not to CA the ratio which a square number has to a square number; {Lemma I after X. 28} let any rational straight line D be set out, and let EF be commensurable in length with D.

Therefore EF is also rational.

Let it be contrived that, as the number BA is to AC, so is the square on EF to the square on FG. {X. 6, Por.} But AB has to AC the ratio which a number has to a number; therefore the square on EF also has to the square on FG the ratio which a number has to a number, so that the square on EF is commensurable with the square on FG. {X. 6} And EF is rational; therefore FG is also rational.

And, since BA has not to AC the ratio which a square number has to a square number. neither, therefore, has the square on EF to the square on FG the ratio which a square number has to a

square number; therefore EF is incommensurable in length with FG. {X. 9} Therefore EF, FG are rational straight lines commensurable in square only; therefore EG is binomial. {X. 36} I say that it is also a first binomial straight line.

For since, as the number BA is to AC, so is the square on EF to the square on FG, while BA is greater than AC, therefore the square on EF is also greater than the square on FG.

Let then the squares on FG, H be equal to the square on EF.

Now since, as BA is to AC, so is the square on EF to the square on FG, therefore, convertendo, as AB is to BC, so is the square on EF to the square on H. {V. 19, Por.} But AB has to BC the ratio which a square number has to a square number; therefore the square on EF also has to the square on H the ratio which a square number has to a square number.

Therefore EF is commensurable in length with H; {X. 9} therefore the square on EF is greater than the square on FG by the square on a straight line commensurable with EF.

And EF, FG are rational, and EF is commensurable in length with D.

Therefore EF is a first binomial straight line. Q. E. D.

PROPOSITION 49. To find the second binomial straight line.

Let two numbers AC, CB be set out such that the sum of them AB has to BC the ratio which a square number has to a square number, but has not to AC the ratio which a square number has to a square number; let a rational straight line D be set out, and let EF be commensurable in length with D; therefore EF is rational.

Let it be contrived then that, as the number CA is to AB, so also is the square on EF to the square on FG; {X. 6, Por.} therefore the square on EF is commensurable with the square on FG. {X. 6}

Therefore FG is also rational.

Now, since the number CA has not to AB the ratio which a square number has to a square number, neither has the square on EF to the square on FG the ratio which a square number has to a square number.

Therefore EF is incommensurable in length with FG; {X. 9} therefore EF, FG are rational straight lines commensurable in square only; therefore EG is binomial. {X. 36} It is next to be proved that it is also a second binomial straight line.

For since, inversely, as the number BA is to AC, so is the square on GF to the square on FE, while BA is greater than AC, therefore the square on GF is greater than the square on FE.

Let the squares on EF, H be equal to the square on GF; therefore, convertendo, as AB is to BC, so is the square on FG to the square on H. {V. 19, Por.} But AB has to BC the ratio which a square number has to a square number; therefore the square on FG also has to the square on H the ratio which a square number has to a square number.

Therefore FG is commensurable in length with H; {X. 9} so that the square on FG is greater than the square on FE by the square on a straight line commensurable with FG.

And FG, FE are rational straight lines commensurable in square only, and EF, the lesser term, is commensurable in length with the rational straight line D set out.

Therefore EG is a second binomial straight line. Q. E. D.

PROPOSITION 50. To find the third binomial straight line.

Let two numbers AC, CB be set out such that the sum of them AB has to BC the ratio which a square number has to a square number, but has not to AC the ratio which a square number has to a square number.

Let any other number D , not square, be set out also, and let it not have to either of the numbers BA . AC the ratio which a square number has to a square number.

Let any rational straight line E be set out, and let it be contrived that, as D is to AB , so is the square on E to the square on FG ; {X. 6, Por.} therefore the square on E is commensurable with the square on FG . {X. 6} And E is rational; therefore FG is also rational.

And, since D has not to AB the ratio which a square number has to a square number, neither has the square on E to the square on FG the ratio which a square number has to a square number; therefore E is incommensurable in length with FG . {X. 9} Next let it be contrived that, as the number BA is to AC , so is the square on FG to the square on GH ; {X. 6, Por.} therefore the square on FG is commensurable with the square on GH . {X. 6} But FG is rational; therefore GH is also rational.

And, since BA has not to AC the ratio which a square number has to a square number, neither has the square on FG to the square on HG the ratio which a square number has to a square number; therefore FG is incommensurable in length with GH . {X. 9} Therefore FG , GH are rational straight lines commensurable in square only; therefore FH is binomial. {X. 36} I say next that it is also a third binomial straight line.

For since, as D is to AB , so is the square on E to the square on FG , and, as BA is to AC , so is the square on FG to the square on GH , therefore, ex aequali, as D is to AC , so is the square on E to the square on GH . {V. 22} But D has not to AC the ratio which a square number has to a square number; therefore neither has the square on E to the square on GH the ratio which a square number has to a square number; therefore E is incommensurable in length with GH . {X. 9} And since, as BA is to AC , so is the square on FG to the square on GH , therefore the

square on FG is greater than the square on GH.

Let then the squares on GH, K be equal to the square on FG; therefore, convertendo, as AB is to BC, so is the square on FG to the square on K. {V. 19, Por.} But AB has to BC the ratio which a square number has to a square number; therefore the square on FG also has to the square on K the ratio which a square number has to a square number; therefore FG is commensurable in length with K. {X. 9} Therefore the square on FG is greater than the square on GH by the square on a straight line commensurable with FG.

And FG, GH are rational straight lines commensurable in square only, and neither of them is commensurable in length with E.

Therefore FH is a third binomial straight line. Q. E. D.

PROPOSITION 51. To find the fourth binomial straight line.

Let two numbers AC, CB be set out such that AB neither has to BC, nor yet to AC, the ratio which a square number has to a square number.

Let a rational straight line D be set out, and let EF be commensurable in length with D; therefore EF is also rational.

Let it be contrived that, as the number BA is to AC, so is the square on EF to the square on FG; {X. 6, Por.} therefore the square on EF is commensurable with the square on FG; {X. 6} therefore FG is also rational.

Now, since BA has not to AC the ratio which a square number has to a square number, neither has the square on EF to the square on FG the ratio which a square number has to a square number; therefore EF is incommensurable in length with FG. {X. 9} Therefore EF, FG are rational straight lines commensurable in square only; so that EG is binomial.

I say next that it is also a fourth binomial straight line.

For since, as BA is to AC, so is the square on EF to the square on FG, therefore the square on EF is greater than the square on FG.

Let then the squares on FG, H be equal to the square on EF; therefore, convertendo, as the number AB is to BC, so is the square on EF to the square on H. {V. 19, Por.} But AB has not to BC the ratio which a square number has to a square number; therefore neither has the square on EF to the square on H the ratio which a square number has to a square number.

Therefore EF is incommensurable in length with H; {X. 9} therefore the square on EF is greater than the square on GF by the square on a straight line incommensurable with EF.

And EF, FG are rational straight lines commensurable in square only, and EF is commensurable in length with D.

Therefore EG is a fourth binomial straight line. Q. E. D.

PROPOSITION 52. To find the fifth binomial straight line.

Let two numbers AC, CB be set out such that AB has not to either of them the ratio which a square number has to a square number; let any rational straight line D be set out, and let EF be commensurable with D; therefore EF is rational.

Let it be contrived that, as CA is to AB, so is the square on EF to the square on FG. {X. 6, Por.} But CA has not to AB the ratio which a square number has to a square number; therefore neither has the square on EF to the square on FG the ratio which a square number has to a square number.

Therefore EF, FG are rational straight lines commensurable in square only; {X. 9} therefore EG is binomial. {X. 36} I say next that it is also a fifth binomial straight line.

For since, as CA is to AB, so is the square on EF to the square on FG, inversely, as BA is to AC, so is the square on FG to the square on FE; therefore the square on GF is greater than the square on

FE.

Let then the squares on EF, H be equal to the square on GF; therefore, convertendo, as the number AB is to BC, so is the square on GF to the square on H. {V. 19, Por.} But AB has not to BC the ratio which a square number has to a square number; therefore neither has the square on FG to the square on H the ratio which a square number has to a square number.

Therefore FG is incommensurable in length with H; {X. 9} so that the square on FG is greater than the square on FE by the square on a straight line incommensurable with FG.

And GF, FE are rational straight lines commensurable in square only, and the lesser term EF is commensurable in length with the rational straight line D set out.

Therefore EG is a fifth binomial straight line. Q. E. D.

PROPOSITION 53. To find the sixth binomial straight line.

Let two numbers AC, CB be set out such that AB has not to either of them the ratio which a square number has to a square number; and let there also be another number D which is not square and which has not to either of the numbers BA, AC the ratio which a square number has to a square number.

Let any rational straight line E be set out, and let it be contrived that, as D is to AB, so is the square on E to the square on FG; {X. 6, Por.} therefore the square on E is commensurable with the square on FG. {X. 6} And E is rational; therefore FG is also rational.

Now, since D has not to AB the ratio which a square number has to a square number, neither has the square on E to the square on FG the ratio which a square number has to a square number; therefore E is incommensurable in length with FG. {X. 9} Again, let it be contrived that, as BA is to AC, so is the square on FG to the square on GH. {X. 6, Por.} Therefore the square on FG is

commensurable with the square on HG. {X. 6} Therefore the square on HG is rational; therefore HG is rational.

And, since BA has not to AC the ratio which a square number has to a square number, neither has the square on FG to the square on GH the ratio which a square number has to a square number; therefore FG is incommensurable in length with GH. {X. 9} Therefore FG, GH are rational straight lines commensurable in square only; therefore FH is binomial. {X. 36} It is next to be proved that it is also a sixth binomial straight line.

For since, as D is to AB, so is the square on E to the square on FG, and also, as BA is to AC, so is the square on FG to the square on GH, therefore, ex aequali, as D is to AC, so is the square on E to the square on GH. {V. 22} But D has not to AC the ratio which a square number has to a square number; therefore neither has the square on E to the square on GH the ratio which a square number has to a square number; therefore E is incommensurable in length with GH. {X. 9} But it was also proved incommensurable with FG; therefore each of the straight lines FG, GH is incommensurable in length with E.

And, since, as BA is to AC, so is the square on FG to the square on GH, therefore the square on FG is greater than the square on GH.

Let then the squares on GH, K be equal to the square on FG; therefore, convertendo, as AB is to BC, so is the square on FG to the square on K. {V. 19, Por.} But AB has not to BC the ratio which a square number has to a square number; so that neither has the square on FG to the square on K the ratio which a square number has to a square number.

Therefore FG is incommensurable in length with K; {X. 9} therefore the square on FG is greater than the square on GH by the square on a straight line incommensurable with FG.

And FG, GH are rational straight lines commensurable in square only, and neither of them is commensurable in length with the rational straight line E set out.

Therefore FH is a sixth binomial straight line. Q. E. D.

LEMMA.

Let there be two squares AB, BC, and let them be placed so that DB is in a straight line with BE; therefore FB is also in a straight line with BG.

Let the parallelogram AC be completed; I say that AC is a square, that DG is a mean proportional between AB, BC, and further that DC is a mean proportional between AC, CB.

For, since DB is equal to BF, and BE to BG, therefore the whole DE is equal to the whole FG.

But DE is equal to each of the straight lines AH, KC, and FG is equal to each of the straight lines AK, HC; {I. 34} therefore each of the straight lines AH, KC is also equal to each of the straight lines AK, HC.

Therefore the parallelogram AC is equilateral.

And it is also rectangular; therefore AC is a square.

And since, as FB is to BG, so is DB to BE, while, as FB is to BG, so is AB to DG, and, as DB is to BE, so is DG to BC, {VI. 1} therefore also, as AB is to DG, so is DG to BC. {V. 11} Therefore DG is a mean proportional between AB, BC.

I say next that DC is also a mean proportional between AC, CB.

For since, as AD is to DK, so is KG to GC — for they are equal respectively — and, componendo, as AK is to KD, so is KC to CG, {V. 18} while, as AK is to KD, so is AC to CD, and, as KC is to CG, so is DC to CB, {VI. 1} therefore also, as AC is to DC, so is DC to BC. {V. 11} Therefore DC is a mean proportional between AC, CB.

Being what it was proposed to prove.

PROPOSITION 54. If an area be contained by a rational straight line and the first binomial, the “side” of the area is the irrational straight line which is called binomial.

For let the area AC be contained by the rational straight line AB and the first binomial AD; I say that the “side” of the area AC is the irrational straight line which is called binomial.

For, since AD is a first binomial straight line, let it be divided into its terms at E, and let AE be the greater term.

It is then manifest that AE, ED are rational straight lines commensurable in square only, the square on AE is greater than the square on ED by the square on a straight line commensurable with AE, and AE is commensurable in length with the rational straight line AB set out. {X. Deff. II. 1} Let ED be bisected at the point F.

Then, since the square on AE is greater than the square on ED by the square on a straight line commensurable with AE, therefore, if there be applied to the greater AE a parallelogram equal to the fourth part of the square on the less, that is, to the square on EF, and deficient by a square figure, it divides it into commensurable parts. {X. 17} Let then the rectangle AG, GE equal to the square on EF be applied to AE; therefore AG is commensurable in length with EG.

Let GH, EK, FL be drawn from G, E, F parallel to either of the straight lines AB, CD; let the square SN be constructed equal to the parallelogram AH, and the square NQ equal to GK, {II. 14} and let them be placed so that MN is in a straight line with NO; therefore RN is also in a straight line with NP.

And let the parallelogram SQ be completed; therefore SQ is a square. {Lemma} Now, since the rectangle AG, GE is equal to the square on EF, therefore, as AG is to EF, so is FE to EG; {VI. 17}

therefore also, as AH is to EL, so is EL to KG; {VI. 1} therefore EL is a mean proportional between AH, GK.

But AH is equal to SN, and GK to NQ; therefore EL is a mean proportional between SN, NQ.

But MR is also a mean proportional between the same SN, NQ; {Lemma} therefore EL is equal to MR, so that it is also equal to PO.

But AH, GK are also equal to SN, NQ; therefore the whole AC is equal to the whole SQ, that is, to the square on MO; therefore MO is the “side” of AC.

I say next that MO is binomial.

For, since AG is commensurable with GE, therefore AE is also commensurable with each of the straight lines AG, GE. {X. 15} But AE is also, by hypothesis, commensurable with AB; therefore AG, GE are also commensurable with AB. {X. 12} And AB is rational; therefore each of the straight lines AG, GE is also rational; therefore each of the rectangles AH, GK is rational, {X. 19} and AH is commensurable with GK.

But AH is equal to SN, and GK to NQ; therefore SN, NQ, that is, the squares on MN, NO, are rational and commensurable.

And, since AE is incommensurable in length with ED, while AE is commensurable with AG, and DE is commensurable with EF, therefore AG is also incommensurable with EF, {X. 13} so that AH is also incommensurable with EL. {VI. 1, X. 11} But AH is equal to SN, and EL to MR; therefore SN is also incommensurable with MR.

But, as SN is to MR, so is PN to NR; {VI. 1} therefore PN is incommensurable with NR. {X. 11} But PN is equal to MN, and NR to NO; therefore MN is incommensurable with NO.

And the square on MN is commensurable with the square on NO, and each is rational; therefore MN, NO are rational straight lines commensurable in square only.

Therefore MO is binomial {X. 36} and the “side” of AC. Q. E. D.

PROPOSITION 55. If an area be contained by a rational straight line and the second binomial, the “side” of the area is the irrational straight line which is called a first bimedral.

For let the area ABCD be contained by the rational straight line AB and the second binomial AD; I say that the “side” of the area AC is a first bimedral straight line.

For, since AD is a second binomial straight line, let it be divided into its terms at E, so that AE is the greater term; therefore AE, ED are rational straight lines commensurable in square only, the square on AE is greater than the square on ED by the square on a straight line commensurable with AE, and the lesser term ED is commensurable in length with AB. {X. Deff. II. 2} Let ED be bisected at F, and let there be applied to AE the rectangle AG, GE equal to the square on EF and deficient by a square figure; therefore AG is commensurable in length with GE. {X. 17} Through G, E, F let GH, EK, FL be drawn parallel to AB, CD, let the square SN be constructed equal to the parallelogram AH, and the square NQ equal to GK, and let them be placed so that MN is in a straight line with NO; therefore RN is also in a straight line with NP.

Let the square SQ be completed.

It is then manifest from what was proved before that MR is a mean proportional between SN, NQ and is equal to EL, and that MO is the “side” of the area AC.

It is now to be proved that MO is a first bimedral straight line.

Since AE is incommensurable in length with ED, while ED is commensurable with AB, therefore AE is incommensurable with

AB. {X. 13} And, since AG is commensurable with EG, AE is also commensurable with each of the straight lines AG, GE. {X. 15} But AE is incommensurable in length with AB; therefore AG, GE are also incommensurable with AB. {X. 13} Therefore BA, AG and BA, GE are pairs of rational straight lines commensurable in square only; so that each of the rectangles AH, GK is medial. {X. 21} Hence each of the squares SN, NQ is medial.

Therefore MN, NO are also medial.

And, since AG is commensurable in length with GE, AH is also commensurable with GK, {VI. 1. X. 11} that is, SN is commensurable with NQ, that is, the square on MN with the square on NO.

And, since AE is incommensurable in length with ED, while AE is commensurable with AG, and ED is commensurable with EF, therefore AG is incommensurable with EF; {X. 13} so that AH is also incommensurable with EL, that is, SN is incommensurable with MR, that is, PN with NR, {VI. 1, X. 11} that is, MN is incommensurable in length with NO.

But MN, NO were proved to be both medial and commensurable in square; therefore MN, NO are medial straight lines commensurable in square only.

I say next that they also contain a rational rectangle.

For, since DE is, by hypothesis, commensurable with each of the straight lines AB, EF, therefore EF is also commensurable with EK. {X. 12} And each of them is rational; therefore EL, that is, MR is rational, {X. 19} and MR is the rectangle MN, NO.

But, if two medial straight lines commensurable in square only and containing a rational rectangle be added together, the whole is irrational and is called a first bimedial straight line. {X. 37} Therefore MO is a first bimedial straight line. Q. E. D.

PROPOSITION 56. If an area be contained by a rational straight line and the third binomial, the “side” of the area is the irrational straight line called a second bimedial.

For let the area ABCD be contained by the rational straight line AB and the third binomial AD divided into its terms at E, of which terms AE is the greater; I say that the “side” of the area AC is the irrational straight line called a second bimedial.

For let the same construction be made as before.

Now, since AD is a third binomial straight line, therefore AE, ED are rational straight lines commensurable in square only, the square on AE is greater than the square on ED by the square on a straight line commensurable with AE, and neither of the terms AE, ED is commensurable in length with AB. {X. Deff. II. 3} Then, in manner similar to the foregoing, we shall prove that MO is the “side” of the area AC, and MN, NO are medial straight lines commensurable in square only; so that MO is bimedial.

It is next to be proved that it is also a second bimedial straight line.

Since DE is incommensurable in length with AB, that is, with EK, and DE is commensurable with EF, therefore EF is incommensurable in length with EK. {X. 13} And they are rational; therefore FE, EK are rational straight lines commensurable in square only.

Therefore EL, that is, MR, is medial. {X. 21} And it is contained by MN, NO; therefore the rectangle MN, NO is medial.

Therefore MO is a second bimedial straight line. {X. 38} Q. E. D.

PROPOSITION 57. If an area be contained by a rational straight line and the fourth binomial, the “side” of the area is the irrational straight line called major.

For let the area AC be contained by the rational straight line AB and the fourth binomial AD divided into its terms at E, of which terms let AE be the greater; I say that the “side” of the area AC is the irrational straight line called major.

For, since AD is a fourth binomial straight line, therefore AE, ED are rational straight lines commensurable in square only, the square on AE is greater than the square on ED by the square on a straight line incommensurable with AE, and AE is commensurable in length with AB. {X. Deff. II. 4} Let DE be bisected at F, and let there be applied to AE a parallelogram, the rectangle AG, GE, equal to the square on EF; therefore AG is incommensurable in length with GE. {X. 18} Let GH, EK, FL be drawn parallel to AB, and let the rest of the construction be as before; it is then manifest that MO is the “side” of the area AC.

It is next to be proved that MO is the irrational straight line called major.

Since AG is incommensurable with EG, AH is also incommensurable with GK, that is, SN with NQ; {VI. 1, X. 11} therefore MN, NO are incommensurable in square.

And, since AE is commensurable with AB, AK is rational; {X. 19} and it is equal to the squares on MN, NO; therefore the sum of the squares on MN, NO is also rational.

And, since DE is incommensurable in length with AB, that is, with EK, while DE is commensurable with EF, therefore EF is incommensurable in length with EK. {X. 13} Therefore EK, EF are rational straight lines commensurable in square only; therefore LE, that is, MR, is medial. {X. 21} And it is contained by MN, NO; therefore the rectangle MN, NO is medial.

And the {sum} of the squares on MN, NO is rational, and MN, NO are incommensurable in square.

But, if two straight lines incommensurable in square and making the sum of the squares on them rational, but the rectangle contained by them medial, be added together, the whole is irrational and is called major. {X. 39} Therefore MO is the irrational straight line called major and is the “side” of the area AC. Q. E. D.

PROPOSITION 58. If an area be contained by a rational straight line and the fifth binomial, the “side” of the area is the irrational straight line called the side of a rational plus a medial area.

For let the area AC be contained by the rational straight line AB and the fifth binomial AD divided into its terms at E, so that AE is the greater term; I say that the “side” of the area AC is the irrational straight line called the side of a rational plus a medial area.

For let the same construction be made as before shown; it is then manifest that MO is the “side” of the area AC.

It is then to be proved that MO is the side of a rational plus a medial area.

For, since AG is incommensurable with GE, {X. 18} therefore AH is also commensurable with HE, {VI. 1, X. 11} that is, the square on MN with the square on NO; therefore MN, NO are incommensurable in square.

And, since AD is a fifth binomial straight line, and ED the lesser segment, therefore ED is commensurable in length with AB. {X. Def. II. 5} But AE is incommensurable with ED; therefore AB is also incommensurable in length with AE. {X. 13} Therefore AK, that is, the sum of the squares on MN, NO, is medial. {X. 21} And, since DE is commensurable in length with AB, that is, with EK, while DE is commensurable with EF, therefore EF is also commensurable with EK. {X. 12} And EK is rational; therefore EL,

that is, MR, that is, the rectangle MN, NO, is also rational. {X. 19} Therefore MN, NO are straight lines incommensurable in square which make the sum of the squares on them medial, but the rectangle contained by them rational.

Therefore MO is the side of a rational plus a medial area {X. 40} and is the “side” of the area AC. Q. E. D.

PROPOSITION 59. If an area be contained by a rational straight line and the sixth binomial, the “side” of the area is the irrational straight line called the side of the sum of two medial areas.

For let the area ABCD be contained by the rational straight line AB and the sixth binomial AD, divided into its terms at E, so that AE is the greater term; I say that the “side” of AC is the side of the sum of two medial areas.

Let the same construction be made as before shown.

It is then manifest that MO is the “side” of AC, and that MN is incommensurable in square with NO.

Now, since EA is incommensurable in length with AB, therefore EA, AB are rational straight lines commensurable in square only; therefore AK, that is, the sum of the squares on MN, NO, is medial. {X. 21} Again, since ED is incommensurable in length with AB, therefore FE is also incommensurable with EK; {X. 13} therefore FE, EK are rational straight lines commensurable in square only; therefore EL, that is, MR, that is, the rectangle MN, NO, is medial. {X. 21} And, since AE is incommensurable with EF, AK is also incommensurable with EL. {VI. 1, X. 11} But AK is the sum of the squares on MN, NO, and EL is the rectangle MN, NO; therefore the sum of the squares on MN, NO is incommensurable with the rectangle MN, NO.

And each of them is medial, and MN, NO are incommensurable in square.

Therefore MO is the side of the sum of two medial areas {X. 41}, and is the “side” of AC. Q. E. D.

{LEMMA.

If a straight line be cut into unequal parts, the squares on the unequal parts are greater than twice the rectangle contained by the unequal parts.

Let AB be a straight line, and let it be cut into unequal parts at C, and let AC be the greater; I say that the squares on AC, CB are greater than twice the rectangle AC, CB.

For let AB be bisected at D.

Since then a straight line has been cut into equal parts at D, and into unequal parts at C, therefore the rectangle AC, CB together with the square on CD is equal to the square on AD, {II. 5} so that the rectangle AC, CB is less than double of the square on AD.

But the squares on AC, CB are double of the squares on AD, DC; {II. 9} therefore the squares on AC, CB are greater than twice the rectangle AC, CB. Q. E. D.} **PROPOSITION 60.** The square on the binomial straight line applied to a rational straight line produces as breadth the first binomial.

Let AB be a binomial straight line divided into its terms at C, so that AC is the greater term; let a rational straight line DE be set out, and let DEFG equal to the square on AB be applied to DE producing DG as its breadth; I say that DG is a first binomial straight line.

For let there be applied to DE the rectangle DH equal to the square on AC, and KL equal to the square on BC; therefore the remainder, twice the rectangle AC, CB, is equal to MF.

Let MG be bisected at N, and let NO be drawn parallel {to ML or GF}.

Therefore each of the rectangles MO, NF is equal to once the rectangle AC, CB.

Now, since AB is a binomial divided into its terms at C, therefore AC, CB are rational straight lines commensurable in square only; {X. 36} therefore the squares on AC, CB are rational and commensurable with one another, so that the sum of the squares on AC, CB is also rational. {X. 15} And it is equal to DL; therefore DL is rational.

And it is applied to the rational straight line DE; therefore DM is rational and commensurable in length with DE. {X. 20} Again, since AC, CB are rational straight lines commensurable in square only, therefore twice the rectangle AC, CB, that is MF, is medial. {X. 21} And it is applied to the rational straight line ML; therefore MG is also rational and incommensurable in length with ML, that is, DE. {X. 22} But MD is also rational and is commensurable in length with DE; therefore DM is incommensurable in length with MG. {X. 13} And they are rational; therefore DM, MG are rational straight lines commensurable in square only; therefore DG is binomial. {X. 36} It is next to be proved that it is also a first binomial straight line.

Since the rectangle AC, CB is a mean proportional between the squares on AC, CB, {cf. Lemma after X. 53} therefore MO is also a mean proportional between DH, KL.

Therefore, as DH is to MO, so is MO to KL, that is, as DK is to MN, so is MN to MK; {VI. 1} therefore the rectangle DK, KM is equal to the square on MN. {VI. 17} And, since the square on AC is commensurable with the square on CB, DH is also commensurable with KL, so that DK is also commensurable with KM. {VI. 1, X. 11} And, since the squares on AC, CB are greater than twice the rectangle AC, CB, {Lemma} therefore DL is also greater than MF, so that DM is also greater than MG. {VI. 1} And the rectangle DK, KM is equal to the square on MN, that is, to

the fourth part of the square on MG, and DK is commensurable with KM.

But, if there be two unequal straight lines, and to the greater there be applied a parallelogram equal to the fourth part of the square on the less and deficient by a square figure, and if it divide it into commensurable parts, the square on the greater is greater than the square on the less by the square on a straight line commensurable with the greater; {X. 17} therefore the square on DM is greater than the square on MG by the square on a straight line commensurable with DM.

And DM, MG are rational, and DM, which is the greater term, is commensurable in length with the rational straight line DE set out.

Therefore DG is a first binomial straight line. {X. Deff. II. 1} Q. E. D.

PROPOSITION 61. The square on the first bimedial straight line applied to a rational straight line produces as breadth the second binomial.

Let AB be a first bimedial straight line divided into its medials at C, of which medials AC is the greater; let a rational straight line DE be set out, and let there be applied to DE the parallelogram DF equal to the square on AB, producing DG as its breadth; I say that DG is a second binominal straight line.

For let the same construction as before be made.

Then, since AB is a first bimedial divided at C, therefore AC, CB are medial straight lines commensurable in square only, and containing a rational rectangle, {X. 37} so that the squares on AC, CB are also medial. {X. 21} Therefore DL is medial. {X. 15 and 23, Por.} And it has been applied to the rational straight line DE; therefore MD is rational and incommensurable in length with DE. {X. 22} Again, since twice the rectangle AC, CB is rational,

MF is also rational.

And it is applied to the rational straight line ML; therefore MG is also rational and commensurable in length with ML, that is, DE; {X. 20} therefore DM is incommensurable in length with MG. {X. 13} And they are rational; therefore DM, MG are rational straight lines commensurable in square only; therefore DG is binomial. {X. 36} It is next to be proved that it is also a second binomial straight line.

For, since the squares on AC, CB are greater than twice the rectangle AC, CB, therefore DL is also greater than MF, so that DM is also greater than MG. {VI. 1} And, since the square on AC is commensurable with the square on CB, DH is also commensurable with KL, so that DK is also commensurable with KM. {VI. 1, X. 11} And the rectangle DK, KM is equal to the square on MN; therefore the square on DM is greater than the square on MG by the square on a straight line commensurable with DM. {X. 17} And MG is commensurable in length with DE.

Therefore DG is a second binomial straight line. {X. Deff. II. 2} **PROPOSITION 62.** The square on the second binomial straight line applied to a rational straight line produces as breadth the third binomial.

Let AB be a second binomial straight line divided into its medials at C, so that AC is the greater segment; let DE be any rational straight line, and to DE let there be applied the parallelogram DF equal to the square on AB and producing DG as its breadth; I say that DG is a third binomial straight line.

Let the same construction be made as before shown.

Then, since AB is a second binomial divided at C, therefore AC, CB are medial straight lines commensurable in square only and containing a medial rectangle, {X. 38} so that the sum of the squares on AC, CB is also medial. {X. 15 and 23 Por.} And it is

equal to DL; therefore DL is also medial.

And it is applied to the rational straight line DE; therefore MD is also rational and incommensurable in length with DE. {X. 22} For the same reason, MG is also rational and incommensurable in length with ML, that is, with DE; therefore each of the straight lines DM, MG is rational and incommensurable in length with DE.

And, since AC is incommensurable in length with CB, and, as AC is to CB, so is the square on AC to the rectangle AC, CB, therefore the square on AC is also incommensurable with the rectangle AC, CB. {X. 11} Hence the sum of the squares on AC, CB is incommensurable with twice the rectangle AC, CB, {X. 12, 13} that is, DL is incommensurable with MF, so that DM is also incommensurable with MG. {VI. 1, X. 11} And they are rational; therefore DG is binomial. {X. 36} It is to be proved that it is also a third binomial straight line.

In manner similar to the foregoing we may conclude that DM is greater than MG, and that DK is commensurable with KM.

And the rectangle DK, KM is equal to the square on MN; therefore the square on DM is greater than the square on MG by the square on a straight line commensurable with DM.

And neither of the straight lines DM, MG is commensurable in length with DE.

Therefore DG is a third binomial straight line. {X. Deff. II. 3} Q. E. D.

PROPOSITION 63. The square on the major straight line applied to a rational straight line produces as breadth the fourth binomial.

Let AB be a major straight line divided at C, so that AC is greater than CB; let DE be a rational straight line, and to DE let there be applied the parallelogram DF equal to the square on AB and

producing DG as its breadth; I say that DG is a fourth binomial straight line.

Let the same construction be made as before shown.

Then, since AB is a major straight line divided at C, AC, CB are straight lines incommensurable in square which make the sum of the squares on them rational, but the rectangle contained by them medial. {X. 39} Since then the sum of the squares on AC, CB is rational, therefore DL is rational; therefore DM is also rational and commensurable in length with DE. {X. 20} Again, since twice the rectangle AC, CB, that is, MF, is medial, and it is applied to the rational straight line ML, therefore MG is also rational and incommensurable in length with DE; {X. 22} therefore DM is also incommensurable in length with MG. {X. 13} Therefore DM, MG are rational straight lines commensurable in square only; therefore DG is binomial. {X. 36} It is to be proved that it is also a fourth binomial straight line.

In manner similar to the foregoing we can prove that DM is greater than MG, and that the rectangle DK, KM is equal to the square on MN.

Since then the square on AC is incommensurable with the square on CB, therefore DH is also incommensurable with KL, so that DK is also incommensurable with KM. {VI. 1, X. 11} But, if there be two unequal straight lines, and to the greater there be applied a parallelogram equal to the fourth part of the square on the less and deficient by a square figure, and if it divide it into incommensurable parts, then the square on the greater will be greater than the square on the less by the square on a straight line incommensurable in length with the greater; {X. 18} therefore the square on DM is greater than the square on MG by the square on a straight line incommensurable with DM.

And DM, MG are rational straight lines commensurable in square only, and DM is commensurable with the rational

straight line DE set out.

Therefore DG is a fourth binomial straight line. {X. Deff. II. 4} Q. E. D.

PROPOSITION 64. The square on the side of a rational plus a medial area applied to a rational straight line produces as breadth the fifth binomial.

Let AB be the side of a rational plus a medial area, divided into its straight lines at C, so that AC is the greater; let a rational straight line DE be set out, and let there be applied to DE the parallelogram DF equal to the square on AB, producing DG as its breadth; I say that DG is a fifth binomial straight line.

Let the same construction as before be made.

Since then AB is the side of a rational plus a medial area, divided at C, therefore AC, CB are straight lines incommensurable in square which make the sum of the squares on them medial, but the rectangle contained by them rational. {X. 40} Since then the sum of the squares on AC, CB is medial, therefore DL is medial, so that DM is rational and incommensurable in length with DE. {X. 22} Again, since twice the rectangle AC, CB, that is MF, is rational, therefore MG is rational and commensurable with DE. {X. 20} Therefore DM is incommensurable with MG; {X. 13} therefore DM, MG are rational straight lines commensurable in square only; therefore DG is binomial. {X. 36} I say next that it is also a fifth binomial straight line.

For it can be proved similarly that the rectangle DK, KM is equal to the square on MN, and that DK is incommensurable in length with KM; therefore the square on DM is greater than the square on MG by the square on a straight line incommensurable with DM. {X. 18} And DM, MG are commensurable in square only, and the less, MG, is commensurable in length with DE.

Therefore DG is a fifth binomial. Q. E. D.

PROPOSITION 65. The square on the side of the sum of two medial areas applied to a rational straight line produces as breadth the sixth binomial.

Let AB be the side of the sum of two medial areas, divided at C, let DE be a rational straight line, and let there be applied to DE the parallelogram DF equal to the square on AB, producing DG as its breadth; I say that DG is a sixth binomial straight line.

For let the same construction be made as before.

Then, since AB is the side of the sum of two medial areas, divided at C, therefore AC, CB are straight lines incommensurable in square which make the sum of the squares on them medial, the rectangle contained by them medial, and moreover the sum of the squares on them incommensurable with the rectangle contained by them, {X. 41} so that, in accordance with what was before proved, each of the rectangles DL, MF is medial.

And they are applied to the rational straight line DE; therefore each of the straight lines DM, MG is rational and incommensurable in length with DE. {X. 22} And, since the sum of the squares on AC, CB is incommensurable with twice the rectangle AC, CB, therefore DL is incommensurable with MF.

Therefore DM is also incommensurable with MG; {VI. 1, X. 11} therefore DM, MG are rational straight lines commensurable in square only; therefore DG is binomial. {X. 36} I say next that it is also a sixth binomial straight line.

Similarly again we can prove that the rectangle DK, KM is equal to the square on MN, and that DK is incommensurable in length with KM; and, for the same reason, the square on DM is greater than the square on MG by the square on a straight line incommensurable in length with DM.

And neither of the straight lines DM, MG is commensurable in length with the rational straight line DE set out.

Therefore DG is a sixth binomial straight line. Q. E. D.

PROPOSITION 66. A straight line commensurable in length with a binomial straight line is itself also binomial and the same in order.

Let AB be binomial, and let CD be commensurable in length with AB; I say that CD is binomial and the same in order with AB.

For, since AB is binomial, let it be divided into its terms at E, and let AE be the greater term; therefore AE, EB are rational straight lines commensurable in square only. {X. 36} Let it be contrived that, as AB is to CD, so is AE to CF; {VI. 12} therefore also the remainder EB is to the remainder FD as AB is to CD. {V. 19} But AB is commensurable in length with CD; therefore AE is also commensurable with CF, and EB with FD. {X. 11} And AE, EB are rational; therefore CF, FD are also rational.

And, as AE is to CF, so is EB to FD. {V. 11} Therefore, alternately, as AE is to EB, so is CF to FD. {V. 16} But AE, EB are commensurable in square only; therefore CF, FD are also commensurable in square only. {X. 11} And they are rational; therefore CD is binomial. {X. 36} I say next that it is the same in order with AB.

For the square on AE is greater than the square on EB either by the square on a straight line commensurable with AE or by the square on a straight line incommensurable with it.

If then the square on AE is greater than the square on EB by the square on a straight line commensurable with AE, the square on CF will also be greater than the square on FD by the square on a straight line commensurable with CF. {X. 14} And, if AE is commensurable with the rational straight line set out, CF will also be commensurable with it, {X. 12} and for this reason each

of the straight lines AB, CD is a first binomial, that is, the same in order. {X. Deff. II. 1} But, if EB is commensurable with the rational straight line set out, FD is also commensurable with it, {X. 12} and for this reason again CD will be the same in order with AB, for each of them will be a second binomial. {X. Deff. II. 2} But, if neither of the straight lines AE, EB is commensurable with the rational straight line set out, neither of the straight lines CF, FD will be commensurable with it, {X. 13} and each of the straight lines AB, CD is a third binomial. {X. Deff. II. 3} But, if the square on AE is greater than the square on EB by the square on a straight line incommensurable with AE, the square on CF is also greater than the square on FD by the square on a straight line incommensurable with CF. {X. 14} And, if AE is commensurable with the rational straight line set out, CF is also commensurable with it, and each of the straight lines AB, CD is a fourth binomial. {X. Deff. II. 4} But, if EB is so commensurable, so is FD also, and each of the straight lines AB, CD will be a fifth binomial. {X. Deff. II. 5} But, if neither of the straight lines AE, EB is so commensurable, neither of the straight lines CF, FD is commensurable with the rational straight line set out, and each of the straight lines AB, CD will be a sixth binomial. {X. Deff. II. 6} Hence a straight line commensurable in length with a binomial straight line is binomial and the same in order. Q. E. D.

PROPOSITION 67. A straight line commensurable in length with a binomial straight line is itself also binomial and the same in order.

Let AB be binomial, and let CD be commensurable in length with AB; I say that CD is binomial and the same in order with AB.

For, since AB is binomial, let it be divided into its medials at E; therefore AE, EB are medial straight lines commensurable in square only. {X. 37, 38} And let it be contrived that, as AB is to CD, so is AE to CF; therefore also the remainder EB is to the remainder FD as AB is to CD. {V. 19} But AB is commensurable in

length with CD; therefore AE, EB are also commensurable with CF, FD respectively. {X. 11} But AE, EB are medial; therefore CF, FD are also medial. {X. 23} And since, as AE is to EB, so is CF to FD, {V. 11} and AE, EB are commensurable in square only, CF, FD are also commensurable in square only. {X. 11} But they were also proved medial; therefore CD is bimedral.

I say next that it is also the same in order with AB.

For since, as AE is to EB, so is CF to FD, therefore also, as the square on AE is to the rectangle AE, EB, so is the square on CF to the rectangle CF, FD; therefore, alternately, as the square on AE is to the square on CF, so is the rectangle AE, EB to the rectangle CF, FD. {V. 16} But the square on AE is commensurable with the square on CF; therefore the rectangle AE, EB is also commensurable with the rectangle CF, FD.

If therefore the rectangle AE, EB is rational, the rectangle CF, FD is also rational, {and for this reason CD is a first bimedral}; {X. 37} but if medial, medial, {X. 23, Por.} and each of the straight lines AB, CD is a second bimedral. {X. 38} And for this reason CD will be the same in order with AB. Q. E. D.

PROPOSITION 68. A straight line commensurable with a major straight line is itself also major.

Let AB be major, and let CD be commensurable with AB; I say that CD is major.

Let AB be divided at E; therefore AE, EB are straight lines incommensurable in square which make the sum of the squares on them rational, but the rectangle contained by them medial. {X. 39} Let the same construction be made as before.

Then since, as AB is to CD, so is AE to CF, and EB to FD, therefore also, as AE is to CF, so is EB to FD. {V. 11} But AB is commensurable with CD; therefore AE, EB are also commensurable with CF, FD respectively. {X. 11} And since, as

AE is to CF, so is EB to FD, alternately also, as AE is to EB, so is CF to FD; {V. 16} therefore also, componendo, as AB is to BE, so is CD to DF; {V. 18} therefore also, as the square on AB is to the square on BE, so is the square on CD to the square on DF. {VI. 20} Similarly we can prove that, as the square on AB is to the square on AE, so also is the square on CD to the square on CF.

Therefore also, as the square on AB is to the squares on AE, EB, so is the square on CD to the squares on CF, FD; therefore also, alternately, as the square on AB is to the square on CD, so are the squares on AE, EB to the squares on CF, FD. {V. 16} But the square on AB is commensurable with the square on CD; therefore the squares on AE, EB are also commensurable with the squares on CF, FD.

And the squares on AE, EB together are rational; therefore the squares on CF, FD together are rational.

Similarly also twice the rectangle AE, EB is commensurable with twice the rectangle CF, FD.

And twice the rectangle AE, EB is medial; therefore twice the rectangle CF, FD is also medial. {X. 23, Por.} Therefore CF, FD are straight lines incommensurable in square which make, at the same time, the sum of the squares on them rational, but the rectangle contained by them medial; therefore the whole CD is the irrational straight line called major. {X. 39} Therefore a straight line commensurable with the major straight line is major. Q. E. D.

PROPOSITION 69. A straight line commensurable with the side of a rational plus a medial area is itself also the side of a rational plus a medial area.

Let AB be the side of a rational plus a medial area, and let CD be commensurable with AB; it is to be proved that CD is also the side of a rational plus a medial area.

Let AB be divided into its straight lines at E; therefore AE, EB are straight lines incommensurable in square which make the sum of the squares on them medial, but the rectangle contained by them rational. {X. 40} Let the same construction be made as before.

We can then prove similarly that CF, FD are incommensurable in square, and the sum of the squares on AE, EB is commensurable with the sum of the squares on CF, FD, and the rectangle AE, EB with the rectangle CF, FD; so that the sum of the squares on CF, FD is also medial, and the rectangle CF, FD rational.

Therefore CD is the side of a rational plus a medial area. Q. E. D.

PROPOSITION 70. A straight line commensurable with the side of the sum of two medial areas is the side of the sum of two medial areas.

Let AB be the side of the sum of two medial areas, and CD commensurable with AB; it is to be proved that CD is also the side of the sum of two medial areas.

For, since AB is the side of the sum of two medial areas, let it be divided into its straight lines at E; therefore AE, EB are straight lines incommensurable in square which make the sum of the squares on them medial, the rectangle contained by them medial, and furthermore the sum of the squares on AE, EB incommensurable with the rectangle AE, EB. {X. 41} Let the same construction be made as before.

We can then prove similarly that CF, FD are also incommensurable in square, the sum of the squares on AE, EB is commensurable with the sum of the squares on CF, FD, and the rectangle AE, EB with the rectangle CF, FD; so that the sum of the squares on CF, FD is also medial, the rectangle CF, FD is medial, and moreover the sum of the squares on CF, FD is

incommensurable with the rectangle CF, FD.

Therefore CD is the side of the sum of two medial areas. Q. E. D.

PROPOSITION 71. If a rational and a medial area be added together, four irrational straight lines arise, namely a binomial or a first bimedial or a major or a side of a rational plus a medial area.

Let. AB be rational, and CD medial; I say that the “side” of the area AD is a binomial or a first bimedial or a major or a side of a rational plus a medial area.

For AB is either greater or less than CD.

First, let it be greater; let a rational straight line EF be set out, let there be applied to EF the rectangle EG equal to AB, producing EH as breadth, and let HI, equal to DC, be applied to EF, producing HK as breadth.

Then, since AB is rational and is equal to EG, therefore EG is also rational.

And it has been applied to EF, producing EH as breadth; therefore EH is rational and commensurable in length with EF. {X. 20} Again, since CD is medial and is equal to HI, therefore HI is also medial.

And it is applied to the rational straight line EF, producing HK as breadth; therefore HK is rational and incommensurable in length with EF {X. 22} And, since CD is medial, while AB is rational, therefore AB is incommensurable with CD, so that EG is also incommensurable with HI.

But, as EG is to HI, so is EH to HK; {VI. 1} therefore EH is also incommensurable in length with HK. {X. 11} And both are rational; therefore EH, HK are rational straight lines commensurable in square only; therefore EK is a binomial straight line, divided at H. {X. 36} And, since AB is greater than

CD, while AB is equal to EG and CD to HI, therefore EG is also greater than HI; therefore EH is also greater than HK.

The square, then, on EH is greater than the square on HK either by the square on a straight line commensurable in length with EH or by the square on a straight line incommensurable with it.

First, let the square on it be greater by the square on a straight line commensurable with itself.

Now the greater straight line HE is commensurable in length with the rational straight line EF set out; therefore EK is a first binomial. {X. Deff. II. 1} But EF is rational; and, if an area be contained by a rational straight line and the first binomial, the side of the square equal to the area is binomial. {X. 54} Therefore the “side” of EI is binomial; so that the “side” of AD is also binomial.

Next, let the square on EH be greater than the square on HK by the square on a straight line incommensurable with EH.

Now the greater straight line EH is commensurable in length with the rational straight line EF set out; therefore EK is a fourth binomial. {X. Deff. II. 4} But EF is rational; and, if an area be contained by a rational straight line and the fourth binomial, the “side” of the area is the irrational straight line called major. {X. 57} Therefore the “side” of the area EI is major; so that the “side” of the area AD is also major.

Next, let AB be less than CD; therefore EG is also less than HI, so that EH is also less than HK.

Now the square on HK is greater than the square on EH either by the square on a straight line commensurable with HK or by the square on a straight line incommensurable with it.

First, let the square on it be greater by the square on a straight line commensurable in length with itself.

Now the lesser straight line EH is commensurable in length with the rational straight line EF set out; therefore EK is a second binomial. {X. Deff. II. 2} But EF is rational, and, if an area be contained by a rational straight line and the second binomial, the side of the square equal to it is a first bimedial; {X. 55} therefore the “side” of the area EI is a first bimedial, so that the “side” of AD is also a first bimedial.

Next, let the square on HK be greater than the square on HE by the square on a straight line incommensurable with HK.

Now the lesser straight line EH is commensurable with the rational straight line EF set out; therefore EK is a fifth binomial. {X. Deff. II. 5} But EF is rational; and, if an area be contained by a rational straight line and the fifth binomial, the side of the square equal to the area is a side of a rational plus a medial area. {X. 58} Therefore the “side” of the area EI is a side of a rational plus a medial area, so that the “side” of the area AD is also a side of a rational plus a medial area.

Therefore etc. Q. E. D.

PROPOSITION 72. If two medial areas incommensurable with one another be added together, the remaining two irrational straight lines arise, namely either a second bimedial or a side of the sum of two medial areas.

For let two medial areas AB, CD incommensurable with one another be added together; I say that the “side” of the area AD is either a second bimedial or a side of the sum of two medial areas.

For AB is either greater or less than CD.

First, if it so chance, let AB be greater than CD.

Let the rational straight line EF be set out, and to EF let there be applied the rectangle EG equal to AB and producing EH as breadth, and the rectangle HI equal to CD and producing HK as

breadth.

Now, since each of the areas AB, CD is medial, therefore each of the areas EG, HI is also medial.

And they are applied to the rational straight line FE, producing EH, HK as breadth; therefore each of the straight lines EH, HK is rational and incommensurable in length with EF. {X. 22} And, since AB is incommensurable with CD, and AB is equal to EG, and CD to HI, therefore EG is also incommensurable with HI.

But, as EG is to HI, so is EH to HK; {VI. 1} therefore EH is incommensurable in length with HK. {X. 11} Therefore EH, HK are rational straight lines commensurable in square only; therefore EK is binomial. {X. 36} But the square on EH is greater than the square on HK either by the square on a straight line commensurable with EH or by the square on a straight line incommensurable with it.

First, let the square on it be greater by the square on a straight line commensurable in length with itself.

Now neither of the straight lines EH, HK is commensurable in length with the rational straight line EF set out; therefore EK is a third binomial. {X. Deff. II. 3} But EF is rational; and, if an area be contained by a rational straight line and the third binomial, the “side” of the area is a second binomial; {X. 56} therefore the “side” of EI, that is, of AD, is a second binomial.

Next, let the square on EH be greater than the square on HK by the square on a straight line incommensurable in length with EH.

Now each of the straight lines EH, HK is incommensurable in length with EF; therefore EK is a sixth binomial. {X. Deff. II. 6} But, if an area be contained by a rational straight line and the sixth binomial, the “side” of the area is the side of the sum of two medial areas; {X. 59} so that the “side” of the area AD is also

the side of the sum of two medial areas.

Therefore etc. Q. E. D.

PROPOSITION 73. If from a rational straight line there be subtracted a rational straight line commensurable with the whole in square only, the remainder is irrational; and let it be called an apotome.

For from the rational straight line AB let the rational straight line BC, commensurable with the whole in square only, be subtracted; I say that the remainder AC is the irrational straight line called apotome.

For, since AB is incommensurable in length with BC, and, as AB is to BC, so is the square on AB to the rectangle AB, BC, therefore the square on AB is incommensurable with the rectangle AB, BC. {X. 11} But the squares on AB, BC are commensurable with the square on AB, {X. 15} and twice the rectangle AB, BC is commensurable with the rectangle AB, BC. {X. 6} And, inasmuch as the squares on AB, BC are equal to twice the rectangle AB, BC together with the square on CA, {II. 7} therefore the squares on AB, BC are also incommensurable with the remainder, the square on AC. {X. 13, 16} But the squares on AB, BC are rational; therefore AC is irrational. {X. Def. 4} And let it be called an apotome. Q. E. D.

PROPOSITION 74. If from a medial straight line there be subtracted a medial straight line which is commensurable with the whole in square only, and which contains with the whole a rational rectangle, the remainder is irrational. And let it be called a first apotome of a medial straight line.

For from the medial straight line AB let there be subtracted the medial straight line BC which is commensurable with AB in square only and with AB makes the rectangle AB, BC rational; I say that the remainder AC is irrational; and let it be called a first

apotome of a medial straight line.

For, since AB, BC are medial, the squares on AB, BC are also medial.

But twice the rectangle AB, BC is rational; therefore the squares on AB, BC are incommensurable with twice the rectangle AB, BC; therefore twice the rectangle AB, BC is also incommensurable with the remainder, the square on AC, {Cf. II. 7} since, if the whole is incommensurable with one of the magnitudes, the original magnitudes will also be incommensurable. {X. 16} But twice the rectangle AB, BC is rational; therefore the square on AC is irrational; therefore AC is irrational. {X. Def. 4} And let it be called a first apotome of a medial straight line.

PROPOSITION 75. If from a medial straight line there be subtracted a medial straight line which is commensurable with the whole in square only, and which contains with the whole a medial rectangle, the remainder is irrational; and let it be called a second apotome of a medial straight line.

For from the medial straight line AB let there be subtracted the medial straight line CB which is commensurable with the whole AB in square only and such that the rectangle AB, BC, which it contains with the whole AB, is medial; {X. 28} I say that the remainder AC is irrational; and let it be called a second apotome of a medial straight line.

For let a rational straight line DI be set out, let DE equal to the squares on AB, BC be applied to DI, producing DG as breadth, and let DH equal to twice the rectangle AB, BC be applied to DI, producing DF as breadth; therefore the remainder FE is equal to the square on AC. {II. 7} Now, since the squares on AB, BC are medial and commensurable, therefore DE is also medial. {X. 15 and 23, Por.} And it is applied to the rational straight line DI, producing DG as breadth; therefore DG is rational and

incommensurable in length with DI. {X. 22} Again, since the rectangle AB, BC is medial, therefore twice the rectangle AB, BC is also medial. {X. 23, Por.} And it is equal to DH; therefore DH is also medial.

And it has been applied to the rational straight line DI, producing DF as breadth; therefore DF is rational and incommensurable in length with DI. {X. 22} And, since AB, BC are commensurable in square only, therefore AB is incommensurable in length with BC; therefore the square on AB is also incommensurable with the rectangle AB, BC. {X. 11} But the squares on AB, BC are commensurable with the square on AB, {X. 15} and twice the rectangle AB, BC is commensurable with the rectangle AB, BC; {X. 6} therefore twice the rectangle AB, BC is incommensurable with the squares on AB, BC. {X. 13} But DE is equal to the squares on AB, BC, and DH to twice the rectangle AB, BC; therefore DE is incommensurable with DH.

But, as DE is to DH, so is GD to DF; {VI. 1} therefore GD is incommensurable with DF. {X. 11} And both are rational; therefore GD, DF are rational straight lines commensurable in square only; therefore FG is an apotome. {X. 73} But DI is rational, and the rectangle contained by a rational and an irrational straight line is irrational, {deduction from X. 20} and its 'side' is irrational.

And AC is the 'side' of FE; therefore AC is irrational.

And let it be called a second apotome of a medial straight line. Q. E. D.

PROPOSITION 76. If from a straight line there be subtracted a straight line which is incommensurable in square with the whole and which with the whole makes the squares on them added together rational, but the rectangle contained by them medial, the remainder is irrational; and let it be called minor.

For from the straight line AB let there be subtracted the straight line BC which is incommensurable in square with the whole and fulfils the given conditions. {X. 33} I say that the remainder AC is the irrational straight line called minor.

For, since the sum of the squares on AB, BC is rational, while twice the rectangle AB, BC is medial, therefore the squares on AB, BC are incommensurable with twice the rectangle AB, BC; and, convertendo, the squares on AB, BC are incommensurable with the remainder, the square on AC. {II. 7, X. 16} But the squares on AB, BC are rational; therefore the square on AC is irrational; therefore AC is irrational.

And let it be called minor.

PROPOSITION 77. If from a straight line there be subtracted a straight line which is incommensurable in square with the whole, and which with the whole makes the sum of the squares on them medial, but twice the rectangle contained by them rational, the remainder is irrational: and let it be called that which produces with a rational area a medial whole.

For from the straight line AB let there be subtracted the straight line BC which is incommensurable in square with AB and fulfils the given conditions; {X. 34} I say that the remainder AC is the irrational straight line aforesaid.

For, since the sum of the squares on AB, BC is medial, while twice the rectangle AB, BC is rational, therefore the squares on AB, BC are incommensurable with twice the rectangle AB, BC; therefore the remainder also, the square on AC, is incommensurable with twice the rectangle AB, BC. {II. 7, X. 16} And twice the rectangle AB, BC is rational; therefore the square on AC is irrational; therefore AC is irrational.

And let it be called that which produces with a rational area a medial whole. Q. E. D.

PROPOSITION 78. If from a straight line there be subtracted a straight line which is incommensurable in square with the whole and which with the whole makes the sum of the squares on them medial, twice the rectangle contained by them medial, and further the squares on them incommensurable with twice the rectangle contained by them, the remainder is irrational; and let it be called that which produces with a medial area a medial whole.

For from the straight line AB let there be subtracted the straight line BC incommensurable in square with AB and fulfilling the given conditions; {X. 35} I say that the remainder AC is the irrational straight line called that which produces with a medial area a medial whole.

For let a rational straight line DI be set out, to DI let there be applied DE equal to the squares on AB, BC, producing DG as breadth, and let DH equal to twice the rectangle AB, BC be subtracted.

Therefore the remainder FE is equal to the square on AC, {II. 7} so that AC is the “side” of FE.

Now, since the sum of the squares on AB, BC is medial and is equal to DE, therefore DE is medial.

And it is applied to the rational straight line DI, producing DG as breadth; therefore DG is rational and incommensurable in length with DI. {X. 22} Again, since twice the rectangle AB, BC is medial and is equal to DH, therefore DH is medial.

And it is applied to the rational straight line DI, producing DF as breadth; therefore DF is also rational and incommensurable in length with DI. {X. 22} And, since the squares on AB, BC are incommensurable with twice the rectangle AB, BC, therefore DE is also incommensurable with DH.

But, as DE is to DH, so also is DG to DF; {VI. 1} therefore DG is incommensurable with DF. {X. 11} And both are rational; therefore GD, DF are rational straight lines commensurable in square only.

Therefore FG is an apotome. {X. 73} And FH is rational; but the rectangle contained by a rational straight line and an apotome is irrational, {deduction from X. 20} and its “side” is irrational.

And AC is the “side” of FE; therefore AC is irrational.

And let it be called that which produces with a medial area a medial whole. Q. E. D.

PROPOSITION 79. To an apotome only one rational straight line can be annexed which is commensurable with the whole in square only.

Let AB be an apotome, and BC an annex to it; therefore AC, CB are rational straight lines commensurable in square only. {X. 73} I say that no other rational straight line can be annexed to AB which is commensurable with the whole in square only.

For, if possible, let BD be so annexed; therefore AD, DB are also rational straight lines commensurable in square only. {X. 73} Now, since the excess of the squares on AD, DB over twice the rectangle AD, DB is also the excess of the squares on AC, CB over twice the rectangle AC, CB, for both exceed by the same, the square on AB, {II. 7} therefore, alternately, the excess of the squares on AD, DB over the squares on AC, CB is the excess of twice the rectangle AD, DB over twice the rectangle AC, CB.

But the squares on AD, DB exceed the squares on AC, CB by a rational area, for both are rational; therefore twice the rectangle AD, DB also exceeds twice the rectangle AC, CB by a rational area: which is impossible, for both are medial {X. 21}, and a medial area does not exceed a medial by a rational area. {X. 26} Therefore no other rational straight line can be annexed

to AB which is commensurable with the whole in square only.

Therefore only one rational straight line can be annexed to an apotome which is commensurable with the whole in square only. Q. E. D.

PROPOSITION 80. To a first apotome of a medial straight line only one medial straight line can be annexed which is commensurable with the whole in square only and which contains with the whole a rational rectangle.

For let AB be a first apotome of a medial straight line, and let BC be an annex to AB; therefore AC, CB are medial straight lines commensurable in square only and such that the rectangle AC, CB which they contain is rational; {X. 74} I say that no other medial straight line can be annexed to AB which is commensurable with the whole in square only and which contains with the whole a rational area.

For, if possible, let DB also be so annexed; therefore AD, DB are medial straight lines commensurable in square only and such that the rectangle AD, DB which they contain is rational. {X. 74} Now, since the excess of the squares on AD, DB over twice the rectangle AD, DB is also the excess of the squares on AC, CB over twice the rectangle AC, CB, for they exceed by the same, the square on AB, {II. 7} therefore, alternately, the excess of the squares on AD, DB over the squares on AC, CB is also the excess of twice the rectangle AD, DB over twice the rectangle AC, CB.

But twice the rectangle AD, DB exceeds twice the rectangle AC, CB by a rational area, for both are rational.

Therefore the squares on AD, DB also exceed the squares on AC, CB by a rational area. which is impossible, for both are medial {X. 15 and 23, Por.}, and a medial area does not exceed a medial by a rational area. {X. 26} Therefore etc. Q. E. D.

PROPOSITION 81. To a second apotome of a medial straight line only one medial straight line can be annexed which is commensurable with the whole in square only and which contains with the whole a medial rectangle.

Let AB be a second apotome of a medial straight line and BC an annex to AB; therefore AC, CB are medial straight lines commensurable in square only and such that the rectangle AC, CB which they contain is medial. {X. 75} I say that no other medial straight line can be annexed to AB which is commensurable with the whole in square only and which contains with the whole a medial rectangle.

For, if possible, let BD also be so annexed; therefore AD, DB are also medial straight lines commensurable in square only and such that the rectangle AD, DB which they contain is medial. {X. 75} Let a rational straight line EF be set out, let EG equal to the squares on AC, CB be applied to EF, producing EM as breadth, and let HG equal to twice the rectangle AC, CB be subtracted, producing HM as breadth; therefore the remainder EL is equal to the square on AB, {II. 7} so that AB is the “side” of EL.

Again, let EI equal to the squares on AD, DB be applied to EF, producing EN as breadth.

But EL is also equal to the square on AB; therefore the remainder HI is equal to twice the rectangle AD, DB. {II. 7} Now, since AC, CB are medial straight lines, therefore the squares on AC, CB are also medial.

And they are equal to EG; therefore EG is also medial. {X. 15 and 23, Por.} And it is applied to the rational straight line EF, producing EM as breadth; therefore EM is rational and incommensurable in length with EF. {X. 22} Again, since the rectangle AC, CB is medial, twice the rectangle AC, CB is also medial. {X. 23, Por.} And it is equal to HG; therefore HG is also medial.

And it is applied to the rational straight line EF, producing HM as breadth; therefore HM is also rational and incommensurable in length with EF. {X. 22} And, since AC, CB are commensurable in square only, therefore AC is incommensurable in length with CB.

But, as AC is to CB, so is the square on AC to the rectangle AC, CB; therefore the square on AC is incommensurable with the rectangle AC, CB. {X. 11} But the squares on AC, CB are commensurable with the square on AC, while twice the rectangle AC, CB is commensurable with the rectangle AC, CB; {X. 6} therefore the squares on AC, CB are incommensurable with twice the rectangle AC, CB. {X. 13} And EG is equal to the squares on AC, CB, while GH is equal to twice the rectangle AC, CB; therefore EG is incommensurable with HG.

But, as EG is to HG, so is EM to HM; {VI. 1} therefore EM is incommensurable in length with MH. {X. 11} And both are rational; therefore EM, MH are rational straight lines commensurable in square only; therefore EH is an apotome, and HM an annex to it. {X. 73} Similarly we can prove that HN is also an annex to it; therefore to an apotome different straight lines are annexed which are commensurable with the wholes in square only: which is impossible. {X. 79} Therefore etc. Q. E. D.

PROPOSITION 82. To a minor straight line only one straight line can be annexed which is incommensurable in square with the whole and which makes, with the whole, the sum of the squares on them rational but twice the rectangle contained by them medial.

Let AB be the minor straight line, and let BC be an annex to AB; therefore AC, CB are straight lines incommensurable in square which make the sum of the squares on them rational, but twice the rectangle contained by them medial. {X. 76} I say that no other straight line can be annexed to AB fulfilling the same

conditions.

For, if possible, let BD be so annexed; therefore AD, DB are also straight lines incommensurable in square which fulfil the aforesaid conditions. {X. 76} Now, since the excess of the squares on AD, DB over the squares on AC, CB is also the excess of twice the rectangle AD, DB over twice the rectangle AC, CB, while the squares on AD, DB exceed the squares on AC, CB by a rational area, for both are rational, therefore twice the rectangle AD, DB also exceeds twice the rectangle AC, CB by a rational area: which is impossible, for both are medial. {X. 26} Therefore to a minor straight line only one straight line can be annexed which is incommensurable in square with the whole and which makes the squares on them added together rational, but twice the rectangle contained by them medial. Q. E. D.

PROPOSITION 83. To a straight line which produces with a rational area a medial whole only one straight line can be annexed which is incommensurable in square with the whole straight line and which with the whole straight line makes the sum of the squares on them medial, but twice the rectangle contained by them rational.

Let AB be the straight line which produces with a rational area a medial whole, and let BC be an annex to AB; therefore AC, CB are straight lines incommensurable in square which fulfil the given conditions. {X. 77} I say that no other straight line can be annexed to AB which fulfils the same conditions.

For, if possible, let BD be so annexed; therefore AD, DB are also straight lines incommensurable in square which fulfil the given conditions. {X. 77} Since then, as in the preceding cases, the excess of the squares on AD, DB over the squares on AC, CB is also the excess of twice the rectangle AD, DB over twice the rectangle AC, CB, while twice the rectangle AD, DB exceeds twice the rectangle AC, CB by a rational area, for both are

rational, therefore the squares on AD, DB also exceed the squares on AC, CB by a rational area: which is impossible, for both are medial. {X. 26} Therefore no other straight line can be annexed to AB which is incommensurable in square with the whole and which with the whole fulfils the aforesaid conditions; therefore only one straight line can be so annexed. Q. E. D.

PROPOSITION 84. To a straight line which produces with a medial area a medial whole only one straight line can be annexed which is incommensurable in square with the whole straight line and which with the whole straight line makes the sum of the squares on them medial and twice the rectangle contained by them both medial and also incommensurable with the sum of the squares on them.

Let AB be the straight line which produces with a medial area a medial whole, and BC an annex to it; therefore AC, CB are straight lines incommensurable in square which fulfil the aforesaid conditions. {X. 78} I say that no other straight line can be annexed to AB which fulfils the aforesaid conditions.

For, if possible, let BD be so annexed, so that AD, DB are also straight lines incommensurable in square which make the squares on AD, DB added together medial, twice the rectangle AD, DB medial, and also the squares on AD, DB incommensurable with twice the rectangle AD, DB. {X. 78} Let a rational straight line EF be set out, let EG equal to the squares on AC, CB be applied to EF, producing EM as breadth, and let HG equal to twice the rectangle AC, CB be applied to EF, producing HM as breadth; therefore the remainder, the square on AB {II. 7}, is equal to EL; therefore AB is the “side” of EL.

Again, let EI equal to the squares on AD, DB be applied to EF, producing EN as breadth.

But the square on AB is also equal to EL; therefore the remainder, twice the rectangle AD, DB {II. 7}, is equal to HI.

Now, since the sum of the squares on AC, CB is medial and is equal to EG, therefore EG is also medial.

And it is applied to the rational straight line EF, producing EM as breadth; therefore EM is rational and incommensurable in length with EF. {X. 22} Again, since twice the rectangle AC, CB is medial and is equal to HG, therefore HG is also medial.

And it is applied to the rational straight line EF, producing HM as breadth; therefore HM is rational and incommensurable in length with EF. {X. 22} And, since the squares on AC, CB are incommensurable with twice the rectangle AC, CB, EG is also incommensurable with HG; therefore EM is also incommensurable in length with MH. {VI. 1, X. 11} And both are rational; therefore EM, MH are rational straight lines commensurable in square only; therefore EH is an apotome, and HM an annex to it. {X. 73} Similarly we can prove that EH is again an apotome and HN an annex to it.

Therefore to an apotome different rational straight lines are annexed which are commensurable with the wholes in square only: which was proved impossible. {X. 79} Therefore no other straight line can be so annexed to AB.

Therefore to AB only one straight line can be annexed which is incommensurable in square with the whole and which with the whole makes the squares on them added together medial, twice the rectangle contained by them medial, and also the squares on them incommensurable with twice the rectangle contained by them. Q. E. D.

DEFINITIONS III.

1. GIVEN A RATIONAL straight line and an apotome, if the square on the whole be greater than the square on the annex by the square on a straight line commensurable in length with the whole, and the whole be commensurable in length with the

rational straight line set out, let the apotome be called a first apotome.

2. But if the annex be commensurable in length with the rational straight line set out, and the square on the whole be greater than that on the annex by the square on a straight line commensurable with the whole, let the apotome be called a second apotome.

3. But if neither be commensurable in length with the rational straight line set out, and the square on the whole be greater than the square on the annex by the square on a straight line commensurable with the whole, let the apotome be called a third apotome.

4. Again, if the square on the whole be greater than the square on the annex by the square on a straight line incommensurable with the whole, then, if the whole be commensurable in length with the rational straight line set out, let the apotome be called a fourth apotome; 5. if the annex be so commensurable, a fifth; 6. and, if neither, a sixth.

PROPOSITIONS 85-115.

PROPOSITION 85. TO FIND THE first apotome.

Let a rational straight line A be set out, and let BG be commensurable in length with A ; therefore BG is also rational.

Let two square numbers DE , EF be set out, and let their difference FD not be square; therefore neither has ED to DF the ratio which a square number has to a square number.

Let it be contrived that, as ED is to DF , so is the square on BG to the square on GC ; {X. 6, Por.} therefore the square on BG is commensurable with the square on GC . {X. 6} But the square on BG is rational; therefore the square on GC is also rational; therefore GC is also rational.

And, since ED has not to DF the ratio which a square number has to a square number, therefore neither has the square on BG to the square on GC the ratio which a square number has to a square number; therefore BG is incommensurable in length with GC. {X. 9} And both are rational; therefore BG, GC are rational straight lines commensurable in square only; therefore BC is an apotome. {X. 73} I say next that it is also a first apotome.

For let the square on H be that by which the square on BG is greater than the square on GC.

Now since, as ED is to FD, so is the square on BG to the square on GC, therefore also, convertendo, {v. 19, Por.} as DE is to EF, so is the square on GB to the square on H.

But DE has to EF the ratio which a square number has to a square number, for each is square; therefore the square on GB also has to the square on H the ratio which a square number has to a square number; therefore BG is commensurable in length with H. {X. 9} And the square on BG is greater than the square on GC by the square on a straight line commensurable in length with BG.

And the whole BG is commensurable in length with the rational straight line A set out.

Therefore BC is a first apotome. {X. Deff. III. 1} Therefore the first apotome BC has been found.

(Being) that which it was required to find.

PROPOSITION 86. To find the second apotome.

Let a rational straight line A be set out, and GC commensurable in length with A; therefore GC is rational.

Let two square numbers DE, EF be set out, and let their difference DF not be square.

Now let it be contrived that, as FD is to DE, so is the square on CG to the square on GB. {X. 6, Por.} Therefore the square on CG is commensurable with the square on GB. {X. 6} But the square on CG is rational; therefore the square on GB is also rational; therefore BG is rational.

And, since the square on GC has not to the square on GB the ratio which a square number has to a square number, CG is incommensurable in length with GB. {X. 9} And both are rational; therefore CG, GB are rational straight lines commensurable in square only; therefore BC is an apotome. {X. 73} I say next that it is also a second apotome.

For let the square on H be that by which the square on BG is greater than the square on GC.

Since then, as the square on BG is to the square on GC, so is the number ED to the number DF, therefore, convertendo, as the square on BG is to the square on H, so is DE to EF. {V. 19, Por.} And each of the numbers DE, EF is square; therefore the square on BG has to the square on H the ratio which a square number has to a square number; therefore BG is commensurable in length with H. {X. 9} And the square on BG is greater than the square on GC by the square on H; therefore the square on BG is greater than the square on GC by the square on a straight line commensurable in length with BG.

And CG, the annex, is commensurable with the rational straight line A set out.

Therefore BC is a second apotome. {X. Deff. III. 2} Therefore the second apotome BC has been found. Q. E. D.

PROPOSITION 87. To find the third apotome.

Let a rational straight line A be set out, let three numbers E, BC, CD be set out which have not to one another the ratio which a square number has to a square number, but let CB have to BD

the ratio which a square number has to a square number.

Let it be contrived that, as E is to BC, so is the square on A to the square on FG, and, as BC is to CD, so is the square on FG to the square on GH. {X. 6, Por.} Since then, as E is to BC, so is the square on A to the square on FG, therefore the square on A is commensurable with the square on FG. {X. 6} But the square on A is rational; therefore the square on FG is also rational; therefore FG is rational.

And, since E has not to BC the ratio which a square number has to a square number, therefore neither has the square on A to the square on FG the ratio which a square number has to a square number; therefore A is incommensurable in length with FG. {X. 9} Again, since, as BC is to CD, so is the square on FG to the square on GH, therefore the square on FG is commensurable with the square on GH. {X. 6} But the square on FG is rational; therefore the square on GH is also rational; therefore GH is rational.

And, since BC has not to CD the ratio which a square number has to a square number, therefore neither has the square on FG to the square on GH the ratio which a square number has to a square number; therefore FG is incommensurable in length with GH. {X. 9} And both are rational; therefore FG, GH are rational straight lines commensurable in square only; therefore FH is an apotome. {X. 73} I say next that it is also a third apotome.

For since, as E is to BC, so is the square on A to the square on FG, and, as BC is to CD, so is the square on FG to the square on HG, therefore, ex aequali, as E is to CD, so is the square on A to the square on HG. {V. 22} But E has not to CD the ratio which a square number has to a square number; therefore neither has the square on A to the square on GH the ratio which a square number has to a square number; therefore A is incommensurable in length with GH. {X. 9} Therefore neither of

the straight lines FG, GH is commensurable in length with the rational straight line A set out.

Now let the square on K be that by which the square on FG is greater than the square on GH.

Since then, as BC is to CD, so is the square on FG to the square on GH, therefore, convertendo, as BC is to BD, so is the square on FG to the square on K. {V. 19, Por.} But BC has to BD the ratio which a square number has to a square number; therefore the square on FG also has to the square on K the ratio which a square number has to a square number.

Therefore FG is commensurable in length with K, {X. 9} and the square on FG is greater than the square on GH by the square on a straight line commensurable with FG.

And neither of the straight lines FG, GH is commensurable in length with the rational straight line A set out; therefore FH is a third apotome. {X. Deff. III. 3} Therefore the third apotome FH has been found. Q. E. D.

PROPOSITION 88. To find the fourth apotome.

Let a rational straight line A be set out, and BG commensurable in length with it; therefore BG is also rational.

Let two numbers DF, FE be set out such that the whole DE has not to either of the numbers DF, EF the ratio which a square number has to a square number.

Let it be contrived that, as DE is to EF, so is the square on BG to the square on GC; {X. 6, Por.} therefore the square on BG is commensurable with the square on GC. {X. 6} But the square on BG is rational; therefore the square on GC is also rational; therefore GC is rational.

Now, since DE has not to EF the ratio which a square number has to a square number, therefore neither has the square on BG

to the square on GC the ratio which a square number has to a square number; therefore BG is incommensurable in length with GC. {X. 9} And both are rational; therefore BG, GC are rational straight lines commensurable in square only; therefore BC is an apotome. {X. 73} Now let the square on H be that by which the square on BG is greater than the square on GC.

Since then, as DE is to EF, so is the square on BG to the square on GC, therefore also, convertendo, as ED is to DF, so is the square on GB to the square on H. {v. 19, Por.} But ED has not to DF the ratio which a square number has to a square number; therefore neither has the square on GB to the square on H the ratio which a square number has to a square number; therefore BG is incommensurable in length with H. {X. 9} And the square on BG is greater than the square on GC by the square on H; therefore the square on BG is greater than the square on GC by the square on a straight line incommensurable with BG.

And the whole BG is commensurable in length with the rational straight line A set out.

Therefore BC is a fourth apotome. {X. Deff. III. 4} Therefore the fourth apotome has been found. Q. E. D.

PROPOSITION 89. To find the fifth apotome.

Let a rational straight line A be set out, and let CG be commensurable in length with A; therefore CG is rational.

Let two numbers DF, FE be set out such that DE again has not to either of the numbers DF, FE the ratio which a square number has to a square number; and let it be contrived that, as FE is to ED, so is the square on CG to the square on GB.

Therefore the square on GB is also rational; {X. 6} therefore BG is also rational.

Now since, as DE is to EF, so is the square on BG to the square on GC, while DE has not to EF the ratio which a square number has

to a square number, therefore neither has the square on BG to the square on GC the ratio which a square number has to a square number; therefore BG is incommensurable in length with GC. {X. 9} And both are rational; therefore BG, GC are rational straight lines commensurable in square only; therefore BC is an apotome. {X. 73} I say next that it is also a fifth apotome.

For let the square on H be that by which the square on BG is greater than the square on GC.

Since then, as the square on BG is to the square on GC, so is DE to EF, therefore, convertendo, as ED is to DF, so is the square on BG to the square on H. {V. 19, Por.} But ED has not to DF the ratio which a square number has to a square number; therefore neither has the square on BG to the square on H the ratio which a square number has to a square number; therefore BG is incommensurable in length with H. {X. 9} And the square on BG is greater than the square on GC by the square on H; therefore the square on GB is greater than the square on GC by the square on a straight line incommensurable in length with GB.

And the annex CG is commensurable in length with the rational straight line A set out; therefore BC is a fifth apotome. {X. Deff. III. 5} Therefore the fifth apotome BC has been found. Q. E. D.

PROPOSITION 90. To find the sixth apotome.

Let a rational straight line A be set out, and three numbers E, BC, CD not having to one another the ratio which a square number has to a square number; and further let CB also not have to BD the ratio which a square number has to a square number.

Let it be contrived that, as E is to BC, so is the square on A to the square on FG, and, as BC is to CD, so is the square on FG to the square on GH. {X. 6, Por.} Now since, as E is to BC, so is the

square on A to the square on FG, therefore the square on A is commensurable with the square on FG. {X. 6} But the square on A is rational; therefore the square on FG is also rational; therefore FG is also rational.

And, since E has not to BC the ratio which a square number has to a square number, therefore neither has the square on A to the square on FG the ratio which a square number has to a square number; therefore A is incommensurable in length with FG. {X. 9} Again, since, as BC is to CD, so is the square on FG to the square on GH, therefore the square on FG is commensurable with the square on GH. {X. 6} But the square on FG is rational; therefore the square on GH is also rational; therefore GH is also rational.

And, since BC has not to CD the ratio which a square number has to a square number, therefore neither has the square on FG to the square on GH the ratio which a square number has to a square number; therefore FG is incommensurable in length with GH. {X. 9} And both are rational; therefore FG, GH are rational straight lines commensurable in square only; therefore FH is an apotome. {X. 73} I say next that it is also a sixth apotome.

For since, as E is to BC, so is the square on A to the square on FG, and, as BC is to CD, so is the square on FG to the square on GH, therefore, ex aequali, as E is to CD, so is the square on A to the square on GH. {v. 22} But E has not to CD the ratio which a square number has to a square number; therefore neither has the square on A to the square on GH the ratio which a square number has to a square number; therefore A is incommensurable in length with GH; {X. 9} therefore neither of the straight lines FG, GH is commensurable in length with the rational straight line A.

Now let the square on K be that by which the square on FG is greater than the square on GH.

Since then, as BC is to CD, so is the square on FG to the square on GH, therefore, convertendo, as CB is to BD, so is the square on FG to the square on K. {v. 19, Por.} But CB has not to BD the ratio which a square number has to a square number; therefore neither has the square on FG to the square on K the ratio which a square number has to a square number; therefore FG is incommensurable in length with K. {X. 9} And the square on FG is greater than the square on GH by the square on K; therefore the square on FG is greater than the square on GH by the square on a straight line incommensurable in length with FG.

And neither of the straight lines FG, GH is commensurable with the rational straight line A set out.

Therefore FH is a sixth apotome. {X. Deff. III. 6} Therefore the sixth apotome FH has been found. Q. E. D.

PROPOSITION 91. If an area be contained by a rational straight line and a first apotome, the “side” of the area is an apotome.

For let the area AB be contained by the rational straight line AC and the first apotome AD; I say that the “side” of the area AB is an apotome.

For, since AD is a first apotome, let DG be its annex; therefore AG, GD are rational straight lines commensurable in square only. {X. 73} And the whole AG is commensurable with the rational straight line AC set out, and the square on AG is greater than the square on GD by the square on a straight line commensurable in length with AG; {X. Deff. III. 1} if therefore there be applied to AG a parallelogram equal to the fourth part of the square on DG and deficient by a square figure, it divides it into commensurable parts. {X. 17} Let DG be bisected at E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG; therefore AF is commensurable with FG.

And through the points E, F, G let EH, FI, GK be drawn parallel to AC.

Now, since AF is commensurable in length with FG, therefore AG is also commensurable in length with each of the straight lines AF, FG. {X. 15} But AG is commensurable with AC; therefore each of the straight lines AF, FG is commensurable in length with AC. {X. 12} And AC is rational; therefore each of the straight lines AF, FG is also rational, so that each of the rectangles AI, FK is also rational. {X. 19} Now, since DE is commensurable in length with EG, therefore DG is also commensurable in length with each of the straight lines DE, EG. {X. 15} But DG is rational and incommensurable in length with AC; therefore each of the straight lines DE, EG is also rational and incommensurable in length with AC; {X. 13} therefore each of the rectangles DH, EK is medial. {X. 21} Now let the square LM be made equal to AI, and let there be subtracted the square NO having a common angle with it, the angle LPM, and equal to FK; therefore the squares LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Since then the rectangle contained by AF, FG is equal to the square on EG, therefore, as AF is to EG, so is EG to FG. {VI. 17} But, as AF is to EG, so is AI to EK, and, as EG is to FG, so is EK to KF; {VI. 1} therefore EK is a mean proportional between AI, KF. {V. 11} But MN is also a mean proportional between LM, NO, as was before proved, {Lemma after X. 53} and AI is equal to the square LM, and KF to NO; therefore MN is also equal to EK.

But EK is equal to DH, and MN to LO; therefore DK is equal to the gnomon UVW and NO.

But AK is also equal to the squares LM, NO; therefore the remainder AB is equal to ST.

But ST is the square on LN; therefore the square on LN is equal to AB; therefore LN is the “side” of AB.

I say next that LN is an apotome.

For, since each of the rectangles AI, FK is rational, and they are equal to LM, NO, therefore each of the squares LM, NO, that is, the squares on LP, PN respectively, is also rational; therefore each of the straight lines LP, PN is also rational.

Again, since DH is medial and is equal to LO, therefore LO is also medial.

Since then LO is medial, while NO is rational, therefore LO is incommensurable with NO.

But, as LO is to NO, so is LP to PN; {VI. 1} therefore LP is incommensurable in length with PN. {X. 11} And both are rational; therefore LP, PN are rational straight lines commensurable in square only; therefore LN is an apotome. {X. 73} And it is the “side” of the area AB; therefore the “side” of the area AB is an apotome.

Therefore etc.

PROPOSITION 92. If an area be contained by a rational straight line and a second apotome, the “side” of the area is a first apotome of a medial straight line.

For let the area AB be contained by the rational straight line AC and the second apotome AD; I say that the “side” of the area AB is a first apotome of a medial straight line.

For let DG be the annex to AD; therefore AG, GD are rational straight lines commensurable in square only, {X. 73} and the annex DG is commensurable with the rational straight line AC set out, while the square on the whole AG is greater than the square on the annex GD by the square on a straight line commensurable in length with AG. {X. Deff. III. 2} Since then the square on AG is greater than the square on GD by the square on a straight line commensurable with AG, therefore, if there be applied to AG a parallelogram equal to the fourth part of the

square on GD and deficient by a square figure, it divides it into commensurable parts. {X. 17} Let then DG be bisected at E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG; therefore AF is commensurable in length with FG.

Therefore AG is also commensurable in length with each of the straight lines AF, FG. {X. 15} But AG is rational and incommensurable in length with AC; therefore each of the straight lines AF, FG is also rational and incommensurable in length with AC; {X. 13} therefore each of the rectangles AI, FK is medial. {X. 21} Again, since DE is commensurable with EG, therefore DG is also commensurable with each of the straight lines DE, EG. {X. 15} But DG is commensurable in length with AC.

Therefore each of the rectangles DH, EK is rational. {X. 19} Let then the square LM be constructed equal to AI, and let there be subtracted NO equal to FK and being about the same angle with LM, namely the angle LPM; therefore the squares LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Since then AI, FK are medial and are equal to the squares on LP, PN, the squares on LP, PN are also medial; therefore LP, PN are also medial straight lines commensurable in square only.

And, since the rectangle AF, FG is equal to the square on EG, therefore, as AF is to EG, so is EG to FG, {VI. 17} while, as AF is to EG, so is AI to EK, and, as EG is to FG, so is EK to FK; {VI. 1} therefore EK is a mean proportional between AI, FK. {V. 11} But MN is also a mean proportional between the squares LM, NO, and AI is equal to LM, and FK to NO; therefore MN is also equal to EK.

But DH is equal to EK, and LO equal to MN; therefore the whole DK is equal to the gnomon UVW and NO.

Since then the whole AK is equal to LM, NO, and, in these, DK is equal to the gnomon UVW and NO, therefore the remainder AB is equal to TS.

But TS is the square on LN; therefore the square on LN is equal to the area AB; therefore LN is the “side” of the area AB.

I say that LN is a first apotome of a medial straight line.

For, since EK is rational and is equal to LO, therefore LO, that is, the rectangle LP, PN, is rational.

But NO was proved medial; therefore LO is incommensurable with NO.

But, as LO is to NO, so is LP to PN; {VI. 1} therefore LP, PN are incommensurable in length. {X. 11} Therefore LP, PN are medial straight lines commensurable in square only which contain a rational rectangle; therefore LN is a first apotome of a medial straight line. {X. 74} And it is the “side” of the area AB.

Therefore the “side” of the area AB is a first apotome of a medial straight line. Q. E. D.

PROPOSITION 93. If an area be contained by a rational straight line and a third apotome, the “side” of the area is a second apotome of a medial straight line.

For let the area AB be contained by the rational straight line AC and the third apotome AD; I say that the “side” of the area AB is a second apotome of a medial straight line.

For let DG be the annex to AD; therefore AG, GD are rational straight lines commensurable in square only, and neither of the straight lines AG, GD is commensurable in length with the rational straight line AC set out, while the square on the whole AG is greater than the square on the annex DG by the square on a straight line commensurable with AG. {X. Deff. III. 3} Since then the square on AG is greater than the square on GD by the

square on a straight line commensurable with AG, therefore, if there be applied to AG a parallelogram equal to the fourth part of the square on DG and deficient by a square figure, it will divide it into commensurable parts. {X. 17} Let then DG be bisected at E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG.

Let EH, FI, GK be drawn through the points E, F, G parallel to AC.

Therefore AF, FG are commensurable; therefore AI is also commensurable with FK. {VI. 1, X. 11} And, since AF, FG are commensurable in length, therefore AG is also commensurable in length with each of the straight lines AF, FG. {X. 15} But AG is rational and incommensurable in length with AC; so that AF, FG are so also. {X. 13} Therefore each of the rectangles AI, FK is medial. {X. 21} Again, since DE is commensurable in length with EG, therefore DG is also commensurable in length with each of the straight lines DE, EG. {X. 15} But GD is rational and incommensurable in length with AC; therefore each of the straight lines DE, EG is also rational and incommensurable in length with AC; {X. 13} therefore each of the rectangles DH, EK is medial. {X. 21} And, since AG, GD are commensurable in square only, therefore AG is incommensurable in length with GD.

But AG is commensurable in length with AF, and DG with EG; therefore AF is incommensurable in length with EG. {X. 13} But, as AF is to EG, so is AI to EK; {VI. 1} therefore AI is incommensurable with EK. {X. 11} Now let the square LM be constructed equal to AI, and let there be subtracted NO equal to FK and being about the same angle with LM; therefore LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Now, since the rectangle AF, FG is equal to the square on EG, therefore, as AF is to EG, so is EG to FG. {VI. 17} But, as AF is to EG, so is AI to EK, and, as EG is to FG, so is EK to FK; {VI. 1} therefore also, as AI is to EK, so is EK to FK; {V. 11} therefore EK is a mean proportional between AI, FK.

But MN is also a mean proportional between the squares LM, NO, and AI is equal to LM, and FK to NO; therefore EK is also equal to MN.

But MN is equal to LO, and EK equal to DH; therefore the whole DK is also equal to the gnomon UVW and NO.

But AK is also equal to LM, NO; therefore the remainder AB is equal to ST, that is, to the square on LN; therefore LN is the “side” of the area AB.

I say that LN is a second apotome of a medial straight line.

For, since AI, FK were proved medial, and are equal to the squares on LP, PN, therefore each of the squares on LP, PN is also medial; therefore each of the straight lines LP, PN is medial.

And, since AI is commensurable with FK, {VI. 1, X. 11} therefore the square on LP is also commensurable with the square on PN.

Again, since AI was proved incommensurable with EK, therefore LM is also incommensurable with MN, that is, the square on LP with the rectangle LP, PN; so that LP is also incommensurable in length with PN; {VI. 1, X. 11} therefore LP, PN are medial straight lines commensurable in square only.

I say next that they also contain a medial rectangle.

For, since EK was proved medial, and is equal to the rectangle LP, PN, therefore the rectangle LP, PN is also medial, so that LP, PN are medial straight lines commensurable in square only which contain a medial rectangle.

Therefore LN is a second apotome of a medial straight line; {X. 75} and it is the “side” of the area AB.

Therefore the “side” of the area AB is a second apotome of a medial straight line. Q. E. D.

PROPOSITION 94. If an area be contained by a rational straight line and a fourth apotome, the “side” of the area is minor.

For let the area AB be contained by the rational straight line AC and the fourth apotome AD; I say that the “side” of the area AB is minor.

For let DG be the annex to AD; therefore AG, GD are rational straight lines commensurable in square only, AG is commensurable in length with the rational straight line AC set out, and the square on the whole AG is greater than the square on the annex DG by the square on a straight line incommensurable in length with AG, {X. Deff. III. 4} Since then the square on AG is greater than the square on GD by the square on a straight line incommensurable in length with AG, therefore, if there be applied to AG a parallelogram equal to the fourth part of the square on DG and deficient by a square figure, it will divide it into incommensurable parts. {X. 18} Let then DG be bisected at E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG; therefore AF is incommensurable in length with FG.

Let EH, FI, GK be drawn through E, F, G parallel to AC, BD.

Since then AG is rational and commensurable in length with AC, therefore the whole AK is rational. {X. 19} Again, since DG is incommensurable in length with AC, and both are rational, therefore DK is medial. {X. 21} Again, since AF is incommensurable in length with FG, therefore AI is also incommensurable with FK. {VI. 1, X. 11} Now let the square LM

be constructed equal to AI, and let there be subtracted NO equal to FK and about the same angle, the angle LPM.

Therefore the squares LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Since then the rectangle AF, FG is equal to the square on EG, therefore, proportionally, as AF is to EG, so is EG to FG. {VI. 17} But, as AF is to EG, so is AI to EK, and, as EG is to FG, so is EK to FK; {VI. 1} therefore EK is a mean proportional between AI, FK. {V. 11} But MN is also a mean proportional between the squares LM, NO, and AI is equal to LM, and FK to NO; therefore EK is also equal to MN.

But DH is equal to EK, and LO is equal to MN; therefore the whole DK is equal to the gnomon UVW and NO.

Since, then, the whole AK is equal to the squares LM, NO, and, in these, DK is equal to the gnomon UVW and the square NO, therefore the remainder AB is equal to ST, that is, to the square on LN; therefore LN is the “side” of the area AB.

I say that LN is the irrational straight line called minor.

For, since AK is rational and is equal to the squares on LP, PN, therefore the sum of the squares on LP, PN is rational.

Again, since DK is medial, and DK is equal to twice the rectangle LP, PN, therefore twice the rectangle LP, PN is medial.

And, since AI was proved incommensurable with FK, therefore the square on LP is also incommensurable with the square on PN.

Therefore LP, PN are straight lines incommensurable in square which make the sum of the squares on them rational, but twice the rectangle contained by them medial.

Therefore LN is the irrational straight line called minor; {X. 76} and it is the “side” of the area AB.

Therefore the “side” of the area AB is minor. Q. E. D.

PROPOSITION 95. If an area be contained by a rational straight line and a fifth apotome, the “side” of the area is a straight line which produces with a rational area a medial whole.

For let the area AB be contained by the rational straight line AC and the fifth apotome AD; I say that the “side” of the area AB is a straight line which produces with a rational area a medial whole.

For let DG be the annex to AD; therefore AG, GD are rational straight lines commensurable in square only, the annex GD is commensurable in length with the rational straight line AC set out, and the square on the whole AG is greater than the square on the annex DG by the square on a straight line incommensurable with AG. {X. Deff. III. 5} Therefore, if there be applied to AG a parallelogram equal to the fourth part of the square on DG and deficient by a square figure, it will divide it into incommensurable parts. {X. 18} Let then DG be bisected at the point E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG; therefore AF is incommensurable in length with FG.

Now, since AG is incommensurable in length with CA, and both are rational, therefore AK is medial. {X. 21} Again, since DG is rational and commensurable in length with AC, DK is rational. {X. 19} Now let the square LM be constructed equal to AI, and let the square NO equal to FK and about the same angle, the angle LPM, be subtracted; therefore the squares LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Similarly then we can prove that LN is the “side” of the area AB.

I say that LN is the straight line which produces with a rational area a medial whole.

For, since AK was proved medial and is equal to the squares on LP, PN, therefore the sum of the squares on LP, PN is medial.

Again, since DK is rational and is equal to twice the rectangle LP, PN, the latter is itself also rational.

And, since AI is incommensurable with FK, therefore the square on LP is also incommensurable with the square on PN; therefore LP, PN are straight lines incommensurable in square which make the sum of the squares on them medial but twice the rectangle contained by them rational.

Therefore the remainder LN is the irrational straight line called that which produces with a rational area a medial whole; {X. 77} and it is the “side” of the area AB.

Therefore the “side” of the area AB is a straight line which produces with a rational area a medial whole. Q. E. D.

PROPOSITION 96. If an area be contained by a rational straight line and a sixth apotome, the “side” of the area is a straight line which produces with a medial area a medial whole.

For let the area AB be contained by the rational straight line AC and the sixth apotome AD; I say that the “side” of the area AB is a straight line which produces with a medial area a medial whole.

For let DG be the annex to AD; therefore AG, GD are rational straight lines commensurable in square only, neither of them is commensurable in length with the rational straight line AC set out, and the square on the whole AG is greater than the square on the annex DG by the square on a straight line incommensurable in length with AG. {X. Deff. III. 6} Since then the square on AG is greater than the square on GD by the square on a straight line incommensurable in length with AG,

therefore, if there be applied to AG a parallelogram equal to the fourth part of the square on DG and deficient by a square figure, it will divide it into incommensurable parts. {X. 18} Let then DG be bisected at E, let there be applied to AG a parallelogram equal to the square on EG and deficient by a square figure, and let it be the rectangle AF, FG; therefore AF is incommensurable in length with FG.

But, as AF is to FG, so is AI to FK. {VI. 1} therefore AI is incommensurable with FK. {X. 11} And, since AG, AC are rational straight lines commensurable in square only, AK is medial. {X. 21} Again, since AC, DG are rational straight lines and incommensurable in length, DK is also medial. {X. 21} Now, since AG, GD are commensurable in square only, therefore AG is incommensurable in length with GD.

But, as AG is to GD, so is AK to KD; {VI. 1} therefore AK is incommensurable with KD. {X. 11} Now let the square LM be constructed equal to AI, and let NO equal to FK, and about the same angle, be subtracted; therefore the squares LM, NO are about the same diameter. {VI. 26} Let PR be their diameter, and let the figure be drawn.

Then in manner similar to the above we can prove that LN is the “side” of the area AB.

I say that LN is a straight line which produces with a medial area a medial whole.

For, since AK was proved medial and is equal to the squares on LP, PN, therefore the sum of the squares on LP, PN is medial.

Again, since DK was proved medial and is equal to twice the rectangle LP, PN, twice the rectangle LP, PN is also medial.

And, since AK was proved incommensurable with DK, the squares on LP, PN are also incommensurable with twice the rectangle LP, PN.

And, since AI is incommensurable with FK, therefore the square on LP is also incommensurable with the square on PN; therefore LP, PN are straight lines incommensurable in square which make the sum of the squares on them medial, twice the rectangle contained by them medial, and further the squares on them incommensurable with twice the rectangle contained by them.

Therefore LN is the irrational straight line called that which produces with a medial area a medial whole; {X. 78} and it is the “side” of the area AB.

Therefore the “side” of the area is a straight line which produces with a medial area a medial whole. Q. E. D.

PROPOSITION 97. The square on an apotome applied to a rational straight line produces as breadth a first apotome.

Let AB be an apotome, and CD rational, and to CD let there be applied CE equal to the square on AB and producing CF as breadth; I say that CF is a first apotome.

For let BG be the annex to AB; therefore AG, GB are rational straight lines commensurable in square only. {X. 73} To CD let there be applied CH equal to the square on AG, and KL equal to the square on BG.

Therefore the whole CL is equal to the squares on AG, GB, and, in these, CE is equal to the square on AB; therefore the remainder FL is equal to twice the rectangle AG, GB. {II. 7} Let FM be bisected at the point N, and let NO be drawn through N parallel to CD; therefore each of the rectangles FO, LN is equal to the rectangle AG, GB.

Now, since the squares on AG, GB are rational, and DM is equal to the squares on AG, GB, therefore DM is rational.

And it has been applied to the rational straight line CD, producing CM as breadth; therefore CM is rational and

commensurable in length with CD. {X. 20} Again, since twice the rectangle AG, GB is medial, and FL is equal to twice the rectangle AG, GB, therefore FL is medial.

And it is applied to the rational straight line CD, producing FM as breadth; therefore FM is rational and incommensurable in length with CD. {X. 22} And, since the squares on AG, GB are rational, while twice the rectangle AG, GB is medial, therefore the squares on AG, GB are incommensurable with twice the rectangle AG, GB.

And CL is equal to the squares on AG, GB, and FL to twice the rectangle AG, GB; therefore DM is incommensurable with FL.

But, as DM is to FL, so is CM to FM; {VI. 1} therefore CM is incommensurable in length with FM. {X. 11} And both are rational; therefore CM, MF are rational straight lines commensurable in square only; therefore CF is an apotome. {X. 73} I say next that it is also a first apotome.

For, since the rectangle AG, GB is a mean proportional between the squares on AG, GB, and CH is equal to the square on AG, KL equal to the square on BG, and NL equal to the rectangle AG, GB, therefore NL is also a mean proportional between CH, KL; therefore, as CH is to NL, so is NL to KL.

But, as CH is to NL, so is CK to NM, and, as NL is to KL, so is NM to KM; {VI. 1} therefore the rectangle CK, KM is equal to the square on NM {VI. 17}, that is, to the fourth part of the square on FM.

And, since the square on AG is commensurable with the square on GB, CH is also commensurable with KL.

But, as CH is to KL, so is CK to KM; {VI. 1} therefore CK is commensurable with KM. {X. 11} Since then CM, MF are two unequal straight lines, and to CM there has been applied the rectangle CK, KM equal to the fourth part of the square on FM

and deficient by a square figure, while CK is commensurable with KM, therefore the square on CM is greater than the square on MF by the square on a straight line commensurable in length with CM. {X. 17} And CM is commensurable in length with the rational straight line CD set out; therefore CF is a first apotome. {X. Deff. III. 1} Therefore etc. Q. E. D.

PROPOSITION 98. The square on a first apotome of a medial straight line applied to a rational straight line produces as breadth a second apotome.

Let AB be a first apotome of a medial straight line and CD a rational straight line, and to CD let there be applied CE equal to the square on AB, producing CF as breadth; I say that CF is a second apotome.

For let BG be the annex to AB; therefore AG, GB are medial straight lines commensurable in square only which contain a rational rectangle. {X. 74} To CD let there be applied CH equal to the square on AG, producing CK as breadth, and KL equal to the square on GB, producing KM as breadth; therefore the whole CL is equal to the squares on AG, GB; therefore CL is also medial. {X. 15 and 23, Por.} And it is applied to the rational straight line CD, producing CM as breadth; therefore CM is rational and incommensurable in length with CD. {X. 22} Now, since CL is equal to the squares on AG, GB, and, in these, the square on AB is equal to CE, therefore the remainder, twice the rectangle AG, GB, is equal to FL. {II. 7} But twice the rectangle AG, GB is rational; therefore FL is rational.

And it is applied to the rational straight line FE, producing FM as breadth; therefore FM is also rational and commensurable in length with CD. {X. 20} Now, since the sum of the squares on AG, GB, that is, CL, is medial, while twice the rectangle AG, GB, that is, FL, is rational, therefore CL is incommensurable with FL.

But, as CL is to FL, so is CM to FM; {VI. 1} therefore CM is incommensurable in length with FM. {X. 11} And both are rational; therefore CM, MF are rational straight lines commensurable in square only; therefore CF is an apotome. {X. 73} I say next that it is also a second apotome.

For let FM be bisected at N, and let NO be drawn through N parallel to CD; therefore each of the rectangles FO, NL is equal to the rectangle AG, GB.

Now, since the rectangle AG, GB is a mean proportional between the squares on AG, GB, and the square on AG is equal to CH, the rectangle AG, GB to NL, and the square on BG to KL, therefore NL is also a mean proportional between CH, KL; therefore, as CH is to NL, so is NL to KL.

But, as CH is to NL, so is CK to NM, and, as NL is to KL, so is NM to MK; {VI. 1} therefore, as CK is to NM, so is NM, so is KM; {V. 11} therefore the rectangle CK, KM is equal to the square on NM {VI. 17}, that is, to the fourth part of the square on FM.

Since the CM, MF are two unequal straight lines, and the rectangle CK, KM equal to the fourth part of the square on MF and deficient by a square figure has been applied to the greater, CM, and divides it into commensurable parts, therefore the square on CM is greater than the square on MF by the square on a straight line commensurable in length with CM. {X. 17} And the annex FM is commensurable in length with the rational straight line CD set out; therefore CF is a second apotome. {X. Deff. III. 2} Therefore etc. Q. E. D.

PROPOSITION 99. The square on a second apotome of a medial straight line applied to a rational straight line produces as breadth a third apotome.

Let AB be a second apotome of a medial straight line, and CD rational, and to CD let there be applied CE equal to the square

on AB, producing CF as breadth; I say that CF is a third apotome.

For let BG be the annex to AB; therefore AG, GB are medial straight lines commensurable in square only which contain a medial rectangle. {X. 75} Let CH equal to the square on AG be applied to CD, producing CK as breadth, and let KL equal to the square on BG be applied to KH, producing KM as breadth; therefore the whole CL is equal to the squares on AG, GB; therefore CL is also medial. {X. 15 and 23, Por.} And it is applied to the rational straight line CD, producing CM as breadth; therefore CM is rational and incommensurable in length with CD. {X. 22} Now, since the whole CL is equal to the squares on AG, GB, and, in these, CE is equal to the square on AB, therefore the remainder LF is equal to twice the rectangle AG, GB. {II. 7} Let then FM be bisected at the point N, and let NO be drawn parallel to CD; therefore each of the rectangles FO, NL is equal to the rectangle AG, GB.

But the rectangle AG, GB is medial; therefore FL is also medial.

And it is applied to the rational straight line EF, producing FM as breadth; therefore FM is also rational and incommensurable in length with CD. {X. 22} And, since AG, GB are commensurable in square only, therefore AG is incommensurable in length with GB; therefore the square on AG is also incommensurable with the rectangle AG, GB. {VI. 1, X. 11} But the squares on AG, GB are commensurable with the square on AG, and twice the rectangle AG, GB with the rectangle AG, GB; therefore the squares on AG, GB are incommensurable with twice the rectangle AG, GB. {X. 13} But CL is equal to the squares on AG, GB, and FL is equal to twice the rectangle AG, GB; therefore CL is also incommensurable with FL.

But, as CL is to FL, so is CM to FM; {VI. 1} therefore CM is incommensurable in length with FM. {X. 11} And both are rational; therefore CM, MF are rational straight lines

commensurable in square only; therefore CF is an apotome. {X. 73} I say next that it is also a third apotome.

For, since the square on AG is commensurable with the square on GB, therefore CH is also commensurable with KL, so that CK is also commensurable with KM. {VI. 1, X. 11} And, since the rectangle AG, GB is a mean proportional between the squares on AG, GB, and CH is equal to the square on AG, KL equal to the square on GB, and NL equal to the rectangle AG, GB, therefore NL is also a mean proportional between CH, KL; therefore, as CH is to NL, so is NL to KL.

But, as CH is to NL, so is CK to NM, and, as NL is to KL, so is NM to KM; {VI. 1} therefore, as CK is to MN, so is MN to KM; {V. 11} therefore the rectangle CK, KM is equal to {the square on MN, that is, to} the fourth part of the square on FM.

Since then CM, MF are two unequal straight lines, and a parallelogram equal to the fourth part of the square on FM and deficient by a square figure has been applied to CM, and divides it into commensurable parts, therefore the square on CM is greater than the square on MF by the square on a straight line commensurable with CM. {X. 17} And neither of the straight lines CM, MF is commensurable in length with the rational straight line CD set out; therefore CF is a third apotome. {X. Deff. III. 3} Therefore etc. Q. E. D.

PROPOSITION 100. The square on a minor straight line applied to a rational straight line produces as breadth a fourth apotome.

Let AB be a minor and CD a rational straight line, and to the rational straight line CD let CE be applied equal to the square on AB and producing CF as breadth; I say that CF is a fourth apotome.

For let BG be the annex to AB; therefore AG, GB are straight lines incommensurable in square which make the sum of the

squares on AG, GB rational, but twice the rectangle AG, GB medial. {X. 76} To CD let there be applied CH equal to the square on AG and producing CK as breadth, and KL equal to the square on BG, producing KM as breadth; therefore the whole CL is equal to the squares on AG, GB.

And the sum of the squares on AG, GB is rational; therefore CL is also rational.

And it is applied to the rational straight line CD, producing CM as breadth; therefore CM is also rational and commensurable in length with CD. {X. 20} And, since the whole CL is equal to the squares on AG, GB, and, in these, CE is equal to the square on AB, therefore the remainder FL is equal to twice the rectangle AG, GB. {II. 7} Let then FM be bisected at the point N, and let NO be drawn through N parallel to either of the straight lines CD, ML; therefore each of the rectangles FO, NL is equal to the rectangle AG, GB.

And, since twice the rectangle AG, GB is medial and is equal to FL, therefore FL is also medial.

And it is applied to the rational straight line FE, producing FM as breadth; therefore FM is rational and incommensurable in length with CD. {X. 22} And, since the sum of the squares on AG, GB is rational, while twice the rectangle AG, GB is medial, the squares on AG, GB are incommensurable with twice the rectangle AG, GB.

But CL is equal to the squares on AG, GB, and FL equal to twice the rectangle AG, GB; therefore CL is incommensurable with FL.

But, as CL is to FL, so is CM to MF; {VI. 1} therefore CM is incommensurable in length with MF. {X. 11} And both are rational; therefore CM, MF are rational straight lines commensurable in square only; therefore CF is an apotome. {X. 73} I say that it is also a fourth apotome.

For, since AG, GB are incommensurable in square, therefore the square on AG is also incommensurable with the square on GB.

And CH is equal to the square on AG, and KL equal to the square on GB; therefore CH is incommensurable with KL.

But, as CH is to KL, so is CK to KM; {VI. 1} therefore CK is incommensurable in length with KM. {X. 11} And, since the rectangle AG, GB is a mean proportional between the squares on AG, GB, and the square on AG is equal to CH, the square on GB to KL, and the rectangle AG, GB to NL, therefore NL is a mean proportional between CH, KL; therefore, as CH is to NL, so is NL to KL.

But, as CH is to NL, so is CK to NM, and, as NL is to KL, so is NM to KM; {VI. 1} therefore, as CK is to MN, so is MN to KM; {V. 11} therefore the rectangle CK, KM is equal to the square on MN {VI. 17}, that is, to the fourth part of the square on FM.

Since then CM, MF are two unequal straight lines, and the rectangle CK, KM equal to the fourth part of the square on MF and deficient by a square figure has been applied to CM and divides it into incommensurable parts, therefore the square on CM is greater than the square on MF by the square on a straight line incommensurable with CM. {X. 18} And the whole CM is commensurable in length with the rational straight line CD set out; therefore CF is a fourth apotome. {X. Deff. III. 4} Therefore etc. Q. E. D.

PROPOSITION 101. The square on the straight line which produces with a rational area a medial whole, if applied to a rational straight line, produces as breadth a fifth apotome.

Let AB be the straight line which produces with a rational area a medial whole, and CD a rational straight line, and to CD let CE be applied equal to the square on AB and producing CF as breadth; I say that CF is a fifth apotome.

For let BG be the annex to AB; therefore AG, GB are straight lines incommensurable in square which make the sum of the squares on them medial but twice the rectangle contained by them rational. {X. 77} To CD let there be applied CH equal to the square on AG, and KL equal to the square on GB; therefore the whole CL is equal to the squares on AG, GB.

But the sum of the squares on AG, GB together is medial; therefore CL is medial.

And it is applied to the rational straight line CD, producing CM as breadth; therefore CM is rational and incommensurable with CD. {X. 22} And, since the whole CL is equal to the squares on AG, GB, and, in these, CE is equal to the square on AB, therefore the remainder FL is equal to twice the rectangle AG, GB. {II. 7} Let then FM be bisected at N, and through N let NO be drawn parallel to either of the straight lines CD, ML; therefore each of the rectangles FO, NL is equal to the rectangle AG, GB: And, since twice the rectangle AG, GB is rational and equal to FL, therefore FL is rational.

And it is applied to the rational straight line EF, producing FM as breadth; therefore FM is rational and commensurable in length with CD. {X. 20} Now, since CL is medial, and FL rational, therefore CL is incommensurable with FL.

But, as CL is to FL, so is CM to MF; {VI. 1} therefore CM is incommensurable in length with MF. {X. 11} And both are rational; therefore CM, MF are rational straight lines commensurable in square only; therefore CF is an apotome. {X. 73} I say next that it is also a fifth apotome.

For we can prove similarly that the rectangle CK, KM is equal to the square on NM, that is, to the fourth part of the square on FM.

And, since the square on AG is incommensurable with the square on GB, while the square on AG is equal to CH, and the square on GB to KL, therefore CH is incommensurable with KL.

But, as CH is to KL, so is CK to KM; {VI. 1} therefore CK is incommensurable in length with KM. {X. 11} Since then CM, MF are two unequal straight lines, and a parallelogram equal to the fourth part of the square on FM and deficient by a square figure has been applied to CM, and divides it into incommensurable parts, therefore the square on CM is greater than the square on MF by the square on a straight line incommensurable with CM. {X. 18} And the annex FM is commensurable with the rational straight line CD set out; therefore CF is a fifth apotome. {X. Deff. III. 5} Q. E. D.

PROPOSITION 102. The square on the straight line which produces with a medial area a medial whole, if applied to a rational straight line, produces as breadth a sixth apotome.

Let AB be the straight line which produces with a medial area a medial whole, and CD a rational straight line, and to CD let CE be applied equal to the square on AB and producing CF as breadth; I say that CF is a sixth apotome.

For let BG be the annex to AB; therefore AG, GB are straight lines incommensurable in square which make the sum of the squares on them medial, twice the rectangle AG, GB medial, and the squares on AG, GB incommensurable with twice the rectangle AG, GB. {X. 78} Now to CD let there be applied CH equal to the square on AG and producing CK as breadth, and KL equal to the square on BG; therefore the whole CL is equal to the squares on AG, GB; therefore CL is also medial.

And it is applied to the rational straight line CD, producing CM as breadth; therefore CM is rational and incommensurable in length with CD. {X. 22} Since now CL is equal to the squares on AG, GB, and, in these, CE is equal to the square on AB, therefore

the remainder FL is equal to twice the rectangle AG, GB. {II. 7} And twice the rectangle AG, GB is medial; therefore FL is also medial.

And it is applied to the rational straight line FE, producing FM as breadth; therefore FM is rational and incommensurable in length with CD. {X. 22} And, since the squares on AG, GB are incommensurable with twice the rectangle AG, GB, and CL is equal to the squares on AG, GB, and FL equal to twice the rectangle AG, GB, therefore CL is incommensurable with FL.

But, as CL is to FL, so is CM to MF; {VI. 1} therefore CM is incommensurable in length with MF. {X. 11} And both are rational.

Therefore CM, MF are rational straight lines commensurable in square only; therefore CF is an apotome. {X. 73} I say next that it is also a sixth apotome.

For, since FL is equal to twice the rectangle AG, GB, let FM be bisected at N, and let NO be drawn through N parallel to CD; therefore each of the rectangles FO, NL is equal to the rectangle AG, GB.

And, since AG, GB are incommensurable in square, therefore the square on AG is incommensurable with the square on GB.

But CH is equal to the square on AG, and KL is equal to the square on GB; therefore CH is incommensurable with KL.

But, as CH is to KL, so is CK to KM; {VI. 1} therefore CK is incommensurable with KM. {X. 11} And, since the rectangle AG, GB is a mean proportional between the squares on AG, GB, and CH is equal to the square on AG, KL equal to the square on GB, and NL equal to the rectangle AG, GB, therefore NL is also a mean proportional between CH, KL; therefore, as CH is to NL, so is NL to KL.

And for the same reason as before the square on CM is greater than the square on MF by the square on a straight line incommensurable with CM. {X. 18} And neither of them is commensurable with the rational straight line CD set out; therefore CF is a sixth apotome. {X. Deff. III. 6} Q. E. D.

PROPOSITION 103. A straight line commensurable in length with an apotome is an apotome and the same in order.

Let AB be an apotome, and let CD be commensurable in length with AB; I say that CD is also an apotome and the same in order with AB.

For, since AB is an apotome, let BE be the annex to it; therefore AE, EB are rational straight lines commensurable in square only. {X. 73} Let it be contrived that the ratio of BE to DF is the same as the ratio of AB to CD; {VI. 12} therefore also, as one is to one, so are all to all; {V. 12} therefore also, as the whole AE is to the whole CF, so is AB to CD.

But AB is commensurable in length with CD.

Therefore AE is also commensurable with CF, and BE with DF. {X. 11} And AE, EB are rational straight lines commensurable in square only; therefore CF, FD are also rational straight lines commensurable in square only. {X. 13} Now since, as AE is to CF, so is BE to DF, alternately therefore, as AE is to EB, so is CF to FD. {V. 16} And the square on AE is greater than the square on EB either by the square on a straight line commensurable with AE or by the square on a straight line incommensurable with it.

If then the square on AE is greater than the square on EB by the square on a straight line commensurable with AE, the square on CF will also be greater than the square on FD by the square on a straight line commensurable with CF. {X. 14} And, if AE is commensurable in length with the rational straight line set out, CF is so also, {X. 12} if BE, then DF also, {id.} and, if neither of the

straight lines AE, EB, then neither of the straight lines CF, FD. {X. 13} But, if the square on AE is greater than the square on EB by the square on a straight line incommensurable with AE, the square on CF will also be greater than the square on FD by the square on a straight line incommensurable with CF. {X. 14} And, if AE is commensurable in length with the rational straight line set out, CF is so also, if BE, then DF also, {X. 12} and, if neither of the straight lines AE, EB, then neither of the straight lines CF, FD. {X. 13} Therefore CD is an apotome and the same in order with AB. Q. E. D.

PROPOSITION 104. A straight line commensurable with an apotome of a medial straight line is an apotome of a medial straight line and the same in order.

Let AB be an apotome of a medial straight line, and let CD be commensurable in length with AB; I say that CD is also an apotome of a medial straight line and the same in order with AB.

For, since AB is an apotome of a medial straight line, let EB be the annex to it.

Therefore AE, EB are medial straight lines commensurable in square only. {X. 74, 75} Let it be contrived that, as AB is to CD, so is BE to DF; {VI. 12} therefore AE is also commensurable with CF, and BE with DF. {V. 12, X. 11} But AE, EB are medial straight lines commensurable in square only; therefore CF, FD are also medial straight lines {X. 23} commensurable in square only; {X. 13} therefore CD is an apotome of a medial straight line. {X. 74, 75} I say next that it is also the same in order with AB.

Since, as AE is to EB, so is CF to FD, therefore also, as the square on AE is to the rectangle AE, EB, so is the square on CF to the rectangle CF, FD.

But the square on AE is commensurable with the square on CF; therefore the rectangle AE, EB is also commensurable with the rectangle CF, FD. {V. 16, X. 11} Therefore, if the rectangle AE, EB is rational, the rectangle CF, FD will also be rational, {X. Def. 4} and if the rectangle AE, EB is medial, the rectangle CF, FD is also medial. {X. 23, Por.} Therefore CD is an apotome of a medial straight line and the same in order with AB. {X. 74, 75} Q. E. D.

PROPOSITION 105. A straight line commensurable with a minor straight line is minor.

Let AB be a minor straight line, and CD commensurable with AB; I say that CD is also minor.

Let the same construction be made as before; then, since AE, EB are incommensurable in square, {X. 76} therefore CF, FD are also incommensurable in square. {X. 13} Now since, as AE is to EB, so is CF to FD, {V. 12, V. 16} therefore also, as the square on AE is to the square on EB, so is the square on CF to the square on FD. {VI. 22} Therefore, componendo, as the squares on AE, EB are to the square on EB, so are the squares on CF, FD to the square on FD. {V. 18} But the square on BE is commensurable with the square on DF; therefore the sum of the squares on AE, EB is also commensurable with the sum of the squares on CF, FD. {V. 16, X. 11} But the sum of the squares on AE, EB is rational; {X. 76} therefore the sum of the squares on CF, FD is also rational. {X. Def. 4} Again, since, as the square on AE is to the rectangle AE, EB, so is the square on CF to the rectangle CF, FD, while the square on AE is commensurable with the square on CF, therefore the rectangle AE, EB is also commensurable with the rectangle CF, FD.

But the rectangle AE, EB is medial; {X. 76} therefore the rectangle CF, FD is also medial; {X. 23, Por.} therefore CF, FD are straight lines incommensurable in square which make the sum of the squares on them rational, but the rectangle contained by

them medial.

Therefore CD is minor. {X. 76} Q. E. D.

PROPOSITION 106. A straight line commensurable with that which produces with a rational area a medial whole is a straight line which produces with a rational area a medial whole.

Let AB be a straight line which produces with a rational area a medial whole, and CD commensurable with AB; I say that CD is also a straight line which produces with a rational area a medial whole.

For let BE be the annex to AB; therefore AE, EB are straight lines incommensurable in square which make the sum of the squares on AE, EB medial, but the rectangle contained by them rational. {X. 77} Let the same construction be made.

Then we can prove, in manner similar to the foregoing, that CF, FD are in the same ratio as AE, EB, the sum of the squares on AE, EB is commensurable with the sum of the squares on CF, FD, and the rectangle AE, EB with the rectangle CF, FD; so that CF, FD are also straight lines incommensurable in square which make the sum of the squares on CF, FD medial, but the rectangle contained by them rational.

Therefore CD is a straight line which produces with a rational area a medial whole. {X. 77} Q. E. D.

PROPOSITION 107. A straight line commensurable with that which produces with a medial area a medial whole is itself also a straight line which produces with a medial area a medial whole.

Let AB be a straight line which produces with a medial area a medial whole, and let CD be commensurable with AB; I say that CD is also a straight line which produces with a medial area a medial whole.

For let BE be the annex to AB, and let the same construction be made; therefore AE, EB are straight lines incommensurable in square which make the sum of the squares on them medial, the rectangle contained by them medial, and further the sum of the squares on them incommensurable with the rectangle contained by them. {X. 78} Now, as was proved, AE, EB are commensurable with CF, FD, the sum of the squares on AE, EB with the sum of the squares on CF, FD, and the rectangle AE, EB with the rectangle CF, FD; therefore CF, FD are also straight lines incommensurable in square which make the sum of the squares on them medial, the rectangle contained by them medial, and further the sum of the squares on them incommensurable with the rectangle contained by them.

Therefore CD is a straight line which produces with a medial area a medial whole. {X. 78} **PROPOSITION 108.** If from a rational area a medial area be subtracted, the “side” of the remaining area becomes one of two irrational straight lines, either an apotome or a minor straight line.

For from the rational area BC let the medial area BD be subtracted; I say that the “side” of the remainder EC becomes one of two irrational straight lines, either an apotome or a minor straight line.

For let a rational straight line FG be set out, to FG let there be applied the rectangular parallelogram GH equal to BC, and let GK equal to DB be subtracted; therefore the remainder EC is equal to LH.

Since then BC is rational, and BD medial, while BC is equal to GH, and BD to GK, therefore GH is rational, and GK medial.

And they are applied to the rational straight line FG; therefore FH is rational and commensurable in length with FG, {X. 20} while FK is rational and incommensurable in length with FG; {X. 22} therefore FH is incommensurable in length with FK. {X. 13}

Therefore FH, FK are rational straight lines commensurable in square only; therefore KH is an apotome {X. 73}, and KF the annex to it.

Now the square on HF is greater than the square on FK by the square on a straight line either commensurable with HF or not commensurable.

First, let the square on it be greater by the square on a straight line commensurable with it.

Now the whole HF is commensurable in length with the rational straight line FG set out; therefore KH is a first apotome. {X. Deff. III. 1} But the “side” of the rectangle contained by a rational straight line and a first apotome is an apotome. {X. 91} Therefore the “side” of LH, that is, of EC, is an apotome.

But, if the square on HF is greater than the square on FK by the square on a straight line incommensurable with HF, while the whole FH is commensurable in length with the rational straight line FG set out, KH is a fourth apotome. {X. Deff. III. 4} But the “side” of the rectangle contained by a rational straight line and a fourth apotome is minor. {X. 94} Q. E. D.

PROPOSITION 109. If from a medial area a rational area be subtracted, there arise two other irrational straight lines, either a first apotome of a medial straight line or a straight line which produces with a rational area a medial whole.

For from the medial area BC let the rational area BD be subtracted.

I say that the “side” of the remainder EC becomes one of two irrational straight lines, either a first apotome of a medial straight line or a straight line which produces with a rational area a medial whole.

For let a rational straight line FG be set out, and let the areas be similarly applied.

It follows then that FH is rational and incommensurable in length with FG, while KF is rational and commensurable in length with FG; therefore FH, FK are rational straight lines commensurable in square only; {X. 13} therefore KH is an apotome, and FK the annex to it. {X. 73} Now the square on HF is greater than the square on FK either by the square on a straight line commensurable with HF or by the square on a straight line incommensurable with it.

If then the square on HF is greater than the square on FK by the square on a straight line commensurable with HF, while the annex FK is commensurable in length with the rational straight line FG set out, KH is a second apotome. {X. Deff. III. 2} But FG is rational; so that the “side” of LH, that is, of EC, is a first apotome of a medial straight line. {X. 92} But, if the square on HF is greater than the square on FK by the square on a straight line incommensurable with HF, while the annex FK is commensurable in length with the rational straight line FG set out, KH is a fifth apotome; {X. Deff. III. 5} so that the “side” of EC is a straight line which produces with a rational area a medial whole. {X. 95} **PROPOSITION 110.** If from a medial area there be subtracted a medial area incommensurable with the whole, the two remaining irrational straight lines arise, either a second apotome of a medial straight line or a straight line which produces with a medial area a medial whole.

For, as in the foregoing figures, let there be subtracted from the medial area BC the medial area BD incommensurable with the whole; I say that the “side” of EC is one of two irrational straight lines, either a second apotome of a medial straight line or a straight line which produces with a medial area a medial whole.

For, since each of the rectangles BC, BD is medial, and BC is incommensurable with BD, it follows that each of the straight lines FH, FK will be rational and incommensurable in length with FG. {X. 22} And, since BC is incommensurable with BD, that

is, GH with GK, HF is also incommensurable with FK; {VI. 1, X. 11} therefore FH, FK are rational straight lines commensurable in square only; therefore KH is an apotome. {X. 73} If then the square on FH is greater than the square on FK by the square on a straight line commensurable with FH, while neither of the straight lines FH, FK is commensurable in length with the rational straight line FG set out, KH is a third apotome. {X. Deff. III. 3} But KL is rational, and the rectangle contained by a rational straight line and a third apotome is irrational, and the “side” of it is irrational, and is called a second apotome of a medial straight line; {X. 93} so that the “side” of LH, that is, of EC, is a second apotome of a medial straight line.

But, if the square on FH is greater than the square on FK by the square on a straight line incommensurable with FH, while neither of the straight lines HF, FK is commensurable in length with FG, KH is a sixth apotome. {X. Deff. III. 6} But the “side” of the rectangle contained by a rational straight line and a sixth apotome is a straight line which produces with a medial area a medial whole. {X. 96} Therefore the “side” of LH, that is, of EC, is a straight line which produces with a medial area a medial whole. Q. E. D.

PROPOSITION 111. The apotome is not the same with the binomial straight line.

Let AB be an apotome; I say that AB is not the same with the binomial straight line.

For, if possible, let it be so; let a rational straight line DC be set out, and to CD let there be applied the rectangle CE equal to the square on AB and producing DE as breadth.

Then, since AB is an apotome, DE is a first apotome. {X. 97} Let EF be the annex to it; therefore DF, FE are rational straight lines commensurable in square only, the square on DF is greater than the square on FE by the square on a straight line

commensurable with DF, and DF is commensurable in length with the rational straight line DC set out. {X. Deff. III. 1} Again, since AB is binomial, therefore DE is a first binomial straight line. {X. 60} Let it be divided into its terms at G, and let DG be the greater term; therefore DG, GE are rational straight lines commensurable in square only, the square on DG is greater than the square on GE by the square on a straight line commensurable with DG, and the greater term DG is commensurable in length with the rational straight line DC set out. {X. Deff. II. 1} Therefore DF is also commensurable in length with DG; {X. 12} therefore the remainder GF is also commensurable in length with DF. {X. 15} But DF is incommensurable in length with EF; therefore FG is also incommensurable in length with EF. {X. 13} Therefore GF, FE are rational straight lines commensurable in square only; therefore EG is an apotome. {X. 73} But it is also rational: which is impossible.

Therefore the apotome is not the same with the binomial straight line. Q. E. D.

PROPOSITION 112. The square on a rational straight line applied to the binomial straight line produces as breadth an apotome the terms of which are commensurable with the terms of the binomial and moreover in the same ratio; and further the apotome so arising will have the same order as the binomial straight line.

Let A be a rational straight line, let BC be a binomial, and let DC be its greater term; let the rectangle BC, EF be equal to the square on A; I say that EF is an apotome the terms of which are commensurable with CD, DB, and in the same ratio, and further EF will have the same order as BC.

For again let the rectangle BD, G be equal to the square on A.

Since then the rectangle BC, EF is equal to the rectangle BD, G, therefore, as CB is to BD, so is G to EF. {VI. 16} But CB is greater than BD; therefore G is also greater than EF. {V. 16, V. 14} Let EH be equal to G; therefore, as CB is to BD, so is HE to EF; therefore, separando, as CD is to BD, so is HF to FE. {V. 17} Let it be contrived that, as HF is to FE, so is FK to KE; therefore also the whole HK is to the whole KF as FK is to KE; for, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents. {V. 12} But, as FK is to KE, so is CD to DB; {V. 11} therefore also, as HK is to KF, so is CD to DB. {id.} But the square on CD is commensurable with the square on DB; {X. 36} therefore the square on HK is also commensurable with the square on KF. {VI. 22, X. 11} And, as the square on HK is to the square on KF, so is HK to KE, since the three straight lines HK, KF, KE are proportional. {V. Def. 9} Therefore HK is commensurable in length with KE, so that HE is also commensurable in length with EK. {X. 15} Now, since the square on A is equal to the rectangle EH, BD, while the square on A is rational, therefore the rectangle EH, BD is also rational.

And it is applied to the rational straight line BD; therefore EH is rational and commensurable in length with BD; {X. 20} so that EK, being commensurable with it, is also rational and commensurable in length with BD.

Since, then, as CD is to DB, so is FK to KE, while CD, DB are straight lines commensurable in square only, therefore FK, KE are also commensurable in square only. {X. 11} But KE is rational; therefore FK is also rational.

Therefore FK, KE are rational straight lines commensurable in square only; therefore EF is an apotome. {X. 73} Now the square on CD is greater than the square on DB either by the square on a straight line commensurable with CD or by the square on a straight line incommensurable with it.

If then the square on CD is greater than the square on DB by the square on a straight line commensurable with CD, the square on FK is also greater than the square on KE by the square on a straight line commensurable with FK. {X. 14} And, if CD is commensurable in length with the rational straight line set out, so also is FK; {X. 11, 12} if BD is so commensurable, so also is KE; {X. 12} but, if neither of the straight lines CD, DB is so commensurable, neither of the straight lines FK, KE is so.

But, if the square on CD is greater than the square on DB by the square on a straight line incommensurable with CD, the square on FK is also greater than the square on KE by the square on a straight line incommensurable with FK. {X. 14} And, if CD is commensurable with the rational straight line set out, so also is FK; if BD is so commensurable, so also is KE; but, if neither of the straight lines CD, DB is so commensurable, neither of the straight lines FK, KE is so; so that FE is an apotome, the terms of which FK, KE are commensurable with the terms CD, DB of the binomial straight line and in the same ratio, and it has the same order as BC. Q. E. D.

PROPOSITION 113. The square on a rational straight line, if applied to an apotome, produces as, breadth the binomial straight line the terms of which are commensurable with the terms of the apotome and in the same ratio; and further the binomial so arising has the same order as the apotome.

Let A be a rational straight line and BD an apotome, and let the rectangle BD, KH be equal to the square on A, so that the square on the rational straight line A when applied to the apotome BD produces KH as breadth; I say that KH is a binomial straight line the terms of which are commensurable with the terms of BD and in the same ratio; and further KH has the same order as BD.

For let DC be the annex to BD; therefore BC, CD are rational straight lines commensurable in square only. {X. 73} Let the

rectangle BC, G be also equal to the square on A.

But the square on A is rational; therefore the rectangle BC, G is also rational.

And it has been applied to the rational straight line BC; therefore G is rational and commensurable in length with BC. {X. 20} Since now the rectangle BC, G is equal to the rectangle BD, KH, therefore, proportionally, as CB is to BD, so is KH to G. {VI. 16} But BC is greater than BD; therefore KH is also greater than G. {V. 16, V. 14} Let KE be made equal to G; therefore KE is commensurable in length with BC.

And since, as CB is to BD, so is HK to KE, therefore, convertendo, as BC is to CD, so is KH to HE. {V. 19, Por.} Let it be contrived that, as KH is to HE, so is HF to FE; therefore also the remainder KF is to FH as KH is to HE, that is, as BC is to CD. {V. 19} But BC, CD are commensurable in square only; therefore KF, FH are also commensurable in square only. {X. 11} And since, as KH is to HE, so is KF to FH, while, as KH is to HE, so is HF to FE, therefore also, as KF is to FH, so is HF to FE, {V. 11} so that also, as the first is to the third, so is the square on the first to the square on the second; {V. Def. 9} therefore also, as KF is to FE, so is the square on KF to the square on FH.

But the square on KF is commensurable with the square on FH, for KF, FH are commensurable in square; therefore KF is also commensurable in length with FE, {X. 11} so that KF is also commensurable in length with KE. {X. 15} But KE is rational and commensurable in length with BC; therefore KF is also rational and commensurable in length with BC. {X. 12} And, since, as BC is to CD, so is KF to FH, alternately, as BC is to KF, so is DC to FH. {V. 16} But BC is commensurable with KF; therefore FH is also commensurable in length with CD. {X. 11} But BC, CD are rational straight lines commensurable in square only; therefore KF, FH are also rational straight lines {X. Def. 3} commensurable

in square only; therefore KH is binomial. {X. 36} If now the square on BC is greater than the square on CD by the square on a straight line commensurable with BC, the square on KF will also be greater than the square on FH by the square on a straight line commensurable with KF. {X. 14} And, if BC is commensurable in length with the rational straight line set out, so also is KF; if CD is commensurable in length with the rational straight line set out, so also is FH, but, if neither of the straight lines BC, CD, then neither of the straight lines KF, FH.

But, if the square on BC is greater than the square on CD by the square on a straight line incommensurable with BC, the square on KF is also greater than the square on FH by the square on a straight line incommensurable with KF. {X. 14} And, if BC is commensurable with the rational straight line set out, so also is KF; if CD is so commensurable, in length with the rational straight line set out, so also is FH; but, if neither of the straight lines BC, CD, then neither of the straight lines KF, FH.

Therefore KH is a binomial straight line, the terms of which KF, FH are commensurable with the terms BC, CD of the apotome and in the same ratio, and further KH has the same order as BD. Q. E. D.

PROPOSITION 114. If an area be contained by an apotome and the binomial straight line the terms of which are commensurable with the terms of the apotome and in the same ratio, the “side” of the area is rational.

For let an area, the rectangle AB, CD, be contained by the apotome AB and the binomial straight line CD, and let CE be the greater term of the latter; let the terms CE, ED of the binomial straight line be commensurable with the terms AF, FB of the apotome and in the same ratio; and let the “side” of the rectangle AB, CD be G; I say that G is rational.

For let a rational straight line H be set out, and to CD let there be applied a rectangle equal to the square on H and producing KL as breadth.

Therefore KL is an apotome.

Let its terms be KM , ML commensurable with the terms CE , ED of the binomial straight line and in the same ratio. {X. 112} But CE , ED are also commensurable with AF , FB and in the same ratio; therefore, as AF is to FB , so is KM to ML .

Therefore, alternately, as AF is to KM , so is BF to LM ; therefore also the remainder AB is to the remainder KL as AF is to KM . {V. 19} But AF is commensurable with KM ; {X. 12} therefore AB is also commensurable with KL . {X. 11} And, as AB is to KL , so is the rectangle CD , AB to the rectangle CD , KL ; {VI. 1} therefore the rectangle CD , AB is also commensurable with the rectangle CD , KL . {X. 11} But the rectangle CD , KL is equal to the square on H ; therefore the rectangle CD , AB is commensurable with the square on H .

But the square on G is equal to the rectangle CD , AB ; therefore the square on G is commensurable with the square on H .

But the square on H is rational; therefore the square on G is also rational; therefore G is rational.

And it is the “side” of the rectangle CD , AB .

Therefore etc.

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And it is made manifest to us by this also that it is possible for a rational area to be contained by irrational straight lines. Q. E. D.

PROPOSITION 115. From a medial straight line there arise irrational straight lines infinite in number, and none of them is the same as any of the preceding.

Let A be a medial straight line; I say that from A there arise irrational straight lines infinite in number, and none of them is the same as any of the preceding.

Let a rational straight line B be set out, and let the square on C be equal to the rectangle B, A; therefore C is irrational; {X. Def. 4} for that which is contained by an irrational and a rational straight line is irrational. {deduction from X. 20} And it is not the same with any of the preceding; for the square on none of the preceding, if applied to a rational straight line produces as breadth a medial straight line.

Again, let the square on D be equal to the rectangle B, C; therefore the square on D is irrational. {deduction from X. 20} Therefore D is irrational; {X. Def. 4} and it is not the same with any of the preceding, for the square on none of the preceding, if applied to a rational straight line, produces C as breadth.

Similarly, if this arrangement proceeds ad infinitum, it is manifest that from the medial straight line there arise irrational straight lines infinite in number, and none is the same with any of the preceding. Q. E. D.

ENDNOTES.

15. But the unit measures the number D; therefore C also measures A. These words are redundant, though they are apparently found in all the MSS.

3, 5, 8, 10. Euclid speaks of the square on the first (third) being greater than the square on the second (fourth) by the square on a straight line commensurable (incommensurable) “with itself (ἑαυτῇ),” and similarly in all like phrases throughout the Book. For clearness’ sake I substitute “the first,” “the third,” or whatever it may be, for “itself” in these cases.

45. After saying literally that “the square on BC is greater than the square on A by the square on DF,” Euclid adds the

equivalent expression with δύναται in its technical sense, ἡ ΒΓ ἄρα τῆς Α μείζον δύναται τῇ ΔΖ. As this is untranslatable in English except by a paraphrase in practically the same words as have preceded, I have not attempted to reproduce it.

3. and which produce the types in question. The Greek is ποιοῦσών τὰ προκείμενα εἶδη, and I have taken εἶδη to mean “types (of irrational straight lines),” though the expression might perhaps mean “satisfying the conditions in question.”

2. “side.” I use the word “side” in the sense explained in the note on X. Def. 4 (P. 13 above), i.e. as short for “side of a square equal to.” The Greek is ἡ τὸ χωρίον δυναμένη.

39. Therefore BA, AG and BA, GE are pairs of rational straight lines commensurable in square only. The text has “Therefore BA, AG, GE are rational straight lines commensurable in square only,” which I have altered because it would naturally convey the impression that any two of the three straight lines are commensurable in square only, whereas AG, GE are commensurable in length (I. 18), and it is only the other two pairs which are commensurable in square only.

BOOK XI. DEFINITIONS.

1. A SOLID IS that which has length, breadth, and depth.
2. An extremity of a solid is a surface.
3. A straight line is at right angles to a plane, when it makes right angles with all the straight lines which meet it and are in the plane.
4. A plane is at right angles to a plane when the straight lines drawn, in one of the planes, at right angles to the common section of the planes are at right angles to the remaining plane.

5. The inclination of a straight line to a plane is, assuming a perpendicular drawn from the extremity of the straight line which is elevated above the plane to the plane, and a straight line joined from the point thus arising to the extremity of the straight line which is in the plane, the angle contained by the straight line so drawn and the straight line standing up.

6. The inclination of a plane to a plane is the acute angle contained by the straight lines drawn at right angles to the common section at the same point, one in each of the planes.

7. A plane is said to be similarly inclined to a plane as another is to another when the said angles of the inclinations are equal to one another.

8. Parallel planes are those which do not meet.

9. Similar solid figures are those contained by similar planes equal in multitude.

10. Equal and similar solid figures are those contained by similar planes equal in multitude and in magnitude.

11. A solid angle is the inclination constituted by more than two lines which meet one another and are not in the same surface, towards all the lines.

Otherwise: A solid angle is that which is contained by more than two plane angles which are not in the same plane and are constructed to one point.

12. A pyramid is a solid figure, contained by planes, which is constructed from one plane to one point.

13. A prism is a solid figure contained by planes two of which, namely those which are opposite, are equal, similar and parallel, while the rest are parallelograms.

14. When, the diameter of a semicircle remaining fixed, the semicircle is carried round and restored again to the same

position from which it began to be moved, the figure so comprehended is a sphere.

15. The axis of the sphere is the straight line which remains fixed and about which the semicircle is turned.

16. The centre of the sphere is the same as that of the semicircle.

17. A diameter of the sphere is any straight line drawn through the centre and terminated in both directions by the surface of the sphere.

18. When, one side of those about the right angle in a right-angled triangle remaining fixed, the triangle is carried round and restored again to the same position from which it began to be moved, the figure so comprehended is a cone.

And, if the straight line which remains fixed be equal to the remaining side about the right angle which is carried round, the cone will be right-angled; if less, obtuse-angled; and if greater, acute-angled.

19. The axis of the cone is the straight line which remains fixed and about which the triangle is turned.

20. And the base is the circle described by the straight line which is carried round.

21. When, one side of those about the right angle in a rectangular parallelogram remaining fixed, the parallelogram is carried round and restored again to the same position from which it began to be moved, the figure so comprehended is a cylinder.

22. The axis of the cylinder is the straight line which remains fixed and about which the parallelogram is turned.

23. And the bases are the circles described by the two sides opposite to one another which are carried round.

24. Similar cones and cylinders are those in which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

27. An icosahedron is a solid figure contained by twenty equal and equilateral triangles.

28. A dodecahedron is a solid figure contained by twelve equal, equilateral, and equiangular pentagons.

PROPOSITIONS.

PROPOSITION 1. A PART OF a straight line cannot be in the plane of reference and a part in a plane more elevated.

For, if possible, let a part AB of the straight line ABC be in the plane of reference, and a part BC in a plane more elevated.

There will then be in the plane of reference some straight line continuous with AB in a straight line.

Let it be BD; therefore AB is a common segment of the two straight lines ABC, ABD: which is impossible, inasmuch as, if we describe a circle with centre B and distance AB, the diameters will cut off unequal circumferences of the circle.

Therefore a part of a straight line cannot be in the plane of reference, and a part in a plane more elevated. Q. E. D.

PROPOSITION 2. If two straight lines cut one another, they are in one plane, and every triangle is in one plane.

For let the two straight lines AB, CD cut one another at the point E; I say that AB, CD are in one plane, and every triangle is in one plane.

For let points F, G be taken at random on EC, EB, let CB, FG be joined, and let FH, GK be drawn across; I say first that the triangle ECB is in one plane.

For, if part of the triangle ECB, either FHC or GBK, is in the plane of reference, and the rest in another, a part also of one of the straight lines EC, EB will be in the plane of reference, and a part in another.

But, if the part FCBG of the triangle ECB be in the plane of reference, and the rest in another, a part also of both the straight lines EC, EB will be in the plane of reference and a part in another: which was proved absurd. {XI. 1} Therefore the triangle ECB is in one plane.

But, in whatever plane the triangle ECB is, in that plane also is each of the straight lines EC, EB, and, in whatever plane each of the straight lines EC, EB is, in that plane are AB, CD also. {XI. 1} Therefore the straight lines AB, CD are in one plane, and every triangle is in one plane. Q. E. D.

PROPOSITION 3. If two planes cut one another, their common section is a straight line.

For let the two planes AB, BC cut one another, and let the line DB be their common section; I say that the line DB is a straight line.

For, if not, from D to B let the straight line DEB be joined in the plane AB, and in the plane BC the straight line DFB.

Then the two straight lines DEB, DFB will have the same extremities, and will clearly enclose an area: which is absurd.

Therefore DEB, DFB are not straight lines.

Similarly we can prove that neither will there be any other straight line joined from D to B except DB the common section of the planes AB, BC.

Therefore etc. Q. E. D.

PROPOSITION 4. If a straight line be set up at right angles to two straight lines which cut one another, at their common point of section, it will also be at right angles to the plane through them.

For let a straight line EF be set up at right angles to the two straight lines AB, CD, which cut one another at the point E, from E; I say that EF is also at right angles to the plane through AB, CD.

For let AE, EB, CE, ED be cut off equal to one another, and let any straight line GEH be drawn across through E, at random; let AD, CB be joined, and further let FA, FG, FD, FC, FH, FB be joined from the point F taken at random $\langle$ on EF $\rangle$.

Now, since the two straight lines AE, ED are equal to the two straight lines CE, EB, and contain equal angles, {I. 15} therefore the base AD is equal to the base CB, and the triangle AED will be equal to the triangle CEB; {I. 4} so that the angle DAE is also equal to the angle EBC.

But the angle AEG is also equal to the angle BEH; {I. 15} therefore AGE, BEH are two triangles which have two angles equal to two angles respectively, and one side equal to one side, namely that adjacent to the equal angles, that is to say, AE to EB; therefore they will also have the remaining sides equal to the remaining sides. {I. 26} Therefore GE is equal to EH, and AG to BH.

And, since AE is equal to EB, while FE is common and at right angles, therefore the base FA is equal to the base FB. {I. 4} For the same reason FC is also equal to FD.

And, since AD is equal to CB, and FA is also equal to FB, the two sides FA, AD are equal to the two sides FB, BC respectively; and the base FD was proved equal to the base FC; therefore the angle

FAD is also equal to the angle FBC. {I. 8} And since, again, AG was proved equal to BH, and further FA also equal to FB, the two sides FA, AG are equal to the two sides FB, BH.

And the angle FAG was proved equal to the angle FBH; therefore the base FG is equal to the base FH. {I. 4} Now since, again, GE was proved equal to EH, and EF is common, the two sides GE, EF are equal to the two sides HE, EF; and the base FG is equal to the base FH; therefore the angle GEF is equal to the angle HEF. {I. 8} Therefore each of the angles GEF, HEF is right.

Therefore FE is at right angles to GH drawn at random through E.

Similarly we can prove that FE will also make right angles with all the straight lines which meet it and are in the plane of reference.

But a straight line is at right angles to a plane when it makes right angles with all the straight lines which meet it and are in that same plane; {XI. Def. 3} therefore FE is at right angles to the plane of reference.

But the plane of reference is the plane through the straight lines AB, CD.

Therefore FE is at right angles to the plane through AB, CD.

Therefore etc. Q. E. D.

PROPOSITION 5. If a straight line be set up at right angles to three straight lines which meet one another, at their common point of section, the three straight lines are in one plane.

For let a straight line AB be set up at right angles to the three straight lines BC, BD, BE, at their point of meeting at B; I say that BC, BD, BE are in one plane.

For suppose they are not, but, if possible, let BD, BE be in the plane of reference and BC in one more elevated; let the plane

through AB, BC be produced; it will thus make, as common section in the plane of reference, a straight line. {XI. 3} Let it make BF.

Therefore the three straight lines AB, BC, BF are in one plane, namely that drawn through AB, BC.

Now, since AB is at right angles to each of the straight lines BD, BE, therefore AB is also at right angles to the plane through BD, BE. {XI. 4} But the plane through BD, BE is the plane of reference; therefore AB is at right angles to the plane of reference.

Thus AB will also make right angles with all the straight lines which meet it and are in the plane of reference. {XI. Def. 3} But BF which is in the plane of reference meets it; therefore the angle ABF is right.

But, by hypothesis, the angle ABC is also right; therefore the angle ABF is equal to the angle ABC.

And they are in one plane: which is impossible.

Therefore the straight line BC is not in a more elevated plane; therefore the three straight lines BC, BD, BE are in one plane.

Therefore, if a straight line be set up at right angles to three straight lines, at their point of meeting, the three straight lines are in one plane. Q. E. D.

PROPOSITION 6. If two straight lines be at right angles to the same plane, the straight lines will be parallel.

For let the two straight lines AB, CD be at right angles to the plane of reference; I say that AB is parallel to CD.

For let them meet the plane of reference at the points B, D, let the straight line BD be joined, let DE be drawn, in the plane of reference, at right angles to BD, let DE be made equal to AB, and let BE, AE, AD be joined.

Now, since AB is at right angles to the plane of reference, it will also make right angles with all the straight lines which meet it and are in the plane of reference. {XI. Def. 3} But each of the straight lines BD, BE is in the plane of reference and meets AB; therefore each of the angles ABD, ABE is right.

For the same reason each of the angles CDB, CDE is also right.

And, since AB is equal to DE, and BD is common, the two sides AB, BD are equal to the two sides ED, DB; and they include right angles; therefore the base AD is equal to the base BE. {I. 4} And, since AB is equal to DE, while AD is also equal to BE, the two sides AB, BE are equal to the two sides ED, DA; and AE is their common base; therefore the angle ABE is equal to the angle EDA. {I. 8} But the angle ABE is right; therefore the angle EDA is also right; therefore ED is at right angles to DA.

But it is also at right angles to each of the straight lines BD, DC; therefore ED is set up at right angles to the three straight lines BD, DA, DC at their point of meeting; therefore the three straight lines BD, DA, DC are in one plane. {XI. 5} But, in whatever plane DB, DA are, in that plane is AB also, for every triangle is in one plane; {XI. 2} therefore the straight lines AB, BD, DC are in one plane.

And each of the angles ABD, BDC is right; therefore AB is parallel to CD. {I. 28} Therefore etc. Q. E. D.

PROPOSITION 7. If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB, CD be two parallel straight lines, and let points E, F be taken at random on them respectively; I say that the straight line joining the points E, F is in the same plane with the parallel straight lines.

For suppose it is not, but, if possible, let it be in a more elevated plane as EGF, and let a plane be drawn through EGF; it will then make, as section in the plane of reference, a straight line. {XI. 3} Let it make it, as EF; therefore the two straight lines EGF, EF will enclose an area: which is impossible.

Therefore the straight line joined from E to F is not in a plane more elevated; therefore the straight line joined from E to F is in the plane through the parallel straight lines AB, CD.

Therefore etc. Q. E. D.

PROPOSITION 8. If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB, CD be two parallel straight lines, and let one of them, AB, be at right angles to the plane of reference; I say that the remaining one, CD, will also be at right angles to the same plane.

For let AB, CD meet the plane of reference at the points B, D, and let BD be joined; therefore AB, CD, BD are in one plane. {XI. 7} Let DE be drawn, in the plane of reference, at right angles to BD, let DE be made equal to AB, and let BE, AE, AD be joined.

Now, since AB is at right angles to the plane of reference, therefore AB is also at right angles to all the straight lines which meet it and are in the plane of reference; {XI. Def. 3} therefore each of the angles ABD, ABE is right.

And, since the straight line BD has fallen on the parallels AB, CD, therefore the angles ABD, CDB are equal to two right angles. {I. 29} But the angle ABD is right; therefore the angle CDB is also right; therefore CD is at right angles to BD.

And, since AB is equal to DE, and BD is common, the two sides AB, BD are equal to the two sides ED, DB; and the angle ABD is equal to the angle EDB, for each is right; therefore the base AD is equal to the base BE.

And, since AB is equal to DE, and BE to AD, the two sides AB, BE are equal to the two sides ED, DA respectively, and AE is their common base; therefore the angle ABE is equal to the angle EDA.

But the angle ABE is right; therefore the angle EDA is also right; therefore ED is at right angles to AD.

But it is also at right angles to DB; therefore ED is also at right angles to the plane through BD, DA. {XI. 4} Therefore ED will also make right angles with all the straight lines which meet it and are in the plane through BD, DA.

But DC is in the plane through BD, DA, inasmuch as AB, BD are in the plane through BD, DA, {XI. 2} and DC is also in the plane in which AB, BD are.

Therefore ED is at right angles to DC, so that CD is also at right angles to DE.

But CD is also at right angles to BD.

Therefore CD is set up at right angles to the two straight lines DE, DB which cut one another, from the point of section at D; so that CD is also at right angles to the plane through DE, DB. {XI. 4} But the plane through DE, DB is the plane of reference; therefore CD is at right angles to the plane of reference.

Therefore etc. Q. E. D.

PROPOSITION 9. Straight lines which are parallel to the same straight line and are not in the same plane with it are also parallel to one another.

For let each of the straight lines AB, CD be parallel to EF, not being in the same plane with it; I say that AB is parallel to CD.

For let a point G be taken at random on EF, and from it let there be drawn GH, in the plane through EF, AB, at right angles to EF, and GK in the plane through FE, CD again at right angles to EF.

Now, since EF is at right angles to each of the straight lines GH, GK, therefore EF is also at right angles to the plane through GH, GK. {XI. 4} And EF is parallel to AB; therefore AB is also at right angles to the plane through HG, GK. {XI. 8} For the same reason CD is also at right angles to the plane through HG, GK; therefore each of the straight lines AB, CD is at right angles to the plane through HG, GK.

But if two straight lines be at right angles to the same plane, the straight lines are parallel; {XI. 6} therefore AB is parallel to CD.

PROPOSITION 10. If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

For let the two straight lines AB, BC meeting one another be parallel to the two straight lines DE, EF meeting one another, not in the same plane; I say that the angle ABC is equal to the angle DEF.

For let BA, BC, ED, EF be cut off equal to one another, and let AD, CF, BE, AC, DF be joined.

Now, since BA is equal and parallel to ED, therefore AD is also equal and parallel to BE. {I. 33} For the same reason CF is also equal and parallel to BE.

Therefore each of the straight lines AD, CF is equal and parallel to BE.

But straight lines which are parallel to the same straight line and are not in the same plane with it are parallel to one another; {XI. 9} therefore AD is parallel and equal to CF.

And AC, DF join them; therefore AC is also equal and parallel to DF. {I. 33} Now, since the two sides AB, BC are equal to the two sides DE, EF, and the base AC is equal to the base DF, therefore the angle ABC is equal to the angle DEF. {I. 8} Therefore etc.

PROPOSITION 11. From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the given elevated point, and the plane of reference the given plane; thus it is required to draw from the point A a straight line perpendicular to the plane of reference.

Let any straight line BC be drawn, at random, in the plane of reference, and let AD be drawn from the point A perpendicular to BC. {I. 12} If then AD is also perpendicular to the plane of reference, that which was enjoined will have been done.

But, if not, let DE be drawn from the point D at right angles to BC and in the plane of reference, {I. 11} let AF be drawn from A perpendicular to DE, {I. 12} and let GH be drawn through the point F parallel to BC. {I. 31} Now, since BC is at right angles to each of the straight lines DA, DE, therefore BC is also at right angles to the plane through ED, DA. {XI. 4} And GH is parallel to it; but, if two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane; {XI. 8} therefore GH is also at right angles to the plane through ED, DA.

Therefore GH is also at right angles to all the straight lines which meet it and are in the plane through ED, DA. {XI. Def. 3} But AF meets it and is in the plane through ED, DA; therefore GH is at right angles to FA, so that FA is also at right angles to GH.

But AF is also at right angles to DE; therefore AF is at right angles to each of the straight lines GH, DE.

But, if a straight line be set up at right angles to two straight lines which cut one another, at the point of section, it will also be at right angles to the plane through them; {XI. 4} therefore FA is at right angles to the plane through ED, GH.

But the plane through ED, GH is the plane of reference; therefore AF is at right angles to the plane of reference.

Therefore from the given elevated point A the straight line AF has been drawn perpendicular to the plane of reference. Q. E. F.

PROPOSITION 12. To set up a straight line at right angles to a given plane from a given point in it.

Let the plane of reference be the given plane, and A the point in it; thus it is required to set up from the point A a straight line at right angles to the plane of reference.

Let any elevated point B be conceived, from B let BC be drawn perpendicular to the plane of reference, {XI. 11} and through the point A let AD be drawn parallel to BC. {I. 31} Then, since AD, CB are two parallel straight lines, while one of them, BC, is at right angles to the plane of reference, therefore the remaining one, AD, is also at right angles to the plane of reference. {XI. 8} Therefore AD has been set up at right angles to the given plane from the point A in it.

PROPOSITION 13. From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

For, if possible, from the same point A let the two straight lines AB, AC be set up at right angles to the plane of reference and on the same side, and let a plane be drawn through BA, AC; it will then make, as section through A in the plane of reference, a straight line. {XI. 3} Let it make DAE; therefore the straight lines AB, AC, DAE are in one plane.

And, since CA is at right angles to the plane of reference, it will also make right angles with all the straight lines which meet it and are in the plane of reference. {XI. Def. 3} But DAE meets it and is in the plane of reference; therefore the angle CAE is right.

For the same reason the angle BAE is also right; therefore the angle CAE is equal to the angle BAE.

And they are in one plane: which is impossible.

Therefore etc. Q. E. D.

PROPOSITION 14. Planes to which the same straight line is at right angles will be parallel.

For let any straight line AB be at right angles to each of the planes CD, EF; I say that the planes are parallel.

For, if not, they will meet when produced.

Let them meet; they will then make, as common section, a straight line. {XI. 3} Let them make GH; let a point K be taken at random on GH, and let AK, BK be joined.

Now, since AB is at right angles to the plane EF, therefore AB is also at right angles to BK which is a straight line in the plane EF produced; {XI. Def. 3} therefore the angle ABK is right.

For the same reason the angle BAK is also right.

Thus, in the triangle ABK, the two angles ABK, BAK are equal to two right angles: which is impossible. {I. 17} Therefore the planes CD, EF will not meet when produced; therefore the planes CD, EF are parallel. {XI. Def. 8} Therefore planes to which the same straight line is at right angles are parallel. Q. E. D.

PROPOSITION 15. If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

For let the two straight lines AB, BC meeting one another be parallel to the two straight lines DE, EF meeting one another, not being in the same plane; I say that the planes produced through AB, BC and DE, EF will not meet one another.

For let BG be drawn from the point B perpendicular to the plane through DE, EF {XI. 11}, and let it meet the plane at the point G; through G let GH be drawn parallel to ED, and GK parallel to EF. {I. 31} Now, since BG is at right angles to the plane through DE, EF, therefore it will also make right angles with all the straight

lines which meet it and are in the plane through DE, EF. {XI. Def. 3} But each of the straight lines GH, GK meets it and is in the plane through DE, EF; therefore each of the angles BGH, BGK is right.

And, since BA is parallel to GH, {XI. 9} therefore the angles GBA, BGH are equal to two right angles. {I. 29} But the angle BGH is right; therefore the angle GBA is also right; therefore GB is at right angles to BA.

For the same reason GB is also at right angles to BC.

Since then the straight line GB is set up at right angles to the two straight lines BA, BC which cut one another, therefore GB is also at right angles to the plane through BA, BC. {XI. 4} But planes to which the same straight line is at right angles are parallel; {XI. 14} therefore the plane through AB, BC is parallel to the plane through DE, EF.

Therefore, if two straight lines meeting one another be parallel to two straight lines meeting one another, not in the same plane, the planes through them are parallel. Q. E. D.

PROPOSITION 16. If two parallel planes be cut by any plane, their common sections are parallel.

For let the two parallel planes AB, CD be cut by the plane EFGH, and let EF, GH be their common sections; I say that EF is parallel to GH.

For, if not, EF, GH will, when produced, meet either in the direction of F, H or of E, G.

Let them be produced, as in the direction of F, H, and let them, first, meet at K.

Now, since EFK is in the plane AB, therefore all the points on EFK are also in the plane AB. {XI. 1} But K is one of the points on the straight line EFK; therefore K is in the plane AB.

For the same reason K is also in the plane CD; therefore the planes AB, CD will meet when produced.

But they do not meet, because they are, by hypothesis, parallel; therefore the straight lines EF, GH will not meet when produced in the direction of F, H.

Similarly we can prove that neither will the straight lines EF, GH meet when produced in the direction of E, G.

But straight lines which do not meet in either direction are parallel. {I. Def. 23} Therefore EF is parallel to GH.

Therefore etc. Q. E. D.

PROPOSITION 17. If two straight lines be cut by parallel planes, they will be cut in the same ratios.

For let the two straight lines AB, CD be cut by the parallel planes GH, KL, MN at the points A, E, B and C, F, D; I say that, as the straight line AE is to EB, so is CF to FD.

For let AC, BD, AD be joined, let AD meet the plane KL at the point O, and let EO, OF be joined.

Now, since the two parallel planes KL, MN are cut by the plane EBDO, their common sections EO, BD are parallel. {XI. 16} For the same reason, since the two parallel planes GH, KL are cut by the plane AOFC, their common sections AC, OF are parallel. {id.} And, since the straight line EO has been drawn parallel to BD, one of the sides of the triangle ABD, therefore, proportionally, as AE is to EB, so is AO to OD. {VI. 2} Again, since the straight line OF has been drawn parallel to AC, one of the sides of the triangle ADC, proportionally, as AO is to OD, so is CF to FD. {id.} But it was also proved that, as AO is to OD, so is AE to EB; therefore also, as AE is to EB, so is CF to FD. {V. 11} Therefore etc. Q. E. D.

PROPOSITION 18. If a straight line be at right angles to any plane, all the planes through it will also be at right angles to the

same plane.

For let any straight line AB be at right angles to the plane of reference; I say that all the planes through AB are also at right angles to the plane of reference.

For let the plane DE be drawn through AB, let CE be the common section of the plane DE and the plane of reference, let a point F be taken at random on CE, and from F let FG be drawn in the plane DE at right angles to CE. {I. 11} Now, since AB is at right angles to the plane of reference, AB is also at right angles to all the straight lines which meet it and are in the plane of reference; {XI. Def. 3} so that it is also at right angles to CE; therefore the angle ABF is right.

But the angle GFB is also right; therefore AB is parallel to FG. {I. 28} But AB is at right angles to the plane of reference; therefore FG is also at right angles to the plane of reference. {XI. 8} Now a plane is at right angles to a plane, when the straight lines drawn, in one of the planes, at right angles to the common section of the planes are at right angles to the remaining plane. {XI. Def. 4} And FG, drawn in one of the planes DE at right angles to CE, the common section of the planes, was proved to be at right angles to the plane of reference; therefore the plane DE is at right angles to the plane of reference.

Similarly also it can be proved that all the planes through AB are at right angles to the plane of reference.

Therefore etc. Q. E. D.

PROPOSITION 19. If two planes which cut one another be at right angles to any plane, their common section will also be at right angles to the same plane.

For let the two planes AB, BC be at right angles to the plane of reference, and let BD be their common section; I say that BD is at right angles to the plane of reference.

For suppose it is not, and from the point D let DE be drawn in the plane AB at right angles to the straight line AD, and DF in the plane BC at right angles to CD.

Now, since the plane AB is at right angles to the plane of reference, and DE has been drawn in the plane AB at right angles to AD, their common section, therefore DE is at right angles to the plane of reference. {XI. Def. 4} Similarly we can prove that DF is also at right angles to the plane of reference.

Therefore from the same point D two straight lines have been set up at right angles to the plane of reference on the same side: which is impossible. {XI. 13} Therefore no straight line except the common section DB of the planes AB, BC can be set up from the point D at right angles to the plane of reference.

Therefore etc. Q. E. D.

PROPOSITION 20. If a solid angle be contained by three plane angles, any two, taken together in any manner, are greater than the remaining one.

For let the solid angle at A be contained by the three plane angles BAC, CAD, DAB; I say that any two of the angles BAC, CAD, DAB, taken together in any manner, are greater than the remaining one.

If now the angles BAC, CAD, DAB are equal to one another, it is manifest that any two are greater than the remaining one.

But, if not, let BAC be greater, and on the straight line AB, and at the point A on it, let the angle BAE be constructed, in the plane through BA, AC, equal to the angle DAB; let AE be made equal to AD, and let BEC, drawn across through the point E, cut the straight lines AB, AC at the points B, C; let DB, DC be joined.

Now, since DA is equal to AE, and AB is common, two sides are equal to two sides; and the angle DAB is equal to the angle BAE; therefore the base DB is equal to the base BE. {I. 4} And, since

the two sides BD, DC are greater than BC, {I. 20} and of these DB was proved equal to BE, therefore the remainder DC is greater than the remainder EC.

Now, since DA is equal to AE, and AC is common, and the base DC is greater than the base EC, therefore the angle DAC is greater than the angle EAC. {I. 25} But the angle DAB was also proved equal to the angle BAE; therefore the angles DAB, DAC are greater than the angle BAC.

Similarly we can prove that the remaining angles also, taken together two and two, are greater than the remaining one.

Therefore etc. Q. E. D.

PROPOSITION 21. Any solid angle is contained by plane angles less than four right angles.

Let the angle at A be a solid angle contained by the plane angles BAC, CAD, DAB; I say that the angles BAC, CAD, DAB are less than four right angles.

For let points B, C, D be taken at random on the straight lines AB, AC, AD respectively, and let BC, CD, DB be joined.

Now, since the solid angle at B is contained by the three plane angles CBA, ABD, CBD, any two are greater than the remaining one; {XI. 20} therefore the angles CBA, ABD are greater than the angle CBD.

For the same reason the angles BCA, ACD are also greater than the angle BCD, and the angles CDA, ADB are greater than the angle CDB; therefore the six angles CBA, ABD, BCA, ACD, CDA, ADB are greater than the three angles CBD, BCD, CDB.

But the three angles CBD, BDC, BCD are equal to two right angles; {I. 32} therefore the six angles CBA, ABD, BCA, ACD, CDA, ADB are greater than two right angles.

And, since the three angles of each of the triangles ABC, ACD, ADB are equal to two right angles, therefore the nine angles of the three triangles, the angles CBA, ACB, BAC, ACD, CDA, CAD, ADB, DBA, BAD are equal to six right angles; and of them the six angles ABC, BCA, ACD, CDA, ADB, DBA are greater than two right angles; therefore the remaining three angles BAC, CAD, DAB containing the solid angle are less than four right angles.

Therefore etc. Q. E. D.

PROPOSITION 22. If there be three plane angles of which two, taken together in any manner, are greater than the remaining one, and they are contained by equal straight lines, it is possible to construct a triangle out of the straight lines joining the extremities of the equal straight lines.

Let there be three plane angles ABC, DEF, GHK, of which two, taken together in any manner, are greater than the remaining one, namely the angles ABC, DEF greater than the angle GHK, the angles DEF, GHK greater than the angle ABC, and, further, the angles GHK, ABC greater than the angle DEF; let the straight lines AB, BC, DE, EF, GH, HK be equal, and let AC, DF, GK be joined; I say that it is possible to construct a triangle out of straight lines equal to AC, DF, GK, that is, that any two of the straight lines AC, DF, GK are greater than the remaining one.

Now, if the angles ABC, DEF, GHK are equal to one another, it is manifest that, AC, DF, GK being equal also, it is possible to construct a triangle out of straight lines equal to AC, DF, GK.

But, if not, let them be unequal, and on the straight line HK, and at the point H on it, let the angle KHL be constructed equal to the angle ABC; let HL be made equal to one of the straight lines AB, BC, DE, EF, GH, HK, and let KL, GL be joined.

Now, since the two sides AB, BC are equal to the two sides KH, HL, and the angle at B is equal to the angle KHL, therefore the

base AC is equal to the base KL. {I. 4} And, since the angles ABC, GHK are greater than the angle DEF, while the angle ABC is equal to the angle KHL, therefore the angle GHL is greater than the angle DEF.

And, since the two sides GH, HL are equal to the two sides DE, EF, and the angle GHL is greater than the angle DEF, therefore the base GL is greater than the base DF. {I. 24} But GK, KL are greater than GL.

Therefore GK, KL are much greater than DF.

But KL is equal to AC; therefore AC, GK are greater than the remaining straight line DF.

Similarly we can prove that AC, DF are greater than GK, and further DF, GK are greater than AC.

Therefore it is possible to construct a triangle out of straight lines equal to AC, DF, GK. Q. E. D.

PROPOSITION 23. To construct a solid angle out of three plane angles two of which, taken together in any manner, are greater than the remaining one: thus the three angles must be less than four right angles.

Let the angles ABC, DEF, GHK be the three given plane angles, and let two of these, taken together in any manner, be greater than the remaining one, while, further, the three are less than four right angles; thus it is required to construct a solid angle out of angles equal to the angles ABC, DEF, GHK.

Let AB, BC, DE, EF, GH, HK be cut off equal to one another, and let AC, DF, GK be joined; it is therefore possible to construct a triangle out of straight lines equal to AC, DF, GK. {XI. 22} Let LMN be so constructed that AC is equal to LM, DF to MN, and further GK to NL, let the circle LMN be described about the triangle LMN, let its centre be taken, and let it be O; let LO, MO, NO be joined; I say that AB is greater than LO.

For, if not, AB is either equal to LO, or less.

First, let it be equal.

Then, since AB is equal to LO, while AB is equal to BC, and OL to OM, the two sides AB, BC are equal to the two sides LO, OM respectively; and, by hypothesis, the base AC is equal to the base LM; therefore the angle ABC is equal to the angle LOM. {I. 8} For the same reason the angle DEF is also equal to the angle MON, and further the angle GHK to the angle NOL; therefore the three angles ABC, DEF, GHK are equal to the three angles LOM, MON, NOL.

But the three angles LOM, MON, NOL are equal to four right angles; therefore the angles ABC, DEF, GHK are equal to four right angles.

But they are also, by hypothesis, less than four right angles: which is absurd.

Therefore AB is not equal to LO.

I say next that neither is AB less than LO.

For, if possible, let it be so, and let OP be made equal to AB, and OQ equal to BC, and let PQ be joined.

Then, since AB is equal to BC, OP is also equal to OQ, so that the remainder LP is equal to QM.

Therefore LM is parallel to PQ, {VI. 2} and LMO is equiangular with PQO; {I. 29} therefore, as OL is to LM, so is OP to PQ; {VI. 4} and alternately, as LO is to OP, so is LM to PQ. {V. 16} But LO is greater than OP; therefore LM is also greater than PQ.

But LM was made equal to AC; therefore AC is also greater than PQ.

Since, then, the two sides AB, BC are equal to the two sides PO, OQ, and the base AC is greater than the base PQ, therefore the

angle ABC is greater than the angle POQ. {I. 25} Similarly we can prove that the angle DEF is also greater than the angle MON, and the angle GHK greater than the angle NOL.

Therefore the three angles ABC, DEF, GHK are greater than the three angles LOM, MON, NOL.

But, by hypothesis, the angles ABC, DEF, GHK are less than four right angles; therefore the angles LOM, MON, NOL are much less than four right angles.

But they are also equal to four right angles: which is absurd.

Therefore AB is not less than LO.

And it was proved that neither is it equal; therefore AB is greater than LO.

Let then OR be set up from the point O at right angles to the plane of the circle LMN, {XI. 12} and let the square on OR be equal to that area by which the square on AB is greater than the square on LO; {Lemma} let RL, RM, RN be joined.

Then, since RO is at right angles to the plane of the circle LMN, therefore RO is also at right angles to each of the straight lines LO, MO, NO.

And, since LO is equal to OM, while OR is common and at right angles, therefore the base RL is equal to the base RM. {I. 4} For the same reason RN is also equal to each of the straight lines RL, RM; therefore the three straight lines RL, RM, RN are equal to one another.

Next, since by hypothesis the square on OR is equal to that area by which the square on AB is greater than the square on LO, therefore the square on AB is equal to the squares on LO, OR.

But the square on LR is equal to the squares on LO, OR, for the angle LOR is right; {I. 47} therefore the square on AB is equal to the square on RL; therefore AB is equal to RL.

But each of the straight lines BC, DE, EF, GH, HK is equal to AB, while each of the straight lines RM, RN is equal to RL; therefore each of the straight lines AB, BC, DE, EF, GH, HK is equal to each of the straight lines RL, RM, RN.

And, since the two sides LR, RM are equal to the two sides AB, BC, and the base LM is by hypothesis equal to the base AC, therefore the angle LRM is equal to the angle ABC. {I. 8} For the same reason the angle MRN is also equal to the angle DEF, and the angle LRN to the angle GHK.

Therefore, out of the three plane angles LRM, MRN, LRN, which are equal to the three given angles ABC, DEF, GHK, the solid angle at R has been constructed, which is contained by the angles LRM, MRN, LRN. Q. E. F.

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But how it is possible to take the square on OR equal to that area by which the square on AB is greater than the square on LO, we can show as follows.

Let the straight lines AB, LO be set out, and let AB be the greater; let the semicircle ABC be described on AB, and into the semicircle ABC let AC be fitted equal to the straight line LO, not being greater than the diameter AB; {IV. 1} let CB be joined. Since then the angle ACB is an angle in the semicircle ACB, therefore the angle ACB is right. {III. 31} Therefore the square on AB is equal to the squares on AC, CB. {I. 47} Hence the square on AB is greater than the square on AC by the square on CB.

But AC is equal to LO.

Therefore the square on AB is greater than the square on LO by the square on CB.

If then we cut off OR equal to BC, the square on AB will be greater than the square on LO by the square on OR. Q. E. F.

PROPOSITION 24. If a solid be contained by parallel planes, the opposite planes in it are equal and parallelogrammic.

For let the solid CDHG be contained by the parallel planes AC, GF, AH, DF, BF, AE; I say that the opposite planes in it are equal and parallelogrammic.

For, since the two parallel planes BG, CE are cut by the plane AC, their common sections are parallel. {XI. 16} Therefore AB is parallel to DC.

Again, since the two parallel planes BF, AE are cut by the plane AC, their common sections are parallel. {XI. 16} Therefore BC is parallel to AD.

But AB was also proved parallel to DC; therefore AC is a parallelogram.

Similarly we can prove that each of the planes DF, FG, GB, BF, AE is a parallelogram.

Let AH, DF be joined.

Then, since AB is parallel to DC, and BH to CF, the two straight lines AB, BH which meet one another are parallel to the two straight lines DC, CF which meet one another, not in the same plane; therefore they will contain equal angles; {XI. 10} therefore the angle ABH is equal to the angle DCF.

And, since the two sides AB, BH are equal to the two sides DC, CF, {I. 34} and the angle ABH is equal to the angle DCF, therefore the base AH is equal to the base DF, and the triangle ABH is equal to the triangle DCF. {I. 4} And the parallelogram BG is double of the triangle ABH, and the parallelogram CE double of the triangle DCF; {I. 34} therefore the parallelogram BG is equal to the parallelogram CE.

Similarly we can prove that AC is also equal to GF, and AE to BF.

Therefore etc. Q. E. D.

PROPOSITION 25. If a parallelepipedal solid be cut by a plane which is parallel to the opposite planes, then, as the base is to the base, so will the solid be to the solid.

For let the parallelepipedal solid ABCD be cut by the plane FG which is parallel to the opposite planes RA, DH; I say that, as the base AEFV is to the base EHCF, so is the solid ABFU to the solid EGCD.

For let AH be produced in each direction, let any number of straight lines whatever, AK, KL, be made equal to AE, and any number whatever, HM, MN, equal to EH; and let the parallelograms LP, KV, HW, MS and the solids LQ, KR, DM, MT be completed.

Then, since the straight lines LK, KA, AE are equal to one another, the parallelograms LP, KV, AF are also equal to one another, KO, KB, AG are equal to one another, and further LX, KQ, AR are equal to one another, for they are opposite. {XI. 24} For the same reason the parallelograms EC, HW, MS are also equal to one another, HG, HI, IN are equal to one another, and further DH, MY, NT are equal to one another.

Therefore in the solids LQ, KR, AU three planes are equal to three planes.

But the three planes are equal to the three opposite; therefore the three solids LQ, KR, AU are equal to one another.

For the same reason the three solids ED, DM, MT are also equal to one another.

Therefore, whatever multiple the base LF is of the base AF, the same multiple also is the solid LU of the solid AU.

For the same reason, whatever multiple the base NF is of the base FH, the same multiple also is the solid NU of the solid HU.

And, if the base LF is equal to the base NF, the solid LU is also equal to the solid NU; if the base LF exceeds the base NF, the solid LU also exceeds the solid NU; and, if one falls short, the other falls short.

Therefore, there being four magnitudes, the two bases AF, FH, and the two solids AU, UH, equimultiples have been taken of the base AF and the solid AU, namely the base LF and the solid LU, and equimultiples of the base HF and the solid HU, namely the base NF and the solid NU, and it has been proved that, if the base LF exceeds the base FN, the solid LU also exceeds the solid NU, if the bases are equal, the solids are equal, and if the base falls short, the solid falls short.

Therefore, as the base AF is to the base FH, so is the solid AU to the solid UH. {V. Def. 5} Q. E. D.

PROPOSITION 26. On a given straight line, and at a given point on it, to construct a solid angle equal to a given solid angle.

Let AB be the given straight line, A the given point on it, and the angle at D, contained by the angles EDC, EDF, FDC, the given solid angle; thus it is required to construct on the straight line AB, and at the point A on it, a solid angle equal to the solid angle at D.

For let a point F be taken at random on DF, let FG be drawn from F perpendicular to the plane through ED, DC, and let it meet the plane at G, {XI. 11} let DG be joined, let there be constructed on the straight line AB and at the point A on it the angle BAL equal to the angle EDC, and the angle BAK equal to the angle EDG, {I. 23} let AK be made equal to DG, let KH be set up from the point K at right angles to the plane through BA, AL, {XI. 12} let KH be made equal to GF, and let HA be joined; I say that the solid angle at A, contained by the angles BAL, BAH, HAL is equal to the solid angle at D contained by the angles EDC, EDF, FDC.

For let AB, DE be cut off equal to one another, and let HB, KB, FE, GE be joined.

Then, since FG is at right angles to the plane of reference, it will also make right angles with all the straight lines which meet it and are in the plane of reference; {XI. Def. 3} therefore each of the angles FGD, FGE is right.

For the same reason each of the angles HKA, HKB is also right.

And, since the two sides KA, AB are equal to the two sides GD, DE respectively, and they contain equal angles, therefore the base KB is equal to the base GE. {I. 4} But KH is also equal to GF, and they contain right angles; therefore HB is also equal to FE. {I. 4} Again, since the two sides AK, KH are equal to the two sides DG, GF, and they contain right angles, therefore the base AH is equal to the base FD. {I. 4} But AB is also equal to DE; therefore the two sides HA, AB are equal to the two sides DF, DE.

And the base HB is equal to the base FE; therefore the angle BAH is equal to the angle EDF. {I. 8} For the same reason the angle HAL is also equal to the angle FDC.

And the angle BAL is also equal to the angle EDC.

Therefore on the straight line AB, and at the point A on it, a solid angle has been constructed equal to the given solid angle at D. Q. E. F.

PROPOSITION 27. On a given straight line to describe a parallelepipedal solid similar and similarly situated to a given parallelepipedal solid.

Let AB be the given straight line and CD the given parallelepipedal solid; thus it is required to describe on the given straight line AB a parallelepipedal solid similar and similarly situated to the given parallelepipedal solid CD.

For on the straight line AB and at the point A on it let the solid angle, contained by the angles BAH, HAK, KAB, be constructed equal to the solid angle at C, so that the angle BAH is equal to the angle ECF, the angle BAK equal to the angle ECG, and the angle KAH to the angle GCF; and let it be contrived that, as EC is to CG, so is BA to AK, and, as GC is to CF, so is KA to AH. {VI. 12} Therefore also, ex aequali, as EC is to CF, so is BA to AH. {V. 22} Let the parallelogram HB and the solid AL be completed.

Now since, as EC is to CG, so is BA to AK, and the sides about the equal angles ECG, BAK are thus proportional, therefore the parallelogram GE is similar to the parallelogram KB.

For the same reason the parallelogram KH is also similar to the parallelogram GF, and further FE to HB; therefore three parallelograms of the solid CD are similar to three parallelograms of the solid AL.

But the former three are both equal and similar to the three opposite parallelograms, and the latter three are both equal and similar to the three opposite parallelograms; therefore the whole solid CD is similar to the whole solid AL. {XI. Def. 9} Therefore on the given straight line AB there has been described AL similar and similarly situated to the given parallelepipedal solid CD. Q. E. F.

PROPOSITION 28. If a parallelepipedal solid be cut by a plane through the diagonals of the opposite planes, the solid will be bisected by the plane.

For let the parallelepipedal solid AB be cut by the plane CDEF through the diagonals CF, DE of opposite planes; I say that the solid AB will be bisected by the plane CDEF.

For, since the triangle CGF is equal to the triangle CFB, {I. 34} and ADE to DEH, while the parallelogram CA is also equal to the parallelogram EB, for they are opposite, and GE to CH, therefore

the prism contained by the two triangles CGF, ADE and the three parallelograms GE, AC, CE is also equal to the prism contained by the two triangles CFB, DEH and the three parallelograms CH, BE, CE; for they are contained by planes equal both in multitude and in magnitude. {XI. Def. 10} Hence the whole solid AB is bisected by the plane CDEF. Q. E. D.

PROPOSITION 29. Parallelepipedal solids which are on the same base and of the same height, and in which the extremities of the sides which stand up are on the same straight lines, are equal to one another.

Let CM, CN be parallelepipedal solids on the same base AB and of the same height, and let the extremities of their sides which stand up, namely AG, AF, LM, LN, CD, CE, BH, BK, be on the same straight lines FN, DK; I say that the solid CM is equal to the solid CN.

For, since each of the figures CH, CK is a parallelogram, CB is equal to each of the straight lines DH, EK, {I. 34} hence DH is also equal to EK.

Let EH be subtracted from each; therefore the remainder DE is equal to the remainder HK.

Hence the triangle DCE is also equal to the triangle HBK, {I. 8, 4} and the parallelogram DG to the parallelogram HN. {I. 36} For the same reason the triangle AFG is also equal to the triangle MLN.

But the parallelogram CF is equal to the parallelogram BM, and CG to BN, for they are opposite; therefore the prism contained by the two triangles AFG, DCE and the three parallelograms AD, DG, CG is equal to the prism contained by the two triangles MLN, HBK and the three parallelograms BM, HN, BN.

Let there be added to each the solid of which the parallelogram AB is the base and GEHM its opposite; therefore the whole

parallelepipedal solid CM is equal to the whole parallelepipedal solid CN.

Therefore etc. Q. E. D.

PROPOSITION 30. Parallelepipedal solids which are on the same base and of the same height, and in which the extremities of the sides which stand up are not on the same straight lines, are equal to one another.

Let CM, CN be parallelepipedal solids on the same base AB and of the same height, and let the extremities of their sides which stand up, namely AF, AG, LM, LN, CD, CE, BH, BK, not be on the same straight lines; I say that the solid CM is equal to the solid CN.

For let NK, DH be produced and meet one another at R, and further let FM, GE be produced to P, Q; let AO, LP, CQ, BR be joined.

Then the solid CM, of which the parallelogram ACBL is the base, and FDHM its opposite, is equal to the solid CP, of which the parallelogram ACBL is the base, and OQRP its opposite; for they are on the same base ACBL and of the same height, and the extremities of their sides which stand up, namely AF, AO, LM, LP, CD, CQ, BH, BR, are on the same straight lines FP, DR. {XI. 29} But the solid CP, of which the parallelogram ACBL is the base, and OQRP its opposite, is equal to the solid CN, of which the parallelogram ACBL is the base and GEKN its opposite; for they are again on the same base ACBL and of the same height, and the extremities of their sides which stand up, namely AG, AO, CE, CQ, LN, LP, BK, BR, are on the same straight lines GQ, NR.

Hence the solid CM is also equal to the solid CN.

Therefore etc. Q. E. D.

PROPOSITION 31. Parallelepipedal solids which are on equal bases and of the same height are equal to one another.

Let the parallelepipedal solids AE, CF, of the same height, be on equal bases AB, CD.

I say that the solid AE is equal to the solid CF.

First, let the sides which stand up, HK, BE, AG, LM, PQ, DF, CO, RS, be at right angles to the bases AB, CD; let the straight line RT be produced in a straight line with CR; on the straight line RT, and at the point R on it, let the angle TRU be constructed equal to the angle ALB, {I. 23} let RT be made equal to AL, and RU equal to LB, and let the base RW and the solid XU be completed.

Now, since the two sides TR, RU are equal to the two sides AL, LB, and they contain equal angles, therefore the parallelogram RW is equal and similar to the parallelogram HL.

Since again AL is equal to RT, and LM to RS, and they contain right angles, therefore the parallelogram RX is equal and similar to the parallelogram AM.

For the same reason LE is also equal and similar to SU; therefore three parallelograms of the solid AE are equal and similar to three parallelograms of the solid XU.

But the former three are equal and similar to the three opposite, and the latter three to the three opposite; {XI. 24} therefore the whole parallelepipedal solid AE is equal to the whole parallelepipedal solid XU. {XI. Def. 10} Let DR, WU be drawn through and meet one another at Y, let aTb be drawn through T parallel to DY, let PD be produced to a, and let the solids YX, RI be completed.

Then the solid XY, of which the parallelogram RX is the base and Yc its opposite, is equal to the solid XU of which the parallelogram RX is the base and UV its opposite, for they are on the same base RX and of the same height, and the extremities of their sides which stand up, namely RY, RU, Tb, TW, Se, Sd, Xc, XV, are on the same straight lines YW, eV. {XI.

29) But the solid XU is equal to AE: therefore the solid XY is also equal to the solid AE.

And, since the parallelogram RUWT is equal to the parallelogram YT for they are on the same base RT and in the same parallels RT, YW, {I. 35} while RUWT is equal to CD, since it is also equal to AB, therefore the parallelogram YT is also equal to CD.

But DT is another parallelogram; therefore, as the base CD is to DT, so is YT to DT. {V. 7} And, since the parallelepipedal solid CI has been cut by the plane RF which is parallel to opposite planes, as the base CD is to the base DT, so is the solid CF to the solid RI. {XI. 25} For the same reason, since the parallelepipedal solid YI has been cut by the plane RX which is parallel to opposite planes, as the base YT is to the base TD, so is the solid YX to the solid RI. {XI. 25} But, as the base CD is to DT, so is YT to DT; therefore also, as the solid CF is to the solid RI, so is the solid YX to RI. {V. 11} Therefore each of the solids CF, YX has to RI the same ratio; therefore the solid CF is equal to the solid YX. {V. 9} But YX was proved equal to AE; therefore AE is also equal to CF.

Next, let the sides standing up, AG, HK, BE, LM, CN, PQ, DF, RS, not be at right angles to the bases AB, CD; I say again that the solid AE is equal to the solid CF.

For from the points K, E, G, M, Q, F, N, S let KO, ET, GU, MV, QW, FX, NY, SI be drawn perpendicular to the plane of reference, and let them meet the plane at the points O, T, U, V, W, X, Y, I, and let OT, OU, UV, TV, WX, WY, YI, IX be joined.

Then the solid KV is equal to the solid QI, for they are on the equal bases KM, QS and of the same height, and their sides which stand up are at right angles to their bases. {First part of this Prop.} But the solid KV is equal to the solid AE, and QI to CF; for they are on the same base and of the same height, while the extremities of their sides which stand up are not on the same

straight lines. {XI. 30} Therefore the solid AE is also equal to the solid CF.

Therefore etc. Q. E. D.

PROPOSITION 32. Parallelepipedal solids which are of the same height are to one another as their bases.

Let AB, CD be parallelepipedal solids of the same height; I say that the parallelepipedal solids AB, CD are to one another as their bases, that is, that, as the base AE is to the base CF, so is the solid AB to the solid CD.

For let FH equal to AE be applied to FG, {I. 45} and on FH as base, and with the same height as that of CD, let the parallelepipedal solid GK be completed.

Then the solid AB is equal to the solid GK; for they are on equal bases AE, FH and of the same height. {XI. 31} And, since the parallelepipedal solid CK is cut by the plane DG which is parallel to opposite planes, therefore, as the base CF is to the base FH, so is the solid CD to the solid DH. {XI. 25} But the base FH is equal to the base AE, and the solid GK to the solid AB; therefore also, as the base AE is to the base CF, so is the solid AB to the solid CD.

Therefore etc. Q. E. D.

PROPOSITION 33. Similar parallelepipedal solids are to one another in the triplicate ratio of their corresponding sides.

Let AB, CD be similar parallelepipedal solids, and let AE be the side corresponding to CF; I say that the solid AB has to the solid CD the ratio triplicate of that which AE has to CF.

For let EK, EL, EM be produced in a straight line with AE, GE, HE, let EK be made equal to CF, EL equal to FN, and further EM equal to FR, and let the parallelogram KL and the solid KP be completed.

Now, since the two sides KE, EL are equal to the two sides CF, FN, while the angle KEL is also equal to the angle CFN, inasmuch as the angle AEG is also equal to the angle CFN because of the similarity of the solids AB, CD, therefore the parallelogram KL is equal <and similar> to the parallelogram CN.

For the same reason the parallelogram KM is also equal and similar to CR, and further EP to DF; therefore three parallelograms of the solid KP are equal and similar to three parallelograms of the solid CD.

But the former three parallelograms are equal and similar to their opposites, and the latter three to their opposites; {XI. 24} therefore the whole solid KP is equal and similar to the whole solid CD. {XI. Def. 10} Let the parallelogram GK be completed, and on the parallelograms GK, KL as bases, and with the same height as that of AB, let the solids EO, LQ be completed.

Then since; owing to the similarity of the solids AB, CD, as AE is to CF, so is EG to FN, and EH to FR, while CF is equal to EK, FN to EL, and FR to EM, therefore, as AE is to EK, so is GE to EL, and HE to EM.

But, as AE is to EK, so is AG to the parallelogram GK, as GE is to EL, so is GK to KL, and, as HE is to EM, so is QE to KM; {VI. 1} therefore also, as the parallelogram AG is to GK, so is GK to KL, and QE to KM.

But, as AG is to GK, so is the solid AB to the solid EO, as GK is to KL, so is the solid OE to the solid QL, and, as QE is to KM, so is the solid QL to the solid KP; {XI. 32} therefore also, as the solid AB is to EO, so is EO to QL, and QL to KP.

But, if four magnitudes be continuously proportional, the first has to the fourth the ratio triplicate of that which it has to the second; {V. Def. 10} therefore the solid AB has to KP the ratio triplicate of that which AB has to EO.

But, as AB is to EO, so is the parallelogram AG to GK, and the straight line AE to EK {VI. 1}; hence the solid AB has also to KP the ratio triplicate of that which AE has to EK.

But the solid KP is equal to the solid CD, and the straight line EK to CF; therefore the solid AB has also to the solid CD the ratio triplicate of that which the corresponding side of it, AE, has to the corresponding side CF.

Therefore etc. Q. E. D.

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From this it is manifest that, if four straight lines be <continuously> proportional, as the first is to the fourth, so will a parallelepipedal solid on the first be to the similar and similarly described parallelepipedal solid on the second, inasmuch as the first has to the fourth the ratio triplicate of that which it has to the second.

PROPOSITION 34. In equal parallelepipedal solids the bases are reciprocally proportional to the heights; and those parallelepipedal solids in which the bases are reciprocally proportional to the heights are equal.

Let AB, CD be equal parallelepipedal solids; I say that in the parallelepipedal solids AB, CD the bases are reciprocally proportional to the heights, that is, as the base EH is to the base NQ, so is the height of the solid CD to the height of the solid AB.

First, let the sides which stand up, namely AG, EF, LB, HK, CM, NO, PD, QR, be at right angles to their bases; I say that, as the base EH is to the base NQ, so is CM to AG.

If now the base EH is equal to the base NQ, while the solid AB is also equal to the solid CD, CM will also be equal to AG.

For parallelepipedal solids of the same height are to one another as the bases; {XI. 32} and, as the base EH is to NQ, so will

CM be to AG, and it is manifest that in the parallelepipedal solids AB, CD the bases are reciprocally proportional to the heights.

Next, let the base EH not be equal to the base NQ, but let EH be greater.

Now the solid AB is equal to the solid CD; therefore CM is also greater than AG.

Let then CT be made equal to AG, and let the parallelepipedal solid VC be completed on NQ as base and with CT as height.

Now, since the solid AB is equal to the solid CD, and CV is outside them, while equals have to the same the same ratio, {V. 7} therefore, as the solid AB is to the solid CV, so is the solid CD to the solid CV.

But, as the solid AB is to the solid CV, so is the base EH to the base NQ, for the solids AB, CV are of equal height; {XI. 32} and, as the solid CD is to the solid CV, so is the base MQ to the base TQ {XI. 25} and CM to CT {VI. 1}; therefore also, as the base EH is to the base NQ, so is MC to CT.

But CT is equal to AG; therefore also, as the base EH is to the base NQ, so is MC to AG.

Therefore in the parallelepipedal solids AB, CD the bases are reciprocally proportional to the heights.

Again, in the parallelepipedal solids AB, CD let the bases be reciprocally proportional to the heights, that is, as the base EH is to the base NQ, so let the height of the solid CD be to the height of the solid AB; I say that the solid AB is equal to the solid CD.

Let the sides which stand up be again at right angles to the bases.

Now, if the base EH is equal to the base NQ, and, as the base EH is to the base NQ, so is the height of the solid CD to the height of the solid AB, therefore the height of the solid CD is also equal to the height of the solid AB.

But parallelepipedal solids on equal bases and of the same height are equal to one another; {XI. 31} therefore the solid AB is equal to the solid CD.

Next, let the base EH not be equal to the base NQ, but let EH be greater; therefore the height of the solid CD is also greater than the height of the solid AB, that is, CM is greater than AG.

Let CT be again made equal to AG, and let the solid CV be similarly completed.

Since, as the base EH is to the base NQ, so is MC to AG, while AG is equal to CT, therefore, as the base EH is to the base NQ, so is CM to CT.

But, as the base EH is to the base NQ, so is the solid AB to the solid CV, for the solids AB, CV are of equal height; {XI. 32} and, as CM is to CT, so is the base MQ to the base QT {VI. 1} and the solid CD to the solid CV. {XI. 25} Therefore also, as the solid AB is to the solid CV, so is the solid CD to the solid CV; therefore each of the solids AB, CD has to CV the same ratio.

Therefore the solid AB is equal to the solid CD. {V. 9} Now let the sides which stand up, FE, BL, GA, HK, ON, DP, MC, RQ, not be at right angles to their bases; let perpendiculars be drawn from the points F, G, B, K, O, M, D, R to the planes through EH, NQ, and let them meet the planes at S, T, U, V, W, X, Y, a, and let the solids FV, Oa be completed; I say that, in this case too, if the solids AB, CD are equal, the bases are reciprocally proportional to the heights, that is, as the base EH is to the base NQ, so is the height of the solid CD to the height of the solid AB.

Since the solid AB is equal to the solid CD, while AB is equal to BT, for they are on the same base FK and of the same height; {XI. 29, 30} and the solid CD is equal to DX, for they are again on the same base RO and of the same height; {id.} therefore the solid BT is also equal to the solid DX.

Therefore, as the base FK is to the base OR, so is the height of the solid DX to the height of the solid BT. {Part 1.} But the base FK is equal to the base EH, and the base OR to the base NQ; therefore, as the base EH is to the base NQ, so is the height of the solid DX to the height of the solid BT.

But the solids DX, BT and the solids DC, BA have the same heights respectively; therefore, as the base EH is to the base NQ, so is the height of the solid DC to the height of the solid AB.

Therefore in the parallelepipedal solids AB, CD the bases are reciprocally proportional to the heights.

Again, in the parallelepipedal solids AB, CD let the bases be reciprocally proportional to the heights, that is, as the base EH is to the base NQ, so let the height of the solid CD be to the height of the solid AB; I say that the solid AB is equal to the solid CD.

For, with the same construction, since, as the base EH is to the base NQ, so is the height of the solid CD to the height of the solid AB, while the base EH is equal to the base FK, and NQ to OR, therefore, as the base FK is to the base OR, so is the height of the solid CD to the height of the solid AB.

But the solids AB, CD and BT, DX have the same heights respectively; therefore, as the base FK is to the base OR, so is the height of the solid DX to the height of the solid BT.

Therefore in the parallelepipedal solids BT, DX the bases are reciprocally proportional to the heights; therefore the solid BT is equal to the solid DX. {Part 1.} But BT is equal to BA, for they

are on the same base FK and of the same height; {XI. 29, 30} and the solid DX is equal to the solid DC. {id.} Therefore the solid AB is also equal to the solid CD. Q. E. D.

PROPOSITION 35. If there be two equal plane angles, and on their vertices there be set up elevated straight lines containing equal angles with the original straight lines respectively, if on the elevated straight lines points be taken at random and perpendiculars be drawn from them to the planes in which the original angles are, and if from the points so arising in the planes straight lines be joined to the vertices of the original angles, they will contain, with the elevated straight lines, equal angles.

Let the angles BAC, EDF be two equal rectilineal angles, and from the points A, D let the elevated straight lines AG, DM be set up containing, with the original straight lines, equal angles respectively, namely, the angle MDE to the angle GAB and the angle MDF to the angle GAC, let points G, M be taken at random on AG, DM, let GL, MN be drawn from the points G, M perpendicular to the planes through BA, AC and ED, DF, and let them meet the planes at L, N, and let LA, ND be joined; I say that the angle GAL is equal to the angle MDN.

Let AH be made equal to DM, and let HK be drawn through the point H parallel to GL.

But GL is perpendicular to the plane through BA, AC; therefore HK is also perpendicular to the plane through. BA, AC. {XI. 8} From the points K, N let KC, NF, KB, NE be drawn perpendicular to the straight lines AC, DF, AB, DE, and let HC, CB, MF, FE be joined.

Since the square on HA is equal to the squares on HK, KA, and the squares on KC, CA are equal to the square on KA, {I. 47} therefore the square on HA is also equal to the squares on HK, KC, CA.

But the square on HC is equal to the squares on HK, KC; {I. 47} therefore the square on HA is equal to the squares on HC, CA.

Therefore the angle HCA is right. {I. 48} For the same reason the angle DFM is also right.

Therefore the angle ACH is equal to the angle DFM.

But the angle HAC is also equal to the angle MDF.

Therefore MDF, HAC are two triangles which have two angles equal to two angles respectively, and one side equal to one side, namely, that subtending one of the equal angles, that is, HA equal to MD; therefore they will also have the remaining sides equal to the remaining sides respectively. {I. 26} Therefore AC is equal to DF.

Similarly we can prove that AB is also equal to DE.

Since then AC is equal to DF, and AB to DE, the two sides CA, AB are equal to the two sides FD, DE.

But the angle CAB is also equal to the angle FDE; therefore the base BC is equal to the base EF, the triangle to the triangle, and the remaining angles to the remaining angles; {I. 4} therefore the angle ACB is equal to the angle DFE.

But the right angle ACK is also equal to the right angle DFN; therefore the remaining angle BCK is also equal to the remaining angle EFN.

For the same reason the angle CBK is also equal to the angle FEN.

Therefore BCK, EFN are two triangles which have two angles equal to two angles respectively, and one side equal to one side, namely, that adjacent to the equal angles, that is, BC equal to EF; therefore they will also have the remaining sides equal to the remaining sides. {I. 26} Therefore CK is equal to FN.

But AC is also equal to DF; therefore the two sides AC, CK are equal to the two sides DF, FN; and they contain right angles.

Therefore the base AK is equal to the base DN. {I. 4} And, since AH is equal to DM, the square on AH is also equal to the square on DM.

But the squares on AK, KH are equal to the square on AH, for the angle AKH is right; {I. 47} and the squares on DN, NM are equal to the square on DM, for the angle DNM is right; {I. 47} therefore the squares on AK, KH are equal to the squares on DN, NM; and of these the square on AK is equal to the square on DN; therefore the remaining square on KH is equal to the square on NM; therefore HK is equal to MN.

And, since the two sides HA, AK are equal to the two sides MD, DN respectively, and the base HK was proved equal to the base MN, therefore the angle HAK is equal to the angle MDN. {I. 8} Therefore etc.

PORISM.

From this it is manifest that, if there be two equal plane angles, and if there be set up on them elevated straight lines which are equal and contain equal angles with the original straight lines respectively, the perpendiculars drawn from their extremities to the planes in which are the original angles are equal to one another. Q. E. D.

PROPOSITION 36. If three straight lines be proportional, the parallelepipedal solid formed out of the three is equal to the parallelepipedal solid on the mean which is equilateral, but equiangular with the aforesaid solid.

Let A, B, C be three straight lines in proportion, so that, as A is to B, so is B to C; I say that the solid formed out of A, B, C is equal to the solid on B which is equilateral, but equiangular with the aforesaid solid.

Let there be set out the solid angle at E contained by the angles DEG, GEF, FED, let each of the straight lines DE, GE, EF be made equal to B, and let the parallelepipedal solid EK be completed, let LM be made equal to A, and on the straight line LM, and at the point L on it, let there be constructed a solid angle equal to the solid angle at E, namely that contained by NLO, OLM, MLN; let LO be made equal to B, and LN equal to C.

Now, since, as A is to B, so is B to C, while A is equal to LM, B to each of the straight lines LO, ED, and C to LN, therefore, as LM is to EF, so is DE to LN.

Thus the sides about the equal angles NLM, DEF are reciprocally proportional; therefore the parallelogram MN is equal to the parallelogram DF. {VI. 14} And, since the angles DEF, NLM are two plane rectilineal angles, and on them the elevated straight lines LO, EG are set up which are equal to one another and contain equal angles with the original straight lines respectively, therefore the perpendiculars drawn from the points G, O to the planes through NL, LM and DE, EF are equal to one another; {XI. 35, Por.} hence the solids LH, EK are of the same height.

But parallelepipedal solids on equal bases and of the same height are equal to one another; {XI. 31} therefore the solid HL is equal to the solid EK.

And LH is the solid formed out of A, B, C, and EK the solid on B; therefore the parallelepipedal solid formed out of A, B, C is equal to the solid on B which is equilateral, but equiangular with the aforesaid solid. Q. E. D.

PROPOSITION 37. If four straight lines be proportional, the parallelepipedal solids on them which are similar and similarly described will also be proportional; and, if the parallelepipedal solids on them which are similar and similarly described be proportional, the straight lines will themselves also be

proportional.

Let AB, CD, EF, GH be four straight lines in proportion, so that, as AB is to CD, so is EF to GH; and let there be described on AB, CD, EF, GH the similar and similarly situated parallelepipedal solids KA, LC, ME, NG; I say that, as KA is to LC, so is ME to NG.

For, since the parallelepipedal solid KA is similar to LC, therefore KA has to LC the ratio triplicate of that which AB has to CD. {XI. 33} For the same reason ME also has to NG the ratio triplicate of that which EF has to GH. {id.} And, as AB is to CD, so is EF to GH.

Therefore also, as KA is to LC, so is ME to NG.

Next, as the solid KA is to the solid LC, so let the solid ME be to the solid NG; I say that, as the straight line AB is to CD, so is EF to GH.

For since, again, KA has to LC the ratio triplicate of that which AB has to CD, {XI. 33} and ME also has to NG the ratio triplicate of that which EF has to GH, {id.} and, as KA is to LC, so is ME to NG, therefore also, as AB is to CD, so is EF to GH.

Therefore etc. Q. E. D.

PROPOSITION 38. If the sides of the opposite planes of a cube be bisected, and planes be carried through the points of section, the common section of the planes and the diameter of the cube bisect one another.

For let the sides of the opposite planes CF, AH of the cube AF be bisected at the points K, L, M, N, O, Q, P, R, and through the points of section let the planes KN, OR be carried; let US be the common section of the planes, and DG the diameter of the cube AF.

I say that UT is equal to TS, and DT to TG.

For let DU, UE, BS, SG be joined.

Then, since DO is parallel to PE, the alternate angles DOU, UPE are equal to one another. {I. 29} And, since DO is equal to PE, and OU to UP, and they contain equal angles, therefore the base DU is equal to the base UE, the triangle DOU is equal to the triangle PUE, and the remaining angles are equal to the remaining angles; {I. 4} therefore the angle OUD is equal to the angle PUE.

For this reason DUE is a straight line. {I. 14} For the same reason, BSG is also a straight line, and BS is equal to SG.

Now, since CA is equal and parallel to DB, while CA is also equal and parallel to EG, therefore DB is also equal and parallel to EG. {XI. 9} And the straight lines DE, BG join their extremities; therefore DE is parallel to BG. {I. 33} Therefore the angle EDT is equal to the angle BGT, for they are alternate; {I. 29} and the angle DTU is equal to the angle GTS. {I. 15} Therefore DTU, GTS are two triangles which have two angles equal to two angles, and one side equal to one side, namely that subtending one of the equal angles, that is, DU equal to GS, for they are the halves of DE, BG; therefore they will also have the remaining sides equal to the remaining sides. {I. 26} Therefore DT is equal to TG, and UT to TS.

Therefore etc. Q. E. D.

PROPOSITION 39. If there be two prisms of equal height, and one have a parallelogram as base and the other a triangle, and if the parallelogram be double of the triangle, the prisms will be equal.

Let ABCDEF, GHKLMN be two prisms of equal height, let one have the parallelogram AF as base, and the other the triangle GHK, and let the parallelogram AF be double of the triangle GHK; I say that the prism ABCDEF is equal to the prism GHKLMN.

For let the solids AO, GP be completed.

Since the parallelogram AF is double of the triangle GHK, while the parallelogram HK is also double of the triangle GHK, {I. 34} therefore the parallelogram AF is equal to the parallelogram HK.

But parallelepipedal solids which are on equal bases and of the same height are equal to one another; {XI. 31} therefore the solid AO is equal to the solid GP.

And the prism ABCDEF is half of the solid AO, and the prism GHKLMN is half of the solid GP; {XI. 28} therefore the prism ABCDEF is equal to the prism GHKLMN.

Therefore etc. Q. E. D.

ENDNOTES.

1. the plane of reference, τὸ ὑποκείμενον ἐπίπεδον, the plane laid down or assumed.

2. more elevated, μετεωροτέρω.

BOOK XII. HISTORICAL NOTE.

The predominant feature of Book XII. is the use of the method of exhaustion, which is applied in Propositions 2, 3-5, 10, 11, 12, and (in a slightly different form) in Propositions 16-18. We conclude therefore that for the content of this Book Euclid was greatly indebted to Eudoxus, to whom the discovery of the method of exhaustion is attributed. The evidence for this attribution comes mainly from Archimedes. (1) In the preface to *On the Sphere and Cylinder I.*, after stating the main results obtained by himself regarding the surface of a sphere or a segment thereof, and the volume and surface of a right cylinder with height equal to its diameter as compared with those of a sphere with the same diameter, Archimedes adds: "Having now discovered that the properties mentioned are true of these figures, I cannot feel any hesitation in setting them side by side

both with my former investigations and with those of the theorems of Eudoxus on solids which are held to be most irrefragably established, namely that any pyramid is one third part of the prism which has the same base with the pyramid and equal height {i.e. Eucl. XII. 7}, and that any cone is one third part of the cylinder which has the same base with the cone and equal height {i.e. Eucl. XII. 10}. For, though these properties also were naturally inherent in the figures all along, yet they were in fact unknown to all the many able geometers who lived before Eudoxus and had not been observed by any one.” (2) In the preface to the treatise known as the Quadrature of the Parabola Archimedes states the “lemma” assumed by him and known as the “Axiom of Archimedes” (see note on X. 1 above) and proceeds: “Earlier geometers (οἱ πρότερον γεωμέτραι) have also used this lemma; for it is by the use of this same lemma that they have shown that circles are to one another in the duplicate ratio of their diameters {Eucl. XII. 2}, and that spheres are to one another in the triplicate ratio of their diameters {Eucl. XII. 18}, and further that every pyramid is one third part of the prism which has the same base with the pyramid and equal height {Eucl. XII. 7}; also, that every cone is one third part of the cylinder which has the same base with the cone and equal height {Eucl. XII. 10} they proved by assuming a certain lemma similar to that aforesaid.” Thus in the first passage two theorems of Eucl. XII. are definitely attributed to Eudoxus; and, when Archimedes says, in the second passage, that “earlier geometers” proved these two theorems by means of the lemma known as the “Axiom of Archimedes” and of a lemma similar to it respectively, we can hardly suppose him to be alluding to any other proof than that given by Eudoxus. As a matter of fact, the lemma used by Euclid to prove both propositions (XII. 3-5 and 7, and XII. 10) is the theorem of Eucl. X. 1. As regards the connexion between the two “lemmas” see note on X. 1.

We are not, however, to suppose that none of the results obtained by the method of exhaustion had been discovered before the time of Eudoxus (fl. about 368-5 B.C.). Two at least are of earlier date, those of Eucl. XII. 2 and XII. 7.

□a) Simplicius (Comment. in Aristot. Phys. p. 61, ed. Diels) quotes Eudemus as saying, in his History of Geometry, that Hippocrates of Chios (fl. say 430 B.C.) first laid it down (ἔθετο) that similar segments of circles are in the ratio of the squares on their bases and that he proved this (ἐδείκνυεν) by proving (ἐκ τοῦ δείξαι) that the squares on the diameters have the same ratio as the (whole) circles. We know nothing of the method by which Hippocrates proved this proposition; but, having regard to the evidence from Archimedes quoted above, it is not permissible to suppose that the method was the fully developed method of exhaustion as we know it.

□b) As regards the two theorems about the volume of a pyramid and of a cone respectively, which Eudoxus was the first to prove, we now have authentic evidence in the short treatise by Archimedes discovered by Heiberg in a MS. at Constantinople in 1906 and published in Hermes the following year (see now Archimedis opera omnia, ed. Heiberg, 2. ed., Vol. II., 1913, pp. 425-507; T. L. Heath, The Method of Archimedes, Cambridge, 1912). The said treatise, complete in all essentials, bears the title Ἀρχιμήδους περὶ τῶν μηχανικῶν θεωρημάτων πρὸς Ἐρατοσθένην ἔφοδος. This “Method” (or “Plan of attack”), addressed to Eratosthenes, is none other than the ἐφόδιον on which, according to Suidas, Theodosius wrote a commentary, and which is several times cited by Heron in his Metrica; its discovery adds a new and important chapter to the history of the integral calculus. In the preface to this work Archimedes alludes to the theorems which he first discovered by means of mechanical considerations, but proved afterwards by geometry, because the investigation by means of mechanics did not

constitute a rigid proof; he observes, however, that the mechanical method is of great use for the discovery of theorems, and it is much easier to provide the rigid proof when the fact to be proved has once been discovered than it would be if nothing were known to begin with. He goes on: "Hence too, in the case of those theorems the proof of which was first discovered by Eudoxus, namely those relating to the cone and the pyramid, that the cone is one third part of the cylinder, and the pyramid one third part of the prism, having the same base and equal height, no small part of the credit will naturally be assigned to Democritus, who was the first to make the statement (of the fact) regarding the said figure {i.e. property}, though without proving it." Hence the discovery of the two theorems must now be attributed to Democritus (fl. towards the end of 5th cent. B.C.). The words "without proving it" (χωρὶς ἀποδείξεως) do not mean that Democritus gave no sort of proof, but only that he did not give a proof on the rigorous lines required later; for the same words are used by Archimedes of his own investigations by means of mechanics, which, however, do constitute a reasoned argument. The character of Archimedes' mechanical arguments combined with a passage of Plutarch about a particular question in infinitesimals said to have been raised by Democritus may perhaps give a clue to the line of Democritus' argument as regards the pyramid. The essential feature of Archimedes' mechanical arguments in this tract is that he regards an area as the sum of an infinite number of straight lines parallel to one another and terminated by the boundary or boundaries of the closed figure the area of which is to be found, and a volume as the sum of an infinite number of plane sections parallel to one another: which is of course the same thing as taking (as we do in the integral calculus) the sum of an infinite number of strips of breadth dx (say), when dx becomes indefinitely small, or the sum of an infinite number of parallel laminae of depth dz (say), when dz becomes indefinitely

small. To give only one instance, we may take the case of the area of a segment of a parabola cut off by a chord.

Let CBA be the parabolic segment, CE the tangent at C meeting the diameter EBD through the middle point of the chord CA in E, so that .

Draw AF parallel to ED meeting CE produced in F. Produce CB to H so that , where K is the point in which CH meets AF; and suppose CH to be a lever.

Let any diameter MNPO be drawn meeting the curve in P and CF, CK, CA in M, N, O respectively.

Archimedes then observes that (“for this is proved in a lemma”), whence , so that, if a straight line TG equal to PO be placed with its middle point at H, the straight line MO with centre of gravity at N, and the straight line TG with centre of gravity at H, will balance about K.

Taking all other parts of diameters like PO intercepted between the curve and CA, and placing equal straight lines with their centres of gravity at H, these straight lines collected at H will balance (about K) all the lines like MO parallel to FA intercepted within the triangle CFA in the positions in which they severally lie in the figure.

Hence Archimedes infers that an area equal to that of the parabolic segment hung at H will balance (about K) the triangle CFA hung at its centre of gravity, the point X (a point on CK such that), and therefore that , from which it follows that .

The same sort of argument is used for solids, plane sections taking the place of straight lines.

Archimedes is careful to state once more that this method of argument does not constitute a proof. Thus, at the end of the above proposition about the parabolic segment, he adds: “This property is of course not proved by what has just been said; but

it has furnished a sort of indication (ἐμφασίν τινα) that the conclusion is true.”

Let us now turn to the passage of Plutarch (De Comm. Not. adv. Stoicos XXXIX 3) about Democritus above referred to. Plutarch speaks of Democritus as having raised the question in natural philosophy (φυσικῶς): “if a cone were cut by a plane parallel to the base {by which is clearly meant a plane indefinitely near to the base}, what must we think of the surfaces of the sections, that they are equal or unequal? For, if they are unequal, they will make the cone irregular, as having many indentations, like steps, and unevennesses; but, if they are equal, the sections will be equal, and the cone will appear to have the property of the cylinder and to be made up of equal, not unequal circles, which is very absurd.” The phrase “made up of equal...circles” (ἐξ ἴσων συγκείμενος...κύκλων) shows that Democritus already had the idea of a solid being the sum of an infinite number of parallel planes, or indefinitely thin laminae, indefinitely near together: a most important anticipation of the same thought which led to such fruitful results in Archimedes. If then one may hazard a conjecture as to Democritus’ argument with regard to a pyramid, it seems probable that he would notice that, if two pyramids of the same height and equal triangular bases are respectively cut by planes parallel to the base and dividing the heights in the same ratio, the corresponding sections of the two pyramids are equal, whence he would infer that the pyramids are equal as being the sum of the same infinite number of equal plane sections or indefinitely thin laminae. (This would be a particular anticipation of Cavalieri’s proposition that the areal or solid contents of two figures are equal if two sections of them taken at the same height, whatever the height may be, always give equal straight lines or equal surfaces respectively.) And Democritus would of course see that the three pyramids into which a prism on the same base and of equal height with the original pyramid is divided (as in Eucl. XII. 7) satisfy this test of

equality, so that the pyramid would be one third part of the prism. The extension to a pyramid with a polygonal base would be easy. And Democritus may have stated the proposition for the cone (of course without an absolute proof) as a natural inference from the result of increasing indefinitely the number of sides in a regular polygon forming the base of a pyramid.

PROPOSITIONS.

PROPOSITION 1. Similar polygons inscribed in circles are to one another as the squares on the diameters.

Let ABC, FGH be circles, let ABCDE, FGHL be similar polygons inscribed in them, and let BM, GN be diameters of the circles; I say that, as the square on BM is to the square on GN, so is the polygon ABCDE to the polygon FGHL.

For let BE, AM, GL, FN be joined.

Now, since the polygon ABCDE is similar to the polygon FGHL, the angle BAE is equal to the angle GFL, and, as BA is to AE, so is GF to FL. {VI. Def. 1} Thus BAE, GFL are two triangles which have one angle equal to one angle, namely the angle BAE to the angle GFL, and the sides about the equal angles proportional; therefore the triangle ABE is equiangular with the triangle FGL. {VI. 6} Therefore the angle AEB is equal to the angle FLG.

But the angle AEB is equal to the angle AMB, for they stand on the same circumference; {III. 27} and the angle FLG to the angle FNG; therefore the angle AMB is also equal to the angle FNG.

But the right angle BAM is also equal to the right angle GFN; {III. 31} therefore the remaining angle is equal to the remaining angle. {I. 32} Therefore the triangle ABM is equiangular with the triangle FGN.

Therefore, proportionally, as BM is to GN, so is BA to GF. {VI. 4} But the ratio of the square on BM to the square on GN is

duplicate of the ratio of BM to GN, and the ratio of the polygon ABCDE to the polygon FGHKL is duplicate of the ratio of BA to GF; {VI. 20} therefore also, as the square on BM is to the square on GN, so is the polygon ABCDE to the polygon FGHKL.

Therefore etc. Q. E. D.

PROPOSITION 2. Circles are to one another as the squares on the diameters.

Let ABCD, EFGH be circles, and BD, FH their diameters; I say that, as the circle ABCD is to the circle EFGH, so is the square on BD to the square on FH.

For, if the square on BD is not to the square on FH as the circle ABCD is to the circle EFGH, then, as the square on BD is to the square on FH, so will the circle ABCD be either to some less area than the circle EFGH, or to a greater.

First, let it be in that ratio to a less area S.

Let the square EFGH be inscribed in the circle EFGH; then the inscribed square is greater than the half of the circle EFGH, inasmuch as, if through the points E, F, G, H we draw tangents to the circle, the square EFGH is half the square circumscribed about the circle, and the circle is less than the circumscribed square; hence the inscribed square EFGH is greater than the half of the circle EFGH.

Let the circumferences EF, FG, GH, HE be bisected at the points K, L, M, N, and let EK, KF, FL, LG, GM, MH, HN, NE be joined; therefore each of the triangles EKF, FLG, GMH, HNE is also greater than the half of the segment of the circle about it, inasmuch as, if through the points K, L, M, N we draw tangents to the circle and complete the parallelograms on the straight lines EF, FG, GH, HE, each of the triangles EKF, FLG, GMH, HNE will be half of the parallelogram about it, while the segment about it is less than the parallelogram; hence each of the

triangles EKF, FLG, GMH, HNE is greater than the half of the segment of the circle about it.

Thus, by bisecting the remaining circumferences and joining straight lines, and by doing this continually, we shall leave some segments of the circle which will be less than the excess by which the circle EFGH exceeds the area S.

For it was proved in the first theorem of the tenth book that, if two unequal magnitudes be set out, and if from the greater there be subtracted a magnitude greater than the half, and from that which is left a greater than the half, and if this be done continually, there will be left some magnitude which will be less than the lesser magnitude set out.

Let segments be left such as described, and let the segments of the circle EFGH on EK, KF, FL, LG, GM, MH, HN, NE be less than the excess by which the circle EFGH exceeds the area S.

Therefore the remainder, the polygon EKFLGMHN, is greater than the area S.

Let there be inscribed, also, in the circle ABCD the polygon AOBPCQDR similar to the polygon EKFLGMHN; therefore, as the square on BD is to the square on FH, so is the polygon AOBPCQDR to the polygon EKFLGMHN. {XII. 1} But, as the square on BD is to the square on FH, so also is the circle ABCD to the area S; therefore also, as the circle ABCD is to the area S, so is the polygon AOBPCQDR to the polygon EKFLGMHN; {V. 11} therefore, alternately, as the circle ABCD is to the polygon inscribed in it, so is the area S to the polygon EKFLGMHN. {V. 16} But the circle ABCD is greater than the polygon inscribed in it; therefore the area S is also greater than the polygon EKFLGMHN.

But it is also less: which is impossible.

Therefore, as the square on BD is to the square on FH, so is not the circle ABCD to any area less than the circle EFGH.

Similarly we can prove that neither is the circle EFGH to any area less than the circle ABCD as the square on FH is to the square on BD.

I say next that neither is the circle ABCD to any area greater than the circle EFGH as the square on BD is to the square on FH.

For, if possible, let it be in that ratio to a greater area S.

Therefore, inversely, as the square on FH is to the square on DB, so is the area S to the circle ABCD.

But, as the area S is to the circle ABCD, so is the circle EFGH to some area less than the circle ABCD; therefore also, as the square on FH is to the square on BD, so is the circle EFGH to some area less than the circle ABCD: {V. 11} which was proved impossible.

Therefore, as the square on BD is to the square on FH, so is not the circle ABCD to any area greater than the circle EFGH.

And it was proved that neither is it in that ratio to any area less than the circle EFGH; therefore, as the square on BD is to the square on FH, so is the circle ABCD to the circle EFGH.

Therefore etc. Q. E. D.

LEMMA.

I say that, the area S being greater than the circle EFGH, as the area S is to the circle ABCD, so is the circle EFGH to some area less than the circle ABCD.

For let it be contrived that, as the area S is to the circle ABCD, so is the circle EFGH to the area T.

I say that the area T is less than the circle ABCD.

For since, as the area S is to the circle $ABCD$, so is the circle $EFGH$ to the area T , therefore, alternately, as the area S is to the circle $EFGH$, so is the circle $ABCD$ to the area T . {V. 16} But the area S is greater than the circle $EFGH$; therefore the circle $ABCD$ is also greater than the area T .

Hence, as the area S is to the circle $ABCD$, so is the circle $EFGH$ to some area less than the circle $ABCD$. Q. E. D.

PROPOSITION 3. Any pyramid which has a triangular base is divided into two pyramids equal and similar to one another, similar to the whole and having triangular bases, and into two equal prisms; and the two prisms are greater than the half of the whole pyramid.

Let there be a pyramid of which the triangle ABC is the base and the point D the vertex; I say that the pyramid $ABCD$ is divided into two pyramids equal to one another, having triangular bases and similar to the whole pyramid, and into two equal prisms; and the two prisms are greater than the half of the whole pyramid.

For let AB , BC , CA , AD , DB , DC be bisected at the points E , F , G , H , K , L , and let HE , EG , GH , HK , KL , LH , KF , FG be joined.

Since AE is equal to EB , and AH to DH , therefore EH is parallel to DB . {VI. 2} For the same reason HK is also parallel to AB .

Therefore $HEBK$ is a parallelogram; therefore HK is equal to EB . {I. 34} But EB is equal to EA ; therefore AE is also equal to HK .

But AH is also equal to HD ; therefore the two sides EA , AH are equal to the two sides KH , HD respectively, and the angle EAH is equal to the angle KHD ; therefore the base EH is equal to the base KD . {I. 4} Therefore the triangle AEH is equal and similar to the triangle HKD .

For the same reason the triangle AHG is also equal and similar to the triangle HLD .

Now, since two straight lines EH, HG meeting one another are parallel to two straight lines KD, DL meeting one another, and are not in the same plane, they will contain equal angles. {XI. 10} Therefore the angle EHG is equal to the angle KDL.

And, since the two straight lines EH, HG are equal to the two KD, DL respectively, and the angle EHG is equal to the angle KDL, therefore the base EG is equal to the base KL; {I. 4} therefore the triangle EHG is equal and similar to the triangle KDL.

For the same reason the triangle AEG is also equal and similar to the triangle HKL.

Therefore the pyramid of which the triangle AEG is the base and the point H the vertex is equal and similar to the pyramid of which the triangle HKL is the base and the point D the vertex. {XI. Def. 10} And, since HK has been drawn parallel to AB, one of the sides of the triangle ADB, the triangle ADB is equiangular to the triangle DHK, {I. 29} and they have their sides proportional; therefore the triangle ADB is similar to the triangle DHK. {VI. Def. 1} For the same reason the triangle DBC is also similar to the triangle DKL, and the triangle ADC to the triangle DLH.

Now, since the two straight lines BA, AC meeting one another are parallel to the two straight lines KH, HL meeting one another, not in the same plane, they will contain equal angles. {XI. 10} Therefore the angle BAC is equal to the angle KHL.

And, as BA is to AC, so is KH to HL; therefore the triangle ABC is similar to the triangle HKL.

Therefore also the pyramid of which the triangle ABC is the base and the point D the vertex is similar to the pyramid of which the triangle HKL is the base and the point D the vertex.

But the pyramid of which the triangle HKL is the base and the point D the vertex was proved similar to the pyramid of which the triangle AEG is the base and the point H the vertex.

Therefore each of the pyramids AEGH, HKLD is similar to the whole pyramid ABCD.

Next, since BF is equal to FC, the parallelogram EBFG is double of the triangle GFC.

And since, if there be two prisms of equal height, and one have a parallelogram as base, and the other a triangle, and if the parallelogram be double of the triangle, the prisms are equal, {XI. 39} therefore the prism contained by the two triangles BKF, EHG, and the three parallelograms EBFG, EBKH, HKFG is equal to the prism contained by the two triangles GFC, HKL and the three parallelograms KFCL, LCGH, HKFG.

And it is manifest that each of the prisms, namely that in which the parallelogram EBFG is the base and the straight line HK is its opposite, and that in which the triangle GFC is the base and the triangle HKL its opposite, is greater than each of the pyramids of which the triangles AEG, HKL are the bases and the points H, D the vertices, inasmuch as, if we join the straight lines EF, EK, the prism in which the parallelogram EBFG is the base and the straight line HK its opposite is greater than the pyramid of which the triangle EBF is the base and the point K the vertex.

But the pyramid of which the triangle EBF is the base and the point K the vertex is equal to the pyramid of which the triangle AEG is the base and the point H the vertex; for they are contained by equal and similar planes.

Hence also the prism in which the parallelogram EBFG is the base and the straight line HK its opposite is greater than the pyramid of which the triangle AEG is the base and the point H the vertex.

But the prism in which the parallelogram EBFG is the base and the straight line HK its opposite is equal to the prism in which the triangle GFC is the base and the triangle HKL its opposite,

and the pyramid of which the triangle AEG is the base and the point H the vertex is equal to the pyramid of which the triangle HKL is the base and the point D the vertex.

Therefore the said two prisms are greater than the said two pyramids of which the triangles AEG, HKL are the bases and the points H, D the vertices.

Therefore the whole pyramid, of which the triangle ABC is the base and the point D the vertex, has been divided into two pyramids equal to one another and into two equal prisms, and the two prisms are greater than the half of the whole pyramid.
Q. E. D.

PROPOSITION 4. If there be two pyramids of the same height which have triangular bases, and each of them be divided into two pyramids equal to one another and similar to the whole, and into two equal prisms, then, as the base of the one pyramid is to the base of the other pyramid, so will all the prisms in the one pyramid be to all the prisms, being equal in multitude, in the other pyramid.

Let there be two pyramids of the same height which have the triangular bases ABC, DEF, and vertices the points G, H, and let each of them be divided into two pyramids equal to one another and similar to the whole and into two equal prisms; {XII. 3} I say that, as the base ABC is to the base DEF, so are all the prisms in the pyramid ABCG to all the prisms, being equal in multitude, in the pyramid DEFH. For, since BO is equal to OC, and AL to LC, therefore LO is parallel to AB, and the triangle ABC is similar to the triangle LOC.

For the same reason the triangle DEF is also similar to the triangle RVF.

And, since BC is double of CO, and EF of FV, therefore, as BC is to CO, so is EF to FV.

And on BC, CO are described the similar and similarly situated rectilineal figures ABC, LOC, and on EF, FV the similar and similarly situated figures DEF, RVF; therefore, as the triangle ABC is to the triangle LOC, so is the triangle DEF to the triangle RVF; {VI. 22} therefore, alternately, as the triangle ABC is to the triangle DEF, so is the triangle LOC to the triangle RVF. {V. 16} But, as the triangle LOC is to the triangle RVF, so is the prism in which the triangle LOC is the base and PMN its opposite to the prism in which the triangle RVF is the base and STU its opposite; {Lemma following} therefore also, as the triangle ABC is to the triangle DEF, so is the prism in which the triangle LOC is the base and PMN its opposite to the prism in which the triangle RVF is the base and STU its opposite.

But, as the said prisms are to one another, so is the prism in which the parallelogram KBOL is the base and the straight line PM its opposite to the prism in which the parallelogram QEVR is the base and the straight line ST its opposite. {XI. 39; cf. XII. 3} Therefore also the two prisms, that in which the parallelogram KBOL is the base and PM its opposite, and that in which the triangle LOC is the base and PMN its opposite, are to the prisms in which QEVR is the base and the straight line ST its opposite and in which the triangle RVF is the base and STU its opposite in the same ratio {V. 12} Therefore also, as the base ABC is to the base DEF, so are the said two prisms to the said two prisms.

And similarly, if the pyramids PMNG, STUH be divided into two prisms and two pyramids, as the base PMN is to the base STU, so will the two prisms in the pyramid PMNG be to the two prisms in the pyramid STUH.

But, as the base PMN is to the base STU, so is the base ABC to the base DEF; for the triangles PMN, STU are equal to the triangles LOC, RVF respectively.

Therefore also, as the base ABC is to the base DEF, so are the four prisms to the four prisms.

And similarly also, if we divide the remaining pyramids into two pyramids and into two prisms, then, as the base ABC is to the base DEF, so will all the prisms in the pyramid ABCG be to all the prisms, being equal in multitude, in the pyramid DEFH. Q. E. D.

LEMMA.

But that, as the triangle LOC is to the triangle RVF, so is the prism in which the triangle LOC is the base and PMN its opposite to the prism in which the triangle RVF is the base and STU its opposite, we must prove as follows.

For in the same figure let perpendiculars be conceived drawn from G, H to the planes ABC, DEF; these are of course equal because, by hypothesis, the pyramids are of equal height.

Now, since the two straight lines GC and the perpendicular from G are cut by the parallel planes ABC, PMN, they will be cut in the same ratios. {XI. 17} And GC is bisected by the plane PMN at N; therefore the perpendicular from G to the plane ABC will also be bisected by the plane PMN.

For the same reason the perpendicular from H to the plane DEF will also be bisected by the plane STU.

And the perpendiculars from G, H to the planes ABC, DEF are equal; therefore the perpendiculars from the triangles PMN, STU to the planes ABC, DEF are also equal.

Therefore the prisms in which the triangles LOC, RVF are bases, and PMN, STU their opposites, are of equal height.

Hence also the parallelepipedal solids described from the said prisms are of equal height and are to one another as their bases; {XI. 32} therefore their halves, namely the said prisms, are to one another as the base LOC is to the base RVF. Q. E. D.

PROPOSITION 5. Pyramids which are of the same height and have triangular bases are to one another as the bases.

Let there be pyramids of the same height, of which the triangles ABC, DEF are the bases and the points G, H the vertices; I say that, as the base ABC is to the base DEF, so is the pyramid ABCG to the pyramid DEFH.

For, if the pyramid ABCG is not to the pyramid DEFH as the base ABC is to the base DEF, then, as the base ABC is to the base DEF, so will the pyramid ABCG be either to some solid less than the pyramid DEFH or to a greater.

Let it, first, be in that ratio to a less solid W, and let the pyramid DEFH be divided into two pyramids equal to one another and similar to the whole and into two equal prisms; then the two prisms are greater than the half of the whole pyramid. {XII. 3} Again, let the pyramids arising from the division be similarly divided, and let this be done continually until there are left over from the pyramid DEFH some pyramids which are less than the excess by which the pyramid DEFH exceeds the solid W. {X. 1} Let such be left, and let them be, for the sake of argument, DQRS, STUH; therefore the remainders, the prisms in the pyramid DEFH, are greater than the solid W.

Let the pyramid ABCG also be divided similarly, and a similar number of times, with the pyramid DEFH; therefore, as the base ABC is to the base DEF, so are the prisms in the pyramid ABCG to the prisms in the pyramid DEFH. {XII. 4} But, as the base ABC is to the base DEF, so also is the pyramid ABCG to the solid W; therefore also, as the pyramid ABCG is to the solid W, so are the prisms in the pyramid ABCG to the prisms in the pyramid DEFH; {V. II} therefore, alternately, as the pyramid ABCG is to the prisms in it, so is the solid W to the prisms in the pyramid DEFH. {V. 16} But the pyramid ABCG is greater than the prisms in it; therefore the solid W is also greater than the prisms in the

pyramid DEFH.

But it is also less: which is impossible.

Therefore the prism ABCG is not to any solid less than the pyramid DEFH as the base ABC is to the base DEF.

Similarly it can be proved that neither is the pyramid DEFH to any solid less than the pyramid ABCG as the base DEF is to the base ABC.

I say next that neither is the pyramid ABCG to any solid greater than the pyramid DEFH as the base ABC is to the base DEF.

For, if possible, let it be in that ratio to a greater solid W; therefore, inversely, as the base DEF is to the base ABC, so is the solid W to the pyramid ABCG.

But, as the solid W is to the solid ABCG, so is the pyramid DEFH to some solid less than the pyramid ABCG, as was before proved; {XII. 2, Lemma} therefore also, as the base DEF is to the base ABC, so is the pyramid DEFH to some solid less than the pyramid ABCG: {V. II} which was proved absurd.

Therefore the pyramid ABCG is not to any solid greater than the pyramid DEFH as the base ABC is to the base DEF.

But it was proved that neither is it in that ratio to a less solid.

Therefore, as the base ABC is to the base DEF, so is the pyramid ABCG to the pyramid DEFH. Q. E. D.

PROPOSITION 6. Pyramids which are of the same height and have polygonal bases are to one another as the bases.

Let there be pyramids of the same height of which the polygons ABCDE, FGHL are the bases and the points M, N the vertices; I say that, as the base ABCDE is to the base FGHL, so is the pyramid ABCDEM to the pyramid FGHLN.

For let AC, AD, FH, FK be joined.

Since then $ABCM$, $ACDM$ are two pyramids which have triangular bases and equal height, they are to one another as the bases; {XII. 5} therefore, as the base ABC is to the base ACD , so is the pyramid $ABCM$ to the pyramid $ACDM$.

And, componendo, as the base $ABCD$ is to the base ACD , so is the pyramid $ABCDM$ to the pyramid $ACDM$. {V. 18} But also, as the base ACD is to the base ADE , so is the pyramid $ACDM$ to the pyramid $ADEM$. {XII. 5} Therefore, ex aequali, as the base $ABCD$ is to the base ADE , so is the pyramid $ABCDM$ to the pyramid $ADEM$. {V. 22} And again componendo, as the base $ABCDE$ is to the base ADE , so is the pyramid $ABCDEM$ to the pyramid $ADEM$. {V. 18} Similarly also it can be proved that, as the base $FGHKL$ is to the base FGH , so is the pyramid $FGHKLN$ to the pyramid $FGHN$.

And, since $ADEM$, $FGHN$ are two pyramids which have triangular bases and equal height, therefore, as the base ADE is to the base FGH , so is the pyramid $ADEM$ to the pyramid $FGHN$. {XII. 5} But, as the base ADE is to the base $ABCDE$, so was the pyramid $ADEM$ to the pyramid $ABCDEM$.

Therefore also, ex aequali, as the base $ABCDE$ is to the base FGH , so is the pyramid $ABCDEM$ to the pyramid $FGHN$. {V. 22} But further, as the base FGH is to the base $FGHKL$, so also was the pyramid $FGHN$ to the pyramid $FGHKLN$.

Therefore also, ex aequali, as the base $ABCDE$ is to the base $FGHKL$, so is the pyramid $ABCDEM$ to the pyramid $FGHKLN$. {V. 22} Q. E. D.

PROPOSITION 7. Any prism which has a triangular base is divided into three pyramids equal to one another which have triangular bases.

Let there be a prism in which the triangle ABC is the base and DEF its opposite; I say that the prism $ABCDEF$ is divided into

three pyramids equal to one another, which have triangular bases.

For let BD, EC, CD be joined.

Since ABED is a parallelogram, and BD is its diameter, therefore the triangle ABD is equal to the triangle EBD; {I. 34} therefore also the pyramid of which the triangle ABD is the base and the point C the vertex is equal to the pyramid of which the triangle DEB is the base and the point C the vertex. {XII. 5} But the pyramid of which the triangle DEB is the base and the point C the vertex is the same with the pyramid of which the triangle EBC is the base and the point D the vertex; for they are contained by the same planes.

Therefore the pyramid of which the triangle ABD is the base and the point C the vertex is also equal to the pyramid of which the triangle EBC is the base and the point D the vertex.

Again, since FCBE is a parallelogram, and CE is its diameter, the triangle CEF is equal to the triangle CBE. {I. 34} Therefore also the pyramid of which the triangle BCE is the base and the point D the vertex is equal to the pyramid of which the triangle ECF is the base and the point D the vertex. {XII. 5} But the pyramid of which the triangle BCE is the base and the point D the vertex was proved equal to the pyramid of which the triangle ABD is the base and the point C the vertex; therefore also the pyramid of which the triangle CEF is the base and the point D the vertex is equal to the pyramid of which the triangle ABD is the base and the point C the vertex; therefore the prism ABCDEF has been divided into three pyramids equal to one another which have triangular bases.

And, since the pyramid of which the triangle ABD is the base and the point C the vertex is the same with the pyramid of which the triangle CAB is the base and the point D the vertex, for they are contained by the same planes, while the pyramid of

which the triangle ABD is the base and the point C the vertex was proved to be a third of the prism in which the triangle ABC is the base and DEF its opposite, therefore also the pyramid of which the triangle ABC is the base and the point D the vertex is a third of the prism which has the same base, the triangle ABC, and DEF as its opposite.

PORISM.

From this it is manifest that any pyramid is a third part of the prism which has the same base with it and equal height. Q. E. D.

PROPOSITION 8. Similar pyramids which have triangular bases are in the triplicate ratio of their corresponding sides.

Let there be similar and similarly situated pyramids of which the triangles ABC, DEF, are the bases and the points G, H the vertices; I say that the pyramid ABCG has to the pyramid DEFH the ratio triplicate of that which BC has to EF.

For let the parallelepipedal solids BGML, EHQP be completed.

Now, since the pyramid ABCG is similar to the pyramid DEFH, therefore the angle ABC is equal to the angle DEF, the angle GBC to the angle HEF, and the angle ABG to the angle DEH; and, as AB is to DE, so is BC to EF, and BG to EH.

And since, as AB is to DE, so is BC to EF, and the sides are proportional about equal angles, therefore the parallelogram BM is similar to the parallelogram EQ.

For the same reason BN is also similar to ER, and BK to EO; therefore the three parallelograms MB, BK, BN are similar to the three EQ, EO, ER.

But the three parallelograms MB, BK, BN are equal and similar to their three opposites, and the three EQ, EO, ER are equal and similar to their three opposites. {XI. 24} Therefore the solids BGML, EHQP are contained by similar planes equal in multitude.

Therefore the solid BGML is similar to the solid EHQP.

But similar parallelepipedal solids are in the triplicate ratio of their corresponding sides. {XI. 33} Therefore the solid BGML has to the solid EHQP the ratio triplicate of that which the corresponding side BC has to the corresponding side EF.

But, as the solid BGML is to the solid EHQP, so is the pyramid ABCG to the pyramid DEFH, inasmuch as the pyramid is a sixth part of the solid, because the prism which is half of the parallelepipedal solid {XI. 28} is also triple of the pyramid. {XII. 7} Therefore the pyramid ABCG also has to the pyramid DEFH the ratio triplicate of that which BC has to EF. Q. E. D.

PORISM.

From this it is manifest that similar pyramids which have polygonal bases are also to one another in the triplicate ratio of their corresponding sides.

For, if they are divided into the pyramids contained in them which have triangular bases, by virtue of the fact that the similar polygons forming their bases are also divided into similar triangles equal in multitude and corresponding to the wholes {VI. 20}, then, as the one pyramid which has a triangular base in the one complete pyramid is to the one pyramid which has a triangular base in the other complete pyramid, so also will all the pyramids which have triangular bases contained in the one pyramid be to all the pyramids which have triangular bases contained in the other pyramid {V. 12}, that is, the pyramid itself which has a polygonal base to the pyramid which has a polygonal base.

But the pyramid which has a triangular base is to the pyramid which has a triangular base in the triplicate ratio of the corresponding sides; therefore also the pyramid which has a polygonal base has to the pyramid which has a similar base the

ratio triplicate of that which the side has to the side.

PROPOSITION 9. In equal pyramids which have triangular bases the bases are reciprocally proportional to the heights; and those pyramids in which the bases are reciprocally proportional to the heights are equal.

For let there be equal pyramids which have the triangular bases ABC, DEF and vertices the points G, H; I say that in the pyramids ABCG, DEFH the bases are reciprocally proportional to the heights, that is, as the base ABC is to the base DEF, so is the height of the pyramid DEFH to the height of the pyramid ABCG.

For let the parallelepipedal solids BGML, EHQP be completed.

Now, since the pyramid ABCG is equal to the pyramid DEFH, and the solid BGML is six times the pyramid ABCG, and the solid EHQP six times the pyramid DEFH, therefore the solid BGML is equal to the solid EHQP.

But in equal parallelepipedal solids the bases are reciprocally proportional to the heights; {XI. 34} therefore, as the base BM is to the base EQ, so is the height of the solid EHQP to the height of the solid BGML.

But, as the base BM is to EQ, so is the triangle ABC to the triangle DEF. {I. 34} Therefore also, as the triangle ABC is to the triangle DEF, so is the height of the solid EHQP to the height of the solid BGML. {V. 11} But the height of the solid EHQP is the same with the height of the pyramid DEFH, and the height of the solid BGML is the same with the height of the pyramid ABCG, therefore, as the base ABC is to the base DEF, so is the height of the pyramid DEFH to the height of the pyramid ABCG.

Therefore in the pyramids ABCG, DEFH the bases are reciprocally proportional to the heights.

Next, in the pyramids ABCG, DEFH let the bases be reciprocally proportional to the heights; that is, as the base ABC is to the

base DEF, so let the height of the pyramid DEFH be to the height of the pyramid ABCG; I say that the pyramid ABCG is equal to the pyramid DEFH.

For, with the same construction, since, as the base ABC is to the base DEF, so is the height of the pyramid DEFH to the height of the pyramid ABCG, while, as the base ABC is to the base DEF, so is the parallelogram BM to the parallelogram EQ, therefore also, as the parallelogram BM is to the parallelogram EQ, so is the height of the pyramid DEFH to the height of the pyramid ABCG. {V. 11} But the height of the pyramid DEFH is the same with the height of the parallelepiped EHQP, and the height of the pyramid ABCG is the same with the height of the parallelepiped BGML; therefore, as the base BM is to the base EQ, so is the height of the parallelepiped EHQP to the height of the parallelepiped BGML.

But those parallelepipedal solids in which the bases are reciprocally proportional to the heights are equal; {XI. 34} therefore the parallelepipedal solid BGML is equal to the parallelepipedal solid EHQP.

And the pyramid ABCG is a sixth part of BGML, and the pyramid DEFH a sixth part of the parallelepiped EHQP; therefore the pyramid ABCG is equal to the pyramid DEFH.

Therefore etc Q. E. D.

PROPOSITION 10. Any cone is a third part of the cylinder which has the same base with it and equal height.

For let a cone have the same base, namely the circle ABCD, with a cylinder and equal height; I say that the cone is a third part of the cylinder, that is, that the cylinder is triple of the cone.

For if the cylinder is not triple of the cone, the cylinder will be either greater than triple or less than triple of the cone.

First let it be greater than triple, and let the square ABCD be inscribed in the circle ABCD; {IV. 6} then the square ABCD is greater than the half of the circle ABCD.

From the square ABCD let there be set up a prism of equal height with the cylinder.

Then the prism so set up is greater than the half of the cylinder, inasmuch as, if we also circumscribe a square about the circle ABCD {IV. 7}, the square inscribed in the circle ABCD is half of that circumscribed about it, and the solids set up from them are parallelepipedal prisms of equal height, while parallelepipedal solids which are of the same height are to one another as their bases; {XI. 32} therefore also the prism set up on the square ABCD is half of the prism set up from the square circumscribed about the circle ABCD; {cf. XI. 28, or XII. 6 and 7, Por.} and the cylinder is less than the prism set up from the square circumscribed about the circle ABCD; therefore the prism set up from the square ABCD and of equal height with the cylinder is greater than the half of the cylinder.

Let the circumferences AB, BC, CD, DA be bisected at the points E, F, G, H, and let AE, EB, BF, FC, CG, GD, DH, HA be joined; then each of the triangles AEB, BFC, CGD, DHA is greater than the half of that segment of the circle ABCD which is about it, as we proved before. {XII. 2} On each of the triangles AEB, BFC, CGD, DHA let prisms be set up of equal height with the cylinder; then each of the prisms so set up is greater than the half part of that segment of the cylinder which is about it, inasmuch as, if we draw through the points E, F, G, H parallels to AB, BC, CD, DA, complete the parallelograms on AB, BC, CD, DA, and set up from them parallelepipedal solids of equal height with the cylinder, the prisms on the triangles AEB, BFC, CGD, DHA are halves of the several solids set up; and the segments of the cylinder are less than the parallelepipedal solids set up; hence also the prisms on the triangles AEB, BFC, CGD, DHA are greater than the half of

the segments of the cylinder about them.

Thus, bisecting the circumferences that are left, joining straight lines, setting up on each of the triangles prisms of equal height with the cylinder, and doing this continually, we shall leave some segments of the cylinder which will be less than the excess by which the cylinder exceeds the triple of the cone. {X. 1} Let such segments be left, and let them be AE, EB, BF, FC, CG, GD, DH, HA; therefore the remainder, the prism of which the polygon AEBFCGDH is the base and the height is the same as that of the cylinder, is greater than triple of the cone.

But the prism of which the polygon AEBFCGDH is the base and the height the same as that of the cylinder is triple of the pyramid of which the polygon AEBFCGDH is the base and the vertex is the same as that of the cone; {XII. 7, Por.} therefore also the pyramid of which the polygon AEBFCGDH is the base and the vertex is the same as that of the cone is greater than the cone which has the circle ABCD as base.

But it is also less, for it is enclosed by it; which is impossible.

Therefore the cylinder is not greater than triple of the cone.

I say next that neither is the cylinder less than triple of the cone, For, if possible, let the cylinder be less than triple of the cone, therefore, inversely, the cone is greater than a third part of the cylinder.

Let the square ABCD be inscribed in the circle ABCD; therefore the square ABCD is greater than the half of the circle ABCD.

Now let there be set up from the square ABCD a pyramid having the same vertex with the cone; therefore the pyramid so set up is greater than the half part of the cone, seeing that, as we proved before, if we circumscribe a square about the circle, the square ABCD will be half of the square circumscribed about the circle, and if we set up from the squares parallelepipedal solids

of equal height with the cone, which are also called prisms, the solid set up from the square ABCD will be half of that set up from the square circumscribed about the circle; for they are to one another as their bases. {XI. 32} Hence also the thirds of them are in that ratio; therefore also the pyramid of which the square ABCD is the base is half of the pyramid set up from the square circumscribed about the circle.

And the pyramid set up from the square about the circle is greater than the cone, for it encloses it.

Therefore the pyramid of which the square ABCD is the base and the vertex is the same with that of the cone is greater than the half of the cone.

Let the circumferences AB, BC, CD, DA be bisected at the points E, F, G, H, and let AE, EB, BF, FC, CG, GD, DH, HA be joined; therefore also each of the triangles AEB, BFC, CGD, DHA is greater than the half part of that segment of the circle ABCD which is about it.

Now, on each of the triangles AEB, BFC, CGD, DHA let pyramids be set up which have the same vertex as the cone; therefore also each of the pyramids so set up is, in the same manner, greater than the half part of that segment of the cone which is about it.

Thus, by bisecting the circumferences that are left, joining straight lines, setting up on each of the triangles a pyramid which has the same vertex as the cone, and doing this continually, we shall leave some segments of the cone which will be less than the excess by which the cone exceeds the third part of the cylinder. {X. 1} Let such be left, and let them be the segments on AE, EB, BF, FC, CG, GD, DH, HA; therefore the remainder, the pyramid of which the polygon AEBFCGDH is the base and the vertex the same with that of the cone, is greater than a third part of the cylinder.

But the pyramid of which the polygon AEBFCGDH is the base and the vertex the same with that of the cone is a third part of the prism of which the polygon AEBFCGDH is the base and the height is the same with that of the cylinder; therefore the prism of which the polygon AEBFCGDH is the base and the height is the same with that of the cylinder is greater than the cylinder of which the circle ABCD is the base.

But it is also less, for it is enclosed by it: which is impossible.

Therefore the cylinder is not less than triple of the cone.

But it was proved that neither is it greater than triple; therefore the cylinder is triple of the cone; hence the cone is a third part of the cylinder.

Therefore etc. Q. E. D.

PROPOSITION 11. Cones and cylinders which are of the same height are to one another as their bases.

Let there be cones and cylinders of the same height, let the circles ABCD, EFGH be their bases, KL, MN their axes and AC, EG the diameters of their bases; I say that, as the circle ABCD is to the circle EFGH, so is the cone AL to the cone EN.

For, if not, then, as the circle ABCD is to the circle EFGH, so will the cone AL be either to some solid less than the cone EN or to a greater.

First, let it be in that ratio to a less solid O, and let the solid X be equal to that by which the solid O is less than the cone EN; therefore the cone EN is equal to the solids O, X.

Let the square EFGH be inscribed in the circle EFGH; therefore the square is greater than the half of the circle.

Let there be set up from the square EFGH a pyramid of equal height with the cone; therefore the pyramid so set up is greater than the half of the cone, inasmuch as, if we circumscribe a

square about the circle, and set up from it a pyramid of equal height with the cone, the inscribed pyramid is half of the circumscribed pyramid, for they are to one another as their bases, {XII. 6} while the cone is less than the circumscribed pyramid.

Let the circumferences EF, FG, GH, HE be bisected at the points P, Q, R, S, and let HP, PE, EQ, QF, FR, RG, GS, SH be joined.

Therefore each of the triangles HPE, EQF, FRG, GSH is greater than the half of that segment of the circle which is about it.

On each of the triangles HPE, EQF, FRG, GSH let there be set up a pyramid of equal height with the cone; therefore, also, each of the pyramids so set up is greater than the half of that segment of the cone which is about it.

Thus, bisecting the circumferences which are left, joining straight lines, setting up on each of the triangles pyramids of equal height with the cone, and doing this continually, we shall leave some segments of the cone which will be less than the solid X. {X. 1} Let such be left, and let them be the segments on HP, PE, EQ, QF, FR, RG, GS, SH; therefore the remainder, the pyramid of which the polygon HPEQFRGS is the base and the height the same with that of the cone, is greater than the solid O.

Let there also be inscribed in the circle ABCD the polygon DTAUBVCW similar and similarly situated to the polygon HPEQFRGS, and on it let a pyramid be set up of equal height with the cone AL.

Since then, as the square on AC is to the square on EG, so is the polygon DTAUBVCW to the polygon HPEQFRGS, {XII. 1} while, as the square on AC is to the square on EG, so is the circle ABCD to the circle EFGH, {XII. 2} therefore also, as the circle ABCD is to the circle EFGH, so is the polygon DTAUBVCW to the polygon

HPEQFRGS.

But, as the circle ABCD is to the circle EFGH, so is the cone AL to the solid O, and, as the polygon DTAUBVCW is to the polygon HPEQFRGS, so is the pyramid of which the polygon DTAUBVCW is the base and the point L the vertex to the pyramid of which the polygon HPEQFRGS is the base and the point N the vertex. {XII. 6} Therefore also, as the cone AL is to the solid O, so is the pyramid of which the polygon DTAUBVCW is the base and the point L the vertex to the pyramid of which the polygon HPEQFRGS is the base and the point N the vertex; {V. 11} therefore, alternately, as the cone AL is to the pyramid in it, so is the solid O to the pyramid in the cone EN. {V. 16} But the cone AL is greater than the pyramid in it; therefore the solid O is also greater than the pyramid in the cone EN.

But it is also less: which is absurd.

Therefore the cone AL is not to any solid less than the cone EN as the circle ABCD is to the circle EFGH.

Similarly we can prove that neither is the cone EN to any solid less than the cone AL as the circle EFGH is to the circle ABCD.

I say next that neither is the cone AL to any solid greater than the cone EN as the circle ABCD is to the circle EFGH.

For, if possible, let it be in that ratio to a greater solid O; therefore, inversely, as the circle EFGH is to the circle ABCD, so is the solid O to the cone AL.

But, as the solid O is to the cone AL, so is the cone EN to some solid less than the cone AL; therefore also, as the circle EFGH is to the circle ABCD, so is the cone EN to some solid less than the cone AL: which was proved impossible.

Therefore the cone AL is not to any solid greater than the cone EN as the circle ABCD is to the circle EFGH.

But it was proved that neither is it in this ratio to a less solid; therefore, as the circle ABCD is to the circle EFGH, so is the cone AL to the cone EN.

But, as the cone is to the cone, so is the cylinder to the cylinder, for each is triple of each; {XII. 10} Therefore also, as the circle ABCD is to the circle EFGH, so are the cylinders on them which are of equal height.

Therefore etc. Q. E. D.

PROPOSITION 12. Similar cones and cylinders are to one another in the triplicate ratio of the diameters in their bases.

Let there be similar cones and cylinders, let the circles ABCD, EFGH be their bases, BD, FH the diameters of the bases, and KL, MN the axes of the cones and cylinders; I say that the cone of which the circle ABCD is the base and the point L the vertex has to the cone of which the circle EFGH is the base and the point N the vertex the ratio triplicate of that which BD has to FH.

For, if the cone ABCDL has not to the cone EFGHN the ratio triplicate of that which BD has to FH, the cone ABCDL will have that triplicate ratio either to some solid less than the cone EFGHN or to a greater.

First, let it have that triplicate ratio to a less solid O.

Let the square EFGH be inscribed in the circle EFGH; {IV. 6} therefore the square EFGH is greater than the half of the circle EFGH.

Now let there be set up on the square EFGH a pyramid having the same vertex with the cone; therefore the pyramid so set up is greater than the half part of the cone.

Let the circumferences EF, FG, GH, HE be bisected at the points P, Q, R, S, and let EP, PF, FQ, QG, GR, RH, HS, SE be joined.

Therefore each of the triangles EPF, FQG, GRH, HSE is also greater than the half part of that segment of the circle EFGH which is about it.

Now on each of the triangles EPF, FQG, GRH, HSE let a pyramid be set up having the same vertex with the cone; therefore each of the pyramids so set up is also greater than the half part of that segment of the cone which is about it.

Thus, bisecting the circumferences so left, joining straight lines, setting up on each of the triangles pyramids having the same vertex with the cone, and doing this continually, we shall leave some segments of the cone which will be less than the excess by which the cone EFGHN exceeds the solid O. {X. 1} Let such be left, and let them be the segments on EP, PF, FQ, QG, GR, RH, HS, SE; therefore the remainder, the pyramid of which the polygon EPFQGRHS is the base and the point N the vertex, is greater than the solid O.

Let there be also inscribed in the circle ABCD the polygon ATBUCVDW similar and similarly situated to the polygon EPFQGRHS, and let there be set up on the polygon ATBUCVDW a pyramid having the same vertex with the cone; of the triangles containing the pyramid of which the polygon ATBUCVDW is the base and the point L the vertex let LBT be one, and of the triangles containing the pyramid of which the polygon EPFQGRHS is the base and the point N the vertex let NFP be one; and let KT, MP be joined.

Now, since the cone ABCDL is similar to the cone EFGHN, therefore, as BD is to FH, so is the axis KL to the axis MN. {XI. Def. 24} But, as BD is to FH, so is BK to FM; therefore also, as BK is to FM, so is KL to MN.

And, alternately, as BK is to KL, so is FM to MN. {V. 16} And the sides are proportional about equal angles, namely the angles BKL, FMN; therefore the triangle BKL is similar to the triangle

FMN. {VI. 6} Again, since, as BK is to KT, so is FM to MP, and they are about equal angles, namely the angles BKT, FMP, inasmuch as, whatever part the angle BKT is of the four right angles at the centre K, the same part also is the angle FMP of the four right angles at the centre M; since then the sides are proportional about equal angles, therefore the triangle BKT is similar to the triangle FMP. {VI. 6} Again, since it was proved that, as BK is to KL, so is FM to MN, while BK is equal to KT, and FM to PM, therefore, as TK is to KL, so is PM to MN; and the sides are proportional about equal angles, namely the angles TKL, PMN, for they are right; therefore the triangle LKT is similar to the triangle NMP. {VI. 6} And since, owing to the similarity of the triangles LKB, NMF, as LB is to BK, so is NF to FM, and, owing to the similarity of the triangles BKT, FMP, as KB is to BT, so is MF to FP, therefore, ex aequali, as LB is to BT, so is NF to FP. {V. 22} Again since, owing to the similarity of the triangles LTK, NPM, as LT is to TK, so is NP to PM, and, owing to the similarity of the triangles TKB, PMF, as KT is to TB, so is MP to PF; therefore, ex aequali, as LT is to TB, so is NP to PF. {V. 22} But it was also proved that, as TB is to BL, so is PF to FN.

Therefore, ex aequali, as TL is to LB, so is PN to NF. {V. 22} Therefore in the triangles LTB, NPF the sides are proportional; therefore the triangles LTB, NPF are equiangular; {VI. 5} hence they are also similar. {VI. Def. 1} Therefore the pyramid of which the triangle BKT is the base and the point L the vertex is also similar to the pyramid of which the triangle FMP is the base and the point N the vertex, for they are contained by similar planes equal in multitude. {XI. Def. 9} But similar pyramids which have triangular bases are to one another in the triplicate ratio of their corresponding sides. {XII. 8} Therefore the pyramid BKTL has to the pyramid FMPN the ratio triplicate of that which BK has to FM.

Similarly, by joining straight lines from A, W, D, V, C, U to K, and from E, S, H, R, G, Q to M, and setting up on each of the triangles pyramids which have the same vertex with the cones, we can prove that each of the similarly arranged pyramids will also have to each similarly arranged pyramid the ratio triplicate of that which the corresponding side BK has to the corresponding side FM, that is, which BD has to FH.

And, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents; {V. 12} therefore also, as the pyramid BKTL is to the pyramid FMPN, so is the whole pyramid of which the polygon ATBUCVDW is the base and the point L the vertex to the whole pyramid of which the polygon EPFQGRHS is the base and the point N the vertex; hence also the pyramid of which ATBUCVDW is the base and the point L the vertex has to the pyramid of which the polygon EPFQGRHS is the base and the point N the vertex the ratio triplicate of that which BD has to FH.

But, by hypothesis, the cone of which the circle ABCD is the base and the point L the vertex has also to the solid O the ratio triplicate of that which BD has to FH; therefore, as the cone of which the circle ABCD is the base and the point L the vertex is to the solid O, so is the pyramid of which the polygon ATBUCVDW is the base and L the vertex to the pyramid of which the polygon EPFQGRHS is the base and the point N the vertex; therefore, alternately, as the cone of which the circle ABCD is the base and L the vertex is to the pyramid contained in it of which the polygon ATBUCVDW is the base and L the vertex, so is the solid O to the pyramid of which the polygon EPFQGRHS is the base and N the vertex. {V. 16} But the said cone is greater than the pyramid in it; for it encloses it.

Therefore the solid O is also greater than the pyramid of which the polygon EPFQGRHS is the base and N the vertex.

But it is also less: which is impossible.

Therefore the cone of which the circle ABCD is the base and L the vertex has not to any solid less than the cone of which the circle EFGH is the base and the point N the vertex the ratio triplicate of that which BD has to FH: Similarly we can prove that neither has the cone EFGHN to any solid less than the cone ABCDL the ratio triplicate of that which FH has to BD.

I say next that neither has the cone ABCDL to any solid greater than the cone EFGHN the ratio triplicate of that which BD has to FH.

For, if possible, let it have that ratio to a greater solid O.

Therefore, inversely, the solid O has to the cone ABCDL the ratio triplicate of that which FH has to BD.

But, as the solid O is to the cone ABCDL, so is the cone EFGHN to some solid less than the cone ABCDL.

Therefore the cone EFGHN also has to some solid less than the cone ABCDL the ratio triplicate of that which FH has to BD: which was proved impossible.

Therefore the cone ABCDL has not to any solid greater than the cone EFGHN the ratio triplicate of that which BD has to FH.

But it was proved that neither has it this ratio to a less solid than the cone EFGHN.

Therefore the cone ABCDL has to the cone EFGHN the ratio triplicate of that which BD has to FH.

But, as the cone is to the cone, so is the cylinder to the cylinder, for the cylinder which is on the same base as the cone and of equal height with it is triple of the cone; {XII. 10} therefore the cylinder also has to the cylinder the ratio triplicate of that which BD has to FH.

Therefore etc. Q. E. D.

PROPOSITION 13. If a cylinder be cut by a plane which is parallel to its opposite planes, then, as the cylinder is to the cylinder, so will the axis be to the axis.

For let the cylinder AD be cut by the plane GH which is parallel to the opposite planes AB, CD, and let the plane GH meet the axis at the point K; I say that, as the cylinder BG is to the cylinder GD, so is the axis EK to the axis KF.

For let the axis EF be produced in both directions to the points L, M, and let there be set out any number whatever of axes EN, NL equal to the axis EK, and any number whatever FO, OM equal to FK; and let the cylinder PW on the axis LM be conceived of which the circles PQ, VW are the bases.

Let planes be carried through the points N, O parallel to AB, CD and to the bases of the cylinder PW, and let them produce the circles RS, TU about the centres N, O.

Then, since the axes LN, NE, EK are equal to one another, therefore the cylinders QR, RB, BG are to one another as their bases. {XII. 11} But the bases are equal; therefore the cylinders QR, RB, BG are also equal to one another.

Since then the axes LN, NE, EK are equal to one another, and the cylinders QR, RB, BG are also equal to one another, and the multitude of the former is equal to the multitude of the latter, therefore, whatever multiple the axis KL is of the axis EK, the same multiple also will the cylinder QG be of the cylinder GB.

For the same reason, whatever multiple the axis MK is of the axis KF, the same multiple also is the cylinder WG of the cylinder GD.

And, if the axis KL is equal to the axis KM, the cylinder QG will also be equal to the cylinder GW, if the axis is greater than the axis, the cylinder will also be greater than the cylinder, and if

less, less.

Thus, there being four magnitudes, the axes EK, KF and the cylinders BG, GD, there have been taken equimultiples of the axis EK and of the cylinder BG, namely the axis LK and the cylinder QG, and equimultiples of the axis KF and of the cylinder GD, namely the axis KM and the cylinder GW; and it has been proved that, if the axis KL is in excess of the axis KM, the cylinder QG is also in excess of the cylinder GW, if equal, equal, and if less, less.

Therefore, as the axis EK is to the axis KF, so is the cylinder BG to the cylinder GD. {V. Def. 5} Q. E. D.

PROPOSITION 14. Cones and cylinders which are on equal bases are to one another as their heights.

For let EB, FD be cylinders on equal bases, the circles AB, CD; I say that, as the cylinder EB is to the cylinder FD, so is the axis GH to the axis KL.

For let the axis KL be produced to the point N, let LN be made equal to the axis GH, and let the cylinder CM be conceived about LN as axis.

Since then the cylinders EB, CM are of the same height, they are to one another as their bases. {XII. 11} But the bases are equal to one another; therefore the cylinders EB, CM are also equal.

And, since the cylinder FM has been cut by the plane CD which is parallel to its opposite planes, therefore, as the cylinder CM is to the cylinder FD, so is the axis LN to the axis KL. {XII. 13} But the cylinder CM is equal to the cylinder EB, and the axis LN to the axis GH; therefore, as the cylinder EB is to the cylinder FD, so is the axis GH to the axis KL.

But, as the cylinder EB is to the cylinder FD, so is the cone ABG to the cone CDK. {XII. 10} Therefore also, as the axis GH is to the axis KL, so is the cone ABG to the cone CDK and the cylinder EB

to the cylinder FD. Q. E. D.

PROPOSITION 15. In equal cones and cylinders the bases are reciprocally proportional to the heights; and those cones and cylinders in which the bases are reciprocally proportional to the heights are equal.

Let there be equal cones and cylinders of which the circles ABCD, EFGH are the bases; let AC, EG be the diameters of the bases, and KL, MN the axes, which are also the heights of the cones or cylinders; let the cylinders AO, EP be completed.

I say that in the cylinders AO, EP the bases are reciprocally proportional to the heights, that is, as the base ABCD is to the base EFGH, so is the height MN to the height KL.

For the height LK is either equal to the height MN or not equal.

First, let it be equal.

Now the cylinder AO is also equal to the cylinder EP.

But cones and cylinders which are of the same height are to one another as their bases; {XII. 11} therefore the base ABCD is also equal to the base EFGH.

Hence also, reciprocally, as the base ABCD is to the base EFGH, so is the height MN to the height KL.

Next, let the height LK not be equal to MN, but let MN be greater; from the height MN let QN be cut off equal to KL, through the point Q let the cylinder EP be cut by the plane TUS parallel to the planes of the circles EFGH, RP, and let the cylinder ES be conceived erected from the circle EFGH as base and with height NQ.

Now, since the cylinder AO is equal to the cylinder EP, therefore, as the cylinder AO is to the cylinder ES, so is the cylinder EP to the cylinder ES. {V. 7} But, as the cylinder AO is to the cylinder ES, so is the base ABCD to the base EFGH, for the

cylinders AO, ES are of the same height; {XII. 11} and, as the cylinder EP is to the cylinder ES, so is the height MN to the height QN, for the cylinder EP has been cut by a plane which is parallel to its opposite planes. {XII. 13} Therefore also, as the base ABCD is to the base EFGH, so is the height MN to the height QN. {V. 11} But the height QN is equal to the height KL; therefore, as the base ABCD is to the base EFGH, so is the height MN to the height KL.

Therefore in the cylinders AO, EP the bases are reciprocally proportional to the heights.

Next, in the cylinders AO, EP let the bases be reciprocally proportional to the heights, that is, as the base ABCD is to the base EFGH, so let the height MN be to the height KL; I say that the cylinder AO is equal to the cylinder EP.

For, with the same construction, since, as the base ABCD is to the base EFGH, so is the height MN to the height KL, while the height KL is equal to the height QN, therefore, as the base ABCD is to the base EFGH, so is the height MN to the height QN. But, as the base ABCD is to the base EFGH, so is the cylinder AO to the cylinder ES, for they are of the same height; {XII. 11} and, as the height MN is to QN, so is the cylinder EP to the cylinder ES; {XII. 13} therefore, as the cylinder AO is to the cylinder ES, so is the cylinder EP to the cylinder ES. {V. 11} Therefore the cylinder AO is equal to the cylinder EP. {V. 9} And the same is true for the cones also. Q. E. D.

PROPOSITION 16. Given two circles about the same centre, to inscribe in the greater circle an equilateral polygon with an even number of sides which does not touch the lesser circle.

Let ABCD, EFGH be the two given circles about the same centre K; thus it is required to inscribe in the greater circle ABCD an equilateral polygon with an even number of sides which does not touch the circle EFGH.

For let the straight line BKD be drawn through the centre K, and from the point G let GA be drawn at right angles to the straight line BD and carried through to C; therefore AC touches the circle EFGH. {III. 16, Por.} Then, bisecting the circumference BAD, bisecting the half of it, and doing this continually, we shall leave a circumference less than AD. {X. 1} Let such be left, and let it be LD; from L let LM be drawn perpendicular to BD and carried through to N, and let LD, DN be joined; therefore LD is equal to DN. {III. 3, I. 4} Now, since LN is parallel to AC, and AC touches the circle EFGH, therefore LN does not touch the circle EFGH; therefore LD, DN are far from touching the circle EFGH.

If then we fit into the circle ABCD straight lines equal to the straight line LD and placed continuously, there will be inscribed in the circle ABCD an equilateral polygon with an even number of sides which does not touch the lesser circle EFGH. Q. E. F.

PROPOSITION 17. Given two spheres about the same centre, to inscribe in the greater sphere a polyhedral solid which does not touch the lesser sphere at its surface.

Let two spheres be conceived about the same centre A; thus it is required to inscribe in the greater sphere a polyhedral solid which does not touch the lesser sphere at its surface.

Let the spheres be cut by any plane through the centre; then the sections will be circles, inasmuch as the sphere was produced by the diameter remaining fixed and the semicircle being carried round it; {XI. Def. 14} hence, in whatever position we conceive the semicircle to be, the plane carried through it will produce a circle on the circumference of the sphere.

And it is manifest that this circle is the greatest possible, inasmuch as the diameter of the sphere, which is of course the diameter both of the semicircle and of the circle, is greater than all the straight lines drawn across in the circle or the sphere.

Let then BCDE be the circle in the greater sphere, and FGH the circle in the lesser sphere; let two diameters in them, BD, CE, be drawn at right angles to one another; then, given the two circles BCDE, FGH about the same centre, let there be inscribed in the greater circle BCDE an equilateral polygon with an even number of sides which does not touch the lesser circle FGH, let BK, KL, LM ME be its sides in the quadrant BE. let KA be joined and carried through to N, let AO be set up from the point A at right angles to the plane of the circle BCDE, and let it meet the surface of the sphere at O, and through AO and each of the straight lines BD, KN let planes be carried; they will then make greatest circles on the surface of the sphere, for the reason stated.

Let them make such, and in them let BOD, KON be the semicircles on BD, KN.

Now, since OA is at right angles to the plane of the circle BCDE, therefore all the planes through OA are also at right angles to the plane of the circle BCDE; {XI. 18} hence the semicircles BOD, KON are also at right angles to the plane of the circle BCDE.

And, since the semicircles BED, BOD, KON are equal, for they are on the equal diameters BD, KN, therefore the quadrants BE, BO, KO are also equal to one another.

Therefore there are as many straight lines in the quadrants BO, KO equal to the straight lines BK, KL, LM, ME as there are sides of the polygon in the quadrant BE.

Let them be inscribed, and let them be BP, PQ, QR, RO and KS, ST, TU, UO, let SP, TQ, UR be joined, and from P, S let perpendiculars be drawn to the plane of the circle BCDE; {XI. 11} these will fall on BD, KN, the common sections of the planes, inasmuch as the planes of BOD, KON are also at right angles to the plane of the circle BCDE. {cf. XI. Def. 4} Let them so fall, and let them be PV, SW, and let WV be joined.

Now since, in the equal semicircles BOD, KON, equal straight lines BP, KS have been cut off, and the perpendiculars PV, SW have been drawn, therefore PV is equal to SW, and BV to KW. {III. 27, I. 26} But the whole BA is also equal to the whole KA; therefore the remainder VA is also equal to the remainder WA; therefore, as BV is to VA, so is KW to WA; therefore WV is parallel to KB. {VI. 2} And, since each of the straight lines PV, SW is at right angles to the plane of the circle BCDE, therefore PV is parallel to SW. {XI. 6} But it was also proved equal to it; therefore WV, SP are also equal and parallel. {I. 33} And, since WV is parallel to SP, while WV is parallel to KB, therefore SP is also parallel to KB. {XI. 9} And BP, KS join their extremities; therefore the quadrilateral KBPS is in one plane, inasmuch as, if two straight lines be parallel, and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallels. {XI. 7} For the same reason each of the quadrilaterals SPQT, TQRU is also in one plane.

But the triangle URO is also in one plane. {XI. 2} If then we conceive straight lines joined from the points P, S, Q, T, R, U to A, there will be constructed a certain polyhedral solid figure between the circumferences BO, KO, consisting of pyramids of which the quadrilaterals KBPS, SPQT, TQRU and the triangle URO are the bases and the point A the vertex.

And, if we make the same construction in the case of each of the sides KL, LM, ME as in the case of BK, and further in the case of the remaining three quadrants, there will be constructed a certain polyhedral figure inscribed in the sphere and contained by pyramids, of which the said quadrilaterals and the triangle URO, and the others corresponding to them, are the bases and the point A the vertex.

I say that the said polyhedron will not touch the lesser sphere at the surface on which the circle FGH is.

Let AX be drawn from the point A perpendicular to the plane of the quadrilateral KBPS, and let it meet the plane at the point X; {XI. 11} let XB, XK be joined.

Then, since AX is at right angles to the plane of the quadrilateral KBPS, therefore it is also at right angles to all the straight lines which meet it and are in the plane of the quadrilateral. {XI. Def. 3} Therefore AX is at right angles to each of the straight lines BX, XK.

And, since AB is equal to AK, the square on AB is also equal to the square on AK.

And the squares on AX, XB are equal to the square on AB, for the angle at X is right; {I. 47} and the squares on AX, XK are equal to the square on AK. {id.} Therefore the squares on AX, XB are equal to the squares on AX, XK.

Let the square on AX be subtracted from each; therefore the remainder, the square on BX, is equal to the remainder, the square on XK; therefore BX is equal to XK.

Similarly we can prove that the straight lines joined from X to P, S are equal to each of the straight lines BX, XK.

Therefore the circle described with centre X and distance one of the straight lines XB, XK will pass through P, S also, and KBPS will be a quadrilateral in a circle.

Now, since KB is greater than WV, while WV is equal to SP, therefore KB is greater than SP.

But KB is equal to each of the straight lines KS, BP; therefore each of the straight lines KS, BP is greater than SP.

And, since KBPS is a quadrilateral in a circle, and KB, BP, KS are equal, and PS less, and BX is the radius of the circle, therefore the square on KB is greater than double of the square on BX.

Let KZ be drawn from K perpendicular to BV.

Then, since BD is less than double of DZ, and, as BD is to DZ, so is the rectangle DB, BZ to the rectangle DZ, ZB, if a square be described upon BZ and the parallelogram on ZD be completed, then the rectangle DB, BZ is also less than double of the rectangle DZ, ZB.

And, if KD be joined, the rectangle DB, BZ is equal to the square on BK, and the rectangle DZ, ZB equal to the square on KZ; {III. 31, VI. 8 and Por.} therefore the square on KB is less than double of the square on KZ.

But the square on KB is greater than double of the square on BX; therefore the square on KZ is greater than the square on BX.

And, since BA is equal to KA, the square on BA is equal to the square on AK.

And the squares on BX, XA are equal to the square on BA, and the squares on KZ, ZA equal to the square on KA; {I. 47} therefore the squares on BX, XA are equal to the squares on KZ, ZA, and of these the square on KZ is greater than the square on BX; therefore the remainder, the square on ZA, is less than the square on XA.

Therefore AX is greater than AZ; therefore AX is much greater than AG.

And AX is the perpendicular on one base of the polyhedron, and AG on the surface of the lesser sphere; hence the polyhedron will not touch the lesser sphere on its surface.

Therefore, given two spheres about the same centre, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere at its surface. Q. E. F.

PORISM.

But if in another sphere also a polyhedral solid be inscribed similar to the solid in the sphere BCDE, the polyhedral solid in

the sphere BCDE has to the polyhedral solid in the other sphere the ratio triplicate of that which the diameter of the sphere BCDE has to the diameter of the other sphere.

For, the solids being divided into their pyramids similar in multitude and arrangement, the pyramids will be similar.

But similar pyramids are to one another in the triplicate ratio of their corresponding sides; {XII. 8, Por.} therefore the pyramid of which the quadrilateral KBPS is the base, and the point A the vertex, has to the similarly arranged pyramid in the other sphere the ratio triplicate of that which the corresponding side has to the corresponding side, that is, of that which the radius AB of the sphere about A as centre has to the radius of the other sphere.

Similarly also each pyramid of those in the sphere about A as centre has to each similarly arranged pyramid of those in the other sphere the ratio triplicate of that which AB has to the radius of the other sphere.

And, as one of the antecedents is to one of the consequents, so are all the antecedents to all the consequents; {V. 12} hence the whole polyhedral solid in the sphere about A as centre has to the whole polyhedral solid in the other sphere the ratio triplicate of that which AB has to the radius of the other sphere, that is, of that which the diameter BD has to the diameter of the other sphere. Q. E. D.

PROPOSITION 18. Spheres are to one another in the triplicate ratio of their respective diameters.

Let the spheres ABC, DEF be conceived, and let BC, EF be their diameters; I say that the sphere ABC has to the sphere DEF the ratio triplicate of that which BC has to EF.

For, if the sphere ABC has not to the sphere DEF the ratio triplicate of that which BC has to EF, then the sphere ABC will

have either to some less sphere than the sphere DEF, or to a greater, the ratio triplicate of that which BC has to EF.

First, let it have that ratio to a less sphere GHK, let DEF be conceived about the same centre with GHK, let there be inscribed in the greater sphere DEF a polyhedral solid which does not touch the lesser sphere GHK at its surface, {XII. 17} and let there also be inscribed in the sphere ABC a polyhedral solid similar to the polyhedral solid in the sphere DEF; therefore the polyhedral solid in ABC has to the polyhedral solid in DEF the ratio triplicate of that which BC has to EF. {XII. 17, Por.} But the sphere ABC also has to the sphere GHK the ratio triplicate of that which BC has to EF; therefore, as the sphere ABC is to the sphere GHK, so is the polyhedral solid in the sphere ABC to the polyhedral solid in the sphere DEF; and, alternately, as the sphere ABC is to the polyhedron in it, so is the sphere GHK to the polyhedral solid in the sphere DEF. {V. 16} But the sphere ABC is greater than the polyhedron in it; therefore the sphere GHK is also greater than the polyhedron in the sphere DEF.

But it is also less, for it is enclosed by it.

Therefore the sphere ABC has not to a less sphere than the sphere DEF the ratio triplicate of that which the diameter BC has to EF.

Similarly we can prove that neither has the sphere DEF to a less sphere than the sphere ABC the ratio triplicate of that which EF has to BC.

I say next that neither has the sphere ABC to any greater sphere than the sphere DEF the ratio triplicate of that which BC has to EF.

For, if possible, let it have that ratio to a greater, LMN; therefore, inversely, the sphere LMN has to the sphere ABC the ratio triplicate of that which the diameter EF has to the

diameter BC.

But, inasmuch as LMN is greater than DEF, therefore, as the sphere LMN is to the sphere ABC, so is the sphere DEF to some less sphere than the sphere ABC, as was before proved. {XII. 2, Lemma} Therefore the sphere DEF also has to some less sphere than the sphere ABC the ratio triplicate of that which EF has to BC: which was proved impossible.

Therefore the sphere ABC has not to any sphere greater than the sphere DEF the ratio triplicate of that which BC has to EF.

But it was proved that neither has it that ratio to a less sphere.

Therefore the sphere ABC has to the sphere DEF the ratio triplicate of that which BC has to EF. Q. E. D.

BOOK XIII. HISTORICAL NOTE.

I have already given, in the note to IV. 10, the evidence upon which the construction of the five regular solids is attributed to the Pythagoreans. Some of them, the cube, the tetrahedron (which is nothing but a pyramid), and the octahedron (which is only a double pyramid with a square base), cannot but have been known to the Egyptians. And it appears that dodecahedra have been found, of bronze or other material, which may belong to periods earlier than Pythagoras' time by some centuries (for references see Cantor's *Geschichte der Mathematik* I{3}, pp. 175-6).

It is true that the author of the scholium No. I to Eucl. XIII. says that the Book is about "the five so-called Platonic figures, which however do not belong to Plato, three of the aforesaid five figures being due to the Pythagoreans, namely the cube, the pyramid and the dodecahedron, while the octahedron and the icosahedron are due to Theaetetus." This statement (taken probably from Geminus) may perhaps rest on the fact that

Theaetetus was the first to write at any length about the two last-mentioned solids. We are told indeed by Suidas (S. V. Θεαιτήτος) that Theaetetus “first wrote on the ‘five solids’ as they are called.” This no doubt means that Theaetetus was the first to write a complete and systematic treatise on all the regular solids; it does not exclude the possibility that Hippasus or others had already written on the dodecahedron. The fact that Theaetetus wrote upon the regular solids agrees very well with the evidence which we possess of his contributions to the theory of irrationals, the connexion between which and the investigation of the regular solids is seen in Euclid’s Book XIII.

Theaetetus flourished about 380 B.C., and his work on the regular solids was soon followed by another, that of Aristaeus, an elder contemporary of Euclid, who also wrote an important book on Solid Loci, i.e. on conics treated as loci. This Aristaeus (known as “the elder”) wrote in the period about 320 B.C. We hear of his Comparison of the five regular solids from Hypsicles (2nd cent. B.C.), the writer of the short book commonly included in the editions of the Elements as Book XIV. Hypsicles gives in this Book some six propositions supplementing Eucl. XIII.; and he introduces the second of the propositions (Heiberg’s Euclid, Vol. v. p. 6) as follows: “The same circle circumscribes both the pentagon of the dodecahedron and the triangle of the icosahedron when both are inscribed in the same sphere. This is proved by Aristaeus in the book entitled Comparison of the five figures.”

Hypsicles proceeds (pp. 7 sqq.) to give a proof of this theorem. Allman pointed out (Greek Geometry from Thales to Euclid, 1889, pp. 201-2) that this proof depends on eight theorems, six of which appear in Euclid’s Book XIII. (in Propositions 8, 10, 12, 15, 16 with Por., 17); two other propositions not mentioned by Allman are also used, namely XIII. 4 and 9. This seems, as Allman says, to confirm the inference of Bretschneider (p. 171) that, as

Aristaeus' work was the newest and latest in which, before Euclid's time, this subject was treated, we have in Eucl. XIII. at least a partial recapitulation of the contents of the treatise of Aristaeus.

After Euclid, Apollonius wrote on the comparison of the dodecahedron and the icosahedron inscribed in one and the same sphere. This we also learn from Hypsicles, who says in the next words following those about Aristaeus above quoted: "But it is proved by Apollonius in the second edition of his Comparison of the dodecahedron with the icosahedron that, as the surface of the dodecahedron is to the surface of the icosahedron {inscribed in the same sphere}, so is the dodecahedron itself {i.e. its volume} to the icosahedron, because the perpendicular is the same from the centre of the sphere to the pentagon of the dodecahedron and to the triangle of the icosahedron."

PROPOSITIONS.

PROPOSITION 1. IF A STRAIGHT line be cut in extreme and mean ratio, the square on the greater segment added to the half of the whole is five times the square on the half.

For let the straight line AB be cut in extreme and mean ratio at the point C, and let AC be the greater segment; let the straight line AD be produced in a straight line with CA, and let AD be made half of AB; I say that the square on CD is five times the square on AD.

For let the squares AE, DF be described on AB, DC, and let the figure in DF be drawn; let FC be carried through to G.

Now, since AB has been cut in extreme and mean ratio at C, therefore the rectangle AB, BC is equal to the square on AC. {VI. Def. 3, VI. 17} And CE is the rectangle AB, BC, and FH the square on AC; therefore CE is equal to FH.

And, since BA is double of AD, while BA is equal to KA, and AD to AH, therefore KA is also double of AH.

But, as KA is to AH, so is CK to CH; {VI. 1} therefore CK is double of CH.

But LH, HC are also double of CH.

Therefore KC is equal to LH, HC.

But CE was also proved equal to HF; therefore the whole square AE is equal to the gnomon MNO.

And, since BA is double of AD, the square on BA is quadruple of the square on AD, that is, AE is quadruple of DH.

But AE is equal to the gnomon MNO; therefore the gnomon MNO is also quadruple of AP; therefore the whole DF is five times AP.

And DF is the square on DC, and AP the square on DA; therefore the square on CD is five times the square on DA.

Therefore etc. Q. E. D.

PROPOSITION 2. If the square on a straight line be five times the square on a segment of it, then, when the double of the said segment is cut in extreme and mean ratio, the greater segment is the remaining part of the original straight line.

For let the square on the straight line AB be five times the square on the segment AC of it, and let CD be double of AC; I say that, when CD is cut in extreme and mean ratio, the greater segment is CB.

Let the squares AF, CG be described on AB, CD respectively, let the figure in AF be drawn, and let BE be drawn through.

Now, since the square on BA is five times the square on AC, AF is five times AH.

Therefore the gnomon MNO is quadruple of AH.

And, since DC is double of CA, therefore the square on DC is quadruple of the square on CA, that is, CG is quadruple of AH.

But the gnomon MNO was also proved quadruple of AH; therefore the gnomon MNO is equal to CG.

And, since DC is double of CA, while DC is equal to CK, and AC to CH, therefore KB is also double of BH. {VI. 1} But LH, HB are also double of HB; therefore KB is equal to LH, HB.

But the whole gnomon MNO was also proved equal to the whole CG; therefore the remainder HF is equal to BG.

And BG is the rectangle CD, DB, for CD is equal to DG; and HF is the square on CB; therefore the rectangle CD, DB is equal to the square on CB.

Therefore, as DC is to CB, so is CB to BD.

But DC is greater than CB; therefore CB is also greater than BD.

Therefore, when the straight line CD is cut in extreme and mean ratio, CB is the greater segment.

Therefore etc. Q. E. D.

LEMMA.

That the double of AC is greater than BC is to be proved thus.

If not, let BC be, if possible, double of CA.

Therefore the square on BC is quadruple of the square on CA; therefore the squares on BC, CA are five times the square on CA.

But, by hypothesis, the square on BA is also five times the square on CA; therefore the square on BA is equal to the squares on BC, CA: which is impossible. {II. 4} Therefore CB is not double of AC.

Similarly we can prove that neither is a straight line less than CB double of CA; for the absurdity is much greater.

Therefore the double of AC is greater than CB. Q. E. D.

PROPOSITION 3. If a straight line be cut in extreme and mean ratio, the square on the lesser segment added to the half of the greater segment is five times the square on the half of the greater segment.

For let any straight line AB be cut in extreme and mean ratio at the point C, let AC be the greater segment, and let AC be bisected at D; I say that the square on BD is five times the square on DC.

For let the square AE be described on AB, and let the figure be drawn double.

Since AC is double of DC, therefore the square on AC is quadruple of the square on DC, that is, RS is quadruple of FG.

And, since the rectangle AB, BC is equal to the square on AC, and CE is the rectangle AB, BC, therefore CE is equal to RS.

But RS is quadruple of FG; therefore CE is also quadruple of FG.

Again, since AD is equal to DC, HK is also equal to KF.

Hence the square GF is also equal to the square HL.

Therefore GK is equal to KL, that is, MN to NE; hence MF is also equal to FE.

But MF is equal to CG; therefore CG is also equal to FE.

Let CN be added to each; therefore the gnomon OPQ is equal to CE.

But CE was proved quadruple of GF; therefore the gnomon OPQ is also quadruple of the square FG.

Therefore the gnomon OPQ and the square FG are five times FG.

But the gnomon OPQ and the square FG are the square DN.

And DN is the square on DB, and GF the square on DC.

Therefore the square on DB is five times the square on DC. Q. E. D.

PROPOSITION 4. If a straight line be cut in extreme and mean ratio, the square on the whole and the square on the lesser segment together are triple of the square on the greater segment.

Let AB be a straight line, let it be cut in extreme and mean ratio at C, and let AC be the greater segment; I say that the squares on AB, BC are triple of the square on CA.

For let the square ADEB be described on AB, and let the figure be drawn.

Since then AB has been cut in extreme and mean ratio at C, and AC is the greater segment, therefore the rectangle AB, BC is equal to the square on AC. {VI. Def. 3, VI. 17} And AK is the rectangle AB, BC, and HG the square on AC; therefore AK is equal to HG.

And, since AF is equal to FE, let CK be added to each; therefore the whole AK is equal to the whole CE; therefore AK, CE are double of AK.

But AK, CE are the gnomon LMN and the square CK; therefore the gnomon LMN and the square CK are double of AK.

But, further, AK was also proved equal to HG; therefore the gnomon LMN and the squares CK, HG are triple of the square HG.

And the gnomon LMN and the squares CK, HG are the whole square AE and CK, which are the squares on AB, BC, while HG is the square on AC.

Therefore the squares on AB, BC are triple of the square on AC. Q. E. D.

PROPOSITION 5. If a straight line be cut in extreme and mean ratio, and there be added to it a straight line equal to the greater segment, the whole straight line has been cut in extreme and mean ratio, and the original straight line is the greater segment.

For let the straight line AB be cut in extreme and mean ratio at the point C, let AC be the greater segment, and let AD be equal to AC.

I say that the straight line DB has been cut in extreme and mean ratio at A, and the original straight line AB is the greater segment.

For let the square AE be described on AB, and let the figure be drawn.

Since AB has been cut in extreme and mean ratio at C, therefore the rectangle AB, BC is equal to the square on AC. {VI. Def. 3, VI. 17} And CE is the rectangle AB, BC, and CH the square on AC; therefore CE is equal to HC.

But HE is equal to CE, and DH is equal to HC; therefore DH is also equal to HE.

Therefore the whole DK is equal to the whole AE.

And DK is the rectangle BD, DA, for AD is equal to DL; and AE is the square on AB; therefore the rectangle BD, DA is equal to the square on AB.

Therefore, as DB is to BA, so is BA to AD. {VI. 17} And DB is greater than BA; therefore BA is also greater than AD. {V. 14} Therefore DB has been cut in extreme and mean ratio at A, and AB is the greater segment. Q. E. D.

PROPOSITION 6. If a rational straight line be cut in extreme and mean ratio, each of the segments is the irrational straight line called apotome.

Let AB be a rational straight line, let it be cut in extreme and mean ratio at C, and let AC be the greater segment; I say that each of the straight lines AC, CB is the irrational straight line called apotome.

For let BA be produced, and let AD be made half of BA.

Since then the straight line AB has been cut in extreme and mean ratio, and to the greater segment AC is added AD which is half of AB, therefore the square on CD is five times the square on DA. {XIII. 1} Therefore the square on CD has to the square on DA the ratio which a number has to a number; therefore the square on CD is commensurable with the square on DA. {X. 6} But the square on DA is rational, for DA is rational, being half of AB which is rational; therefore the square on CD is also rational; {X. Def. 4} therefore CD is also rational.

And, since the square on CD has not to the square on DA the ratio which a square number has to a square number, therefore CD is incommensurable in length with DA; {X. 9} therefore CD, DA are rational straight lines commensurable in square only; therefore AC is an apotome. {X. 73} Again, since AB has been cut in extreme and mean ratio, and AC is the greater segment, therefore the rectangle AB, BC is equal to the square on AC. {VI. Def. 3, VI. 17} Therefore the square on the apotome AC, if applied to the rational straight line AB, produces BC as breadth.

But the square on an apotome, if applied to a rational straight line, produces as breadth a first apotome; {X. 97} therefore CB is a first apotome.

And CA was also proved to be an apotome.

Therefore etc. Q. E. D.

PROPOSITION 7. If three angles of an equilateral pentagon, taken either in order or not in order, be equal, the pentagon will be equiangular.

For in the equilateral pentagon ABCDE let, first, three angles taken in order, those at A, B, C, be equal to one another; I say that the pentagon ABCDE is equiangular.

For let AC, BE, FD be joined.

Now, since the two sides CB, BA are equal to the two sides BA, AE respectively, and the angle CBA is equal to the angle BAE, therefore the base AC is equal to the base BE, the triangle ABC is equal to the triangle ABE, and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend, {I. 4} that is, the angle BCA to the angle BEA, and the angle ABE to the angle CAB; hence the side AF is also equal to the side BF. {I. 6} But the whole AC was also proved equal to the whole BE; therefore the remainder FC is also equal to the remainder FE.

But CD is also equal to DE.

Therefore the two sides FC, CD are equal to the two sides FE, ED; and the base FD is common to them; therefore the angle FCD is equal to the angle FED. {I. 8} But the angle BCA was also proved equal to the angle AEB; therefore the whole angle BCD is also equal to the whole angle AED.

But, by hypothesis, the angle BCD is equal to the angles at A, B; therefore the angle AED is also equal to the angles at A, B.

Similarly we can prove that the angle CDE is also equal to the angles at A, B, C; therefore the pentagon ABCDE is equiangular.

Next, let the given equal angles not be angles taken in order, but let the angles at the points A, C, D be equal; I say that in this case too the pentagon ABCDE is equiangular.

For let BD be joined.

Then, since the two sides BA, AE are equal to the two sides BC, CD, and they contain equal angles, therefore the base BE is equal

to the base BD, the triangle ABE is equal to the triangle BCD, and the remaining angles will be equal to the remaining angles, namely those which the equal sides subtend; {I. 4} therefore the angle AEB is equal to the angle CDB.

But the angle BED is also equal to the angle BDE, since the side BE is also equal to the side BD. {I. 5} Therefore the whole angle AED is equal to the whole angle CDE.

But the angle CDE is, by hypothesis, equal to the angles at A, C; therefore the angle AED is also equal to the angles at A, C.

For the same reason the angle ABC is also equal to the angles at A, C, D.

Therefore the pentagon ABCDE is equiangular. Q. E. D.

PROPOSITION 8. If in an equilateral and equiangular pentagon straight lines subtend two angles taken in order, they cut one another in extreme and mean ratio, and their greater segments are equal to the side of the pentagon.

For in the equilateral and equiangular pentagon ABCDE let the straight lines AC, BE, cutting one another at the point H, subtend two angles taken in order, the angles at A, B; I say that each of them has been cut in extreme and mean ratio at the point H, and their greater segments are equal to the side of the pentagon.

For let the circle ABCDE be circumscribed about the pentagon ABCDE. {IV. 14} Then, since the two straight lines EA, AB are equal to the two AB, BC, and they contain equal angles, therefore the base BE is equal to the base AC, the triangle ABE is equal to the triangle ABC, and the remaining angles will be equal to the remaining angles respectively, namely those which the equal sides subtend. {I. 4} Therefore the angle BAC is equal to the angle ABE; therefore the angle AHE is double of the angle BAH. {I. 32} But the angle EAC is also double of the angle BAC,

inasmuch as the circumference EDC is also double of the circumference CB; {III. 28, VI. 33} therefore the angle HAE is equal to the angle AHE; hence the straight line HE is also equal to EA, that is, to AB. {I. 6} And, since the straight line BA is equal to AE, the angle ABE is also equal to the angle AEB. {I. 5} But the angle ABE was proved equal to the angle BAH; therefore the angle BEA is also equal to the angle BAH.

And the angle ABE is common to the two triangles ABE and ABH; therefore the remaining angle BAE is equal to the remaining angle AHB; {I. 32} therefore the triangle ABE is equiangular with the triangle ABH; therefore, proportionally, as EB is to BA, so is AB to BH. {VI. 4} But BA is equal to EH; therefore, as BE is to EH, so is EH to HB.

And BE is greater than EH; therefore EH is also greater than HB. {V. 14} Therefore BE has been cut in extreme and mean ratio at H, and the greater segment HE is equal to the side of the pentagon.

Similarly we can prove that AC has also been cut in extreme and mean ratio at H, and its greater segment CH is equal to the side of the pentagon. Q. E. D.

PROPOSITION 9. If the side of the hexagon and that of the decagon inscribed in the same circle be added together, the whole straight line has been cut in extreme and mean ratio, and its greater segment is the side of the hexagon.

Let ABC be a circle; of the figures inscribed in the circle ABC let BC be the side of a decagon, CD that of a hexagon, and let them be in a straight line; I say that the whole straight line BD has been cut in extreme and mean ratio, and CD is its greater segment.

For let the centre of the circle, the point E, be taken, let EB, EC, ED be joined, and let BE be carried through to A.

Since BC is the side of an equilateral decagon, therefore the circumference ACB is five times the circumference BC; therefore the circumference AC is quadruple of CB.

But, as the circumference AC is to CB, so is the angle AEC to the angle CEB; {VI. 33} therefore the angle AEC is quadruple of the angle CEB.

And, since the angle EBC is equal to the angle ECB, {I. 5} therefore the angle AEC is double of the angle ECB. {I. 32} And, since the straight line EC is equal to CD, for each of them is equal to the side of the hexagon inscribed in the circle ABC, {IV. 15, Por.} the angle CED is also equal to the angle CDE; {I. 5} therefore the angle ECB is double of the angle EDC. {I. 32} But the angle AEC was proved double of the angle ECB; therefore the angle AEC is quadruple of the angle EDC.

But the angle AEC was also proved quadruple of the angle BEC; therefore the angle EDC is equal to the angle BEC.

But the angle EBD is common to the two triangles BEC and BED; therefore the remaining angle BED is also equal to the remaining angle ECB; {I. 32} therefore the triangle EBD is equiangular with the triangle EBC.

Therefore, proportionally, as DB is to BE, so is EB to BC. {VI. 4} But EB is equal to CD.

Therefore, as BD is to DC, so is DC to CB.

And BD is greater than DC; therefore DC is also greater than CB.

Therefore the straight line BD has been cut in extreme and mean ratio, and DC is its greater segment. Q. E. D.

PROPOSITION 10. If an equilateral pentagon be inscribed in a circle, the square on the side of the pentagon is equal to the squares on the side of the hexagon and on that of the decagon inscribed in the same circle.

Let $ABCDE$ be a circle, and let the equilateral pentagon $ABCDE$ be inscribed in the circle $ABCDE$.

I say that the square on the side of the pentagon $ABCDE$ is equal to the squares on the side of the hexagon and on that of the decagon inscribed in the circle $ABCDE$.

For let the centre of the circle, the point F , be taken, let AF be joined and carried through to the point G , let FB be joined, let FH be drawn from F perpendicular to AB and be carried through to K , let AK , KB be joined, let FL be again drawn from F perpendicular to AK , and be carried through to M , and let KN be joined.

Since the circumference $ABCG$ is equal to the circumference $AEDG$, and in them ABC is equal to AED , therefore the remainder, the circumference CG , is equal to the remainder GD .

But CD belongs to a pentagon; therefore CG belongs to a decagon.

And, since FA is equal to FB , and FH is perpendicular, therefore the angle AFK is also equal to the angle KFB . {I. 5, I. 26} Hence the circumference AK is also equal to KB ; {III. 26} therefore the circumference AB is double of the circumference BK ; therefore the straight line AK is a side of a decagon.

For the same reason AK is also double of KM .

Now, since the circumference AB is double of the circumference BK , while the circumference CD is equal to the circumference AB , therefore the circumference CD is also double of the circumference BK .

But the circumference CD is also double of CG ; therefore the circumference CG is equal to the circumference BK .

But BK is double of KM , since KA is so also; therefore CG is also double of KM .

But, further, the circumference CB is also double of the circumference BK, for the circumference CB is equal to BA.

Therefore the whole circumference GB is also double of BM; hence the angle GFB is also double of the angle BFM. {VI. 33} But the angle GFB is also double of the angle FAB, for the angle FAB is equal to the angle ABF.

Therefore the angle BFN is also equal to the angle FAB.

But the angle ABF is common to the two triangles ABF and BFN; therefore the remaining angle AFB is equal to the remaining angle BNF; {I. 32} therefore the triangle ABF is equiangular with the triangle BFN.

Therefore, proportionally, as the straight line AB is to BF, so is FB to BN; {VI. 4} therefore the rectangle AB, BN is equal to the square on BF. {VI. 17} Again, since AL is equal to LK, while LN is common and at right angles, therefore the base KN is equal to the base AN; {I. 4} therefore the angle LKN is also equal to the angle LAN.

But the angle LAN is equal to the angle KBN; therefore the angle LKN is also equal to the angle KBN.

And the angle at A is common to the two triangles AKB and AKN.

Therefore the remaining angle AKB is equal to the remaining angle KNA; {I. 32} therefore the triangle KBA is equiangular with the triangle KNA.

Therefore, proportionally, as the straight line BA is to AK, so is KA to AN; {VI. 4} therefore the rectangle BA, AN is equal to the square on AK. {VI. 17} But the rectangle AB, BN was also proved equal to the square on BF; therefore the rectangle AB, BN together with the rectangle BA, AN, that is, the square on BA {II. 2}, is equal to the square on BF together with the square on AK.

And BA is a side of the pentagon, BF of the hexagon {IV. 15, Por.}, and AK of the decagon.

Therefore etc. Q. E. D.

PROPOSITION 11. If in a circle which has its diameter rational an equilateral pentagon be inscribed, the side of the pentagon is the irrational straight line called minor.

For in the circle ABCDE which has its diameter rational let the equilateral pentagon ABCDE be inscribed; I say that the side of the pentagon is the irrational straight line called minor.

For let the centre of the circle, the point F, be taken, let AF, FB be joined and carried through to the points, G, H, let AC be joined, and let FK be made a fourth part of AF.

Now AF is rational; therefore FK is also rational.

But BF is also rational; therefore the whole BK is rational.

And, since the circumference ACG is equal to the circumference ADG, and in them ABC is equal to AED, therefore the remainder CG is equal to the remainder GD.

And, if we join AD, we conclude that the angles at L are right, and CD is double of CL.

For the same reason the angles at M are also right, and AC is double of CM.

Since then the angle ALC is equal to the angle AMF, and the angle LAC is common to the two triangles ACL and AMF, therefore the remaining angle ACL is equal to the remaining angle MFA; {I. 32} therefore the triangle ACL is equiangular with the triangle AMF; therefore, proportionally, as LC is to CA, so is MF to FA.

And the doubles of the antecedents may be taken; therefore, as the double of LC is to CA, so is the double of MF to FA.

But, as the double of MF is to FA, so is MF to the half of FA; therefore also, as the double of LC is to CA, so is MF to the half of FA.

And the halves of the consequents may be taken; therefore, as the double of LC is to the half of CA, so is MF to the fourth of FA.

And DC is double of LC, CM is half of CA, and FK a fourth part of FA; therefore, as DC is to CM, so is MF to FK.

Componendo also, as the sum of DC, CM is to CM, so is MK to KF; {V. 18} therefore also, as the square on the sum of DC, CM is to the square on CM, so is the square on MK to the square on KF.

And since, when the straight line subtending two sides of the pentagon, as AC, is cut in extreme and mean ratio, the greater segment is equal to the side of the pentagon, that is, to DC, {XIII. 8} while the square on the greater segment added to the half of the whole is five times the square on the half of the whole, {XIII. 1} and CM is half of the whole AC, therefore the square on DC, CM taken as one straight line is five times the square on CM.

But it was proved that, as the square on DC, CM taken as one straight line is to the square on CM, so is the square on MK to the square on KF; therefore the square on MK is five times the square on KF.

But the square on KF is rational, for the diameter is rational; therefore the square on MK is also rational; therefore MK is rational And, since BF is quadruple of FK, therefore BK is five times KF; therefore the square on BK is twenty-five times the square on KF.

But the square on MK is five times the square on KF; therefore the square on BK is five times the square on KM; therefore the square on BK has not to the square on KM the ratio which a square number has to a square number; therefore BK is incommensurable in length with KM. {X. 9} And each of them is

rational.

Therefore BK, KM are rational straight lines commensurable in square only.

But, if from a rational straight line there be subtracted a rational straight line which is commensurable with the whole in square only, the remainder is irrational, namely an apotome; therefore MB is an apotome and MK the annex to it. {X. 73} I say next that MB is also a fourth apotome.

Let the square on N be equal to that by which the square on BK is greater than the square on KM; therefore the square on BK is greater than the square on KM by the square on N.

And, since KF is commensurable with FB, componendo also, KB is commensurable with FB. {X. 15} But BF is commensurable with BH; therefore BK is also commensurable with BH. {X. 12} And, since the square on BK is five times the square on KM, therefore the square on BK has to the square on KM the ratio which 5 has to 1.

Therefore, convertendo, the square on BK has to the square on N the ratio which 5 has to 4 {V. 19, Por.}, and this is not the ratio which a square number has to a square number; therefore BK is incommensurable with N; {X. 9} therefore the square on BK is greater than the square on KM by the square on a straight line incommensurable with BK.

Since then the square on the whole BK is greater than the square on the annex KM by the square on a straight line incommensurable with BK, and the whole BK is commensurable with the rational straight line, BH, set out, therefore MB is a fourth apotome. {X. Deff. III. 4} But the rectangle contained by a rational straight line and a fourth apotome is irrational, and its square root is irrational, and is called minor. {X. 94} But the square on AB is equal to the rectangle HB, BM, because, when

AH is joined, the triangle ABH is equiangular with the triangle ABM, and, as HB is to BA, so is AB to BM.

Therefore the side AB of the pentagon is the irrational straight line called minor. Q. E. D.

PROPOSITION 12. If an equilateral triangle be inscribed in a circle, the square on the side of the triangle is triple of the square on the radius of the circle.

Let ABC be a circle, and let the equilateral triangle ABC be inscribed in it; I say that the square on one side of the triangle ABC is triple of the square on the radius of the circle.

For let the centre D of the circle ABC be taken, let AD be joined and carried through to E, and let BE be joined.

Then, since the triangle ABC is equilateral, therefore the circumference BEC is a third part of the circumference of the circle ABC.

Therefore the circumference BE is a sixth part of the circumference of the circle; therefore the straight line BE belongs to a hexagon; therefore it is equal to the radius DE. {IV. 15, Por.} And, since AE is double of DE, the square on AE is quadruple of the square on ED, that is, of the square on BE.

But the square on AE is equal to the squares on AB, BE; {III. 31, I. 47} therefore the squares on AB, BE are quadruple of the square on BE.

Therefore, separando, the square on AB is triple of the square on BE.

But BE is equal to DE; therefore the square on AB is triple of the square on DE.

Therefore the square on the side of the triangle is triple of the square on the radius. Q. E. D.

PROPOSITION 13. To construct a pyramid, to comprehend it in a given sphere, and to prove that the square on the diameter of the sphere is one and a half times the square on the side of the pyramid.

Let the diameter AB of the given sphere be set out, and let it be cut at the point C so that AC is double of CB; let the semicircle ADB be described on AB, let CD be drawn from the point C at right angles to AB, and let DA be joined; let the circle EFG which has its radius equal to DC be set out, let the equilateral triangle EFG be inscribed in the circle EFG, {IV. 2} let the centre of the circle, the point H, be taken, {III. 1} let EH, HF, HG be joined; from the point H let HK be set up at right angles to the plane of the circle EFG, {XI. 12} let HK equal to the straight line AC be cut off from HK, and let KE, KF, KG be joined.

Now, since KH is at right angles to the plane of the circle EFG, therefore it will also make right angles with all the straight lines which meet it and are in the plane of the circle EFG. {XI. Def. 3} But each of the straight lines HE, HF, HG meets it: therefore HK is at right angles to each of the straight lines HE, HF, HG.

And, since AC is equal to HK, and CD to HE, and they contain right angles, therefore the base DA is equal to the base KE. {I. 4} For the same reason each of the straight lines KF, KG is also equal to DA; therefore the three straight lines KE, KF, KG are equal to one another.

And, since AC is double of CB, therefore AB is triple of BC.

But, as AB is to BC, so is the square on AD to the square on DC, as will be proved afterwards.

Therefore the square on AD is triple of the square on DC.

But the square on FE is also triple of the square on EH, {XIII. 12} and DC is equal to EH; therefore DA is also equal to EF.

But DA was proved equal to each of the straight lines KE, KF, KG; therefore each of the straight lines EF, FG, GE is also equal to each of the straight lines KE, KF, KG; therefore the four triangles EFG, KEF, KFG, KEG are equilateral.

Therefore a pyramid has been constructed out of four equilateral triangles, the triangle EFG being its base and the point K its vertex.

It is next required to comprehend it in the given sphere and to prove that the square on the diameter of the sphere is one and a half times the square on the side of the pyramid.

For let the straight line HL be produced in a straight line with KH, and let HL be made equal to CB.

Now, since, as AC is to CD, so is CD to CB, {VI. 8, Por.} while AC is equal to KH, CD to HE, and CB to HL, therefore, as KH is to HE, so is EH to HL; therefore the rectangle KH, HL is equal to the square on EH. {VI. 17} And each of the angles KHE, EHL is right; therefore the semicircle described on KL will pass through E also. {cf. VI. 8, III. 31.} If then, KL remaining fixed, the semicircle be carried round and restored to the same position from which it began to be moved, it will also pass through the points F, G, since, if FL, LG be joined, the angles at F, G similarly become right angles; and the pyramid will be comprehended in the given sphere.

For KL, the diameter of the sphere, is equal to the diameter AB of the given sphere, inasmuch as KH was made equal to AC, and HL to CB.

I say next that the square on the diameter of the sphere is one and a half times the square on the side of the pyramid For, since AC is double of CB, therefore AB is triple of BC; and, convertendo, BA is one and a half times AC.

But, as BA is to AC, so is the square on BA to the square on AD.

Therefore the square on BA is also one and a half times the square on AD.

And BA is the diameter of the given sphere, and AD is equal to the side of the pyramid.

Therefore the square on the diameter of the sphere is one and a half times the square on the side of the pyramid. Q. E. D.

LEMMA.

It is to be proved that, as AB is to BC, so is the square on AD to the square on DC.

For let the figure of the semicircle be set out, let DB be joined, let the square EC be described on AC, and let the parallelogram FB be completed.

Since then, because the triangle DAB is equiangular with the triangle DAC, as BA is to AD, so is DA to AC, {VI. 8, VI. 4} therefore the rectangle BA, AC is equal to the square on AD. {VI. 17} And since, as AB is to BC, so is EB to BF, {VI. 1} and EB is the rectangle BA, AC, for EA is equal to AC, and BF is the rectangle AC, CB, therefore, as AB is to BC, so is the rectangle BA, AC to the rectangle AC, CB.

And the rectangle BA, AC is equal to the square on AD, and the rectangle AC, CB to the square on DC, for the perpendicular DC is a mean proportional between the segments AC, CB of the base, because the angle ADB is right. {VI. 8, Por.} Therefore, as AB is to BC, so is the square on AD to the square on DC. Q. E. D.

PROPOSITION 14. To construct an octahedron and comprehend it in a sphere, as in the preceding case; and to prove that the square on the diameter of the sphere is double of the square on the side of the octahedron.

Let the diameter AB of the given sphere be set out, and let it be bisected at C; let the semicircle ADB be described on AB, let CD

be drawn from C at right angles to AB, let DB be joined; let the square EFGH, having each of its sides equal to DB, be set out, let HF, EG be joined, from the point K let the straight line KL be set up at right angles to the plane of the square EFGH {XI. 12}, and let it be carried through to the other side of the plane, as KM; from the straight lines KL, KM let KL, KM be respectively cut off equal to one of the straight lines EK, FK, GK, HK, and let LE, LF, LG, LH, ME, MF, MG, MH be joined.

Then, since KE is equal to KH, and the angle EKH is right, therefore the square on HE is double of the square on EK. {I. 47} Again, since LK is equal to KE, and the angle LKE is right, therefore the square on EL is double of the square on EK. {id.} But the square on HE was also proved double of the square on EK; therefore the square on LE is equal to the square on EH; therefore LE is equal to EH.

For the same reason LH is also equal to HE; therefore the triangle LEH is equilateral.

Similarly we can prove that each of the remaining triangles of which the sides of the square EFGH are the bases, and the points L, M the vertices, is equilateral; therefore an octahedron has been constructed which is contained by eight equilateral triangles.

It is next required to comprehend it in the given sphere, and to prove that the square on the diameter of the sphere is double of the square on the side of the octahedron.

For, since the three straight lines LK, KM, KE are equal to one another, therefore the semicircle described on LM will also pass through E.

And for the same reason, if, LM remaining fixed, the semicircle be carried round and restored to the same position from which it began to be moved, it will also pass through the points F, G, H,

and the octahedron will have been comprehended in a sphere.

I say next that it is also comprehended in the given sphere.

For, since LK is equal to KM, while KE is common, and they contain right angles, therefore the base LE is equal to the base EM. {I. 4} And, since the angle LEM is right, for it is in a semicircle, {III. 31} therefore the square on LM is double of the square on LE. {I. 47} Again, since AC is equal to CB, AB is double of BC.

But, as AB is to BC, so is the square on AB to the square on BD; therefore the square on AB is double of the square on BD.

But the square on LM was also proved double of the square on LE.

And the square on DB is equal to the square on LE, for EH was made equal to DB.

Therefore the square on AB is also equal to the square on LM; therefore AB is equal to LM.

And AB is the diameter of the given sphere; therefore LM is equal to the diameter of the given sphere.

Therefore the octahedron has been comprehended in the given sphere, and it has been demonstrated at the same time that the square on the diameter of the sphere is double of the square on the side of the octahedron. Q. E. D.

PROPOSITION 15. To construct a cube and comprehend it in a sphere, like the pyramid; and to prove that the square on the diameter of the sphere is triple of the square on the side of the cube.

Let the diameter AB of the given sphere be set out, and let it be cut at C so that AC is double of CB; let the semicircle ADB be described on AB, let CD be drawn from C at right angles to AB, and let DB be joined; let the square EFGH having its side equal to

DB be set out, from E, F, G, H let EK, FL, GM, HN be drawn at right angles to the plane of the square EFGH, from EK, FL, GM, HN let EK, FL, GM, HN respectively be cut off equal to one of the straight lines EF, FG, GH, HE, and let KL, LM, MN, NK be joined; therefore the cube FN has been constructed which is contained by six equal squares.

It is then required to comprehend it in the given sphere, and to prove that the square on the diameter of the sphere is triple of the square on the side of the cube.

For let KG, EG be joined.

Then, since the angle KEG is right, because KE is also at right angles to the plane EG and of course to the straight line EG also, {XI. Def. 3} therefore the semicircle described on KG will also pass through the point E.

Again, since GF is at right angles to each of the straight lines FL, FE, GF is also at right angles to the plane FK; hence also, if we join FK, GF will be at right angles to FK; and for this reason again the semicircle described on GK will also pass through F.

Similarly it will also pass through the remaining angular points of the cube.

If then, KG remaining fixed, the semicircle be carried round and restored to the same position from which it began to be moved, the cube will be comprehended in a sphere.

I say next that it is also comprehended in the given sphere.

For, since GF is equal to FE, and the angle at F is right, therefore the square on EG is double of the square on EF.

But EF is equal to EK; therefore the square on EG is double of the square on EK; hence the squares on GE, EK, that is the square on GK {I. 47}, is triple of the square on EK.

And, since AB is triple of BC, while, as AB is to BC, so is the square on AB to the square on BD, therefore the square on AB is triple of the square on BD.

But the square on GK was also proved triple of the square on KE.

And KE was made equal to DB; therefore KG is also equal to AB.

And AB is the diameter of the given sphere; therefore KG is also equal to the diameter of the given sphere.

Therefore the cube has been comprehended in the given sphere; and it has been demonstrated at the same time that the square on the diameter of the sphere is triple of the square on the side of the cube. Q. E. D.

PROPOSITION 16. To construct an icosahedron and comprehend it in a sphere, like the aforesaid figures; and to prove that the side of the icosahedron is the irrational straight line called minor.

Let the diameter AB of the given sphere be set out, and let it be cut at C so that AC is quadruple of CB, let the semicircle ADB be described on AB, let the straight line CD be drawn from C at right angles to AB, and let DB be joined; let the circle EFGHK be set out and let its radius be equal to DB, let the equilateral and equiangular pentagon EFGHK be inscribed in the circle EFGHK, let the circumferences EF, FG, GH, HK, KE be bisected at the points L, M, N, O, P, and let LM, MN, NO, OP, PL, EP be joined.

Therefore the pentagon LMNOP is also equilateral, and the straight line EP belongs to a decagon.

Now from the points E, F, G, H, K let the straight lines EQ, FR, GS, HT, KU be set up at right angles to the plane of the circle, and let them be equal to the radius of the circle EFGHK, let QR, RS, ST, TU, UQ, QL, LR, RM, MS, SN, NT, TO, OU, UP, PQ be joined.

Now, since each of the straight lines EQ, KU is at right angles to the same plane, therefore EQ is parallel to KU. {XI. 6} But it is also equal to it; and the straight lines joining those extremities of equal and parallel straight lines which are in the same direction are equal and parallel. {I. 33} Therefore QU is equal and parallel to EK.

But EK belongs to an equilateral pentagon; therefore QU also belongs to the equilateral pentagon inscribed in the circle EFGHK.

For the same reason each of the straight lines QR, RS, ST, TU also belongs to the equilateral pentagon inscribed in the circle EFGHK; therefore the pentagon QRSTU is equilateral.

And, since QE belongs to a hexagon, and EP to a decagon, and the angle QEP is right, therefore QP belongs to a pentagon; for the square on the side of the pentagon is equal to the square on the side of the hexagon and the square on the side of the decagon inscribed in the same circle. {XIII. 10} For the same reason PU is also a side of a pentagon.

But QU also belongs to a pentagon; therefore the triangle QPU is equilateral.

For the same reason each of the triangles QLR, RMS, SNT, TOU is also equilateral.

And, since each of the straight lines QL, QP was proved to belong to a pentagon, and LP also belongs to a pentagon, therefore the triangle QLP is equilateral.

For the same reason each of the triangles LRM, MSN, NTO, OUP is also equilateral.

Let the centre of the circle EFGHK the point V, be taken; from V let VZ be set up at right angles to the plane of the circle, let it be produced in the other direction, as VX, let there be cut off VW, the side of a hexagon, and each of the straight lines VX, WZ,

being sides of a decagon, and let QZ, QW, UZ, EV, LV, LX, XM be joined.

Now, since each of the straight lines VW, QE is at right angles to the plane of the circle, therefore VW is parallel to QE. {XI. 6} But they are also equal; therefore EV, QW are also equal and parallel. {I. 33} But EV belongs to a hexagon; therefore QW also belongs to a hexagon.

And, since QW belongs to a hexagon, and WZ to a decagon, and the angle QWZ is right, therefore QZ belongs to a pentagon. {XIII. 10} For the same reason UZ also belongs to a pentagon, inasmuch as, if we join VK, WU, they will be equal and opposite, and VK, being a radius, belongs to a hexagon; {IV. 15, Por.} therefore WU also belongs to a hexagon.

But WZ belongs to a decagon, and the angle UWZ is right; therefore UZ belongs to a pentagon. {XIII. 10} But QU also belongs to a pentagon; therefore the triangle QUZ is equilateral.

For the same reason each of the remaining triangles of which the straight lines QR, RS, ST, TU are the bases, and the point Z the vertex, is also equilateral.

Again, since VL belongs to a hexagon, and VX to a decagon, and the angle LVX is right, therefore LX belongs to a pentagon. {XIII. 10} For the same reason, if we join MV, which belongs to a hexagon, MX is also inferred to belong to a pentagon.

But LM also belongs to a pentagon; therefore the triangle LMX is equilateral.

Similarly it can be proved that each of the remaining triangles of which MN, NO, OP, PL are the bases, and the point X the vertex, is also equilateral.

Therefore an icosahedron has been constructed which is contained by twenty equilateral triangles.

It is next required to comprehend it in the given sphere, and to prove that the side of the icosahedron is the irrational straight line called minor.

For, since VW belongs to a hexagon, and WZ to a decagon, therefore VZ has been cut in extreme and mean ratio at W, and VW is its greater segment; {XIII. 9} therefore, as ZV is to VW, so is VW to WZ.

But VW is equal to VE, and WZ to VX; therefore, as ZV is to VE, so is EV to VX.

And the angles ZVE, EVX are right; therefore, if we join the straight line EZ, the angle XEZ will be right because of the similarity of the triangles XEZ, VEZ.

For the same reason, since, as ZV is to VW, so is VW to WZ, and ZV is equal to XW, and VW to WQ, therefore, as XW is to WQ, so is QW to WZ.

And for this reason again, if we join QX, the angle at Q will be right; {VI. 8} therefore the semicircle described on XZ will also pass through Q. {III. 31} And if, XZ remaining fixed, the semicircle be carried round and restored to the same position from which it began to be moved, it will also pass through Q and the remaining angular points of the icosahedron, and the icosahedron will have been comprehended in a sphere.

I say next that it is also comprehended in the given sphere.

For let VW be bisected at A'.

Then, since the straight line VZ has been cut in extreme and mean ratio at W, and ZW is its lesser segment, therefore the square on ZW added to the half of the greater segment, that is WA', is five times the square on the half of the greater segment; {XIII. 3} therefore the square on ZA' is five times the square on .

And ZX is double of ZA', and VW double of ; therefore the square on ZX is five times the square on WV.

And, since AC is quadruple of CB, therefore AB is five times BC.

But, as AB is to BC, so is the square on AB to the square on BD; {VI. 8, V. Def. 9} therefore the square on AB is five times the square on BD.

But the square on ZX was also proved to be five times the square on VW.

And DB is equal to VW, for each of them is equal to the radius of the circle EFGHK; therefore AB is also equal to XZ.

And AB is the diameter of the given sphere; therefore XZ is also equal to the diameter of the given sphere.

Therefore the icosahedron has been comprehended in the given sphere I say next that the side of the icosahedron is the irrational straight line called minor.

For, since the diameter of the sphere is rational, and the square on it is five times the square on the radius of the circle EFGHK, therefore the radius of the circle EFGHK is also rational; hence its diameter is also rational.

But, if an equilateral pentagon be inscribed in a circle which has its diameter rational, the side of the pentagon is the irrational straight line called minor. {XIII. 11} And the side of the pentagon EFGHK is the side of the icosahedron.

Therefore the side of the icosahedron is the irrational straight line called minor.

PORISM.

From this it is manifest that the square on the diameter of the sphere is five times the square on the radius of the circle from which the icosahedron has been described, and that the

diameter of the sphere is composed of the side of the hexagon and two of the sides of the decagon inscribed in the same circle.
Q. E. D.

PROPOSITION 17. To construct a dodecahedron and comprehend it in a sphere, like the aforesaid figures, and to prove that the side of the dodecahedron is the irrational straight line called apotome.

Let ABCD, CBEF, two planes of the aforesaid cube at right angles to one another, be set out, let the sides AB, BC, CD, DA, EF, EB, FC be bisected at G, H, K, L, M, N, O respectively, let GK, HL, MH, NO be joined, let the straight lines NP, PO, HQ be cut in extreme and mean ratio at the points R, S, T respectively, and let RP, PS, TQ be their greater segments; from the points R, S, T let RU, SV, TW be set up at right angles to the planes of the cube towards the outside of the cube, let them be made equal to RP, PS, TQ, and let UB, BW, WC, CV, VU be joined.

I say that the pentagon UBWCV is equilateral, and in one plane, and is further equiangular.

For let RB, SB, VB be joined.

Then, since the straight line NP has been cut in extreme and mean ratio at R, and RP is the greater segment, therefore the squares on PN, NR are triple of the square on RP. {XIII. 4} But PN is equal to NB, and PR to RU; therefore the squares on BN, NR are triple of the square on RU.

But the square on BR is equal to the squares on BN, NR; {I. 47} therefore the square on BR is triple of the square on RU; hence the squares on BR, RU are quadruple of the square on RU.

But the square on BU is equal to the squares on BR, RU; therefore the square on BU is quadruple of the square on RU; therefore BU is double of RU.

But VU is also double of UR, inasmuch as SR is also double of PR, that is, of RU; therefore BU is equal to UV.

Similarly it can be proved that each of the straight lines BW, WC, CV is also equal to each of the straight lines BU, UV.

Therefore the pentagon BUVCW is equilateral.

I say next that it is also in one plane.

For let PX be drawn from P parallel to each of the straight lines RU, SV and towards the outside of the cube, and let XH, HW be joined; I say that XHW is a straight line.

For, since HQ has been cut in extreme and mean ratio at T, and QT is its greater segment, therefore, as HQ is to QT, so is QT to TH.

But HQ is equal to HP, and QT to each of the straight lines TW, PX; therefore, as HP is to PX, so is WT to TH.

And HP is parallel to TW, for each of them is at right angles to the plane BD; {XI. 6} and TH is parallel to PX, for each of them is at right angles to the plane BF. {id.} But if two triangles, as XPH, HTW, which have two sides proportional to two sides be placed together at one angle so that their corresponding sides are also parallel, the remaining straight lines will be in a straight line; {VI. 32} therefore XH is in a straight line with HW.

But every straight line is in one plane; {XI. 1} therefore the pentagon UBWCV is in one plane.

I say next that it is also equiangular.

For, since the straight line NP has been cut in extreme and mean ratio at R, and PR is the greater segment, while PR is equal to PS, therefore NS has also been cut in extreme and mean ratio at P, and NP is the greater segment; {XIII. 5} therefore the squares on NS, SP are triple of the square on NP. {XIII. 4} But NP is equal to NB, and PS to SV; therefore the squares on NS, SV are

triple of the square on NB; hence the squares on VS, SN, NB are quadruple of the square on NB.

But the square on SB is equal to the squares on SN, NB; therefore the squares on BS, SV, that is, the square on BV — for the angle VSB is right — is quadruple of the square on NB; therefore VB is double of BN.

But BC is also double of BN; therefore BV is equal to BC.

And, since the two sides BU, UV are equal to the two sides BW, WC, and the base BV is equal to the base BC, therefore the angle BUV is equal to the angle BWC. {I. 8} Similarly we can prove that the angle UVC is also equal to the angle BWC; therefore the three angles BWC, BUV, UVC are equal to one another.

But if in an equilateral pentagon three angles are equal to one another, the pentagon will be equiangular, {XIII. 7} therefore the pentagon BUVCW is equiangular.

And it was also proved equilateral; therefore the pentagon BUVCW is equilateral and equiangular, and it is on one side BC of the cube.

Therefore, if we make the same construction in the case of each of the twelve sides of the cube, a solid figure will have been constructed which is contained by twelve equilateral and equiangular pentagons, and which is called a dodecahedron.

It is then required to comprehend it in the given sphere, and to prove that the side of the dodecahedron is the irrational straight line called apotome.

For let XP be produced, and let the produced straight line be XZ; therefore PZ meets the diameter of the cube, and they bisect one another, for this has been proved in the last theorem but one of the eleventh book. {XI. 38} Let them cut at Z; therefore Z is the centre of the sphere which comprehends the cube, and ZP is half of the side of the cube.

Let UZ be joined.

Now, since the straight line NS has been cut in extreme and mean ratio at P, and NP is its greater segment, therefore the squares on NS, SP are triple of the square on NP. {XIII. 4} But NS is equal to XZ, inasmuch as NP is also equal to PZ, and XP to PS.

But further PS is also equal to XU, since it is also equal to RP; therefore the squares on ZX, XU are triple of the square on NP.

But the square on UZ is equal to the squares on ZX, XU; therefore the square on UZ is triple of the square on NP.

But the square on the radius of the sphere which comprehends the cube is also triple of the square on the half of the side of the cube, for it has previously been shown how to construct a cube and comprehend it in a sphere, and to prove that the square on the diameter of the sphere is triple of the square on the side of the cube. {XIII. 15} But, if whole is so related to whole, so is half to half also; and NP is half of the side of the cube; therefore UZ is equal to the radius of the sphere which comprehends the cube.

And Z is the centre of the sphere which comprehends the cube; therefore the point U is on the surface of the sphere.

Similarly we can prove that each of the remaining angles of the dodecahedron is also on the surface of the sphere; therefore the dodecahedron has been comprehended in the given sphere.

I say next that the side of the dodecahedron is the irrational straight line called apotome.

For since, when NP has been cut in extreme and mean ratio, RP is the greater segment, and, when PO has been cut in extreme and mean ratio, PS is the greater segment, therefore, when the whole NO is cut in extreme and mean ratio, RS is the greater segment.

{Thus, since, as NP is to PR, so is PR to RN, the same is true of the doubles also, for parts have the same ratio as their equimultiples; {V. 15} therefore as NO is to RS, so is RS to the sum of NR, SO.

But NO is greater than RS; therefore RS is also greater than the sum of NR, SO; therefore NO has been cut in extreme and mean ratio, and RS is its greater segment.} But RS is equal to UV; therefore, when NO is cut in extreme and mean ratio, UV is the greater segment.

And, since the diameter of the sphere is rational, and the square on it is triple of the square on the side of the cube, therefore NO, being a side of the cube, is rational.

{But if a rational line be cut in extreme and mean ratio, each of the segments is an irrational apotome.} Therefore UV, being a side of the dodecahedron, is an irrational apotome. {XIII. 6} PORISM.

From this it is manifest that, when the side of the cube is cut in extreme and mean ratio, the greater segment is the side of the dodecahedron. Q. E. D.

PROPOSITION 18. To set out the sides of the five figures and to compare them with one another.

Let AB, the diameter of the given sphere, be set out, and let it be cut at C so that AC is equal to CB, and at D so that AD is double of DB; let the semicircle AEB be described on AB, from C, D let CE, DF be drawn at right angles to AB, and let AF, FB, EB be joined.

Then, since AD is double of DB, therefore AB is triple of BD.

Convertendo, therefore, BA is one and a half times AD.

But, as BA is to AD, so is the square on BA to the square on AF, {V. Def. 9, VI. 8} for the triangle AFB is equiangular with the triangle AFD; therefore the square on BA is one and a half times

the square on AF.

But the square on the diameter of the sphere is also one and a half times the square on the side of the pyramid. {XIII. 13} And AB is the diameter of the sphere; therefore AF is equal to the side of the pyramid.

Again, since AD is double of DB, therefore AB is triple of BD.

But, as AB is to BD, so is the square on AB to the square on BF; {VI. 8, V. Def. 9} therefore the square on AB is triple of the square on BF.

But the square on the diameter of the sphere is also triple of the square on the side of the cube. {XIII. 15} And AB is the diameter of the sphere; therefore BF is the side of the cube.

And, since AC is equal to CB, therefore AB is double of BC.

But, as AB is to BC, so is the square on AB to the square on BE; therefore the square on AB is double of the square on BE.

But the square on the diameter of the sphere is also double of the square on the side of the octahedron. {XIII. 14} And AB is the diameter of the given sphere; therefore BE is the side of the octahedron.

Next, let AG be drawn from the point A at right angles to the straight line AB, let AG be made equal to AB, let GC be joined, and from H let HK be drawn perpendicular to AB.

Then, since GA is double of AC, for GA is equal to AB, and, as GA is to AC, so is HK to KC, therefore HK is also double of KC.

Therefore the square on HK is quadruple of the square on KC; therefore the squares on HK, KC, that is, the square on HC, is five times the square on KC.

But HC is equal to CB; therefore the square on BC is five times the square on CK.

And, since AB is double of CB, and, in them, AD is double of DB, therefore the remainder BD is double of the remainder DC.

Therefore BC is triple of CD; therefore the square on BC is nine times the square on CD.

But the square on BC is five times the square on CK; therefore the square on CK is greater than the square on CD; therefore CK is greater than CD.

Let CL be made equal to CK, from L let LM be drawn at right angles to AB, and let MB be joined.

Now, since the square on BC is five times the square on CK, and AB is double of BC, and KL double of CK, therefore the square on AB is five times the square on KL.

But the square on the diameter of the sphere is also five times the square on the radius of the circle from which the icosahedron has been described. {XIII. 16, Por.} And AB is the diameter of the sphere; therefore KL is the radius of the circle from which the icosahedron has been described; therefore KL is a side of the hexagon in the said circle. {IV. 15, Por.} And, since the diameter of the sphere is made up of the side of the hexagon and two of the sides of the decagon inscribed in the same circle, {XIII. 16, Por.} and AB is the diameter of the sphere, while KL is a side of the hexagon, and AK is equal to LB, therefore each of the straight lines AK, LB is a side of the decagon inscribed in the circle from which the icosahedron has been described.

And, since LB belongs to a decagon, and ML to a hexagon, for ML is equal to KL, since it is also equal to HK, being the same distance from the centre, and each of the straight lines HK, KL is double of KC, therefore MB belongs to a pentagon. {XIII. 10} But the side of the pentagon is the side of the icosahedron; {XIII. 16} therefore MB belongs to the icosahedron.

Now, since FB is a side of the cube, let it be cut in extreme and mean ratio at N, and let NB be the greater segment; therefore NB is a side of the dodecahedron. {XIII. 17, Por.} And, since the square on the diameter of the sphere was proved to be one and a half times the square on the side AF of the pyramid, double of the square on the side BE of the octahedron and triple of the side FB of the cube, therefore, of parts of which the square on the diameter of the sphere contains six, the square on the side of the pyramid contains four, the square on the side of the octahedron three, and the square on the side of the cube two.

Therefore the square on the side of the pyramid is fourthirds of the square on the side of the octahedron, and double of the square on the side of the cube; and the square on the side of the octahedron is one and a half times the square on the side of the cube.

The said sides, therefore, of the three figures, I mean the pyramid, the octahedron and the cube, are to one another in rational ratios.

But the remaining two, I mean the side of the icosahedron and the side of the dodecahedron, are not in rational ratios either to one another or to the aforesaid sides; for they are irrational, the one being minor {XIII. 16} and the other an apotome {XIII. 17}.

That the side MB of the icosahedron is greater than the side NB of the dodecahedron we can prove thus.

For, since the triangle FDB is equiangular with the triangle FAB, {VI. 8} proportionally, as DB is to BF, so is BF to BA. {VI. 4} And, since the three straight lines are proportional, as the first is to the third, so is the square on the first to the square on the second; {V. Def. 9, VI. 20, Por.} therefore, as DB is to BA, so is the square on DB to the square on BF; therefore, inversely, as AB is to BD, so is the square on FB to the square on BD.

But AB is triple of BD; therefore the square on FB is triple of the square on BD.

But the square on AD is also quadruple of the square on DB, for AD is double of DB; therefore the square on AD is greater than the square on FB; therefore AD is greater than FB; therefore AL is by far greater than FB.

And, when AL is cut in extreme and mean ratio, KL is the greater segment, inasmuch as LK belongs to a hexagon, and KA to a decagon; {XIII. 9} and, when FB is cut in extreme and mean ratio, NB is the greater segment; therefore KL is greater than NB.

But KL is equal to LM; therefore LM is greater than NB.

Therefore MB, which is a side of the icosahedron, is by far greater than NB which is a side of the dodecahedron. Q. E. D.

I say next that no other figure, besides the said five figures, can be constructed which is contained by equilateral and equiangular figures equal to one another.

For a solid angle cannot be constructed with two triangles, or indeed planes.

With three triangles the angle of the pyramid is constructed, with four the angle of the octahedron, and with five the angle of the icosahedron; but a solid angle cannot be formed by six equilateral and equiangular triangles placed together at one point, for, the angle of the equilateral triangle being two-thirds of a right angle, the six will be equal to four right angles: which is impossible, for any solid angle is contained by angles less than four right angles. {XI. 21} For the same reason, neither can a solid angle be constructed by more than six plane angles.

By three squares the angle of the cube is contained, but by four it is impossible for a solid angle to be contained, for they will again be four right angles.

By three equilateral and equiangular pentagons the angle of the dodecahedron is contained; but by four such it is impossible for any solid angle to be contained, for, the angle of the equilateral pentagon being a right angle and a fifth, the four angles will be greater than four right angles: which is impossible.

Neither again will a solid angle be contained by other polygonal figures by reason of the same absurdity.

Therefore etc. Q. E. D.

LEMMA.

But that the angle of the equilateral and equiangular pentagon is a right angle and a fifth we must prove thus.

Let ABCDE be an equilateral and equiangular pentagon, let the circle ABCDE be circumscribed about it, let its centre F be taken, and let FA, FB, FC, FD, FE be joined.

Therefore they bisect the angles of the pentagon at A, B, C, D, E.

And, since the angles at F are equal to four right angles and are equal, therefore one of them, as the angle AFB, is one right angle less a fifth; therefore the remaining angles FAB, ABF consist of one right angle and a fifth.

But the angle FAB is equal to the angle FBC; therefore the whole angle ABC of the pentagon consists of one right angle and a fifth. Q. E. D.